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06

RISK MANAGEMENT

Question – 01

You hold worth 2 crore shares of X Ltd. whose market price standard deviation is 2% per day. Assuming 252 trading days a year, determine maximum loss level over the period of 1 trading day and 10 trading days with 99% confidence level.

Solution:

X Ltd. Shares = ₹ 200 lacs

S.D. = 2% per day

1 year = 252 days

At 99% confidence level.

(i) 1 day VAR

(ii) 10 days VAR

1 Day VAR

$$\begin{aligned}\text{VAR} &= x \sigma z \\ &= 200 \times 2\% \times 2.33 \\ &= 4 \times 2.33 \\ &= ₹ 9.32 \text{ lacs}\end{aligned}$$

10 Days VAR

$$\begin{aligned}10 \text{ Days S.D.} &= 4 \sqrt{10} = ₹ 12.65 \\ &= 12.65 \times 2.33 \\ &= ₹ 29.47 \text{ lacs}\end{aligned}$$

Question – 02

Consider a portfolio consisting of a ₹ 20,000,000 investment in share XYZ and a ₹ 20,000,000 investment in share ABC. The daily standard deviation of both shares is 1% and that the coefficient of correlation between them is 0.3. You are required to determine the 10-day 99% value at risk for the portfolio?

Solution:

XYZ = 200 lacs 0.5 1%

ABC = 200 lacs 0.5 1%

400 lacs

$$\begin{aligned}\sigma_p &= \sqrt{\sigma_A^2 W_A^2 + \sigma_B^2 W_B^2 + 2 \times W_A \times W_B \times \sigma_A \times \sigma_B \times r_{AB}} \\ &= \sqrt{1^2 0.50^2 + 1^2 \times 0.50^2 + 2 \times 0.5 \times 0.5 \times 1 \times 1 \times 0.30} \\ &= 0.8062\%\end{aligned}$$



1 Day VAR

$$1 \text{ Day S.D.} = ₹ 400 \times 0.8062\%$$

$$= ₹ 3.22 \text{ Lacs.}$$

10 Days VAR

$$10 \text{ Days S.D.} = ₹ 3.22 \sqrt{10}$$

$$= ₹ 10.18 \text{ Lacs.}$$

$$10 \text{ Days VAR} = 10.18 \times 2.33$$

$$= ₹ 23.72 \text{ Lacs.}$$

Question – 03

On Tuesday morning (before opening of the capital market) an investor, while going through his bank statement, has observed that an amount of ₹ 7 lakhs is lying in his bank account. This amount is available for use from Tuesday till Friday. The Bank requires a minimum balance of ₹ 1000 all the time. The investor desires to make a maximum possible investment where Value at Risk (VaR) should not exceed the balance lying in his bank account. The standard deviation of market price of the security is 1.5 per cent per day. The required confidence level is 99 per cent.

Given

Standard Normal probabilities										
Z	0.00	.01	.02	.03	0.04	.05	.06	.07	.08	.09
2.2	.9861	.9864	.9868	.09871	.9875	.9878	.9881	.9884	.9987	.9890
2.3	.9893	.9896	.9998	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9923	.9925	.9929	.9931	.9932	.9934	.9936

You are required to determine the maximum possible investment.

Solution:

Maximum possible investment

$$\text{Amount available} = ₹ 7,00,000$$

$$(-) \text{ Minimum Balance} = ₹ 1,000$$

$$\text{Possible loss for 4 days} = ₹ 6,99,000$$

$$4 \text{ Days VAR} = ₹ 6,99,000$$

$$4 \text{ Days VAR} = 4 \text{ days S.D.} \times 2.33$$

$$₹ 6,99,000 = 4 \text{ days S.D.} \times 2.33$$

$$4 \text{ days S.D. (₹)} = \frac{₹ 6,99,000}{2.33}$$

$$= ₹ 3,00,000$$

$$1 \text{ Day S.D. (₹)} = ₹ 3,00,000 \sqrt{\frac{1}{4}}$$

$$= ₹ 1,50,000$$

$$1 \text{ day S.D. (\%)} = 1.5\%$$

$$\text{Maximum possible Investment} = \frac{₹ 1,50,000}{1.5\%} = ₹ 1,00,00,000$$



Question – 04

Mr. Bull is a rational risk taker. He takes his position in a single stock for 4 days in a week. He does not take a position on Friday to avoid weekend effect and takes position only for four days in a week i.e. Monday to Thursday. He transfers the amount on Monday morning and withdraws the balance on Friday morning. He desires to make a maximum investment where Value At Risk (VAR) should not exceed the balance lying in his bank account. The position by his manager, as per standing instructions, is taken on the free balance lying in the bank account in the morning on each Monday.

On Monday morning (before opening of the capital market) he has transferred an amount of ₹ 11 Crore to his bank account. A fixed deposit also matured on this Monday. The maturity amount of ₹ 63,42,560 was also credited to his account by the bank in the morning of the Monday. However, Mr. Bull received the intimation of the same in the evening. The bank needs a minimum balance of ₹ 1,000 all the time. The value of Z score, at the required confidence level of 99 percent is 2.33.

The other information with respect to stocks X and Y, which are under consideration for this week, is as under:

X		Y	
Return	Probability	Return	Probability
6	0.10	4	0.10
7	0.25	6	0.20
8	0.30	8	0.40
9	0.25	10	0.20
10	0.10	12	0.10

You are required to recommend a single stock, where maximum investment can be made.

(Exam May – 2023) (8 Marks)

Solution:

Working Notes:

(1) Security X

P	x	P (x)	(x - $\bar{x}$)	(x - $\bar{x}$) ² × Prob.
0.10	6	0.60	-2	0.40
0.25	7	1.75	1	0.25
0.30	8	2.40	0	0
0.25	9	2.25	1	0.25
0.10	10	1.00	2	0.40
		8.00		1.30

Expected Return (R_x) = 8.00%

Variance (σ_x^2) = 1.30

Standard Deviation (σ_x) = $\sqrt{1.30}$
= 1.14

(2) Security Y

P	y	P (y)	(y - $\bar{y}$)	(y - $\bar{y}$) ² × Prob.
0.10	4	0.40	-4	1.60
0.20	6	1.20	-2	0.80
0.40	8	3.20	0	0
0.20	10	2.00	2	0.80
0.10	12	1.20	4	1.60
		8.00		4.80

Expected Return (R_y) = 8.00%

Variance (σ_y^2) = 4.80

$$\text{Standard Deviation } (\sigma_Y) = \sqrt{4.80}$$

$$= 2.19$$

	Amount
Amount Transferred	₹ 1,10,000,000
Maturity Proceeds of Fixed Deposit	₹ 63,42,560
Amount available in bank account	₹ 1,16,342,560
Minimum balance to be kept	₹ 1,000
Available amount which can be used for potential investment for 4 days	₹ 1,16,341,560

$$4 \text{ Days VAR} = ₹ 1,16,341,560$$

$$4 \text{ Days VAR} = 4 \text{ Days S.D.} \times 2.33$$

$$₹ 1,16,341,560 = 4 \text{ Days S.D.} \times 2.33$$

$$4 \text{ Days S.D. (₹)} = \frac{₹ 1,16,341,560}{2.33}$$

$$= ₹ 49,932,000$$

$$1 \text{ Day S.D. (₹)} = ₹ 49,932,000 \sqrt{\frac{1}{4}}$$

$$= ₹ 24,966,000$$

Stock X

$$1 \text{ day S.D. (\%)} = 1.14\%$$

$$\text{Maximum possible Investment} = \frac{₹ 24,966,000}{1.14\%}$$

$$= ₹ 2,190,000,000$$

Stock Y

$$1 \text{ day S.D. (\%)} = 2.19\%$$

$$\text{Maximum possible Investment} = \frac{₹ 24,966,000}{2.19\%}$$

$$= ₹ 1,140,000,000$$

Recommendation: Position should be taken in X.

MULTIPLE CHOICE QUESTIONS**Case Scenario – 01**

Mr. B is a rational risk taker. He takes his position in a single stock for 4 days in a week. He does not take a position on Friday to avoid weekend effect and takes position only for four days in a week i.e. Monday to Thursday. He transfers the amount on Monday morning and withdraws the balance on Friday morning. He desires to take a maximum exposure in the single stock (not the portfolio) where Value at Risk (VAR) should not exceed the balance lying in his bank account. The position by his manager, as per standing instructions, is taken on the free balance lying in the bank account in the morning on each Monday.

On Monday morning (before opening of the capital market) he has transferred an amount of ₹ 11 Crore to his bank account. A fixed deposit also matured on this Monday. The maturity amount of ₹ 63,42,560 was also credited to his account by the bank in the morning of the Monday. However, Mr. B received the intimation of the same in the evening. The bank needs a minimum balance of ₹ 1,000 all the time. The other information with respect to stocks X and Y, which are under consideration for this week, is as under:

X		Y	
Return	Probability	Return	Probability
6	0.10	4	0.10
7	0.25	6	0.20
8	0.30	8	0.40
9	0.25	10	0.20
10	0.10	12	0.10

From the information given above, choose the correct answer to the following questions:

- I. Available amount which can be used by Mr. B for potential exposure for 4 days on Monday morning shall be.....
 - (a) 11,00,00,000
 - (b) 11,63,41,560
 - (c) 11,00,01,000
 - (d) 11,63,42,560
- II. The Z-score at a 99% confidence level for Mr. B's Value at Risk (VAR) is.....
 - (a) 1.64
 - (b) 1.96
 - (c) 2.33
 - (d) 2.58
- III. The expected return for the stocks X is.....
 - (a) 7%
 - (b) 8%
 - (c) 9%
 - (d) 10%
- IV. The expected return for the stocks Y is.....
 - (a) 7%
 - (b) 8%
 - (c) 9%
 - (d) 10%
- V. In which stock should Mr. B invest in to maximize his returns while maintaining his Value at Risk (VAR) within acceptable limits?
 - (a) Stock X
 - (b) Stock Y
 - (c) Both stocks are equally good
 - (d) Neither stock is suitable

(RTP May – 2025)

Answer: Case Scenario – 01

- I. (b) 11,63,41,560
 II. (c) 2.33
 III. (b) 8%
 IV. (c) 9%
 V. (a) Stock X

Case Scenario – 02

Mr. Y is a rational risk taker. He takes his position in derivative market of a single stock through margin trading for 4 days in a week. He does not take a position on Friday to avoid weekend effect and takes position only for four days in a week i.e. Monday to Thursday. He transfers the amount on Monday morning and withdraws the balance on Friday morning. He desires to take a maximum exposure in the derivative market where Value at Risk (VAR) should not exceed the balance lying in his bank account. The position by his manager, as per standing instructions, is taken on the free balance lying in the bank account in the morning on each Monday.

On Monday morning (before opening of the capital market) he has transferred an amount of ₹ 11 Crore to his bank account. A fixed deposit also matured on this Monday. The maturity amount of ₹ 63,42,560 was also credited to his account by the bank in the morning of the Monday. However, Mr. Y received the intimation of the same in the evening. The bank needs a minimum balance of ₹ 1,000 all the time.

The other information with respect to stocks A and B, which are under consideration for this week, is as under:

A		B	
Return (%)	Probability	Return (%)	Probability
6	0.10	4	0.10
7	0.25	6	0.20
8	0.30	8	0.40
9	0.25	10	0.20
10	0.10	12	0.10

From the information given above, choose the correct answer to the following questions:

- I. The amount that will not be considered for taking position on Monday morning is.....
- (a) ₹ 63,41,560 (b) ₹ 63,42,560
 (c) ₹ 11,00,00,000 (d) ₹ 10,99,99,000
- II. Which stock has a wider dispersion of returns based on the given probability distribution?
- (a) Stock A
 (b) Stock B
 (c) Both have equal dispersion
 (d) Cannot be determined

- III.** In Value at Risk (VaR) analysis, which factor directly increases VaR for a given investment?
- (a) Higher expected return
 - (b) Lower confidence level
 - (c) Higher standard deviation
 - (d) Shorter holding period
- IV.** Since Mr. Y invests in derivative of a single stock (not a portfolio), the risk considered for VaR is.....
- (a) Diversifiable risk
 - (b) Unsystematic risk only
 - (c) Total risk
 - (d) Market risk only

(RTP May – 2026)

Answer Case Scenario – 02

- I. (b) ₹ 63,42,560**
- II. (b) Stock B**
- III. (c) Higher standard deviation**
- IV. (c) Total risk**



Student Notes



07

SECURITY ANALYSIS

Question – 01

Closing Values of NIFTY Index from 3rd to 12th day of the month of January 2022 were as follows:

Days	Date	Closing Values of NIFTY Index
1	03/01/2022	17626
2	04/01/2022	17805
3	05/01/2022	17925
4	06/01/2022	17746
5	07/01/2022	17813
6	10/01/2022	18003
7	11/01/2022	18056
8	12/01/2022	18212

The simple moving average of NIFTY Index for the month of December 2021 was 17174.

You are required to calculate

- The value of exponent for 15 days EMA.
- The exponential moving average (EMA) of NIFTY during the above period. (Calculations to be done up to 2 decimals only)
- Analyze the buy & sell signal on the basis of your calculations.

(Exam May – 2022) (8 Marks)

Solution:

- (i) Value of exponent of 15 days EMA

$$= \frac{2}{n+1}$$

$$= \frac{2}{15+1} = 0.125$$

- (ii) Calculation of EMA

Date	Closing Price	EMA= Previous EMA + (Price – Previous EMA) AF	EMA
03/1/22	17,626	17,174 + (17,626 – 17,174) 0.125	17,230.50
04/1/22	17,805	17,230.5 + (17,805 – 17,230.5) 0.125	17,302.31
05/1/22	17,925	17,302.31 + (17,925 – 17,302.31) 0.125	17,380.15
06/1/22	17,746	17,380.15 + (17,746 – 17,380.15) 0.125	17,425.88
07/1/22	17,813	17,425.88 + (17,813 – 17,425.88) 0.125	17,474.27
10/1/22	18,003	17,474.27 + (18,003 – 17,474.27) 0.125	17,540.36
11/1/22	18,056	17,540.36 + (18,056 – 17,540.36) 0.125	17,604.82
12/1/22	18,212	17,604.82 + (18,212 – 17,604.82) 0.125	17,680.71

- (iii) Since EMA is upward trend, hence market is bullish, it is buy signal.

Question – 02

Closing values of BSE Sensex from 6th to 17th day of the month of January of the year 20XX were as follows:

Days	Date	Day	Sensex
1	6	THU	29,522
2	7	FRI	29,925



3	8	SAT	No Trading
4	9	SUN	No Trading
5	10	MON	30,222
6	11	TUE	31,000
7	12	WED	31,400
8	13	THU	32,000
9	14	FRI	No Trading
10	15	SAT	No Trading
11	16	SUN	No Trading
12	17	MON	33,000

Compute Exponential moving Average (EMA) of Sensex during the above period. The 30 days simple moving average of Sensex can be assumed as 30,000. The value of exponent for 30 days EMA is 0.062.

Provide detailed analysis on the basis of your calculations.

(Exam May – 2018) (8 Marks)

Solution:

Date	Closing Price	EMA= Previous EMA + (Price – Previous EMA) AF	EMA
6	29522	$30,000 + (29,522 - 30,000) 0.062$	29,970.36
7	29925	$29,970.36 + (29,925 - 29,970.36) 0.062$	29,967.55
10	30222	$29,967.55 + (30,222 - 29,967.55) 0.062$	29,983.33
11	31000	$29,983.33 + (31,000 - 29,983.33) 0.062$	30,046.36
12	31400	$30,046.36 + (31,400 - 30,046.36) 0.062$	30,130.29
13	32000	$30,130.29 + (32,000 - 30,130.29) 0.062$	30,246.21
17	33000	$30,246.21 + (33,000 - 30,246.21) 0.062$	30,416.94

On the basis of EMA it is expected that market is bullish hence investor should take long position on Sensex.

MULTIPLE CHOICE QUESTIONS

Case Scenario – 01

An American institutional investor is exploring investment opportunities in different countries. Before proceeding, they believe a thorough analysis of options in the securities available to ensure a higher return while minimizing risk.

To achieve this objective, it formed a team consisting of following persons with respective assigned tasks:

Mr. A – He is entrusted with the task of analysing various Macro-economic factors e.g. historical performance of the economies in the past/ present and expectations in future, growth of different sectors of the economies in future with signs of stagnation/degradation at present. In addition to that he also analysed the trends in peoples' income and expenditure.

Ms. B – After receiving inputs/ recommendations from Mr. A she is entrusted with the task of assessment regarding all the conditions and factors relating to demand of the particular product, cost structure of the industry and other economic and Government constraints in the same country.

Mr. C – After receiving inputs/ recommendations from Ms. B he is entrusted with the task of careful examination of the company's quantitative and qualitative fundamentals. Which includes a comparison of price earning ratios of different companies. Further, In addition to examine the financial solvency, liquidity of the company he is also advised for the evaluation of future growth prospects of the company identified.

Based on the above case scenario, choose the correct answer to the following questions:

- I.** If Mr. A want to evaluate the impact of macroeconomic trends on their potential investment. Which of the following factors is least likely to influence their decision?
- (A) Growth rates of national income
(B) Inflation rates
(C) Market speculation trends
(D) Barometer indicators
- II.** The investor learns that inflation is expected to rise. Based on economic analysis, how might this affect their stock investment decision?
- (A) Stock prices are expected to decline due to reduced consumer demand
(B) Stock prices are expected to rise as stocks act as a hedge against inflation
(C) Stock prices will remain unaffected as inflation only affects bond markets
(D) Stock prices will become highly volatile, but long-term growth remains unchanged
- III.** Which of the techniques shall be primarily used by Ms. B to carry out the required analysis at his part?
- (A) Anticipatory Surveys
(B) Indicator Approach
(C) Input-Output Analysis

(D) Decision Tree Analysis

- IV.** Mr. A while analyzing industry growth, finds that certain indicators tend to peak before the economy's overall growth. These indicators are best classified as.....
- (A) Lagging indicators
(B) Leading indicators
(C) Coincidental indicators
(D) Random indicators
- V.** Specifically the team of Mr. A, Ms. B, and Mr. C are entrusted with the task of carrying out.....
- (A) Fundamental Analysis
(B) Technical Analysis
(C) Market Analysis
(D) Security Analysis

(MTP APRIL: 2025)

Answer: Case Scenario – 01

- I. (C) Market speculation trends**
II. (B) Stock prices are expected to rise as stocks act as a hedge against inflation
III. (C) Input-Output Analysis
IV. (B) Leading indicators
V. (A) Fundamental Analysis





Student Notes



Student Notes



08

FINANCIAL POLICY

Question – 01

The Balance Sheet of M/s. Sundry Ltd. as on 31-03-2023 is follows:

(₹ in lakhs)

Liabilities	₹	Assets	₹
Share Capital	3,000	Fixed Assets	6,000
Reserves	2,000	Inventory	5,000
Long Term Loan	4,000	Receivables	2,400
Short Term Loan	3,000	Cash	600
Payables & Provisions	2,000		
Total	14,000	Total	14,000

Sales for the year was ₹ 6,000 lakhs. The sales are expected to grow by 20% during the year. The profit margin and dividend pay-out ratio are expected to be 4% and 50% respectively.

The company further desires that during the current year Sales to Short Term Loan and Payables and Provision should be in the ratio of 4 : 3. Ratio of fixed assets to Long Term Loans should be 1.5. Debt Equity Ratio should not exceed 1.5.

You are required to determine:

- The amount of External Fund Requirement (EFR)
- The amount to be raised from Short Term, Long Term and Equity funds.

(SM May – 2025)

Solution: (i) External Funds Requirement (EFR) :

	(₹ in lakhs)
Expected sales (₹ 6,000 + 20% of ₹ 6,000)	7,200.00
Profit margin @ 4%	288.00
Dividend payout ratio @ 50%	144.00
Balance to be ploughed back (A)	144.00
Additional funds required (₹ 14,000 – ₹ 2,000) × 0.20 (B)	2,400.00
Balance to be met from external source (B – A)	2,256.00

* As current liabilities shall also be increased proportionately with increase in sales.

(ii) Amount to be raised from different sources with following conditions:

- Sales to short term loans and payables & provisions 4:3
- Ratio of fixed assets to long term loans 1.5
- Debt equity ratio should not exceed 1.5



(1) Amount to be raised from short term funds;

	(₹ in lakhs)
New amount of short-term loans and payable & provision $\left(\frac{3}{4} \times 7,200\right)$	5,400
Less: Existing Amount of short-term loans and payables & provision $(2,000 \times 1.20 + 3,000)$	5,400
Amount to be raised from short term funds.	Nil

(2) Amount to be raised from long term funds:

	(₹ in lakhs)
New fixed assets $(₹ 6,000 + 20\% \text{ of } ₹ 6,000)$	7,200
New long-term loans $(₹ 7,200 / 1.5)$	4,800
Less: Existing long-term loans	4,000
Amount to be raised from long term funds	800

(3) Amount to be raised from equity funds:

	(₹ in lakhs)
Amount to be raised from external sources	2,256.00
Less: Amount to be raised from short term funds	-----
Less: Amount to be raised from long term funds	800.00
Balance amount to be raised from equity funds	1,456.00

$$\text{New DER} = \frac{\text{Debt}}{\text{Shareholder's Fund}}$$

$$= \frac{4,800}{3,000 + 1,456 + 2,000 + 144}$$

$$= 0.727$$

Thus, required condition is satisfied.

Question – 02

MNC Limited company's financial statements for FY 2024-25 are provided:

Income Statement	(₹ in Crore)
Sales revenues	7,500
Costs and expenses	7,300
Income before taxes	200
Taxes (30%)	60
Net income	140

MNC Limited's Balance Sheet as at 31st March, 2025

Liabilities	(₹ in Crore)	Assets	(₹ in Crore)
Equity	2,000	Net Fixed Assets	4,000
Long term Debt	2,500	Current Assets	2,000
Current Liabilities	1,500		
	6,000		6,000

Additional Information:

- The company expects a 40% sales growth next financial year.
- The company will have a 25% dividend payout ratio next year.
- All costs, current assets and current liabilities are expected to increase with sales.
- Except retained earnings no new Equity is to be raised.

Required:

Compute External Funding Requirement through raising Long-term Debt:

(1) If the company is operating at 65% capacity usage for fixed assets.

(2) If the company is operating at 95% capacity usage for fixed assets.

(Exam Jan. – 2026)

Solution:

- (i)** External Funding Requirement in case of 65% capacity utilization
With only 65% Utilization, growth by 40% can be achieved without any corresponding increase in Fixed Assets (65*1.4 times = 91%)

External Funding Requirement Through Long Term Debt	(₹ in Crore)
Expected Sales Revenue (7,500 × 1.4)	10,500
Costs and Expenses (7,300 × 1.4)	10,220
Income before taxes	280
Taxes (30%)	84
Net Income	196
Dividend Payout@25% of ₹ 196	49
Retained Earnings/ Internal Sources of Funds	147
Additional Funds Required (2,000 – 1,500)*0.4	200
Balance to be met from Long Term Debt	53

- (ii)** External Funding Requirement in case of 95% capacity utilization

With only 95% Utilization, growth by 40% can be achieved by corresponding increase in Fixed Assets of ₹ 1320 Crore (95%*1.4 times = 1.33 times of existing Fixed Assets of ₹ 4000 Crore). Projected Fixed Assets increase to ₹ 5,320 Crore.

External Funding Requirement Through Long Term Debt	(₹ in Crore)
Expected Sales Revenue (7,500 × 1.4)	10,500
Costs and Expenses (7,300 × 1.4)	10,220
Income before taxes	280
Taxes (30%)	84
Net Income	196
Dividend Payout@25% of ₹ 196	49
Retained Earnings/ Internal Sources of Funds	147
Additional Funds Required (8,120 – 2,100 – 2,000 – 2500)	1,520
Balance to be met from Long Term Debt	1,373

Alternative Solution:

- (i)** Computation of EFR if company is operating at 65% capacity usage of Fixed Assets.

Full Capacity Sales

$$= \frac{\text{Actual Sales}}{\% \text{ of capacity at which fixed assets were operated}}$$

$$= \frac{₹ 7,500 \text{ Crores}}{0.65} = ₹ 11,538.4615 \text{ Crore}$$

$$\text{Actual Fixed Assets Ratio should be} = \frac{₹ 4,000 \text{ Crores}}{₹ 11,538.4615 \text{ Crores}} = 0.3467$$

$$\text{Revised Fixed Assets} = ₹ 10,500 \times 0.3467$$

= ₹ 3640.35 crore

Proforma Income Statement

	₹ in Crore
Sales Revenue	10,500
Less: Cost and Expenses	10,220
Income Before Tax	280
Tax @ 30%	84
Profit after Tax	196
Less: Dividend paid	49
Retained Earning	147

Proforma Balance Sheet

	₹ in Crores		₹ in Crores
Equity	2,000.00	Fixed Assets	3,640.35
Retained Earnings	147.00	Current Assets	2,800.00
Long Term Debt (Bal. Figure)	2,193.35		
Current Liabilities	2,100.00		
	6,440.35		6,440.35

EFR = ₹ 2,193.35 Crore - ₹ 2,500 Crore = - ₹ 306.65 Crore

Alternatively, it can also be computed using the formula as follows:

$$= \left[\frac{F_0}{S_0} \times \text{Revised Sales} - F_0 \right] + \left[\frac{CA}{S} - \frac{CL}{S} \right] \times \Delta S - \text{Net Profit (1 - d)}$$

$$= (3,640.35 - 4,000) + \left[\frac{2,000}{7,500} - \frac{1,500}{7,500} \right] \times 3,000 - (196 - 49)$$

$$= - 359.65 + 200 - 147 = - ₹ 306.65 \text{ Crore}$$

(ii) Computation of EFR if company is operating at 95% capacity usage of Fixed Assets.

Full Capacity Sales

$$= \frac{\text{Actual Sales}}{\% \text{ of capacity at which fixed assets were operated}}$$

$$= \frac{₹ 7,500 \text{ Crore}}{0.95} = ₹ 7,894.7368 \text{ Crore}$$

$$\text{Actual fixed asset ratio should be} = \frac{₹ 4,000 \text{ Crore}}{₹ 7,894.7368 \text{ Crore}} = 0.5067$$

$$\text{Revised fixed asset} = ₹ 10,500 \times 0.5067 = ₹ 5,320.35 \text{ Crore}$$

Proforma Income Statement

	₹ in Crores
Sales Revenue	10,500
Less: Cost and Expenses	10,220
Income Before Tax	280
Tax @ 30%	84
Profit after Tax	196
Less: Dividend Paid	49
Retained Earning	147

Proforma Balance Sheet

	₹ in Crores		₹ in Crores
Equity	2,000.00	Fixed Assets	5,320.35
Retained Earnings	147.00	Current Assets	2,800.00

Long Term Debt (Bal. Figure)	3,873.35		
Current Liabilities	2,100.00		
	8,120.35		8,120.35

$$\text{EFR} = ₹ 3,873.35 \text{ Crore} - ₹ 2,500 \text{ Crore} = - ₹ 1,373.35 \text{ Crore}$$

Alternatively, it can also be computed using the formula as follows:

$$= \left[\frac{F_0}{S_0} \times \text{Revised Sales} - F_0 \right] + \left[\frac{CA}{S} - \frac{CL}{S} \right] \times \Delta S - \text{Net Profit} (1 - d)$$

$$= (5,320.35 - 4,000) + \left[\frac{2,000}{7,500} - \frac{1,500}{7,500} \right] \times 3,000 - (196 - 49)$$

$$= 1,320.35 + 200 - 147 = - ₹ 1,373.35 \text{ Crore}$$

MULTIPLE CHOICE QUESTIONS

Case Scenario – 01

In a recent Board Meeting of N Ltd. following financials of N Ltd. for the year ending 31st March 2025 were presented:

Balance Sheet as on 31.03.2025

₹'000			
Liabilities	Amount	Assets	Amount
Equity Capital	4,80,000	Fixed Assets	2,42,000
10% Bonds	92,000	Cash	88,000
Sundry Creditors	66,000	Sundry Debtors	1,10,000
Bills Payable	88,000	Closing Stock	3,30,000
Other Current Liabilities	44,000		
Total Liabilities	7,70,000	Total Assets	7,70,000

Income Statement for the Year ending 31.03.2025

Particular	(₹'000)	(₹'000)
Sales		11,77,000
Less: Cost of Goods Sold		
Material	4,18,000	
Wages	2,64,000	
Factory Overheads	1,29,800	8,11,800
Gross Profit		3,65,200
Less: Selling & Distribution Cost	1,10,000	
Administrative Cost		2,32,800

Earnings Before Interest and Taxes (EBIT)	1,22,800	1,32,400
Less: Interest Charges		9,200
Earnings Before Tax		1,23,200
Less: Taxes @ 50%		61,600
Net Profit (PAT)		61,600

During the Board Meeting:

- (i) Director A said that the company can maintain a certain growth even though the net profit margin remains constant, and assets increases proportionately to sales and it distributes its 30% of its net profit. To maintain this growth rate, it will not require any external funds.
- (ii) Director B proposed that just by maintaining a target capital structure and without issuing additional equity and maintaining target dividend pay-out ratio as proposed by Director A, more growth rate can be achieved.
- (iii) Director C though agreed with views of Director A and Director B, but is of the view that in the coming year it is expected that sales is likely rise by 15%, hence if required we can go for issue of equity shares, bonds or debentures to achieve the same growth in sales.

From the information given above, choose the correct answer to the following questions:

- I.** The Director A is talking about_____
- (a) Internal Growth Rate

- (b) Sustainable Growth Rate
- (c) External Funding Requirements
- (d) External Growth Rate

II. The Director B is talking about

- (a) Internal Growth Rate
- (b) Sustainable Growth Rate
- (c) External Funding Requirements
- (d) External Growth Rate

III. The Director C is talking about_____

- (a) Internal Growth Rate
- (b) Sustainable Growth Rate
- (c) External Funding Requirements
- (d) External Growth Rate

- IV.** If we go by the proposal of Director C, then approximately.....funds shall be raised from in form of equity or debt, assuming that dividend as proposed by Director A is paid out and assets and current liabilities are increased in the same proportion as increase in sales.

- (a) ₹ 1,15,500 thousand
- (b) ₹ 85,800 thousand
- (c) ₹ 79,332 thousand
- (d) ₹ 36,212 thousand

(RTP SEP – 2025)

Answer Case Scenario – 01

- I. (a) Internal Growth Rate**
- II. (b) Sustainable Growth Rate**
- III. (c) External Funding Requirements**
- IV. (d) ₹ 36,212 thousand**



Student Notes



Student Notes



10

STARTUP FINANCE

Question – 01

The ABC Startup has the following expected profits under different scenarios along respective probabilities:

Year	Best Case		Base Case		Worst Case	
	Revenue (₹)	Expenses (₹)	Revenue (₹)	Expenses (₹)	Revenue (₹)	Expenses (₹)
1	1,00,00,000	80,00,000	100,00,000	90,00,000	100,00,000	95,00,000
2	1,20,00,000	92,40,000	110,00,000	95,70,000	102,00,000	98,94,000
3	1,44,00,000	1,08,00,000	121,00,000	1,02,85,000	1,04,04,000	1,01,95,920
	Probability 30%		60%		10%	

You are required to suggest the value of ABC Startup using First Chicago Method assuming that:

- Applicable discounting rate is 20%.
- Startup is located in Tax-free Zone.
- The multiple for Terminal is 10.
- No depreciable assets are held by the ABC Startup.

Note: 1. Present Value Factor (PVF)

Year	1	2	3
PVF @ 20%	0.8333	0.6944	0.5787

2. Round off the calculation to whole numbers.

(MTP October – 2024)

Solution:

Valuation of Startup under different scenarios:

(i) Best Case Scenario

	Year 1	Year 2	Year 3	
Revenue	₹ 1,00,00,000	₹ 120,00,000	₹ 144,00,000	
Expenses	₹ 80,00,000	₹ 92,40,000	₹ 108,00,000	
Cash Flow/ Earnings	₹ 20,00,000	₹ 27,60,000	₹ 36,00,000	
Terminal Value				₹ 3,60,00,000
PVF @ 20%	0.8333	0.6944	0.5787	0.5787
PV	₹ 16,66,600	₹ 19,16,544	₹ 20,83,320	₹ 2,08,33,200
Value of Startup				₹ 2,64,99,664

(ii) Base Case Scenario

	Year 1	Year 2	Year 3	
Revenue	₹ 1,00,00,000	₹ 110,00,000	₹ 121,00,000	
Expenses	₹ 90,00,000	₹ 95,70,000	₹ 1,02,85,000	
Cash Flow/ Earnings	₹ 10,00,000	₹ 14,30,000	₹ 18,15,000	
Terminal Value				₹ 1,81,50,000
PVF @ 20%	0.8333	0.6944	0.5787	0.5787
PV	₹ 8,33,300	₹ 9,92,992	₹ 10,50,341	₹ 1,05,03,405
Value of Startup				₹ 1,33,80,038



(iii) Worst Case Scenario

	Year 1	Year 2	Year 3	
Revenue	₹ 1,00,00,000	₹ 102,00,000	₹ 104,04,000	
Expenses	₹ 95,00,000	₹ 98,94,000	₹ 101,95,920	
Cash Flow/ Earnings	₹ 5,00,000	₹ 3,06,000	₹ 2,08,080	
Terminal Value				₹ 20,80,800
PVF @ 20%	0.8333	0.6944	0.5787	0.5787
PV	₹ 4,16,650	₹ 2,12,486	₹ 1,20,416	₹ 12,04,159
Value of Startup				₹ 19,53,711

Value of ABC Startup as per First Chicago Method

$$= 0.30 \times ₹ 2,64,99,664 + 0.60 \times ₹ 133,80,038 + 0.10 \times ₹ 19,53,711$$

$$= ₹ 79,49,899 + ₹ 80,28,023 + ₹ 1,95,371$$

$$= ₹ 1,61,73,293$$



Student Notes



Student Notes

11

SECURITY VALUATION

(I) BOND PRICING OR BOND VALUATION**Question – 01**

M/s Agfa Industries is planning to issue a debenture series on the following terms:

Face Value	₹ 100
Term of maturity	10 years

Yearly coupon rate

Years	
1 – 4	9%
5 – 8	10%
9 – 10	14%

The current market rate on similar debentures is 15 per cent per annum. The Company proposes to price the issue in such a manner that it can yield 16 per cent compounded rate of return to the investors. The Company also proposes to redeem the debentures at 5 per cent premium on maturity. Determine the issue price of the debentures.

Solution:**Issue Price of Bond**

Years	CF	PVF (16%)	PV
1 – 4	₹ 9	2.798	25.18
5 – 8	₹ 10	1.545	15.45
9 – 10	₹ 14	0.490	6.86

10	105	0.227	23.84
Issue Price			71.33

Question – 02

Nominal value of 10% bonds issued by a company is ₹ 100. The bonds are redeemable at ₹ 110 at the end of year 5. Determine the value of the bond if required yield is (i) 5%, (ii) 5.1%, (iii) 10% and (iv) 10.1%.

(SM TYK – 19)

Solution:**Case 1: Required yield rate = 5%**

Year	Cash flow ₹	DF (5%)	Present Value ₹
1 – 5	10	4.3295	43.295
5	110	0.7835	86.185
Value of bond			129.48

Case 2: Required yield rate = 5.1%

Year	Cash flow ₹	DF (5.1%)	Present Value ₹
1-5	10	4.3175	43.175
5	110	0.7798	85.778
Value of bond			128.95

Case 3: Required yield rate = 10%

Year	Cash flow ₹	DF (10%)	Present Value ₹
1-5	10	3.7908	37.908
5	110	0.6209	68.299
Value of bond			106.207



Case 4: Required yield rate = 10.1%

Year	Cash flow ₹	DF (10.1%)	Present Value ₹
1-5	10	3.7811	37.811
5	110	0.6181	67.991
Value of bond			105.802

Question – 03

John inherited the following securities on his uncle's death:

Types of Security	Nos.	Annual Coupon %	Maturity Years	Yield %
Bond A (₹ 1,000)	10	9	3	12
Bond B (₹ 1,000)	10	10	5	12
Preference shares C (₹ 100)	100	11	*	13*
Preference shares D (₹ 100)	100	12	*	13*

*Likelihood of being called at a premium over par.

Compute the current value of his uncle's portfolio.

Solution:

Value of Portfolio

$$\text{Bond A} = (900 \times 2.402) + (10,000 \times 0.712) = ₹ 9,282$$

$$\text{Bond B} = (1,000 \times 3.605) + (10,000 \times 0.567) = ₹ 9,275$$

$$\text{Preference shares C} = 10,000 \times 11\% = \frac{₹ 1,100}{13\%} = ₹ 8,462$$

$$\text{Preference shares D} = 10,000 \times 12\% = \frac{₹ 1,200}{13\%} = ₹ 9,231$$

$$\text{Value of portfolio} = ₹ 36,250$$

(II) BOND YIELD

QUESTION – 04

Based on the credit rating of bonds, Mr. Z has decided to apply the following discount rates for valuing bonds:

Credit Rating	Discount Rate
AAA	364 day T bill rate + 3% spread
AA	AAA + 2% spread
A	AAA + 3% spread

He is considering to invest in AA rated, ₹ 1,000 face value bond currently selling at ₹ 1,025.86. The bond has five years to maturity and the coupon rate on the bond is 15% p.a. payable annually. The next interest payment is due one year from today and the bond is redeemable at par. (Assume the 364 day T-bill rate to be 9%). You are required to calculate the intrinsic value of the bond for Mr. Z. Should he invest in the bond? Also calculate the current yield and the Yield to Maturity (YTM) of the bond.

Solution:

AA Rated:

$$\text{Discounting Rate} = 9 + 3 + 2 = 14\%$$

(1) Calculation of IV₀

$$= (150 \times \text{PVAF}, 14\%, 5) + (1,000 \times \text{PVF}, 14\%, 5)$$

$$= (150 \times 3.433) + (1,000 \times 0.5194)$$



$$= ₹ 1,034.35$$

Sine bond is under priced, hence it should be purchased.

(2) Current Yield

$$\begin{aligned} &= \frac{\text{Coupon}}{\text{CMP}} \times 100 \\ &= \frac{₹ 150}{₹ 1,025.86} \times 100 \\ &= 14.62\% \end{aligned}$$

(3) YTM

$$\begin{aligned} &= \frac{I + \left(\frac{F-P}{n} \right)}{\frac{F+P}{2}} \times 100 \\ &= \frac{150 + \left(\frac{1,000 - 1,025.86}{5} \right)}{\frac{1,000 + 1,025.86}{2}} \times 100 \\ &= 14.30\% \text{ p.a.} \end{aligned}$$

Note: Alternative IRR method can be used.

Question – 05

Calculate Market Price of:

- (i) 10% Government of India security currently quoted at ₹ 110, but yield is expected to go up by 1%.
- (ii) A bond with 7.5% coupon interest, Face Value ₹ 10,000 & term to maturity of 2 years, presently yielding 6% Interest payable half yearly.

Solution:

(i) Yield:

Assumed FV of bond is ₹ 100

Interest Amount = ₹ 10

Assumed Yield = Current Yield

$$\text{Existing Current Yield} = \frac{₹ 10}{₹ 110} \times 100 = 9.09\%$$

$$\text{Revised Yield} = 9.09\% + 1\% = 10.09\%$$

$$\text{Bond price} = \frac{₹ 10}{₹ 10.09\%} = ₹ 99.11$$

Note: Discounting rate के बढ़ने से Bond Price कम हो जाता है।

(ii) Calculation of Value of Bond:

Note: YTM Formula के Method से CMP Answer approx. में आता है & therefore don't do from this formula for calculating CMP.

$$\begin{aligned} \text{CMP} &= (₹ 375 \times \text{PVAF}, 4, 3\%) + (1,000 \times \text{PVF}, 4, 3\%) \\ &= (375 \times 3.717) + (1,000 \times 0.888) \\ &= ₹ 10,274 \end{aligned}$$

Question – 06

An investor is considering the purchase of the following bond:

Face value	₹ 100
Coupon rate	11%
Maturity	3 years

- (i) If he wants a yield of 13% what is the maximum price he should be ready to pay for?
- (ii) If the Bond is selling for ₹ 97.60, what would be his yield?

(SM TYK – 20)**Solution:****(i) Calculation of Maximum Price**

$$\begin{aligned}
 IV_0 &= ₹ 11 \times PVIFA (13\%,3) + ₹ 100 \times PVIF (13\%,3) \\
 &= (₹ 11 \times 2.361) + (₹ 100 \times 0.693) \\
 &= ₹ 25.97 + ₹ 69.30 \\
 &= ₹ 95.27
 \end{aligned}$$

(ii) Calculation of Yield

$$\begin{aligned}
 \text{At 12\% the value} &= ₹ 11 \times PVIFA (12\%,3) + 100 \times PVIF (12\%,3) \\
 &= (₹ 11 \times 2.402) + (₹ 100 \times 0.712) \\
 &= ₹ 26.42 + ₹ 71.20
 \end{aligned}$$

$$= ₹ 97.62$$

It the bond is selling at ₹ 97.60 which is more than the fair value, the YTM of the bond would be less than 13%. This value is almost equal to the amount price of ₹ 97.60.

Therefore, the YTM of the bond would be 12%.

Alternatively

$$\begin{aligned}
 YTM &= \frac{₹ 11 + \frac{(₹ 100 - ₹ 97.60)}{3}}{\frac{(₹ 100 + ₹ 97.60)}{2}} \\
 &= 0.1194 \text{ or } 11.94\% \text{ say } 12\%
 \end{aligned}$$

Note: अगर Question में Factors दिया है तो YTM निकालने के लिए IRR Method ही Use होगा, अगर Factor नहीं दिया है तो किसी भी Method से YTM निकाला जा सकता है।

Question – 07

There is a 9% 5-year bond issue in the market. The issue price is ₹ 90 and the redemption price ₹ 105. For an investor with marginal income tax rate of 30% and capital gains tax rate of 10% (assuming no indexation), what is the post-tax yield to maturity?

Solution:

$$\begin{aligned}
 YTM &= \frac{I(1-t) + \left(\frac{RV^* - P}{n}\right)}{\frac{RV^* + P}{2}} \times 100 \\
 &= \frac{9(1-0.30) + \left(\frac{103.50-90}{5}\right)}{\frac{103.50+90}{2}} \times 100
 \end{aligned}$$



$$= 9.30\% \text{ p.a.}$$

$$\text{RV} = 105$$

$$\text{Capital gain tax } [105 - 90] \times 10\% = 1.50$$

$$\text{RV}^* = 105 - 1.5 = 103.50$$

Question – 08

On 31st March, 2013, the following information about Bonds is available:

Name of Security	Face Value ₹	Maturity Date	Coupon Rate	Coupon Date (s)
Zero coupon T-Bill	10,000	31 st March, 2023	N.A.	N.A.
10.71% GOI 2023	1,00,000	20 th June, 2013	N.A.	N.A.
10.71% GOI 2023	100	31 st March, 2023	10.71	31 st March
10% GOI 2018	100	31 st March, 2018	10.00	31 st March & 30 th September

Calculate:

- If 10 years yield is 7.5% p.a. what price the Zero Coupon Bond would fetch on 31st March, 2013?
- What will be the annualized yield if the T-Bill is traded @ 98500?
- If 10.71% GOI 2023 Bond having yield to maturity is 8%, what price would it fetch on April 1, 2013 (after coupon payment on 31st March)
- If 10% GOI 2018 Bond having yield to maturity is 8%, what price would it fetch on April 1, 2013 (after coupon payment on 31st March)?

Solution:

(i) Price of ZCB

$$\begin{aligned} \text{Price} &= ₹ 10,000 \times \text{PVF}, 7.5\%, 10 \\ &= 10,000 \times 0.485 \\ &= ₹ 4,850 \end{aligned}$$

(ii) Annualized yield

$$\begin{aligned} \text{Annualized yield} &= \frac{1,00,000 - 98,500}{98,500} \times 100 \times \frac{365}{81} \\ &= 6.86\% \text{ p.a.} \end{aligned}$$

(iii) Price of Bond

$$\begin{aligned} \text{Price} &= (₹ 10.71 \times \text{PVAF}, 8\%, 10) + (₹ 100 \times \text{PVF}, 8\%, 10) \\ &= (₹ 10.71 \times 6.710) + (₹ 100 \times 0.463) \\ &= ₹ 118.16 \end{aligned}$$

(iv) Price of Bond

$$\begin{aligned} \text{Price} &= (₹ 5 \times \text{PVAF}, 4\%, 10) + (₹ 100 \times \text{PVF}, 4\%, 10) \\ &= (₹ 5 \times 8.111) + (₹ 100 \times 0.676) \\ &= ₹ 108.16 \end{aligned}$$

Question – 09

Today being 1st January 2019, Ram is considering to purchase an outstanding Corporate Bond having a face value of ₹ 1,000 that was issued on 1st January 2017 which has 9.5% Annual Coupon and 20



years of original maturity (i.e. maturing on 31st December 2027). Since the bond was issued, the interest rates have been on downside and it is now selling at a premium of ₹ 125.75 per bond.

Determine the prevailing interest on the similar type of Bonds if it is held till the maturity which shall be at Par.

PV Factors:

	1	2	3	4	5	6	7	8	9
6 %	0.943	0.890	0.840	0.792	0.747	0.705	0.665	0.627	0.592
8 %	0.926	0.857	0.794	0.735	0.681	0.630	0.583	0.540	0.500

(RTP November – 2020)

Solution:

Calculation of YTM

$$(6\%) \text{ Price} = (\text{₹ } 95 \times 6.801) + (1,000 \times 0.592) = \text{₹ } 1,238.10$$

$$(8\%) \text{ Price} = (\text{₹ } 95 \times 6.246) + (1,000 \times 0.500) = \text{₹ } 1,093.37$$

Interpolation

6%	-----	₹ 1,238.10
8%	-----	₹ 1,093.37
2%	-----	₹ 144.73

$$\begin{aligned} \text{YTM} &= 6 + \left(\frac{2}{144.73} \times (1,238.10 - 1,125.75) \right) \\ &= 7.55\% \text{ p.a.} \end{aligned}$$

Prevailing invest rate of similar debenture should be 7.55% p.a.

Question – 10

Mr. X wants to invest ₹ 1,00,000 in the 7 years 8% bonds in the market (Face Value ₹ 100) which were issued 2 years ago.

- (i) You are requested to advise him what is the maximum price for bonds to be paid in the following scenarios:
 - (1) If Mr. X is expecting minimum 9% return on the bonds
 - (2) If Mr. X is expecting minimum 7% return on the bonds
 - (3) If the present rate of similar bonds issued is 8.25%
 - (4) If the present rate of similar bonds issued is 7.75%
- (ii) If the bonds are available at par and 1% is the transaction cost, what is the effective yield?
- (iii) Find the number of days required to breakeven transaction cost if the bonds are available at par and 2% is the transaction cost.

(Exam Nov – 2022) (8 Marks)

Solution:

(1) Value of Bond

$$(i) \text{ Discount Rate} = 9\%$$

$$\begin{aligned} IV_0 &= (\text{₹ } 8 \times 3.890) + (100 \times 0.650) \\ &= \text{₹ } 96.12 \end{aligned}$$

$$(ii) \text{ Discount Rate} = 7\%$$



$$\begin{aligned} IV_0 &= (\text{₹ } 8 \times 4.100) + (100 \times 0.713) \\ &= \text{₹ } 104.10 \end{aligned}$$

(iii) Discount Rate = 8.25%

$$\begin{aligned} IV_0 &= (\text{₹ } 8 \times 3.967) + (100 \times 0.673) \\ &= \text{₹ } 99.04 \end{aligned}$$

(iv) Discount Rate = 7.75%

$$\begin{aligned} IV_0 &= (\text{₹ } 8 \times 4.019) + (100 \times 0.689) \\ &= \text{₹ } 101.05 \end{aligned}$$

(2) Effective Yield

$$\begin{aligned} \text{Price of Bond} &= \text{₹ } 100 + 1\% \\ &= \text{₹ } 101 \end{aligned}$$

YTM is a rate at which price of bond is ₹ 101 hence YTM = 7.75%

(3) Transaction Cost

$$= \text{₹ } 1,00,000 \times 2\% = \text{₹ } 2,000$$

$$\text{Income per day} = \text{₹ } 1,00,000 \times 8\% = \text{₹ } 8,000$$

$$= \frac{8,000}{365} = 21.92$$

$$\text{No. of days} = \frac{2,000}{\text{₹ } 21.92} = 91 \text{ days}$$

DIRTY PRICE & CLEAN PRICE

Question – 11

MP Ltd. issued a new series of bonds on January 1, 2010. The bonds were sold at par (₹ 1,000), having a coupon rate 10% p.a. and mature on 31st December, 2025. Coupon payments are made semiannually on June 30th and December 31st each year. Assume that you purchase an outstanding MP Ltd. bond on 1st March, 2018 when the going interest rate was 12%. Required:

- (i) What was the YTM of MP Ltd. bonds as on January 1, 2010?
- (ii) What amount you should pay to complete the transaction? Of that amount how much should be accrued interest and how much would represent bonds basic value.

Solution:

(i) YTM as (01/01/2010)

$$\text{YTM} = \frac{100}{1,000} \times 100 = 10\%$$

or

$$\begin{aligned} \text{YTM} &= \frac{\text{₹ } 50 + \left(\frac{1,000 - 1,000}{32} \right)}{\frac{1,000 + 1,000}{2}} \times 100 \\ &= \frac{\text{₹ } 50}{1,000} \times 100 \end{aligned}$$

$$= 5\% \times \frac{12}{6} = 10\%$$

(ii) Full Price (Dirty Price)

Bond price as on 30/06/18

$$= (\text{₹ } 50 \times \text{PVAF}, 6\%, 15) + (1,000 \times \text{PVF}, 6\%, 15)$$

$$= (\text{₹ } 50 \times 9.712) + (1,000 \times 0.417)$$

$$= \text{₹ } 902.60$$

$$= \text{₹ } 902.60 + 50 = \text{₹ } 952.60$$

Bond price as on 01/03/18

$$\text{Full Price} = \frac{952.60}{\left[1 + 0.12 \times \frac{4}{12}\right]} = \text{₹ } 915.96$$

$$\text{Accrued Interest} = \left(\text{₹ } 1,000 \times 10\% \times \frac{2}{12}\right) = 16.67$$

$$\text{Bond Basic Value} = \text{₹ } 915.96 - 16.67 = \text{₹ } 899.29$$

(II) BOND RISK

Question – 12

The following data is available for a bond:

Face Value ₹ 1,000

Coupon Rate 11%

Years to Maturity 6

SECURITY VALUATION

Redemption Value ₹ 1,000

Yield to Maturity 15%

(Round-off your answers to 3 decimals)

Calculate the following in respect of the bond:

- Current Market Price.
- Duration of the Bond.
- Volatility of the Bond.
- Expected market price if increase in required yield is by 100 basis points.
- Expected market price if decrease in required yield is by 75 basis points.

Solution:

(i) & (ii) Current Market Price & Duration of the Bond

Year	CF	YTM (15%)	P.V.	Weights	W × Year
1	110	0.870	95.70	0.113	0.113
2	110	0.756	83.16	0.098	0.196
3	110	0.658	72.38	0.085	0.255
4	110	0.572	62.92	0.074	0.296
5	110	0.497	54.67	0.064	0.32
6	1,110	0.432	479.52	0.565	3.39
		CMP	848.35	Duration	4.570

(iii) Volatility of Bond

$$\text{Volatility} = \frac{D}{1 + \text{YTM}}$$



$$= \frac{4.570}{1.15} = 3.974$$

(iv) If Yield Increases by 100 BP

$$\begin{aligned}\text{Effect} &= - \text{MD} \times \frac{\text{BP}}{100} \\ &= - 3.974 \times \frac{100}{100} \\ &= - 3.974\%\end{aligned}$$

$$\begin{aligned}\text{Bond Price} &= 848.35 - 3.974\% \\ &= ₹ 814.64\end{aligned}$$

(v) Yield Decreases by 75 BP

$$\begin{aligned}\text{Effective Duration} &= - 3.974 \times \frac{-75}{100} \\ &= 2.9805\%\end{aligned}$$

$$\begin{aligned}\text{Bond Price} &= 848.35 + 2.9805\% \\ &= ₹ 873.64\end{aligned}$$

Question – 13

Find the current market price of a bond having face value ₹ 1,00,000 redeemable after 6 year maturity with YTM at 16% payable annually and duration 4.3202 years. Given $1.16^6 = 2.4364$.

Solution:

Let assume interest amount be x

SECURITY VALUATION

Year	CF	PVF (16%)	PVCF	Year × PV
1	x	0.862	0.862 x	0.862 x
2	x	0.743	0.743 x	1.486 x
3	x	0.641	0.641 x	1.923 x
4	x	0.552	0.552 x	2.208 x
5	x	0.476	0.472 x	2.380 x
6	1,00,000 + x	0.410	0.410 x + 41,000	2.480 x + 2,46,000
			3.684 x + 41,000	11.319 x + 2,46,000

$$4.3202 = \frac{11.319 x + 2,46,000}{3.684 x + 41,000}$$

$$15.9156 x + 1,77,128 = 11.319 x + 2,46,000$$

$$4.5966 x = 68,872$$

$$x = \frac{68,872}{4.5966} = 14,983$$

$$\text{Coupon (\%)} = \frac{14,983}{1,00,000} \times 100 = 14.98\% \text{ i.e. } 15\%$$

$$\text{CMP} = (15,000 \times \text{PVAF}, 16\%, 6) + (1,00,000 \times \text{PVF}, 16\%)$$

$$= (15,000 \times 3.685) + (1,00,000 \times 0.410)$$

$$= ₹ 96,275$$

Question – 14

XL Ispat Ltd. has made an issue of 14 per cent non-convertible debentures on January 1, 2007. These debentures have a face value of ₹ 100 and is currently traded in the market at a price of ₹ 90.



Interest on these NCDs will be paid through post-dated cheques dated June 30 and December 31st. Interest payments for the first 3 years will be paid in advance through post-dated cheques while for the last 2 years post-dated cheques will be issued at the third year. The bond is redeemable at par on December 31, 2011 at the end of 5 years.

Required:

- (i) Estimate the current yield and YTM of the bond.
- (ii) Calculate the duration of the NCD.
- (iii) Assuming that intermediate coupon payments are, not available for reinvestment calculate the realized yield on the NCD.

Solution:

(i) Current Yield

$$\begin{aligned}\text{Current Yield} &= \frac{I}{\text{CMP}} \times 100 \times \frac{12}{6} \\ &= \frac{\text{₹ } 7}{90} \times 100 \times \frac{12}{6} \\ &= 15.55\%\end{aligned}$$

$$\begin{aligned}\text{YTM} &= \frac{I + \left(\frac{F-P}{n}\right)}{\frac{F+P}{n}} \times 100 \\ &= \frac{\text{₹ } 7 + \left(\frac{100-90}{10}\right)}{\frac{100+90}{2}} = 8.42\% \\ &= 8.42 \times \frac{12}{6} = 16.84\% \text{ p.a.}\end{aligned}$$

(ii) Duration of NCD

Year	CF	YTM (8.42%)	PV	Weights	W × Year
1	7	0.922	6.454	0.071	0.071
2	7	0.851	5.957	0.066	0.132
3	7	0.785	5.495	0.061	0.183
4	7	0.724	5.068	0.056	0.224
5	7	0.667	4.669	0.051	0.255
6	7	0.616	4.312	0.047	0.282
7	7	0.568	3.976	0.044	0.308
8	7	0.524	3.668	0.040	0.320
9	7	0.483	3.381	0.037	0.333
10	107	0.446	47.722	0.526	5.260
			₹ 90.7		7.368

$$\text{Duration} = \frac{7.368}{2} = 3.684 \text{ Years}$$

(iii) Realized YTM

$$\text{Cash outflows} = \text{₹ } 90$$

$$\text{Cash inflows}$$

$$(\text{₹ } 7 \times 10) + 100 = \text{₹ } 170$$

$$90 (1 + r)^{10} = 170$$

$$\begin{aligned}r &= \left[\left(\frac{170}{90} \right)^{1/10} - 1 \right] \times 100 = \\ 6.57\%\end{aligned}$$

$$\begin{aligned}\text{Realized YTM} &= 6.57 \times \frac{12}{6} \\ &= 13.14\% \text{ p.a.}\end{aligned}$$



Question – 15

- (a) Consider two bonds, one with 5 years to maturity and the other with 20 years to maturity. Both the bonds have a face value of ₹ 1,000 and coupon rate of 8% (with annual interest payments) and both are selling at par. Assume that the yields of both the bonds fall to 6%, whether the price of bond will increase or decrease? What percentage of this increase/decrease comes from a change in the present value of bond's principal amount and what percentage of this increase/decrease comes from a change in the present value of bond's interest payments?
- (b) Consider a bond selling at its par value of ₹ 1,000, with 6 years to maturity and a 7% coupon rate (with annual interest payment), what is bond's duration?
- (c) If the YTM of the bond in (b) above increases to 10%, how it affects the bond's duration? And why?

Solution:

- (a) **It bonds are selling at par & redeemed at par**

Then coupon = Current Yield YTM

5 Years Bond

P.V. of Interest + P.V. of Principal = Market Price

YTM (8%)

Present Value of Interest (80 × 3.993) = ₹ 319

Present Value of Principal (1,000 × 0.681) = ₹ 681

YTM (6%)

Present Value of Interest (80 × 4.212) = ₹ 337

Present Value of Principal (1,000 × 0.747) = ₹ 747

= ₹ 1084

P.V. of Interest = ₹ 319 – ₹ 337 = ₹ 18

P.V. of Principal = ₹ 681 – 747 = ₹ 66

Market = ₹ 1,000 – 1,084 = ₹ 84

If yield decreases to 6% then price of bond will Increase by ₹ 84 due to

Change in P.V. of Interest = $\frac{₹ 18}{84} \times 100 = 21.43\%$

Change in P.V. of Principal = $\frac{₹ 66}{84} \times 100 = 78.57\%$

20 Years Bond

	P.V. of Interest	P.V. Principal	Market Price
Yield 8 %	(80 × 9.818) = 786	(1,000 × 0.214) = 214	1,000
Yield 6 %	(80 × 11.470) = 918	(1,000 × 0.312) = 312	1,230
	132	98	230

If yield decreases to 6% then bond price will increase by ₹ 230



SECURITY VALUATION

$$\text{Due to P.V. of Interest} = \frac{132}{230} \times 100 = 57.39\%$$

$$\text{Due to P.V. of Principal} = \frac{98}{230} \times 100 = 42.61\%$$

(b) Duration in the average time taken to recollect back the investment

Year s (A)	Coupon Payme nt (₹)	Redempti on (₹)	Tota l (₹) (B)	PVIF (₹) (C)	(A)×(B)×(C) (₹)
1	70	-	70	0.935	65.45
2	70	-	70	0.873	122.22
3	70	-	70	0.816	171.36
4	70	-	70	0.763	213.64
5	70	-	70	0.713	249.55
6	70	1000	1070	0.666	4,275.72
			0	Σ ABC	5,097.94

$$\text{Duration} = \frac{\Sigma ABC}{\text{Purchase Price}} = \frac{₹ 5,097.94}{₹ 1,000} = 5.098 \text{ Years}$$

(c) If YTM goes up to 10%, current price of the bond will decrease to

$$₹ 70 \times \text{PVIFA} (10\%, 6) + ₹ 1000 \text{ PVIF} (10\%, 6)$$

$$₹ 304.85 + ₹ 564.00 = ₹ 868.85$$

Years (A)	Inflow (₹) (B)	PVIF (₹) (C)	(A) × (B) × (C) (₹)
1	70	0.909	65.45
2	70	0.826	115.64
3	70	0.751	157.71
4	70	0.683	191.24

5	70	0.621	217.35
6	1070	0.564	3,620.88
		Σ ABC	4,366.45

$$\text{New Duration} ₹ 4,366.45 / ₹ 868.85 = 5.025 \text{ years}$$

The duration of bond decreases, reason being the receipt of slightly higher portion of one's investment on the same intervals.

Question – 16

Mr. A is planning for making investment in bonds of one of the two companies X Ltd. and Y Ltd. The detail of these bonds is as follows:

Company	Face Value	Coupon Rate	Maturity Period
X Ltd.	₹ 10,000	6%	5 Years
Y Ltd.	₹ 10,000	4%	5 Years

The current market price of X Ltd.'s bond is ₹10,796.80 and both bonds have same Yield to Maturity (YTM). Since Mr. A considers duration of bonds as the basis of decision making, you are required to calculate the duration of each bond and you decision.

Solution:

To calculate duration of bond we need YTM, which shall be calculated as follows:

Let us try NPV of Bond @ 5%

$$= \frac{600}{(1.05)^1} + \frac{600}{(1.05)^2} + \frac{600}{(1.05)^3} + \frac{600}{(1.05)^4} + \frac{10,600}{(1.05)^5} - 10,796.80$$

$$= ₹ 571.43 + ₹ 544.22 + ₹ 518.30 + ₹ 493.62 + ₹ 8,305.38 - ₹ 10,796.80$$



$$= - ₹ 363.85$$

Let us now try NPV @ 4%

$$= \frac{600}{(1.04)^1} + \frac{600}{(1.04)^2} + \frac{600}{(1.04)^3} + \frac{600}{(1.04)^4} + \frac{10,600}{(1.05)^5} - 10,796.80$$

$$= ₹ 576.92 + ₹ 554.73 + ₹ 533.40 + ₹ 512.88 + ₹ 8,712.43 - ₹ 10,796.80$$

$$= ₹ 93.56$$

Let us now interpolation formula

$$= 4\% + \frac{93.56}{93.56 - (-363.85)} \times (5\% - 4\%)$$

$$= 4\% + \frac{93.56}{93.56 - (-363.85)}$$

$$= 4\% + \frac{93.56}{457.41} = 4.20\%$$

Duration of X Ltd.' s Bond

Year	Cash flows	P.V. @ 4.2%		Proportion bond value	Proportion bond value × time (Years)
1	600	0.9597	575.82	0.0533	0.0533
2	600	0.9210	552.60	0.0512	0.1024
3	600	0.8839	530.34	0.0491	0.1473
4	600	0.8483	508.98	0.0472	0.1888
5	10600	0.8141	<u>8,629.46</u>	<u>0.7992</u>	<u>3.9960</u>
			<u>10,797.20</u>	<u>1.0000</u>	<u>4.4878</u>

Duration of the Bond is 4.4878 years say 4.49 years.

Duration of Y Ltd.'s Bond

Year	Cash flows	P.V. @ 4.2%		Proportion bond value	Proportion bond value × time (Years)
1	400	0.9597	383.88	0.0387	0.0387
2	400	0.9210	368.40	0.0372	0.0744
3	400	0.8839	253.56	0.0357	0.1071
4	400	0.8483	339.32	0.0342	0.1368
5	10400	0.8141	<u>8,466.64</u>	<u>0.8542</u>	<u>4.2710</u>
			<u>9,911.80</u>	<u>1.0000</u>	<u>4.6280</u>

Duration of the Bond is 4.6280 years say 4.63 years.

Decision: Since the duration of Bond of X Ltd. is lower and also carrying higher interest rate hence it should be preferred.

Question – 17

The following data are available for a bond:

Face Value ₹ 10,000 to be redeemed at par on maturity

Coupon rate 8.5%

Years to Maturity 5 years

Yield to Maturity (YTM) 10%

EVALUATE the change in the expected market price of the Bond, if there is a decrease in the YTM by 200 basis points based on

(i) By Macaulay's Duration after making Convexity Adjustment.

(ii) By Intrinsic Value Method.

Given



Years	1	2	3	4	5
PVIF (10%, n)	0.909	0.826	0.751	0.683	0.621
PVIF (8%, n)	0.926	0.857	0.794	0.735	0.681

(MTP April – 2022)

Solution:

Macaulay's Duration

Year	CF	YTM (10%)	P.V.	Weights	W × year
1	850	0.909	772.65	0.082	0.082
2	850	0.826	702.10	0.074	0.148
3	850	0.751	638.35	0.068	0.204
4	850	0.683	580.55	0.062	0.248
5	10,850	0.621	6737.85	0.714	3.570
			9431.50		4.252

$$\begin{aligned}\text{Macaulay's Duration} &= \frac{D}{1 + \text{YTM}} \\ &= \frac{4.252}{1.10} = 3.865\end{aligned}$$

Intrinsic Value

Yield = 12%

$$\begin{aligned}\text{IV}_0 &= (850 \times 3.605) + (10,000 \times 0.567) \\ &= ₹ 8,734.25\end{aligned}$$

Yield = 8%

$$\text{IV}_0 = (850 \times 3.993) + (10,000 \times 0.681)$$

$$= ₹ 10,204.05$$

Calculation of Convexity

$$\begin{aligned}C^* &= \frac{P_2 + P_1 - 2 P_0}{2 P_0 \times \Delta Y^2} \\ &= \frac{10,204.05 + 8,734.25 - 2 \times 9,431.50}{2 \times 9,431.0 \times (0.02)^2}\end{aligned}$$

$$= 9.979$$

$$\begin{aligned}\text{C.A} &= C^* \times 100 \times \Delta Y^2 \\ &= 9.979 \times 100 \times (0.02)^2 \\ &= 0.399\end{aligned}$$

Bond price after CA

Yield ↓ 200 BP

$$\begin{aligned}\text{Effective Duration} &= \left(-3.865 \times \frac{-200}{100} \right) + 0.399 \\ &= 8.129\end{aligned}$$

$$\text{Price} = 9,431.50 + 8.129\% = 10,198$$

Yield ↓ 200 BP

* On the basis of Macaulay Duration after C.A.

$$\text{Price} = ₹ 10,198$$

* On the basis of Intrinsic Value



Price = 10,204.05

Question – 18

An investor, in the beginning of 2022, has purchased substantial number of 8 year 7.50% ₹ 1,000 bond with 5% premium on maturity at a required Yield to Maturity (YTM) of 8.50%. However, due to the continuing war in Europe, the inflation is running very high in the economies of the countries. The yield on the bonds is decreasing. The risk averse investor wants to protect himself from further loss and decides to sell the bonds in 2023. He has got a proposal from another investor who is willing to purchase these bonds by shelling out a maximum amount of ₹ 797.50 per bond.

Investor follows intrinsic value method for valuation of the Bonds.

You are required to determine

- (i) The market price, duration and volatility of the bond.
- (ii) Will it be a right decision of the new investor if he is looking for Required Yield to Maturity (YTM) as 12% p.a.?

Period	1	2	3	4	5	6	7
PVIF (8.50 %, n)	0.921 7	0.849 5	0.782 9	0.721 6	0.665 0	0.612 9	0.564 9

(Exam May – 2023) (9 Marks)

Solution:

(i) (A) Market Price of Bond

$$= 1,000 \times 7.50\% \times (\text{PVIAF } 8.50\%, 7) + 1,050 \times (\text{PVIF } 8.5\%, 7)$$

$$= 75 \times 5.1185 + 1050 \times 0.5649$$

$$= 383.89 + 593.15 = ₹ 977.04$$

(B) Duration of Bond

Year	Cash Flow	P.V. @ 8.5%		Proportion of Bond Value	Proportion of Bond Value × Time (years)
1	75	0.9217	69.128	0.071	0.071
2	75	0.8495	63.713	0.065	0.130
3	75	0.7829	58.718	0.060	0.180
4	75	0.7216	54.120	0.055	0.220
5	75	0.6650	49.875	0.051	0.255
6	75	0.6129	45.968	0.047	0.282
7	1125	0.5649	635.513	0.651	4.557
			977.035		5.695

Duration of the Bond is 5.695 years.

Alternatively, it can also be calculated as follows:

Year (1)	Cash flow (2)	PVF (3)	PV (4)	(1) × (4)
1	75	0.9217	69.13	69.13
2	75	0.8495	63.71	127.42
3	75	0.7829	58.72	176.16
4	75	0.7216	54.12	216.48
5	75	0.6650	49.88	249.40
6	75	0.6129	45.97	275.82
7	1125	0.5649	635.51	4448.57
			977.04	5562.98

$$\text{Duration of the Bond} = \frac{5,562.98}{977.04} = 5.69 \text{ years}$$

(C) Volatility of Bond

$$\text{Volatility} = \text{Duration} / (1 + \text{YTM})$$



SECURITY VALUATION

$$= 5.695 / (1 + 0.085) = 5.249$$

$$\text{Or } = 5.69 / (1 + 0.085) = 5.24$$

(ii) PV of Bond @ 12% YTM

$$= ₹ 75 \text{ PVIAF } (12\%, 7) + ₹ 1050 \times \text{PVIF } (12\%, 7)$$

$$= ₹ 75 \times 4.5637 + ₹ 1050 \times 0.4523$$

$$= ₹ 342.28 + ₹ 474.92 = ₹ 817.20$$

Since, Intrinsic Value of Bond is ₹ 817.20 the decision of new investor is right at purchase price of ₹ 797.50.

Question – 19

The investment portfolio of a fund is as follows:

Government Bond	Coupon Rate (%)	Purchase Rate (FV ₹ 100 per Bond)	Duration (Years)
GOI 2026	11.68	106.50	3.50
GOI 2030	7.55	105.00	6.50
GOI 2035	7.38	105.00	7.50
GOI 2042	8.35	110.00	8.75
GOI 2052	7.95	101.00	13.00

Face value of total Investment is ₹ 5 crores in each Government Bond.

Required:

- (i) Identify the type of the Fund.

- (ii) Calculate NAV per unit of the Fund if number of units is 2.50 crore.
- (iii) Suggest a suitable action to reduce risk by churning out investment portfolio in the following scenario:
- A. Interest rates are expected to lower by 25 basis points.
- B. Interest rates are expected to raise by 75 basis points.

Also calculate the revised duration of investment portfolio in each scenario.

Note: Use simple average to make calculations.

(MTP April – 2026)

Solution:

- (i) This type of Fund is a Gilt Fund as the amount of fund is invested in dated Government Securities.
- (ii) To calculate NAV per unit first we shall calculate of actual investment of Funds as follows:

Security	Purchase Price	Investment (₹ in lakhs)
GOI 2026	106.50	532.50*
GOI 2030	105.00	525.00
GOI 2035	105.00	525.00
GOI 2042	110.00	550.00
GOI 2052	101.00	505.00
	Total	2,637.50

$$* \frac{₹ 5 \text{ crores}}{₹ 100 \times 1,00,000} \times ₹ 106.50$$



Accordingly, the NAV per units shall be:

$$\frac{\text{₹ } 2,637.50 \text{ lakhs}}{250 \text{ lakhs}} = \text{₹ } 10.55$$

- (iii) Suitable action to churn out investment portfolio in following scenario is to reduce risk and to maximize profit or minimize losses.

$$\text{Average Duration} = \frac{3.5 + 6.5 + 7.5 + 8.75 + 13.00}{5} = \frac{39.25}{5} = 7.85 \text{ years}$$

- (A) Interest rates are expected to be lower by 25 basis points in such case increase the average duration by purchasing GOI 2052 and disposing of GOI 2026.

$$\begin{aligned} \text{Revised average duration shall be} &= \frac{39.25 - 3.5 + 13}{5} \\ &= \frac{48.75}{5} = 9.75 \text{ years} \end{aligned}$$

- (B) Interest rates are expected to rise by 75 basis points in such case reduce the average duration by (*) Purchasing GOI 2030 and disposing of GOI 2052.

$$\begin{aligned} \text{Revised average duration shall be} &= \frac{39.25 - 13 + 6.5}{5} \\ &= \frac{32.75}{5} = 6.55 \text{ years} \end{aligned}$$

(*) Purchasing of GOI 2026 is not beneficial as maturity period is very short and 75 basis points is comparatively higher change.

Thus, it can be said that the in first case the duration of portfolio is revised from 7.85 years to 9.75 years. In second case it stands revised from 7.85 years to 6.55 years.

BOND IMMUNIZATION

Question – 20

Mr. A will need ₹ 1,00,000 after two years for which he wants to make one time necessary investment now. He has a choice of two types of bonds. Their details are as below:

	Bond X	Bond Y
Face value	₹ 1,000	₹ 1,000
Coupon	7% payable annually	8% payable annually
Years to maturity	1	4
Current price	₹ 972.73	₹ 936.52
Current yield	10%	10%

Advice Mr. A whether he should invest all his money in one type of bond or he should buy both the bonds and, if so, in which quantity? Assume that there will not be any call risk or default risk.

(SM TYK – 27, MTP March – 2021 & RTP November – 2021)

Solution:

Bond Duration [Bond X]

YEAR	CF	PVF (10%)	PV	WEIGHT	YEAR × W
1	1,070	0.909	972.63	1	1
			972.63	Bond Duration = 1	



Bond Duration [Bond Y]

YEAR	CF	PVF (10%)	PV	WEIGHT	YEAR × W
1	80	0.909	72.72	0.078	0.078
2	80	0.826	66.08	0.071	0.142
3	80	0.751	60.08	0.064	0.192
4	1080	0.683	737.64	0.788	3.152
			936.52	Bond Duration = 3.564	

Duration of Liability = 2 years, hence we have to invest in bond having duration is 2 years but such bond is not available

In this situation, we have to invest in both bonds in a proportion so that duration of bonds should be 2 years

Immunization is a period at which

$$D_L = D_A$$

$$2 = (1 \times W_A) + (3.564 \times W_B)$$

$$2 = W_A + 3.564 (1 - W_A)$$

$$2 = W_A + 3.564 - 3.564 W_A$$

$$W_A = 0.61$$

$$W_B = 0.39$$

$$\text{Investment in x} = ₹ 1,00,000 \times 0.61$$

$$= ₹ 61,000$$

$$= ₹ 61,000 \times 0.826$$

$$= ₹ 50,386$$

$$\text{No.} = \frac{₹ 50,386}{972.73} = 51.80$$

$$= 52 \text{ Bonds}$$

$$\text{Investment in y} = ₹ 1,00,000 \times 0.39$$

$$= ₹ 39,000$$

$$= ₹ 39,000 \times 0.826$$

$$= ₹ 32,214$$

$$\text{No.} = \frac{₹ 32,214}{936.52} = 34.40$$

$$= 34 \text{ Bonds}$$

Question – 21

The following data are available for three bonds A, B and C. These bonds are used by a bond portfolio manager to fund an outflow scheduled in 6 years. Current yield is 9%. All bonds have face value of ₹ 100 each and will be redeemed at par. Interest is payable annually.

Bond	Maturity (Years)	Coupon rate
A	10	10%
B	8	11%
C	5	9%

- (i) Calculate the duration of each bond.

SECURITY VALUATION

- (ii) The bond portfolio manager has been asked to keep 45% of the portfolio money in Bond A. Calculate the percentage amount to be invested in bonds B and C that need to be purchased to immunize the portfolio.
- (iii) After the portfolio has been formulated, an interest rate change occurs, increasing the yield to 11%. The new duration of these bonds are: Bond A = 7.15 years, Bond B = 6.03 years and Bond C = 4.27 years.

Is the portfolio still immunized? Why or why not?

- (iv) Determine the new percentage of B and C bonds that are needed to immunize the portfolio. Bond A remaining at 45% of the portfolio.

Present values be used as follows:

Present Values	t_1	t_2	t_3	t_4	t_5
$PVIF_{0.09,t}$	0.917	0.842	0.772	0.708	0.650

Present Values	t_6	t_7	t_8	t_9	t_{10}
$PVIF_{0.09,t}$	0.596	0.547	0.502	0.460	0.4224

(MTP March – 2021)

Solution:

- (i) **Calculation of Bond Duration**

Bond A

Year	Cash flow	P.V. @ 9%		Proportion of bond value	Proportion of bond value × time (years)
1	10	0.917	9.17	0.086	0.086
2	10	0.842	8.42	0.079	0.158
3	10	0.772	7.72	0.073	0.219
4	10	0.708	7.08	0.067	0.268
5	10	0.650	6.50	0.061	0.305
6	10	0.596	5.96	0.056	0.336
7	10	0.547	5.47	0.051	0.357
8	10	0.502	5.02	0.047	0.376
9	10	0.460	4.60	0.043	0.387
10	110	0.4224	46.46	0.437	4.370
			106.40	1.000	6.862

Duration of the bond is 6.862 years or 6.86 year

Bond B

Year	Cash flow	P.V. @ 9%		Proportion of bond value	Proportion of bond value × time (years)
1	11	0.917	10.087	0.091	0.091
2	11	0.842	9.262	0.083	0.166
3	11	0.772	8.492	0.076	0.228
4	11	0.708	7.788	0.070	0.280
5	11	0.650	7.150	0.064	0.320
6	11	0.596	6.556	0.059	0.354
7	11	0.547	6.017	0.054	0.378
8	111	0.502	55.772	0.502	4.016
			111.224	1.000	5.833



Duration of the bond B is 5.833 years or 5.84 years.

Bond C

Year	Cash flow	P.V. @ 9%		Proportion of bond value	Proportion of bond value × time (years)
1	9	0.917	8.253	0.082	0.082
2	9	0.842	7.578	0.076	0.152
3	9	0.772	6.948	0.069	0.207
4	9	0.708	6.372	0.064	0.056
5	109	0.650	70.850	0.709	3.545
			100.00	1.000	4.242

Duration of the bond C is 4.242 years or 4.24 years

(ii) Amount of Investment required in Bond B and C

$$DL = DA$$

$$6 = (6.86 \times 0.45) + 5.84 W_B + 4.24 (0.55 - W_B)$$

$$6 = 3.087 + 5.84 W_B + 2.332 - 4.24 W_B$$

$$0.581 = 1.6 W_B$$

$$W_B = 0.3631 \text{ (36.31\%)}$$

$$W_C = 1 - 0.45 - 0.3631$$

$$= 0.1869 \text{ (18.69\%)}$$

(iii) With revised yield the Revised Duration of Bond stands

$$0.45 \times 7.15 + 0.36 \times 6.03 + 0.19 \times 4.27 = 6.20 \text{ year}$$

No portfolio is not immunized as the duration of the portfolio has been increased from 6 years to 6.20 years.

(iv) New percentage of B and C bonds that are needed to immunize the portfolio.

$$DL = DA$$

$$6.00 = (7.15 \times 0.45) + 6.03 W_B + 4.27 (0.55 - W_B)$$

$$6.00 = 3.218 + 6.03 W_B + 2.348 - 4.27 W_B$$

$$0.434 = 1.76 W_B$$

$$W_B = 0.2466 \text{ (24.66\%)}$$

$$W_C = 1 - 0.45 - 0.2466$$

$$= 0.3034 \text{ (30.34\%)}$$

(IV) OPTION EMBEDDED BONDS

(1) CONVERTIBLE BONDS

Question – 22

The following data is related to 8.5% Fully Convertible (into Equity shares) Debentures issued by JAC Ltd. at ₹ 1,000.

Market Price of Debenture	₹ 900
Conversion Ratio	30



Straight Value of Debenture	₹ 700
Market Price of Equity share on the date of Conversion	₹ 25
Expected Dividend Per Share	₹ 1

You are required to calculate:

- Conversion Value of Debenture
- Market Conversion Price
- Conversion Premium per share
- Ratio of Conversion Premium
- Premium over Straight Value of Debenture
- Favorable income differential per share
- Premium pay back period

(RTP November – 2021)

Solution:

- Conversion Value of Debenture

$$= \text{Market Price of one Equity Share} \times \text{Conversion Ratio}$$

$$= ₹ 25 \times 30 = ₹ 750$$
- Market Conversion Price

$$= \frac{\text{Market Price of Convertible Debenture}}{\text{Conversion Ratio}}$$

$$= \frac{₹ 900}{30} = ₹ 30$$

- Conversion Premium per share

$$\text{Market Conversion Price} - \text{Market Price of Equity Share}$$

$$= ₹ 30 - ₹ 25 = ₹ 5$$

- Ratio of Conversion Premium

$$\frac{\text{Conversion premium per share}}{\text{Market Price of Equity Share}} = \frac{₹ 5}{₹ 25} = 20\%$$

- Premium over Straight Value of Debenture

$$\frac{\text{Market Price of Convertible Bond}}{\text{Straight Value of Bond}} - 1 = \frac{₹ 900}{₹ 700} - 1 = 28.6 \%$$

- Favorable income differential per share

$$\frac{\text{Coupon Interest from Debenture} - \text{Conversion Ratio} \times \text{Dividend Per Share}}{\text{Conversion Ratio}}$$

$$\frac{₹ 85 - 30 \times ₹ 1}{30} = ₹ 1.833$$

- Premium pay back period

$$\frac{\text{Conversion premium per share}}{\text{Favourable Income Differential per Share}} = \frac{5}{1.833} = 2.73 \text{ years}$$

Question – 23

Following information is related to the Convertible Bond of A Ltd. which is currently priced at ₹ 1060 per Bond:

- Conversion Parity Price - ₹ 53



- (2) Conversion Premium – 10.41667%
- (3) Percentage of Downside Risk with respect to Straight Value of Bond – 12.766%

Calculate:

- (i) No. of shares on Conversion.
- (ii) Current Market Price Per Share of A Ltd.
- (iii) Straight Value of Bond

(MTP October – 2023)

Solution:

- (i) The No. of share on Conversion shall be computed as follows:**

$$\text{Conversion Parity Price} = \frac{\text{Bond Price}}{\text{No. of Shares on Conversion}}$$

$$₹ 53 = \frac{1060}{\text{No. of Shares on Conversion}}$$

Accordingly, No. of shares on Conversion = 20

- (ii) To determine current Market Price Per Share of A Ltd. we shall use Conversion Premium as follows:**

$$\text{Conversion Premium} = \frac{\text{Market Price of Bond} - \text{Conversion Value of Bond}}{\text{Conversion Value of Bond}}$$

$$0.1041667 = \frac{1060 - \text{Conversion Value of Bond}}{\text{Conversion Value of Bond}}$$

Conversion Value of Bond = ₹ 960

Since the No. of share on Conversion = 20

The current market price of share of A Ltd. shall be = ₹ 960 / 20 = ₹ 48 per share

- (iii) To determine the Straight Value of Bond we shall use Percentage of Downside Risk as follows:**

Percentage of Downside Risk

$$= \frac{\text{Market Price of Bond} - \text{Straight Value of Bond}}{\text{Straight Value of Bond}}$$

$$0.12766 = \frac{1060 - \text{Straight Value of Bond}}{\text{Straight Value of Bond}}$$

Straight Value of Bond = ₹ 940 per Bond

Question – 24

Following information is related to the 7.50% Convertible bond of S Ltd. which is currently priced at ₹ 5,300 per bond:

- Conversion Parity Price = ₹ 265
- Conversion Premium (Based on Market Price) = 10.41667%
- Percentage of Downside Risk based on Straight Value of Bond = 12.766%

Required:

- (i) Calculate No. of shares on Conversion.
- (ii) Analyze Current Market Price Per Share of S Ltd.
- (iii) Assess the Straight Value of Bond.

SECURITY VALUATION

- (iv) Based on straight value of bond computed above, determine the approximate required rate of return by an investor on similar category of bonds.

Note: Use following Present Value Factors (PVFs) for various calculations:

	1	2	3	4	5
PVF @ 8%	0.9259	0.8573	0.7938	0.7350	0.6806
PVF @ 10%	0.9091	0.8264	0.7513	0.6830	0.6209

(MTP April – 2025)

Solution:

$$(i) \text{ Conversion Parity Price} = \frac{\text{Bond Price}}{\text{No. of Shares on Conversion}}$$

$$₹ 265 = \frac{5,300}{\text{No. of Shares on Conversion}}$$

No. of Shares on Conversion = 20 Shares

$$(ii) \text{ Conversion Premium} = \frac{(\text{Conversion Parity Price} - \text{Market Price})}{\text{Market Price}} \times 100$$

$$10.41667\% = \frac{(\text{₹ } 265 - \text{Market Price})}{\text{Market Price}} \times 100$$

Market Price = ₹ 240

- (iii) Percentage of Downside Risk

$$= \frac{\text{Market Price of Bond} - \text{Straight Value of Bond}}{\text{Straight Value of Bond}} \times 100$$

$$12.766\% = \frac{(\text{₹ } 5,300 - \text{Straight Value of Bond})}{\text{Straight Value of Bond}} \times 100$$

Straight Value of Bond = ₹ 4,700

- (iv) To determine the required return, we shall discount related cash flows as follows:

PV@8%

Year	Cash Flow	PVF	PV
0	-4,700	1	-4,700
1	375	0.9259	347.21
2	375	0.8573	321.49
3	375	0.7938	297.68
4	375	0.7350	275.63
5	5,375	0.6806	3,658.23
			200.23

PV@10%

Year	Cash Flow	PVF	PV
0	-4,700	1	-4,700
1	375	0.9091	340.91
2	375	0.8264	309.90
3	375	0.7513	281.74
4	375	0.6830	256.13
5	5,375	0.6209	3,337.34
			-173.99

Calculation of Required return using IRR

$$= 8\% + \frac{200.23}{200.23 + 173.99} \times 2\%$$

$$= 8\% + \frac{200.23}{374.22} \times 2\% = 8\% + 1.07\% = 9.07\%$$



Question – 25

Saranam Ltd. has issued convertible debentures with coupon rate 12%. Each debenture has an option to convert to 20 equity shares at any time until the date of maturity. Debentures will be redeemed at ₹ 100 on maturity of 5 years. An investor generally required a rate of return of 8% p.a. on a 5-year security. As an investor when will you exercise conversion for given market prices of the equity share of (i) ₹ 4, (ii) ₹ 5 and (iii) ₹ 6.

Cumulative PV factor for 8% for 5 years : 3.993

PV factor for 8% for year 5 : 0.681

(SM TYK – 23)**Solution:****Intrinsic Value of Debenture**

$$IV_0 = (12 \times 3.993) + (100 \times 0.681)$$

$$= ₹ 116.016$$

Conversion Value

$$\text{Price} = ₹ 4 [20 \times 4] = ₹ 80$$

$$\text{Price} = ₹ 5 [20 \times 5] = ₹ 100$$

$$\text{Price} = ₹ 6 [20 \times 6] = ₹ 120$$

Investor will exercise option when price of share is ₹ 6

Question – 26

XYZ company has current earnings of ₹ 3 per share with 5,00,000 shares outstanding. The company plans to issue 40,000, 7%

convertible preference shares of ₹ 50 each at par. The preference shares are convertible into 2 shares for each preference shares held. The equity share has a current market price of ₹ 21 per share.

- What is preference share's conversion value?
- What is conversion premium?
- Assuming that total earnings remain the same, calculate the effect of the issue on the basic earning per share (a) before conversion (b) after conversion.
- If profits after tax increases by ₹ 1 million what will be the basic EPS (a) before conversion and (b) on a fully diluted basis?

Solution:

$$(i) \quad \text{Conversion Value} = ₹ 21 \times 2 = ₹ 42$$

$$(ii) \quad \text{Conversion Premium} = \frac{₹ 50 - 42}{42} \times 100 = 19.05\%$$

(iii) EPS

	Before Conversion	After Conversion
EAT	15,00,000	15,00,000
(-) PD [40,000 × 50 × 7%]	1,40,000	-
Earnings	13,60,000	15,00,000
÷ No.	5,00,000	5,80,000 [5,00,000 + 40,000 × 2]
EPS	2.72	2.586



SECURITY VALUATION

EPS Reduced by	0.28 (3 – 2.72)	0.414
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(iv) Revise EPS

	Before Conversion	After Conversion
EAT	25,00,000	25,00,000
(-) PD [40,000 × 7%]	1,40,000	-
Earnings	23,60,000	25,00,000
÷ No.	5,00,000	5,80,000 [5,00,000 + 40,000 × 2]
EPS	4.72	4.31

Question – 27

A Ltd. has issued convertible bonds, which carries a coupon rate of 14%. Each bond is convertible into 20 equity shares of the company A Ltd. The prevailing interest rate for similar credit rating bond is 8%. The convertible bond has 5 years maturity. It is redeemable at par at ₹ 100. The relevant present value table is as follows.

Present Values	t ₁	t ₂	t ₃	t ₄	t ₅
PVIF _{0.14,t}	0.877	0.769	0.675	0.592	0.519
PVIF _{0.08,t}	0.926	0.857	0.794	0.735	0.681

You are required to estimate:

(Calculations be made upto 3 decimal places)

- (i) Current market price of the bond, assuming it being equal to its fundamental value,

- (ii) Minimum market price of equity share at which bond holder should exercise conversion option; and
- (iii) Duration of the bond.

Solution:

- (i) **Current Market Price of Bond**

Time	CF	PVIF 8% PV (CF)	PV (CF)
1	14	0.926	12.964
2	14	0.857	11.998
3	14	0.794	11.116
4	14	0.735	10.290
5	114	0.681	77.634
		P ₀ =	<u>124.002</u>

Say

₹ 124.00

- (ii) Minimum Market Price of Equity Shares at which Bondholder should exercise conversion option:

$$\frac{124.00}{20.00} = ₹ 6.20$$

- (iii) **Duration of the Bond**

Year	Cash flow	P.V. @ 8%		Proportion of bond value	Proportion of bond value × time (years)
1	14	0.926	12.964	0.105	0.105
2	14	0.857	11.998	0.097	0.194
3	14	0.794	11.116	0.089	0.267
4	14	0.735	10.290	0.083	0.332



5	114	0.681	$\frac{77.634}{124.002}$	$\frac{0.626}{1.000}$	$\frac{3.130}{4.028}$
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Question – 28

A hypothetical company ABC Ltd. issued a 10% Debenture (Face Value of ₹ 1000) of the duration of 10 years, currently trading at ₹ 850 per debenture. The bond is convertible into 50 equity shares being currently quoted at ₹ 17 per share.

If yield on equivalent comparable bond is 11.80%, then calculate the spread of yield of the above bond from this comparable bond.

The relevant present value table is as follows.

Present Values	t ₁	t ₂	t ₃	t ₄	t ₅	t ₆	t ₇	t ₈	t ₉	t ₁₀
PVIF _{0.11, t}	0.901	0.812	0.731	0.659	0.593	0.535	0.482	0.434	0.391	0.352
PVIF _{0.13, t}	0.885	0.783	0.693	0.613	0.543	0.480	0.425	0.376	0.333	0.295

(RTP November – 2019)

Solution:

(1) Yield of Bond (YTM)

$$11\% = (\text{₹ } 100 \times 5.580) + (1,000 \times 0.352) = \text{₹ } 941$$

$$13\% = (\text{₹ } 100 \times 5.426) + (1,000 \times 0.295) = \text{₹ } 887.60$$

$$2\% \qquad \qquad \qquad \text{₹ } 103.40$$

SECURITY VALUATION

$$\text{YTM} = 11 + \left(\frac{2}{103.40} \times 91 \right)$$

$$\text{Spread of yield} = 12.76 - 11.80$$

$$= 0.96\%$$

(2) CONVERTIBLE BONDS OR BOND REFUNDING

Question – 29

M/s Trans India Ltd. is contemplating calling ₹ 3 crores of 30 years, ₹ 1,000 bond issued 5 years ago with a coupon interest rate of 14 per cent. The bonds have a call price of ₹ 1,140 and had initially collected proceeds of ₹ 2.91 crores due to a discount of ₹ 30 per bond. The initial floating cost was ₹ 3,60,000. The Company intends to sell ₹ 3 crores of 12 per cent coupon rate, 25 years bonds to raise funds for retiring the old bonds. It proposes to sell the new bonds at their par value of ₹ 1,000. The estimated floatation cost is ₹ 4,00,000. The company is paying 40% tax and its after tax cost of debt is 8 per cent. As the new bonds must first be sold and their proceeds, then used to retire old bonds, the company expects a two months period of overlapping interest during which interest must be paid on both the old and new bonds. What is the feasibility of refunding bonds?

Solution:

Incremental Cash Outflows

Repayment of old bond (30,000 Bonds × 1,140) = ₹ 3,42,00,000

Issue of new bonds = (₹ 3,00,00,000)

Floatation cost of new bonds = ₹ 4,00,000



SECURITY VALUATION

Tax savings on call premium $(42,00,000 \times 40\%) = (16,80,000)$

Tax saving on unamortized expenses of old bonds

$$\left[\frac{9,60,000 + 3,00,000}{30} \times 25 \right] = 10,50,000 \times 40\% = (4,20,000)$$

Interest on overlapping period [After Tax]

$$\left(3,00,00,000 \times 14\% \times \frac{2}{12} \right) (1 - 0.40) = 4,20,000$$

Incremented cash outflows = ₹ 29,20,000

Incremented Cash Inflows

	Old (14%)	New (12%)
Interest on bonds [After Tax]	$(3,00,00,000 \times 14\%) (1 - 0.40) = 25,20,000$	$(3,00,00,000 \times 12\%) (1 - 0.40) = 21,60,000$
Tax savings on floatation cost	$\left(\frac{12,60,000}{30} \times 40\% \right) (16,800)$	$\left(\frac{4,00,000}{25} \right) \times 40\% (6,400)$
	<u>25,03,200</u>	<u>21,53,600</u>

Incremented Cash Inflows $(25,03,200 - 21,53,600) = 3,49,600$ p.a.

PVCI $= 3,49,600 \times \text{PVAF}, 8\%, 25$

$$= 3,49,600 \times 10.675 = ₹ 37,31,980$$

(-) PVCO $= ₹ 29,20,000$

NPV = ₹ 8,11,980

Since NPV is positive, hence bond should be refunded.

Question – 30

ABC Ltd. has ₹ 300 million, 12 per cent bonds outstanding with six years remaining to maturity. Since interest rates are falling, ABC Ltd. is contemplating of refunding these bonds with a ₹ 300 million issue of 6 year bonds carrying a coupon rate of 10 per cent. Issue cost of the new bond will be ₹ 6 million and the call premium is 4 per cent. ₹ 9 million being the unamortized portion of issue cost of old bonds can be written off no sooner the old bonds are called off. Marginal tax rate of ABC Ltd. is 30 per cent. You are required to analyze the bond refunding decision.

(SM TYK – 25 & RTP May – 2020)

Solution:

Incremental Cash Outflows (Millions)

Repayment of old bond $(₹ 300 \text{ million} \times 1.04)$	= ₹ 312
Issue of new bonds	= (₹ 300)
Issue cost of new bonds	= ₹ 6
Tax savings on call premium $(300 \times 4\% \times 30\%)$	= (₹ 3.60)
Tax saving on unamortized expenses of old bonds $(9 \times 30\%)$	= (₹ 2.70)
Incremental cash outflows	= 11.70 Million

Incremented Cash Flows

	Old (12%)	New (10%)
Interest [After Tax]	$(300 \times 12\%) (1 - 0.30)$	$(300 \times 10\%) (1 - 0.30)$



	= 25.20	= 21.00
Tax Savings on Amortization expenses	$\left(\frac{9}{6} \times 30\%\right) (0.45)$	$\left(\frac{6}{6} \times 30\%\right) (0.30)$
	24.75	20.70

Incremented cash Inflows $24.75 - 20.70 = 4.05$ Million p.a.

$$K_d = I(1 - t)$$

$$= 10(1 - 0.30)$$

$$= 7\%$$

$$PVC I = 4.05 \times PVA F, 6, 7\%$$

$$= 4.05 \times 4.767 = 19.31$$

$$(-) PVCO = 11.70$$

$$NPV = 7.61$$

Since NPV is positive, hence bond should be refunded.

Question – 31

TK Ltd. has ₹ 600 Lakh 12% Debenture outstanding with 5 years remaining to redemption. Since interest rates are decreasing, company is planning to redeem these debentures with a ₹ 600 Lakh issue of 5 years 10% Debenture at par. Issued Cost of 10% Debenture will be ₹ 10 Lakh. Premium paid on redemption of 12% Debenture is 5% Tax rate applicable to company 15, is 20%

You are required to advise on the 12% Debenture Redemption Decision.

SECURITY VALUATION

PVF @ 10% & 8% are as under —

Rate	1	2	3	4	5
8%	0.93	0.86	0.79	0.74	0.68
10%	0.91	0.83	0.75	0.68	0.62

Calculation up to 2 decimal points.

(Exam September – 2025)

Solution:

(i)	Calculation of initial outlay:-	₹ (Lakh)
a.	Face value	600
	Add: Call premium	30
	Less: Tax on Premium of Redemption	<u>6</u>
	Cost of calling old bonds	<u>624</u>
b.	Gross proceed of new issue	600
	Less: Issue costs	<u>10</u>
	Net proceeds of new issue	<u>590</u>
	∴ Initial outlay = ₹ 624 Lakh – ₹ 590 Lakh = ₹ 34 Lakh	
(ii)	Calculation of net present value of refunding the bond:	
	Saving in annual interest expenses	₹ (Lakh)
	$[600 \times (0.12 - 0.10)]$	12.00



SECURITY VALUATION

Less: Tax saving on interest (0.20×12)	2.40
Add: Tax Saving on Issue Exp. $(10/5) \times 0.20$	<u>0.40</u>
Annual net cash saving	<u>10.00</u>
PVIFA (8%, 5 years)	4.00
$\therefore$ Present value of net annual cash saving	₹ 40.00 Lakh
Less: Initial outlay	<u>₹ 34.00 Lakh</u>
Net present value of refunding the bond	<u>₹ 6.00 Lakh</u>

Decision: 12% Debentures should be redeemed and new 10% Debentures should be issued because NPV of Bond Refunding decision is positive.

Alternative Solution: Since in the Question specifically nothing has been mentioned about the writing off Issue Expenses for 10% Debentures, if students have assumed it to be written off at the time of issue in one go then solution will be as follows:

(i) Calculation of initial outlay:-	₹ (Lakh)
a. Face value	600
Add: Call premium	30
Less: Tax on Premium of Redemption	6
Cost of calling old bonds	624
b. Gross proceed of new issue	600
Less: Issue costs	10

Add: Tax Saving on Issue Expenses	2
Net proceeds of new issue	592
$\therefore$ Initial outlay = ₹ 624 Lakh – ₹ 592 Lakh = ₹ 32 Lakh	
(ii) Calculation of net present value of refunding the bond:-	
Saving in annual interest expenses	₹ (Lakh)
$[600 \times (0.12 - 0.10)]$	12.00
Less: Tax saving on interest (0.20×12)	<u>2.40</u>
Annual net cash saving	<u>9.60</u>
PVIFA (8%, 5 years)	4.0
$\therefore$ Present value of net annual cash saving	₹ 38.40 Lakh
Less: Initial outlay	<u>₹ 32.00 Lakh</u>
Net present value of refunding the bond	<u>₹ 6.40 Lakh</u>

Decision: 12% Debentures should be redeemed and new 10% Debentures should be issued because NPV of Bond Refunding decision is positive.

(3) EXTENDABLE BONDS

QUESTION – 32

Pet feed plc has outstanding, a high yield Bond with following features:



Face Value	£ 10,000
Coupon	10%
Maturity Period	6 Years
Special Feature	Company can extend the life of Bond to 12 years.

Presently the interest rate on equivalent Bond is 8%.

- (a) If an investor expects that interest will be 8%, six years from now then how much he should pay for this bond now.
- (b) Now suppose, on the basis of that expectation, he invests in the Bond, but interest rate turns out to be 12%, six years from now, then what will be his potential loss/ gain if the company extends the life of Bond for another 6 years.

Solution:

Investor expects that after 6 years, interest rate will be 8%. In this situation, company will not extend [Market में कम Rate पर company को पैसे मिल जाएगा]

$$\begin{aligned}
 \text{PVC I} &= (£ 1,000 \times \text{PVAF}, 8\%, 6) (£ 1,000 \times \text{PVF}, 8\%, 6) \\
 &= (£ 1,000 \times 4.623) \times (£ 1,000 \times 0.630) \\
 &= £ 10,923
 \end{aligned}$$

Value of bond at the end of 6th year if yield (12%)

$$\begin{aligned}
 &= (£ 1,000 \times 4.111) \times (£ 1,000 \times 0.507) \\
 &= £ 9,181
 \end{aligned}$$

$$\begin{aligned}
 \text{Expected loss} &= £ 10,923 - £ 9,181 \\
 &= £ 1,742
 \end{aligned}$$

(V) YIELD STRUCTURE OR TERM STRUCTURE OF INTERESR RATE

Question – 33

From the following data for Government securities, calculate the forward rates:

Face Value (₹)	Interest Rate	Maturity (Year)	Current Price (₹)
1,00,000	0%	1	91,500
1,00,000	10%	2	98,500
1,00,000	10.5%	3	99,000

Solution:

1 Year Rate

$$91,500 = \frac{1,00,000}{(1 + r)^1}$$

$$r = \left(\frac{1,00,000}{91,500} - 1 \right) \times 100 = 9.29\%$$

1 Year FR after 1 Year

$$98,500 = \frac{10,000}{(1.0929)} + \frac{1,10,000}{(1.0929)(1 + r)}$$

$$r = 12.65\%$$



1 Year FR after 2 Years

$$99,000 = \frac{10,500}{1.0929} + \frac{10,500}{(1.0929)(1.1265)} + \frac{1,10,500}{(1.0929)(1.1265)(1+r)}$$

$$r = 11\% \text{ p.a.}$$

Question – 34

ABC Ltd. issued 9%, 5 year bonds of ₹ 1,000/- each having a maturity of 3 years. The present rate of interest is 12% for one year tenure. It is expected that Forward rate of interest for one year tenure is going to fall by 75 basis points and further by 50 basis points for every next year in further for the same tenure. This bond has a beta value of 1.02 and is more popular in the market due to less credit risk.

Calculate:

- (i) Intrinsic value of bond.
- (ii) Expected price of bond in the market.

Solution:

(i) Forward Rate

$$1 \text{ year} = 12\%$$

$$2^{\text{nd}} \text{ Year} = 12 - 0.75 = 11.25\%$$

$$3^{\text{rd}} \text{ year} = 11.25 - 0.5 = 10.75\%$$

$$\begin{aligned} IV_0 &= \frac{₹ 90}{1.12} + \frac{₹ 90}{(1.12)(1.1125)} + \frac{₹ 1,090}{(1.12)(1.1125)(1.1075)} \\ &= ₹ 942.47 \end{aligned}$$

(ii) Expected Price of Bond

$$= ₹ 942.47 \times 1.02$$

$$= 961.32$$

Question – 35

ABC Ltd. wants to issue 9% Bonds redeemable in 5 years at its face value of ₹ 1,000 each. The annual spot yield curve for similar risk class of Bond is as follows:

Year	Interest Rate
1	12%
2	11.62%
3	11.33%
4	11.06%
5	10.80%

- (i) Evaluate the expected market price of the Bond if it has a Beta value of 1.10 due to its popularity because of lesser risk.
- (ii) Interpret the nature of the above yield curve and reasons for the same.

Note: Use PV Factors upto 4 decimal points and value in ₹ upto 2 decimal points.

(MTP April – 2021)

Solution:

(i) Market price

$$= \frac{₹ 90}{(1.12)^1} + \frac{₹ 90}{(1.1162)^2} + \frac{₹ 90}{(1.1133)^3} + \frac{₹ 90}{(1.1106)^4} + \frac{₹ 1,090}{(1.1080)^5}$$



SECURITY VALUATION

$$= ₹ 929.70$$

$$\begin{aligned}\text{Expected Price} &= ₹ 929.70 \times 1.10 \\ &= ₹ 1,022.67\end{aligned}$$

(ii) Inverted yield curve due to upcoming recession.

Question – 36

The following is the Yield structure of AAA rated debenture:

Period	Yield (%)
3 Months	8.5%
6 Months	9.25
1 Year	10.50
2 Years	11.25
3 Years and above	12.00

- (i) Based on the expectation theory calculate the implicit one-year forward rates in year 2 and year 3.
- (ii) If the interest rate increases by 50 basis points, what will be the percentage change in the price of the bond having a maturity of 5 years? Assume that the bond is fairly priced at the moment at ₹ 1,000.

Solution:

(i) Forward Rate

1 year FR in year 2

$$= \frac{(1.1125)^2}{1.1050} = 12\%$$

1 year FR in year 3

$$= \left[\frac{(1.12)^3}{(1.1125)^2} - 1 \right] \times 100 = 13.52\%$$

$$(ii) \text{ R.V.} = 1,000 (1.12)^5 = 1,762.34$$

$$\text{If yield } (12 + 0.5) = 12.5\%$$

$$= \frac{1762.34}{(1.125)^5} = ₹ 977.97$$

% Decrease in Bond Price

$$= \frac{1,000 - 977.97}{1,000} \times 100 = 2.20\%$$

Question – 37

Following are the yields on Zero Coupon Bonds (ZCB) having a face value of ₹ 1,000 :

Maturity (Years)	Yield to Maturity (YTM)
1	10%
2	11%
3	12%

Assume that the term structure of interest rate will remain the same.

You are required to

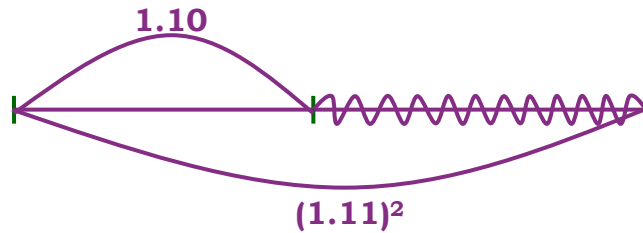
- (i) Calculate the implied one year forward rates
- (ii) Expected Yield to Maturity and prices of one year and two year Zero Coupon bonds at the end of the first year.

(Exam January – 2021) (4 Marks)



Solution:

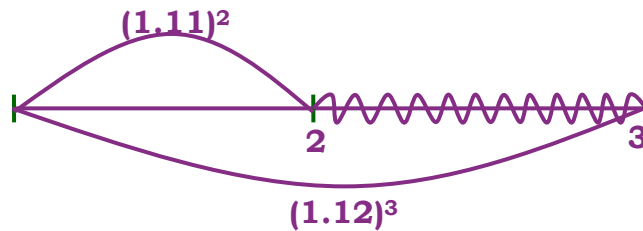
(i) 1 year FR after 1 year



$$= \left[\frac{(1.11)^2}{1.10} - 1 \right] \times 100$$

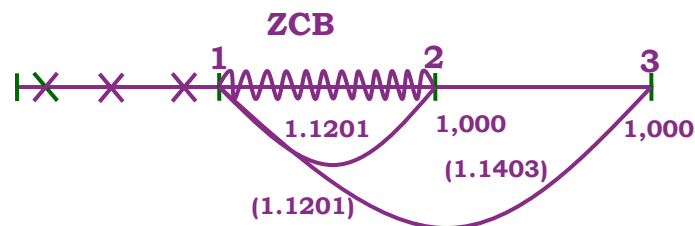
$$= 12.01\%$$

(ii) 1 year FR after 2 years



$$= \left[\frac{(1.12)^3}{(1.11)^2} - 1 \right] \times 100$$

$$= 14.03\%$$



1 year ZCB

Price at the end of year 1

$$= \frac{\text{₹ } 1,000}{1.1201}$$

$$= \text{₹ } 892.78$$

Expected yield = 12.01%

1 year ZCB

$$\text{Price} = \frac{\text{₹ } 1,000}{(1.1403)(1.1201)}$$

$$= 782.93$$

$$= 782.93 (1 + r)^2$$

$$= 1,000$$

$$= \left(\frac{\text{₹ } 1,000}{782.93} \right)^{1/2}$$

$$= 13.02\%$$

DIVIDEND GROWTH MODEL OR, DIVIDEND DISCOUNT MODEL OR, GORDEN'S MODEL

Question – 38

A company has a book value per share of ₹ 137.80. Its return on equity is 15% and it follows a policy of retaining 60% of its earnings. If the Opportunity Cost of Capital is 18%, compute the price of the share today using both Dividend Growth Model and Walter's Model.

(SM TYK – 01)

Solution:

$$BVPS = 137.80$$

$$\begin{aligned} EPS &= 137.80 \times 15\% \\ &= 20.67 \text{ [Expected EPS]} \end{aligned}$$

$$\begin{aligned} D_1 &= 20.67 \times 40\% \\ &= ₹ 8.268 \end{aligned}$$

$$\begin{aligned} g &= b \times r \\ &= 0.60 \times 0.15 = 0.09 \end{aligned}$$

Gordon's Model

$$\begin{aligned} P_0 &= \frac{D_1}{K_e - g} \\ &= \frac{8.268}{0.18 - 0.09} \end{aligned}$$

$$= ₹ 91.87$$

Walter's Model

$$P/E = \frac{MPS}{EPS}$$

No Growth Model

$$K_e = \frac{EPS}{MPS}$$

$$K_e = \frac{EPS}{EPS \times P/E}$$

$$K_e = \frac{1}{P/E \text{ Ratio}}$$

$$\begin{aligned} P_0 &= \frac{8.268 + (20.67 - 8.268)^{0.15}}{0.18} \\ &= ₹ 103.35 \end{aligned}$$

Question – 39

ABC Ltd. has been maintaining a growth rate of 10 percent in dividends. The company has paid dividend @ ₹3 per share. The rate of return on market portfolio is 12 percent and the risk free rate of return in the market has been observed as 8 percent. The Beta coefficient of company's share is 1.5.

You are required to calculate the expected rate of return on company's shares as per CAPM model and equilibrium price per share by dividend growth model.

(SM TYK – 08)



Solution:**CAPM Equation**

$$\begin{aligned}
 K_e &= R_f + (R_m - R_f) \beta \\
 &= 8 + (12 - 8) 1.5 \\
 &= 14\%
 \end{aligned}$$

EQUILIBRIUM PRICE

$$\begin{aligned}
 P_0 &= \frac{D_1}{K_e - g} \\
 &= \frac{3 (1.10)}{0.14 - 0.10} \\
 &= ₹ 82.50
 \end{aligned}$$

Question – 40

A Company pays a dividend of ₹ 2.00 per share with a growth rate of 7%. The risk-free rate is 9% and the market rate of return is 13%. The Company has a beta factor of 1.50. However, due to a decision of the Finance Manager, beta is likely to increase to 1.75. Find out the present as well as the likely value of the share after the decision.

(SM TYK – 09)**Solution:****Present Value**

$$\begin{aligned}
 K_e &= R_f + (R_m - R_f) \beta \\
 &= 9 + (13 - 9) 1.5
 \end{aligned}$$

$$= 15\%$$

$$\begin{aligned}
 P_0 &= \frac{D_1}{K_e - g} \\
 &= \frac{2 (1.07)}{0.15 - 0.07} \\
 &= ₹ 26.75
 \end{aligned}$$

Likely Value of Shares

$$\begin{aligned}
 K_e &= R_f + (R_m - R_f) \beta \\
 &= 9 + (13 - 9) 1.75 \\
 &= 16\%
 \end{aligned}$$

$$\begin{aligned}
 P_0 &= \frac{D_1}{K_e - g} \\
 &= \frac{2 (1.07)}{0.16 - 0.07} \\
 &= ₹ 23.78
 \end{aligned}$$

Question – 41

Shares of Voyage Ltd. are being quoted at a price-earning ratio of 8 times. The company retains 45% of its earnings which are ₹ 5 per share.

You are required to compute

- (1) The cost of equity to the company if the market expects a growth rate of 15% p.a.



- (2) If the anticipated growth rate is 16% per annum, calculate the indicative market price with the same cost of capital.
- (3) If the company's cost of capital is 20% p.a. & the anticipated growth rate is 19% p.a., calculate the market price per share.

(SM TYK – 11)**Solution:****(1) Cost of Equity**

Retention Ratio (b) = 45%

$$\text{EPS} = \frac{5}{45\%} = ₹ 11.11$$

$$D_1 = ₹ 11.11 - ₹ 5 = ₹ 6.11$$

$$\begin{aligned} P_0 \text{ (MPS)} &= \text{EPS} \times P/E \\ &= 11.11 \times 8 = 88.88 \end{aligned}$$

$$\begin{aligned} K_e &= \frac{D_1}{P_0} + g \\ &= \frac{6.11}{88.88} + 0.15 \\ &= 21.87\% \end{aligned}$$

(2) Market Price

$$\begin{aligned} P_0 &= \frac{D_1}{K_e - g} + g \\ &= \frac{6.11}{0.2187 - 0.16} = ₹ 104.09 \end{aligned}$$

(3) Market Price

$$\begin{aligned} P_0 &= \frac{D_1}{K_e - g} + g \\ &= \frac{6.11}{0.20 - 0.19} = ₹ 611 \end{aligned}$$

Question – 42

M/s X Ltd. has paid a dividend of ₹ 2.5 per share on a face value of ₹ 10 in the financial year ending on 31st March, 2009. The details are as follows:

Current market price of share ₹ 60

Growth rate of earnings and dividends 10%

Beta of share 0.75

Average market return 15%

Risk free rate of return 9%

Calculate the intrinsic value of the share.

(SM TYK – 13)**Solution:**

$$\begin{aligned} K_e &= R_f + (R_m - R_f) \beta \\ &= 9 + (15 - 9) 0.75 \\ &= 13.5\% \end{aligned}$$

$$P_0 = \frac{D_1}{K_e - g}$$



$$= \frac{2.5 (1.10)}{0.135 - 0.10}$$

$$= ₹ 78.57$$

Question – 43

A company has an EPS of ₹ 2.5 for the last year and the DPS of ₹ 1. The earnings is expected to grow at 2% a year in long run. Currently it is trading at 7 times its earnings. If the required rate of return is 14%, compute the following:

- An estimate of the P/E ratio using Gordon growth model.
- The Long-term growth rate implied by the current P/E ratio.

(MTP March – 2021)

Solution:

(i) Implicit P/E Ratio

$$P_0 = \frac{D_1}{K_e - g}$$

$$= \frac{1 (1.02)}{0.14 - 0.02} = ₹ 8.50$$

$$P/E = \frac{MPS}{EPS} = \frac{8.50}{2.50} = 3.4 \text{ times}$$

(ii) Implicit Growth Rate

$$MPS (P_0) = EPS \times P/E$$

$$= 2.50 \times 7 = 17.50$$

$$17.50 = \frac{1 (1 + g)}{0.14 - g} = ₹ 8.50$$

$$2.45 - 17.50 g = 1 + 1g$$

$$g = 7.84\%$$

Question – 44

Following are the details of X Ltd. and Y Ltd.:

Particulars	X Ltd.	Y Ltd.
Dividend per Share	₹ 4	₹ 4
Growth Rate	10%	10%
Beta	0.9	1.2
Current Market Price per Share	₹ 150	₹ 70

Other Information:

Risk Free Rate of Return	7%
Market Rate of Return	14%

- Calculate the price of shares of both the companies.
- Write the comment on the valuation on the basis of price calculated and current market price.
- As an investor what course of action should be followed?

(Exam December – 2021) (8 Marks)

Solution:

(i) Calculation of Prices of Shares of both Companies

	X Ltd.	Y Ltd.
Beta	0.9	1.20

Cost of Equity using CAPM	$7\% + 0.9[14\% - 7\%]$ $= 13.30\%$	$7\% + 1.20 [14\% - 7\%]$ $= 15.40\%$
Growth Rate	10%	10%
Price of Share	$\frac{4 \times 1.10}{0.133 - 0.10} = \frac{4.40}{0.033}$ $= ₹ 133.33$	$\frac{4 \times 1.10}{0.154 - 0.10} = \frac{4.40}{0.054}$ $= ₹ 81.48$

(ii) and (iii)

Name of Company	Current Market Price	Value of the Share	Valuation	Action of the Investor
X Ltd.	₹ 150.00	₹ 133.33	Overvalued/overpriced	Not to Invest/to be sold
Y Ltd.	₹ 70.00	₹ 81.48	Undervalued/underpriced	Invest/to be purchased

Question – 45

On the basis of the following information:

Current dividend (D₀) = ₹ 2.50

Discount rate (k) = 10.5%

Growth rate (g) = 2%

(i) Calculate the present value of stock of ABC Ltd.

(ii) Is its stock overvalued if stock price is ₹ 35, ROE = 9% and EPS = ₹ 2.25? Show detailed calculation.

(SM TYK – 04 & MTP – 2022)

Solution:

(1) Present Value of Stock

$$P_0 = \frac{D_1}{K_e - g}$$

$$= \frac{2.50 (1.02)}{0.105 - 0.02}$$

$$= ₹ 30$$

(2) Present Value of Stock

(a) P/E Model

$$P/E = \frac{1}{K_e}$$

$$= \frac{1}{9\%} = 11.11$$

MPS = EPS × P/E Ratio

$$= ₹ 2.25 \times 11.11$$

$$= ₹ 25$$

Actual price is more than intrinsic value, hence share is overpriced.

(b) Earning Growth Model

$$P_0 = \frac{\text{EPS} (1 + g)}{K_e - g}$$

$$= \frac{2.25 (1.02)}{0.09 - 0.02} = ₹ 32.79$$

Stock is overpriced

Question – 46

The risk free rate of return R_f is 9 percent. The expected rate of return on the market portfolio R_m is 13 percent. The expected rate of growth for the dividend of Platinum Ltd. is 7 percent. The last dividend paid on the equity stock of firm A was ₹ 2.00. The beta of Platinum Ltd. equity stock is 1.2.

- (i) What is the equilibrium price of the equity stock of Platinum Ltd.?
- (ii) How would the equilibrium price change when
- The inflation premium increases by 2 percent?
 - The expected growth rate increases by 3 percent?
 - The beta of Platinum Ltd. equity rises to 1.3?

(SM TYK – 15)**Solution:****(1) Equilibrium Price**

$$P_0 = \frac{D_1}{K_e - g}$$

$$K_e = R_f + \beta (R_m - R_f)$$

$$= 9 + 1.2 (13 - 9)$$

$$= 13.8\%$$

$$P_0 = \frac{2 (1.07)}{0.138 - 0.07}$$

$$= ₹ 31.47$$

(2) Assumption 1

Each factor independently

* Inflation premium $\uparrow$ 2%

$$K_e = 11 + 1.2 (15 - 11) = 15.8\%$$

$$P_0 = \frac{2 (1.07)}{0.158 - 0.07} = ₹ 24.32$$

* Growth Rate $\uparrow$ 3%

$$P_0 = \frac{2 (1.10)}{0.138 - 0.10} = ₹ 57.89$$

* Beta = 1.30

$$K_e = 9 + 1.30 (13 - 09) = 14.2\%$$

$$P_0 = \frac{2 (1.07)}{0.142 - 0.07} = ₹ 29.72$$

Assumption II [All Factors Change Simultaneously]

$$K_e = 11 + 1.30 (15 - 11) = 16.20\%$$



SECURITY VALUATION

$$P_0 = \frac{2(1.10)}{0.162 - 0.10} = ₹ 35.48$$

Question – 47

Mr. X has submitted the following data:

Particulars	(₹) in Lakhs
Total Assets	250
Total Liabilities	220
Net Income	12
Dividend Paid	4.5
Sales	100

Mr. X wants to know to what extent sales can be increased without going for additional borrowings by using Sustainable Growth Rate (SGR) Concept?

(Exam Nov – 2022) (4 Marks)

Solution:

SGR

$$\text{Equity} = 250 - 220 = ₹ 30 \text{ lacs}$$

$$\begin{aligned}\text{Retain Earning} &= 12 - 4.50 \\ &= 7.50 \text{ lacs}\end{aligned}$$

$$\text{Retention Ratio (b)} = \frac{7.50}{12} \times 100 = 62.5\%$$

$$\text{ROE (r)} = \frac{\text{NI}}{\text{Equity}} \times 100$$

$$= \frac{12}{30} \times 100 = 40\%$$

$$\text{SGR} = b \times r$$

$$= 0.625 \times 0.40 = 25\%$$

Sales will increment by 25% i.e. $100 \times 25\% = ₹ 25 \text{ lacs}$

MULTIPEL GROWTH MODEL

Question – 48

MNP Ltd. has declared and paid annual dividend of ₹ 4 per share. It is expected to grow @ 20% for the next two years and 10% thereafter. The required rate of return of equity investors is 15%. Compute the current price at which equity shares should sell.

Note: Present Value Interest Factor (PVIF) @ 15%:

For year 1 = 0.8696;

For year 2 = 0.7561

(SM TYK – 03)

Solution:

$$D_0 = ₹ 4$$

$$D_1 = ₹ 4 (1.20) = ₹ 4.80$$

$$D_2 = ₹ 4 (1.20)^2 = ₹ 5.76$$

$$D_3 = ₹ 4 (1.20)^2 (1.10) = ₹ 6.336$$



$$TV = \frac{D_3}{(k_e - g)} = \frac{6.336}{0.15 - 0.10} = 126.72$$

$$P = \frac{4.80}{(1 + 0.15)} + \frac{5.76}{(1 + 0.15)^2} + \frac{126.72}{(1 + 0.15)^2}$$

$$= 4.80 \times 0.8696 + 5.76 \times 0.7561 + 126.72 \times 0.7561$$

$$= 104.34$$

Question – 49

M/s. B Ltd. has declared dividend of ₹ 2.50 per share on the EPS of ₹ 7. Earnings of the company are expected to grow at the rate of 10% for the next 3 years and to be stabilized at 3% thereafter.

The pay-out ratio is expected to remain at the same level during 3 years and then will increase to 60%. If required rate of return is 16% calculate:

- (i) The current price of the share.
- (ii) The expected price of share of B Ltd. At the end of 3rd year.

Following table may be used for calculations.

Present Value	t ₁	t ₂	t ₃	t ₄	t ₅
PVIF _{0.16,t}	0.862	0.743	0.641	0.553	0.477

(Exam Jan – 2021) (5 Marks)

Solution:

EPS & DPS

	1	2	3	4
EPS (₹ 27)	7.70	8.47	9.317	9.597
DPS (2.50)	2.75	3.025	3.328	5.758

Stage I

$$P.V. = (2.75 \times 0.862) + (3.025 \times 0.743) + (3.328 \times 0.641)$$

$$= ₹ 6.751$$

Stage II

$$P_3 = \frac{D_4}{K_e - g}$$

$$= \frac{5.758}{0.16 - 0.03}$$

$$= ₹ 44.29$$

$$= ₹ 44.29 \times 0.641 = 28.39$$

$$P_0 = 6.751 + 28.39$$

$$= ₹ 35.141$$

Question – 50

X Limited, just declared a dividend of ₹14.00 per share Mr. B is planning to purchase the share of X Limited, anticipating increase in growth rate from 8% to 9%, which will continue for three years. He also expects the market price of this share to be ₹ 360.00 after three years.

You are required to determine:

- (i) the maximum amount Mr. B should pay for shares, if he requires a rate of return of 13% per annum.



- (ii) the maximum price Mr. B will be willing to pay for share, if he is of the opinion that the 9% growth can be maintained indefinitely and require 13% rate of return per annum.
- (iii) the price of share at the end of three years, if 9% growth rate is achieved and assuming other conditions remaining same as in (ii) above.

Calculate rupee amount up to two decimal points.

	Year-1	Year-2	Year-3
FVIF @ 9%	1.090	1.188	1.295
FVIF @ 13%	1.130	1.277	1.443
PVIF @ 13%	0.885	0.783	0.693

(SM TYK – 05)

Solution:

(i) Expected dividend for next 3 years.

$$\text{Year 1 (D}_1\text{)} \quad ₹ 14.00 (1.09) \quad = ₹ 15.26$$

$$\text{Year 2 (D}_2\text{)} \quad ₹ 14.00 (1.09)^2 \quad = ₹ 16.63$$

$$\text{Year 3 (D}_3\text{)} \quad ₹ 14.00 (1.09)^3 \quad = ₹ 18.13$$

Required rate of return = 13% (K_e)

Market price of share after 3 years = (P₃) = ₹ 360

The present value of share

$$P_0 = \frac{15.26}{(1 + 0.13)} + \frac{16.63}{(1 + 0.13)^2} + \frac{18.13}{(1 + 0.13)^3} + \frac{360}{(1 + 0.13)^3}$$

$$P_0 = 15.26(0.885) + 16.63(0.783) + 18.13(0.693) + 360(0.693)$$

$$P_0 = 13.50 + 13.02 + 12.56 + 249.48$$

$$P_0 = ₹ 288.56$$

- (ii)** If growth rate 9% is achieved for indefinite period, then maximum price of share should Mr. A willing be to pay is

$$P_0 = \frac{D_1}{(1 - k_e)} = \frac{₹ 15.26}{(0.13 - 0.09)} = \frac{₹ 15.26}{0.04} = ₹ 381.50$$

- (iii)** Assuming that conditions mentioned above remain same, the price expected after 3 years will be:

$$P_3 = \frac{D_4}{k_e - g} = \frac{D_3(1.09)}{0.13 - 0.09} = \frac{18.13 \times 1.09}{0.04} = \frac{19.76}{0.04} = ₹ 494$$

Question – 51

XYZ Ltd. paid a dividend of ₹ 2 for the current year. The dividend is expected to grow at 40% for the next 5 years and at 15% per annum thereafter. The return on 182 days T-bills is 11% per annum and the market return is expected to be around 18% with a variance of 24%.

The co-variance of XYZ's return with that of the market is 30%. You are required to calculate the required rate of return and intrinsic value of the stock.

(SM TYK – 17)

Solution:

$$\text{Beta} = \frac{\text{Cov}_{xm}}{\sigma_m^2}$$

$$= \frac{30}{24} = 1.25$$

Required Rate of Return

$$\begin{aligned} K_e &= R_f + \beta (R_m - R_f) \\ &= 11 + 1.25 (18 - 11) \\ &= 19.75\% \end{aligned}$$

	Year	PVF (19.75%)	Amount	P.V.
Dividend				
2 (1.40) ¹	1	0.835	2.80	2.34
2.80 × 1.40	2	0.697	3.92	2.73
3.92 × 1.40	3	0.582	5.488	3.19
5.488 × 1.40	4	0.486	7.683	3.73
7.683 × 1.40	5	0.406	10.756	4.37
Terminal value (W.N.1)	5	0.406	260.41	105.77
			P₀	122.13

Terminal Value

$$\begin{aligned} P_5 &= \frac{D_6}{K_e - g} \\ &= \frac{10.756 (1.15)}{0.1975 - 0.15} \\ &= ₹ 260.41 \end{aligned}$$

Question – 52

Mr. A is thinking of buying shares at ₹ 500 each having face value of ₹ 100. He is expecting a bonus at the ratio of 1:5 during the fourth year. Annual expected dividend is 20% and the same rate is expected to be maintained on the expanded capital base. He intends to sell the shares at the end of seventh year at an expected price of ₹ 900 each. Incidental expenses for purchase and sale of shares are estimated to be 5% of the market price. He expects a minimum return of 12% per annum.

Should Mr. A buy the share? If so, what maximum price should he pay for each share? Assume no tax on dividend income and capital gain.

(SM TYK – 14 & RTP November – 2019)

Solution:

NPV

	Year	PVF (12%)	Amount	P.V.
Dividend				
₹ 100 × 20%	1	0.893	20.00	17.86
	2	0.797	20.00	15.94
	3	0.712	20.00	14.24
₹ 100 × 20% × 1.2 shares	4	0.636	24.00	15.26
	5	0.567	24.00	13.61
	6	0.507	24.00	12.17
	7	0.452	24.00	10.85
Sell Share 900 × 1.20 × 0.95	7	0.452	1026.00	463.75
			P.V. CI =	₹ 563.68

$$NPV = ₹ 563.68 - 500 \times 1.05$$

$$= ₹ 38.68$$



Since NPV is positive, hence share should be purchased.

$$\begin{aligned}\text{Maximum price} &= \frac{\text{₹ } 563.68}{1.05} \\ &= \text{₹ } 536.84\end{aligned}$$

Question – 53

Seawell Corporation, a manufacturer of do-it-yourself hardware and house wares, reported earnings per share of € 2.10 in 2003, on which it paid dividends per share of €0.69. Earnings are expected to grow 15% a year from 2004 to 2008, during this period the dividend payout ratio is expected to remain unchanged. After 2008, the earnings growth rate is expected to drop to a stable rate of 6%, and the payout ratio is expected to increase to 65% of earnings. The firm has a beta of 1.40 currently, and is expected to have a beta of 1.10 after 2008. The market risk premium is 5.5%. The Treasury bond rate is 6.25%.

- (a) What is the expected price of the stock at the end of 2008?
- (b) What is the value of the stock, using the two-stage dividend discount model?

(RTP May – 2019)

Solution:

Working Note 1: Calculation of EPS & EPS

	2004	2005	2006	2007	2008	2009
EPS (2.10)	2.415	2.777	3.194	3.673	4.224	4.477
DPS (0.69)	0.794	0.913	1.049	1.207	1.388	2.910

$$K_e = R_f + \beta (R_m - R_f)$$

Before 2008

$$K_e = 6.25 + 1.40 \times 5.5 = 13.95$$

After 2008

$$K_e = 6.25 + 1.10 \times 5.5 = 12.30$$

(1) Expected price at the end of 2008

$$\begin{aligned}P_5 &= \frac{D_6}{K_e - g} \\ &= \frac{2.910}{0.1230 - 0.06} = \text{€ } 46.19\end{aligned}$$

(2) Present Value of stock

Stage I

$$\begin{aligned}\text{P.V.} &= (0.794 \times 0.878) + (0.913 \times 0.770) + (1.049 \times 0.676) + (1.207 \times 0.593) + (1.388 \times 0.521) \\ &= \text{€ } 3.548\end{aligned}$$

Stage II

$$\begin{aligned}&= \text{€ } 46.19 \times 0.521 \\ &= \text{€ } 24.065 \\ P_0 &= \text{€ } 3.548 + \text{€ } 24.065 \\ &= \text{€ } 27.613\end{aligned}$$



SECURITY VALUATION

Question – 54

The current EPS of M/s VEE Ltd. is ₹ 4. The company has shown an extraordinary growth of 40% in its earnings in the last few years this high growth rate is likely to continue for the next 5 years after which growth rate in earnings will decline from 40% to 10% during the next 5 years and remain stable at 10% thereafter. The decline in the growth rate during the 5 years transition period will be equal and linear. Currently, the company's pay-out ratio is 10%. It is likely to remain the same for the next five years and from the beginning of the sixth year till the end of the 10th year, the pay-out will linearly increase and stabilize at 50% at the end of the 10th year. The post tax cost of capital is 17% and the PV factors are given below:

Years	1	2	3	4	5	6	7	8	9	10
PVIF @17%	0.855	0.731	0.625	0.534	0.456	0.390	0.333	0.285	0.244	0.209

You are required to Calculate the intrinsic value of the company's stock based on expected dividend. if the current market price of the stock is ₹ 125, suggest if it is advisable for the investor to invest in the company's stock or not.

(Exam November – 2019) (8 Marks)

Solution:

Calculation of EPS & DPS

	1	2	3	4	5	6	7	8	9	10
Growth	40%	40%	40%	40%	40%	34%	28%	22%	16%	10%

EPS (₹)	5.60	7.84	10.976	15.366	21.513	28.827	36.899	45.017	52.220	57.441
DPR	10%	10%	10%	10%	10%	18%	26%	34%	42%	50%
DPS	0.56	0.784	1.098	1.537	2.151	5.189	9.594	15.306	21.932	28.721
PVF	0.855	0.731	0.625	0.534	0.456	0.390	0.333	0.285	0.244	0.209

Stage I: Present value of dividend = ₹ 24.47

Stage II: Present value of terminal value

$$\begin{aligned}
 P_{10} &= \frac{D_{11}}{K_e - g} \\
 &= \frac{28.721 (1.10)}{0.17 - 0.10} \\
 &= ₹ 451.33 \times 0.209 \\
 &= 94.328 \\
 IV_0 &= 24.47 + 94.328 \\
 &= ₹ 118.798
 \end{aligned}$$

As Intrinsic Value of the share is lower than its selling price of ₹ 125, it is overpriced and can not be acquired.

Question – 55

An investor is considering to purchase the equity shares of LX Ltd., whose current market price (CMP) is ₹ 112. The company is proposing a dividend of ₹ 4 for the next year. LX Ltd. is expected to grow @ 20 per cent per annum for the next four years. The growth will decline linearly to 16 per cent per annum after first four years.



Thereafter, it will stabilize at 16 per cent per annum infinitely. The investor requires a return of 20 per cent per annum.

You are required

- To calculate the intrinsic value of the share of LX Ltd.
- Whether it is worth to purchase the share at this price.

Period	1	2	3	4	5	6	7
PVIF (20%,n)	0.833	0.694	0.579	0.482	0.402	0.335	0.279

(Exam November – 2020) (8 Marks)

Solution:

$$D_1 = ₹ 4$$

$$D_2 = ₹ 4 (1.20) = ₹ 4.80$$

$$D_3 = ₹ 4 (1.20)^2 = ₹ 5.76$$

$$D_4 = ₹ 4 (1.20)^3 = ₹ 6.91$$

$$D_5 = ₹ 6.91 (1.19) = ₹ 8.22$$

$$D_6 = ₹ 6.91 (1.19) (1.18) = ₹ 9.70$$

$$D_7 = ₹ 6.91 (1.19) (1.18) (1.17) = ₹ 11.35$$

$$D_8 = ₹ 6.91 (1.19) (1.18) (1.17) (1.16) = ₹ 13.17$$

$$TV_7 = \frac{D_8}{k_e - g} = \frac{13.17}{0.20 - 0.16} = ₹ 329.25$$

SECURITY VALUATION

$$P = \frac{4.00}{(1 + 0.20)} + \frac{4.80}{(1 + 0.20)^2} + \frac{5.76}{(1 + 0.20)^3} + \frac{6.91}{(1 + 0.20)^4} + \frac{8.22}{(1 + 0.20)^5} + \frac{9.70}{(1 + 0.20)^6} + \frac{11.35}{(1 + 0.20)^7} + \frac{329.25}{(1 + 0.20)^7}$$

$$= 4.00 \times 0.833 + 4.80 \times 0.694 + 5.76 \times 0.579 + 6.91 \times 0.482 + 8.22 \times 0.402 + 9.70 \times 0.335 + 11.35 \times 0.279 + 329.25 \times 0.279$$

$$(i) \text{ Intrinsic Value} = ₹ 114.91$$

- As Intrinsic Value of the share is higher than its selling price of ₹ 112, it is underpriced and can be acquired. However, other factors need to be taken into consideration since difference is only slightly higher.

Question – 56

SAM Ltd. has just paid a dividend of ₹ 2 per share and it is expected to grow @ 6% p.a. After paying dividend, the Board declared to take up a project by retaining the next three annual dividends. It is expected that this project is of same risk as the existing projects. The results of this project will start coming from the 4th year onward from now. The dividends will then be ₹ 2.50 per share and will grow @ 7% p.a.

An investor has 1,000 shares in SAM Ltd. and wants a receipt of at least ₹ 2,000 p.a. from this investment.

Show that the market value of the share is affected by the decision of the Board. Also show as to how the investor can maintain his target receipt from the investment for first 3 years and improved income thereafter, given that the cost of capital of the firm is 8%.



(SM TYK – 16)

Solution:

Market Price (P_0)

(i) If not accept the project

$$P_0 = \frac{D_1}{K_e - g}$$

$$= \frac{₹ 2 (1.06)}{0.08 - 0.06}$$

$$= ₹ 106$$

(ii) If accept the project

$$P_3 = \frac{D_4}{K_e - g}$$

$$= \frac{₹ 2.50}{0.08 - 0.07} = ₹ 250$$

$$P_0 = \frac{₹ 250}{(1.08)^3} = ₹ 198.46$$

Since value of share will increase hence accept the project.

If investor wants receipt of ₹ 2,000 p.a., then he should sell shares.

1st Year

$$P_1 = \frac{₹ 250}{(1.08)^2} = ₹ 214.33$$

$$\text{No. of shares} = \frac{₹ 2,000}{214.33} = 9.33 \text{ shares i.e. [10 shares]}$$

2nd Year

$$P_2 = \frac{₹ 250}{(1.08)^1} = ₹ 231.48$$

$$\text{No. of shares} = \frac{₹ 2,000}{₹ 231.48} = 8.64 \text{ shares i.e. [9 shares]}$$

3rd Year

$$P_3 = ₹ 250$$

$$\text{No. of shares} = \frac{₹ 2,000}{250} = 8 \text{ shares}$$

$$\text{Remaining shares} = 1,000 - 10 - 9 - 8$$

$$= 973 \text{ shares}$$

At the end of 3rd year, he would be having 973 shares valued @ ₹ 250 each i.e. ₹ 2,43,250. On these 973 shares, his dividend income for year 4 would be @ ₹ 2.50 i.e. ₹ 2,432.50.

So, if the project is taken up by the company, the investor would be able to maintain his receipt of at least ₹ 2,000 for first three years and would be getting increased income thereafter.

Question – 57

Piyush Loonker and Associates presently pay a dividend of Re. 1.00 per share and has a share price of ₹ 20.00.

- (i) If this dividend were expected to grow at a rate of 12% per annum forever, what is the firm's expected or required return on equity using a dividend-discount model approach?



SECURITY VALUATION

- (ii) Instead of this situation in part (i), suppose that the dividends were expected to grow at a rate of 20% per annum for 5 years and 10% per year thereafter. Now what is the firm's expected, or required, return on equity?

(SM TYK – 06)

Solution:

(1) Cost of Equity

$$\begin{aligned}K_e &= \frac{D_1}{P_0} + g \\&= \frac{1(1.12)}{20} + 0.12 \\&= 17.60\%\end{aligned}$$

(2) Required Rate of Return

Assume $K_e = 18\%$

$$\begin{aligned}P_0 &= \frac{1.20}{(1.18)^1} + \frac{1.440}{(1.18)^2} + \frac{1.728}{(1.18)^3} + \frac{2.074}{(1.18)^4} + \frac{2.488}{(1.18)^5} + \frac{2.488(1.10)}{0.18-0.10} \\&\quad \times \frac{1}{(1.18)^5} \\&= ₹ 20.21\end{aligned}$$

Assume $K_e = 20\%$

$$\begin{aligned}P_0 &= \frac{1.20}{(1.20)^1} + \frac{1.440}{(1.20)^2} + \frac{1.728}{(1.20)^3} + \frac{2.074}{(1.20)^4} + \frac{2.488}{(1.20)^5} + \frac{2.488(1.10)}{0.20-0.10} \\&\quad \times \frac{1}{(1.20)^5} \\&= ₹ 16.26\end{aligned}$$

18%	20.21.
20%	16.00
2%	4.21

$$K_e = 18 + \left(\frac{2}{4.21}\right) \times 0.21 = 18.10\%$$

Question – 58

An investor is considering purchasing equity shares of Alpha Ltd., whose current Market price is ₹ 172.45. The company is proposing a dividend of ₹ 6 for the year ending 31st march, 2024. Alpha Ltd. is expected to grow @ 20 percent per annum for the next four years. Thereafter, the growth, over the next three years, will decline linearly by 100 basis points per annum. Thereafter, it will stabilize at a certain growth rate per annum infinitely. The required rate of return for the investor is 20%.

Dividend value is to be taken in 2 decimal points only.

You are required:

- (i) To calculate the stable growth rate of Alpha Ltd. after the end of 7 years.
- (ii) To advise whether it is worth to purchase the share at this price if the investor has a stable target growth rate of 15% per annum.

Period	1	2	3	4	5	6	7
PVIF (20%, n)	0.83 33	0.694 4	0.578 7	0.482 3	0.401 9	0.334 9	0.279 1

(Exam May – 2023) (8 Marks)



Solution:

(i) Working Notes:

$$D_1 = ₹ 6$$

$$D_2 = 6 (1.20) = ₹ 7.20$$

$$D_3 = 6 (1.20)^2 = ₹ 8.64$$

$$D_4 = 6 (1.20)^3 = ₹ 10.37$$

$$D_5 = 10.37(1.19) = ₹ 12.34$$

$$D_6 = 10.37 (1.19) (1.18) = ₹ 14.56$$

$$D_7 = 10.37 (1.19) (1.18) (1.17) = ₹ 17.04$$

Price at the end of 7th year

Year	Dividend (₹)	PVF @ 20%	PV (₹)
1	6.00	0.8333	5.00
2	7.20	0.6944	5.00
3	8.64	0.5787	5.00
4	10.37	0.4823	5.00
5	12.34	0.4019	4.96
6	14.56	0.3349	4.88
7	17.04	0.2791	4.76
Total			34.60

Current Market Price	₹ 172.45
Less: PV of Dividends upto the year ending 7 th year	₹ 34.60
PV of Expected Market Price at the end of 7 th year	₹ 137.85

$$\text{Let } g \text{ be growth rate then: } 137.85 = \frac{17.04 (1 + g)}{0.20 - g} \times 0.2791$$

$$g = 0.16 \text{ i.e. } 16\%$$

Thus, the stable growth rate after the end of the 7 years shall be 16%.

(ii) Since growth rate is more than target growth rate it is worth to purchase the share.

BUY BACK DECISION

Question – 59

Rahul Ltd. has surplus cash of ₹ 100 lakhs and wants to distribute 27% of it to the shareholders. The company decides to buy back shares. The Finance Manager of the company estimates that its share price after re-purchase is likely to be 10% above the buyback price-if the buyback route is taken. The number of shares outstanding at present is 10 lakhs and the current EPS is ₹ 3.

You are required to determine:

- The price at which the shares can be re-purchased, if the market capitalization of the company should be ₹ 210 lakhs after buyback,
- The number of shares that can be re-purchased, and
- The impact of share re-purchase on the EPS, assuming that net income is the same.

(SM TYK – 18)

Solution:

(i) Amount Available For Dividend



$$= ₹ 100 \text{ lakh} \times 27\%$$

$$= ₹ 27 \text{ lakh}$$

Let Assume buy back price be x

No. of shares $\times$ MPS = Market Capitalization

$$\left(10 \text{ lakh} - \frac{27 \text{ lakh}}{x}\right) \times 1.10x = 210$$

$$11x - 29.70 = 210$$

$$x = ₹ 21.79$$

(ii) No. of Shares Bought Back

$$\text{No.} = \frac{27,00,000}{21.79} = 1,23,910 \text{ shares}$$

(iii) Impact on EPS Due to Buy Back

$$\text{EPS before buy back} = ₹ 3$$

$$\begin{aligned} \text{After buy back} &= \frac{₹ 30,00,000}{10,00,000 - 1,23,910} \\ &= ₹ 3.424 \end{aligned}$$

EPS increases to ₹ 3.424

Question – 60

Eager Ltd. has a market capitalization of ₹ 1,500 crores and the current market price of its share is ₹ 1,500. It made a PAT of 200 crores and the Board is considering a proposal to buy back 20% of the shares at a premium of 10% to the current market price. It plans to fund this through a 16% bank loan. You are required to calculate

SECURITY VALUATION

the post buy back Earnings Per Share (EPS). The company's corporate tax rate is 30%.

(MTP March – 2021)

Solution:

$$\text{No. of Shares} = \frac{1,500 \text{ Cr.}}{1,500}$$

$$= 1 \text{ Cr.}$$

$$\text{No. of shares buy back} = 1 \text{ Cr.} \times 20\%$$

$$= 0.20 \text{ Cr.}$$

$$\text{Buy Back Price} = ₹ 1,500 \times 1.10$$

$$= ₹ 1,650$$

$$\text{Amount of buy back} = ₹ 1,650 \times 0.2 \text{ Cr.}$$

$$= ₹ 330 \text{ Cr.}$$

$$\text{Interest} = ₹ 330 \text{ Cr.} \times 16\%$$

$$= 52.80 \text{ Cr.}$$

$$\text{Post Buy Back EPS} = \frac{200 \text{ Cr.} - 52.80 (1 - 0.30)}{0.8 \text{ Cr.}}$$

$$= ₹ 203.80$$

Question – 61

High Growth Ltd. (HGL) was having an excellent growth over a number of years. The Board of Directors is considering a proposal to reward its shareholders by buying back 20% shares at a premium.



The premium is to be paid by raising a loan from the Bank. The interest on loan is to be serviced by internal accruals as supported by the financials of HGL. The company has a market capitalization of ₹ 15,000 crore and the current Earnings Per Share (EPS) is ₹ 600 with a Price Earning Ratio (PER) of 25. The Board expects a post buy back Market Price per Share (MPS) of ₹ 10,000. The PER, post buy back will remain the same. The loan can be availed at an interest rate of 16% p.a.

Applicable corporate tax rate is 30%.

You are required to calculate:

- The interest amount which can be paid for availing the bank loan.
- The loan amount to be raised.
- Buy back premium per share.

(Exam May – 2023) (10 Marks)

Solution:

(i) Interest Amount

$$\begin{aligned}\text{MPS before buy back} &= \text{EPS} \times \text{P/E} \\ &= 600 \times 25 = ₹ 15,000\end{aligned}$$

No. of shares before buy back

$$\frac{₹ 15,000 \text{ Cr.}}{15,000} = 1 \text{ Cr.}$$

$$\text{No. of shares buy back} = 1 \text{ Cr.} \times 20\% = 0.2 \text{ Cr.}$$

SECURITY VALUATION

$$\begin{aligned}\text{Post buy back EPS} &= \frac{\text{MPS}}{\text{P/E}} \\ &= \frac{10,000}{25} = 400\end{aligned}$$

$$\begin{aligned}\text{PAT before buy back} &= \text{EPS} \times \text{No.} \\ &= 600 \times 1 \text{ Cr.} = ₹ 600 \text{ Cr.}\end{aligned}$$

Let assume interest amount be x

$$= \frac{600 - x(1 - 0.3)}{0.8} = 400 \text{ Cr.}$$

$$\text{Interest amount} = 400 \text{ Cr.}$$

$$\text{(ii) Loan Amount} = \frac{400}{16\%} = ₹ 2,500 \text{ Cr.}$$

(iii) Buy Back Premium per Share

$$\begin{aligned}\text{Premium per share} &= \frac{2,500 \text{ Cr.}}{0.2 \text{ Cr.}} \\ &= ₹ 12,500\end{aligned}$$

$$\text{Buy back price} = 12,500 + 15,000 = 27,500$$

ICAI SOLUTION:

- The interest amount which can be paid for availing the bank loan

$$\text{Current Market Price per Share} = ₹ 600 \times 25 = ₹ 15,000$$

$$\text{No. of Shares before Buyback} = \frac{\text{Market Capitalization}}{\text{Market Price of Share}}$$



$$= \frac{15,000 \text{ Crore}}{15,000} = 1 \text{ Crore}$$

No. of Shares proposed to Buyback = 20% of 1 crore = 20 lakh

Total No. of Share after Buyback = 1 crore – 20 lakh = 80 lakh

Post Buy back Market Price per Share = ₹ 10,000

PE Ratio = 25

$$\text{Post Buyback EPS} = \frac{10,000}{25} = ₹ 400$$

EAT before Buyback = ₹ 600 × 1 crore = ₹ 600 crore

$$\text{EBT before Buyback} = \frac{600}{(1 - 0.30)} = ₹ 857.1429 \text{ crore}$$

EAT after Buyback = ₹ 400.00 × 80 lakh = ₹ 320 crore

$$\text{EBT after Buyback} = \frac{320}{(1 - 0.30)} = ₹ 457.1429 \text{ crore}$$

Interest which can be paid for availing bank loan:

EBT before Buyback ₹ 857.1429 crore

(-) EBT after Buyback ₹ 457.1429 crore

₹ 400.0000 crore

Alternatively, it can also be computed as follows:

SECURITY VALUATION

Pre Buy back Market Capitalization (A)	₹ 15,000
Pre Buy back EPS (B)	crore
Pre Buy back PER (C)	₹ 600
Pre Buy back Market Price Per Share (₹ 600 × 5) D = B × C	25 ₹15,000
Pre Buy back No. of Shares (A)/ (D)	1 Crore
Post Buy back EPS (A) (₹ 10,000/ 25)	₹ 400
Post Buy back No. of shares (B)	80 Lakh
Post Buy back Earning (C) = (A) × (B)	₹ 320 crore
Pre Buy back Earning 1 Crore × ₹ 600 (D)	₹ 600 crore
Post Tax Earning available for interest payment (D) – (C)	₹ 280 Crore
Pre- Tax amount of Interest $\frac{280 \text{ Crore}}{1 - 0.30}$	₹ 400 Crore

(ii) Loan Amount raised = $\frac{400 \text{ Crore}}{0.16} = ₹ 2,500 \text{ Crore}$

(iii) Buyback Premium per Share

Amount of Loan for Buyback of 20 % Shares = ₹ 2,500 crore

No. of Shares Buyback = 20 Lakh

Buyback price per Share = ₹ 2,500 Crore/ 20 Lakh = ₹ 12,500

Market Price after Buyback = ₹ 10,000

Buyback Premium Per Share = ₹ 12,500 – ₹ 10,000 = ₹ 2,500

Alternatively, it can also be computed as follows:



Amount of Loan (A)	₹ 2,500 crore
No. of Shares to be bought back (B)	20 Lakh
Price Per Share to be paid (C) = (A) / (B)	₹ 12,500
Post Buy back Share Price (D)	₹ 10,000
Buy Back Premium per share (C) – (D)	₹ 2,500

VALUATION OF RIGHT

Question – 62

ABC Limited's shares are currently selling at ₹ 13 per share. There are 10,00,000 shares outstanding. The firm is planning to raise ₹ 20 lakhs to Finance a new project.

Required:

What are the ex-right price of shares and the value of a right, if

- The firm offers one right share for every two shares held.
- The firm offers one right share for every four shares held.
- How does the shareholders' wealth change from (i) to (ii)? How does right issue increases shareholders' wealth?

(SM TYK – 02)

Solution:

(i) One share for every 2 shares

$$\text{Right shares} = 10,00,000 \times \frac{1}{2} = 5,00,000 \text{ shares}$$

SECURITY VALUATION

$$\text{Offer price} = \frac{₹ 2,00,00,000}{5,00,000} = ₹ 4$$

$$\text{Ex-Right price} = \frac{(10,00,000 \times 13) + (5,00,000 \times 4)}{15,00,000} = ₹ 10$$

$$\text{Value of right per shares} = 13 - 10 = 3$$

$$\text{Value of right} = 3 \times 2 = ₹ 6$$

(ii) One share for every 4 shares

$$\text{One of shares} = 10,00,000 \times \frac{1}{4} = 2,50,000 \text{ shares}$$

$$\text{Offer price} = \frac{₹ 20,00,000}{2,50,000} = ₹ 8$$

$$\text{Ex-Right} = \frac{(10,00,000 \times 13) + (2,50,000 \times 8)}{12,50,000} = 12$$

$$\text{Value of right per shares} = 13 - 12 = ₹ 1$$

$$\text{Value of right} = ₹ 1 \times 4 \text{ shares} = ₹ 4$$

$$\text{Before right} = 100 \text{ share} \times 13 = ₹ 1,300$$

One share for 2 shares

$$\text{Value of shares } [150 \times 10] = ₹ 1,500$$

$$\begin{aligned} (-) \text{ Buy right shares } [50 \times 4] &= ₹ 200 \\ \hline &= ₹ 1,300 \end{aligned}$$



One share for 4 shares

$$\begin{array}{rcl}
 \text{Value of shares } (125 \times 12) & = & ₹ 1,500 \\
 (-) \text{ Buy right shares } (25 \times 8) & = & ₹ 200 \\
 \hline
 & = & ₹ 1,300
 \end{array}$$

No change in wealth.

Question – 63

AMKO Limited has issued 75,000 equity shares of ₹ 10 each. The current market price per share is ₹ 36. The company has a plan to make a rights issue of one new equity share at a price of ₹ 24 for every four shares held.

You are required to:

- (i) Calculate the theoretical post-rights price per share.
- (ii) Calculate the theoretical value of the right alone.

(Exam November – 2018) (4 Marks)

Solution:**(i) Calculation of theoretical Post-rights (ex-right) price per share:**

$$\text{Ex-right value} = \left[\frac{MN + SR}{N + R} \right]$$

Where,

M = Market price,

N = Number of old shares for a right share

S = Subscription price

R = Right share offer

$$\begin{aligned}
 &= \left[\frac{₹ 36 \times 4 + ₹ 24 \times 1}{4 + 1} \right] \\
 &= ₹ 33.60
 \end{aligned}$$

(ii) Calculation of theoretical value of the rights alone:

= Ex-right price – Cost of rights share

$$= ₹ 33.60 - ₹ 24 = ₹ 9.60$$

Or,

$$= \frac{₹ 33.60 - ₹ 24}{4} = ₹ 2.40$$

Question – 64

Aggressive Ltd. is proposing to fund its expansion plan of ₹ 12 crore by making a rights issue. The current market price (CMP) is ₹ 40. The Board is willing to offer a discount of 20% on the CMP for the rights issue. The Board is also desirous that the fall in Ex-right price of the shares be restricted to 10% of CMP.

CALCULATE:

- (1) The number of new equity shares to be offered for each rights held,
- (2) Theoretical value of right and
- (3) The total number of equity shares to be issued.

(MTP Oct – 2022)



Solution:**(1) Let assume number of right shares be x**

$$\frac{(1 \times 40) + (x \times 32)}{x + 1} = 36$$

$$36 + 36x = 40 + 32x$$

$$x = 1$$

It means 1:1

(2) Value of right

$$= \text{CMP} - \text{Ex right}$$

$$= 40 - 36 = 4$$

(3) No. of shares to be issued

$$= \frac{1,20,00,000}{32} = 0.375 \text{ Cr.}$$

MONEY MARKET INSTRUMENTS**Question – 65**

Suppose Govt. Pays ₹ 5,000 at maturity for 91 days Treasury bill. If Mr. Y is desirous to earn an annualized discount rate of 3.5%, then how he can pay for it.

Solution:

Suppose X be the maximum amount Mr. Y can pay for Treasury bill. Then,

$$= \frac{₹ 5,000 - X}{₹ 5,000} \times \frac{360}{91} = 0.035$$

$$= ₹ 5,000 - X$$

$$= ₹ 44.24$$

$$X = ₹ 4,955.76$$

Question – 66

RBI sold a 91-day T-bill of face value of ₹ 100 at an yield of 6%. What was the issue price?

(SM TYK – 28)**Solution:**

Let the issue price be X

By the terms of the issue of the T-bills:

$$6\% = \frac{100 - x}{x} \times \frac{365}{91} \times 100$$

$$\frac{6 \times 91 \times X}{36,500} = (100 - x)$$

$$0.01496x = 100 - x$$

$$x = \frac{100}{1.01496} = ₹ 98.53$$

Question – 67

Wonderland Limited has excess cash of ₹ 20 lakhs, which it wants to invest in short term marketable securities. Expenses relating to investment will be ₹ 50,000.



The securities invested will have an annual yield of 9%

The company seeks your advice

- (i) as to the period of investment so as to earn a pre-tax income of 5%. (discuss)
- (ii) the minimum period for the company to breakeven its investment expenditure overtime value of money.

(SM TYK – 29)

Solution:

(i) Pre-tax Income required on investment of ₹ 20,00,000

Let the period of Investment be 'P' and return required on investment ₹ 1,00,000 (₹ 20,00,000 × 5%)

Accordingly,

$$(\text{₹ } 20,00,000 \times \frac{9}{100} \times \frac{P}{12}) - \text{₹ } 50,000 = \text{₹ } 1,00,000$$

$$P = 10 \text{ months}$$

(ii) Break-Even its investment expenditure

$$(\text{₹ } 20,00,000 \times \frac{9}{100} \times \frac{P}{12}) - \text{₹ } 50,000 = 0$$

$$P = 3.33 \text{ months}$$

Question – 68

A bond is held for period of 45 days. The current discount yield is 6 per cent per annum. It is expected that current yield will increase by 200 basis points and current market price will come down by ₹ 2.50.

Calculate :

- i Face value of the Bond and
- ii. Bond Equivalent yield

(Exam May – 2019) (4 Marks)

Solution:

(i) Face Value of the Bond

$$(6\%) \text{ 45 days discount yield} = 6 \times \frac{45}{360} = 0.75$$

$$(8\%) \text{ 45 days discount yield} = 8 \times \frac{45}{360} = 1.00$$

$$\text{Change in discount yield} = 1 - 0.75 = 0.25\%$$

$$\text{Change in bond price} = \text{₹ } 2.50$$

$$\text{Hence, face value of bond} = \frac{\text{Change in Price}}{\text{Change in Discount Yield}}$$

$$= \frac{\text{₹ } 2.50}{2.5\%}$$

$$= \text{₹ } 1,000$$

(ii) Bond Equivalent Yield

Current market price of bond

$$\text{If discount yield is 6\% p.a. then } \text{₹ } 1,000 - 0.75\% = \text{₹ } 992.50$$

$$\text{If discount yield is 8\% p.a. then } \text{₹ } 1,000 - 1.00\% = \text{₹ } 900.00$$



At the rate of 6%	$\frac{1,000 - 992.50}{992.50} \times \frac{360}{45} \times 100 \uparrow$	6.05
At the rate of 8%	$\frac{1,000 - 990.00}{990.00} \times \frac{360}{45} \times 100 \uparrow$	8.08

Question – 69

From the following particulars, calculate the effective rate of interest p.a. as well as the total cost of funds to Bhaskar Ltd., which is planning a CP issue:

Issue Price of CP	₹ 97,550
Face Value	₹ 1,00,000
Maturity Period	3 Months
Issue Expenses:	
Brokerage	0.15% for 3 months
Rating Charges	0.50% p.a.
Stamp Duty	0.175% for 3 months

(MTP October – 2020)

Solution:

$$\text{Effective Interest} = \left(\frac{F - P}{P} \right) \times \frac{12}{M} \times 100$$

Substituting the given values of F, P and M we get,

$$\text{Effective Interest} = \left(\frac{1,00,000 - 97,550}{97,550} \right) \times \frac{12}{3} \times 100 = 10.05\%$$

SECURITY VALUATION

Cost of funds to the company

Effective interest rate	= 10.05%
Brokerage (0.15 × 4)	= 0.6%
Rating charges	= 0.5%
Stamp duty (0.175 × 4)	= 0.75%
Total cost of Funds to Bhaskar Ltd.	= 11.9% p.a.

Question – 70

Bank A enter into a Repo for 14 days with Bank B in 10% Government of India Bonds 2018 @ 5.65% for ₹ 8 crore. Assuming that clean price be ₹ 99.42 and initial margin be 2% and days of accrued interest be 262 days. You are required to determine.

- Dirty Price
- Repayment at maturity. (Consider 360 days in a year)

(SM TYK – 31, MTP March & April – 2021)

Solution:

(i) Dirty Price

= Clean Price + Interest Accrued

$$= 99.42 + 100 \times \frac{10}{100} \times \frac{262}{360} = 106.70$$

(ii) First Leg (Start Proceed)



$$= \text{Nominal Value} \times \frac{\text{Dirty Price}}{100} \times \frac{100 - \text{Initial Margin}}{100}$$

$$= ₹ 8,00,00,000 \times \frac{106.70}{100} \times \frac{100 - 2}{100} = ₹ 8,36,52,800$$

Second Leg (Repayment at Maturity)

$$= \text{Start Proceed} \times \left(1 + \text{Repo Rate} \times \frac{\text{No. of Days}}{360} \right)$$

$$= ₹ 8,36,52,800 \times \left(1 + 0.0565 \times \frac{14}{360} \right)$$

$$= ₹ 8,38,36,604$$

Question – 71

The Bank BK enters into a Repo for 9 days with Bank NE in 6% Government bonds 2022 for an amount of ₹ 2 crore. The other relevant details are as follows:

First Leg Payment (Start Proceed)	₹ 2,00,06,750
Second Leg Payment (Repayment Proceed)	₹ 2,00,31,759
Initial Margin	1.25%
Days of accrued interest	240

Assume 360 days in a year.

CALCULATE:

- (1) Repo Rate
- (2) Dirty Price and
- (3) Clean Price

(MTP Oct – 2022)

Solution:

(1) Second Leg = Start Proceed $\times \left(1 + \text{Repo Rate} \times \frac{\text{No. of days}}{360} \right)$

$$₹ 2,00,31,759 = ₹ 2,00,06,750 \times \left(1 + \text{Repo Rate} \times \frac{9}{360} \right)$$

$$1.00125 = \left(1 + \text{Repo Rate} \times \frac{9}{360} \right)$$

$$\text{Repo Rate} = 0.05 = 5\%$$

(2) First Leg (Start Proceed)

$$= \text{Nominal Value} \times \frac{\text{Dirty Price}}{100} \times \frac{100 - \text{Initial Margin}}{100}$$

$$₹ 2,00,06,750 = ₹ 2,00,00,000 \times \frac{\text{Dirty Price}}{100} \times \frac{100 - 1.25}{100}$$

$$10003.375 = 98.75 \times \text{Dirty Price}$$

$$\text{Dirty Price} = ₹ 101.30$$

(3) Dirty Price = Clean Price + Interest Accrued

$$101.30 = \text{Clean Price} + 100 \times \frac{240}{360} \times 6\%$$



Clean Price = ₹ 97.30

RESIDUAL

Question – 72

Following Financial data are available for PQR Ltd. for the year 2008:

	(₹ in lakh)
8% debentures	125
10% bonds (2007)	50
Equity shares (₹ 10 each)	100
Reserves and Surplus	300
Total Assets	600
Assets Turnovers ratio	1.1
Effective interest rate	8%
Effective tax rate	40%
Operating margin	10%
Dividend payout ratio	16.67
Current market Price of Share	14
Required rate of return of investors	15%

You are required to:

- Draw income statement for the year
- Calculate its sustainable growth rate of earnings
- Calculate the fair price of the Company's share using dividend discount model, and
- What is your opinion on investment in the company's share at current price?

(SM TYK – 12 & RTP May – 2020)

Solution:

(i) Income Statement

Sales (600 × 1.10)	₹ 660
(-) Operating Ratio (660 × 90%)	₹ 594
EBIT	₹ 66
(-) Interest [(125 + 50) × 8%]	₹ 14
EBT	₹ 52
(-) Tax 40%	₹ 20.80
EAT	31.20
(÷) No. of Shares	10
EPS	3.12
DPS (16.67%)	0.52

(ii) Sustainable Growth Rate

$$g = b \times r$$

$$b = 1 - 0.1667 = 0.8333$$

$$\begin{aligned} \text{ROE (r)} &= \frac{\text{EAT}}{\text{Equity}} \times 100 \\ &= \frac{31.20}{300 + 100} \times 100 \end{aligned}$$



$$= 7.8 \%$$

$$g = 0.8333 \times 0.078 = 0.065$$

$$= 6.5\%$$

(iii) Price (P_0)

$$\begin{aligned} P_0 &= \frac{D_1}{K_e - g} \\ &= \frac{0.52 \times (1.065)}{0.15 - 0.065} \\ &= ₹ 6.52 \end{aligned}$$

(iv) Actual Price = 14

Share is overpriced hence should not be purchased.

Question – 73

Tiger Ltd. is presently working with an Earning Before Interest and Taxes (EBIT) of ₹ 90 lakhs. Its present borrowings are as follows:

	₹ in lakhs
12% term loan	300
Working capital borrowings:	
From Bank at 15%	200
Public Deposit at 11%	100

The sales of the company are growing and to support this, the company proposes to obtain additional borrowing of ₹ 100 lakhs expected to cost 16%. The increase in EBIT is expected to be 15%.

Calculate the change in interest coverage ratio after the additional borrowing is effected and comment on the arrangement made.

Solution:

$$\text{Interest Coverage Ratio} = \frac{\text{EBIT}}{\text{Interest}}$$

$$\begin{aligned} \text{Present} &= \frac{90}{77} \\ &= 1.69 \end{aligned}$$

$$\begin{aligned} \text{Proposed} &= 90 \times 1.15 \\ &= 103.50 \end{aligned}$$

$$\begin{aligned} \text{Interest} &= 77 + 100 \times 16 \% \\ &= 93 \end{aligned}$$

$$\text{Interest Coverage} = \frac{103.50}{93} = 1.113$$

Since interest coverage ratio is decreasing hence proposal should be accepted.

Question – 74

Capital structure of Sun Ltd., as at 31.3.2003 was as under:

	(₹ lakhs)
Equity share capital	80
8% Preference share capital	40
12% Debentures	64

Reserves

32

Sun Ltd., earns a profit of ₹ 32 lakhs annually on an average before deduction of income-tax, which works out to 35%, and interest on debentures.

Normal return on equity shares of companies similarly placed is 9.6% provided:

- (a) Profit after tax covers fixed interest and fixed dividends at least 3 times.
- (b) Capital gearing ratio is 0.75.
- (c) Yield on share is calculated at 50% of profits distributed and at 5% on undistributed profits.

Sun Ltd., has been regularly paying equity dividend of 8%.

Compute the value per equity share of the company.

- (i) 1% for every time of difference for interest and fixed dividend coverage ratio.
- (ii) 2% for every time of difference for capital gearing ratio.

(SM TYK – 07)

Solution:

Income Statement

EBIT	32,00,000
(-) Interest (64,00,000 × 12 %)	<u>7,68,000</u>
EBT	24,32,000

SECURITY VALUATION

(-) Tax @ 35%	8,51,200
EAT	<u>15,80,800</u>
(-) PD (40,00,000 × 8 %)	<u>3,20,000</u>
Earning	12,60,800
Dividend (8,000,000 × 8%) (Distributed Profit)	<u>6,40,000</u>
Undistributed Profit	6,20,800
Yield (Income)	= (6,40,000 × 50%) + (6,20,800 × 5%) = 3,51,040

Required Rate of Return after Risk Adjustment

(1) Coverage Ratio

$$\begin{aligned} &= \frac{\text{PAT} + \text{INTEREST}}{\text{INTEREST} + \text{PD}} \\ &= \frac{15,50,800 + 7,68,000}{7,68,000 + 3,20,000} \\ &= 2.16 \end{aligned}$$

Coverage ratio is less than 3 times it means higher risk.

(2) Capital Gearing Ratio

$$\begin{aligned} &= \frac{\text{PSC} + \text{Debt}}{\text{ESC} + \text{R \& S}} \\ &= \frac{40,00,000 + 64,00,000}{80,00,000 + 32,00,000} \end{aligned}$$



$$= 0.93$$

Capital gearing ratio is more than 0.75. It means higher risk.

Required Yield

$$\text{Similar Yield} \quad 9.6\%$$

(+) Risk Adjustment

Coverage Ratio

$$(3 - 2.16) = 0.84 \times 1 \quad 0.84\%$$

(+) Risk Adjustment for Capital Gearing Ratio

$$(0.93 - 0.75) = 0.18 \times 2 \quad 0.36\%$$

$$10.80\%$$

Value per Share

$$P_0 = \frac{\text{Yield on Equity}}{\text{Required Yield}} \times \text{per share}$$

$$\text{Yield} = \frac{3,51,040}{8,00,000} \times 100 = 4.368\%$$

$$P_0 = \frac{4.388\%}{10.80\%} \times 100 = ₹ 40.63 \quad \text{Assume} = ₹ 100 \text{ (FV)}$$

Question – 75

AB Industries has Equity Capital of ₹ 12 Lakhs, total Debt of ₹ 8 Lakhs, and annual sales of ₹ 30 Lakhs. Two mutually exclusive proposals are under consideration for the next year. The details of the proposals are as under:

SECURITY VALUATION

Particulars	Proposal No. 1	Proposal No. 2
Target Assets to Sales Ratio	0.65	0.62
Target Net Profit Margin (%)	4	5
Target Debt Equity Ratio (DER)	2:3	4:1
Target Retention Ratio (of Earnings) (%)	75	-
Annual Dividend (₹ In Lakhs)	-	0.30
New Equity Raised (₹ in Lakhs)	-	1

You are required to calculate sustainable growth rate for both the proposals.

(Exam November – 2020) (8 Marks)

Solution:

Proposal I:

$$\text{ROE} = 4 \times \frac{1}{0.65} \times \frac{5}{3}$$

$$= 10.26\%$$

$$g = b \times r$$

$$= 0.75 \times 0.1026$$

$$= 0.0770$$

$$= 7.70\%$$



Proposal II:

$$\text{ROE} = 5 \times \frac{1}{0.65} \times \frac{5}{1}$$

$$= 40.32\%$$

$$\text{Target Equity} = 12 + 1 = 13 \text{ lakh}$$

$$\text{Debt} = 13 \times 4 = 52 \text{ lakh}$$

$$\text{Total Asset} = 13 \text{ lakh} + 52 \text{ lakh} = 65 \text{ lakh}$$

$$\text{TA to Sales} = \frac{65}{0.62} = 104.84$$

$$\text{Net Profit} = 104.84 \times 5\%$$

$$= 5.242$$

$$b = \frac{5.242 - 0.30}{5.242} \times 100$$

$$= 0.943$$

$$g = 0.943 \times 0.4032$$

$$= 38\%$$

MULTIPLE CHOICE QUESTIONS

Case Scenario – 01

Bank A is in need of fund for a period of 14 days. To meet this financial need on 20th September 2023 Bank A enters into an agreement with Bank B under which it will sell 10% Government of India Bonds issued on 1st January 2023 @ 5.65% for ₹ 8 crore (Face value is ₹ 10,000 per Bond).

The clean price of same Bond is ₹ 9,942 and the Initial Margin be 2% and the maturity date of Bond is 31st December 2028. Consider 360 days in a year and interest is payable annually.

Based on above Case Scenario, answer the following questions:

I. The arrangement entered between Bank A and Bank B will be called

- (a) Call Money Arrangement
- (b) Commercial Bill Arrangement
- (c) Commercial Paper
- (d) Repurchase Option

II. Dirty Price of the Bond will approximately be.....

- (a) ₹ 10,353
- (b) ₹ 10,670
- (c) ₹ 10,499
- (d) ₹ 10,816



III. The start proceeds of the transaction shall be approximately

- | | |
|-------------------|-------------------|
| (a) ₹ 8,38,36,804 | (b) ₹ 8,36,53,000 |
| (c) ₹ 8,36,52,800 | (d) ₹ 8,48,52,585 |

IV. The second leg of the transaction shall be approximately.....

- | | |
|-------------------|-------------------|
| (a) ₹ 8,38,36,604 | (b) ₹ 8,36,53,000 |
| (c) ₹ 8,58,36,804 | (d) ₹ 8,48,52,585 |

V. The amount of Accrued Interest per Bond shall be approximately

- | | |
|-----------|-----------|
| (a) ₹ 728 | (b) ₹ 720 |
| (c) ₹ 734 | (d) ₹ 714 |

(MTP September – 2024)

Answer: Case Scenario – 01

- I. (d) Repurchase Option**
- II. (b) ₹ 10,670**
- III. (c) ₹ 8,36,52,800**
- IV. (a) ₹ 8,38,36,604**
- V. (a) ₹ 728**

Case Scenario – 02

Suppose you are a financial consultant and following 3 clients have approached to you seeking advise on the investment to be made in securities. All these clients have different background and risk appetite as well as perception to the market.

- ❖ Client A wants to invest in Fixed income avenues and therefore he is looking at the credit rating of the securities as well as financial ratios such as interest coverage, earning power etc and the general prospect of the industry.
- ❖ Client B wants to earn a fixed income over a period of time by holding the security till its maturity.
- ❖ Client C wants to earn more by taking more risk. Therefore, he is more interested to invest in stocks. He believes that Price reflects all information found in the record of past prices and volumes.

On the basis of above information, choose the most appropriate answer to the MCQs.

- I.** The main factor to be considered in selecting fixed income avenue for client A shall be.....
- | | |
|-----|-------------------|
| (a) | Yield to maturity |
| (b) | Risk of Default |
| (c) | Tax Shield |
| (d) | Liquidity |

SECURITY VALUATION

II. The main factor that have to be evaluated in the selection of Bond for Client B shall be.....

- (a) Yield to maturity
- (b) Risk of Default
- (c) Tax Shield
- (d) Liquidity

III. If Weak form efficiency is prevailing in the market then which approach is best for selection of Equity Shares?

- (a) Technical Analysis
- (b) Fundamental Analysis
- (c) Random selection Analysis
- (d) None of the above.

(MTP March – 2024)

Answer: Case Scenario – 02

- I. (b) Risk of Default**
- II. (a) Yield to maturity**
- III. (b) Fundamental Analysis**

Case Scenario – 03

The data given below relates to a convertible bond of x Ltd:

Face value	₹ 450
Coupon rate	15%

No. of share per bond	25
Market price of share	₹ 20
Straight value of bond	₹ 400
Market price of convertible bond	₹ 550

Based on the above information answer the following questions:

I. The stock value of bond would be _____

Answer-1 : ₹ 500

Answer-2 : ₹ 400

Answer-3 : ₹ 550

Answer-4: ₹ 450

II. The percentage of downside risk based on market price of convertible bond is _____

Answer-1 : 10 %

Answer-2 : 27.27%

Answer-3 : 18.18%

Answer-4: 11.11%

III. The conversion premium is _____

Answer-1 : 10 %

Answer-2 : 27.27%

Answer-3 : 18.18%

Answer-4: 11.11%

IV. The conversion parity price of the stock is _____

Answer-1 : ₹ 25

Answer-2 : ₹ 20

Answer-3: ₹ 22

Answer-4: ₹ 24



(MTP December – 2025)**Answer: Case Scenario – 03**

- I. Answer-1: ₹ 500**
- II. Answer-2: 27.27%**
- III. Answer-1: 10 %**
- IV. Answer-3: ₹ 22**

Case Scenario – 04

You are an investment analyst working for a financial advisory firm. You have been asked to analyze the bond market's yield curve to assist your clients in making investment decisions. The yield curve represents the relationship between the interest rates (yield) and the time to maturity for debt securities, usually government bonds.

For simplicity, assume the following yield data for government bonds over various maturities (measured in years):

Yield Curve Table

Maturity (Years)	Yield (%)
1 Years	3.00%
2 Years	4.00%
3 Years	5.00%
5 Years	6.00%
7 Years	6.40%
10 Years	7.00%
15 Years	7.40%
30 Years	7.60%

Based on above case scenario answer the following questions:

- I.** The main characteristic of a normal yield curve is.....

- (a) Short-term yields are higher than long-term yields.
- (b) Short-term yields are lower than long-term yields.
- (c) Yields remain the same across all maturities.
- (d) Yields fluctuate randomly over different maturities.

- II.** Based on the revised yield data, what is the yield spread between the 10- year bond and the 1-year bond?

- (a) 2.0% (b) 3.5%
- (c) 4.0% (d) 5.0%

- III.** An inverted yield curve typically indicates.....

- (a) Economic growth
- (b) Economic uncertainty
- (c) An upcoming recession
- (d) Inflationary pressure

- IV.** If an investor is looking to invest for 2 years starting 3 years from now, the forward rate he would expect shall be.....

- (a) 7.41% (b) 7.52%
- (c) 7.76% (d) 7.93%



V. If an investor is looking to invest for 2 years starting 5 years from now, the forward rate he would expect shall be.....

- | | |
|-----------|-----------|
| (a) 7.41% | (b) 7.52% |
| (c) 7.76% | (d) 7.93% |

(MTP October – 2024 & July – 2025)

Answer: Case Scenario – 04

- | | | |
|-------------|------------|---|
| I. | (b) | Short-term yields are lower than long-term yields. |
| II. | (c) | 4.0% |
| III. | (c) | An upcoming recession |
| IV. | (b) | 7.52% |
| V. | (a) | 7.41% |

Case Scenario – 05

XYZ Ltd. is in need of funds for a short tenure. Some functional level manager suggested about the Bank Loan option. On conforming from Finance Department, it was found that company exhausted its bank loan limited due to recent huge Capex. Then CA X, CFO suggested the idea of floating Commercial papers by XYZ Ltd.

Accordingly, XYZ Ltd. is planning to issue Commercial Paper (CP), the details of which is given below:

Issue Price of CP	₹ 97,550
Face Value	₹ 1,00,000

SECURITY VALUATION

Maturity Period	3 Months
<u>Issue Expense</u>	
Brokerage	0.15%for 3 months
Rating charges	0.50% p.a.
Stamp Duty	0.175% for 3 months

Based on above case scenario answer the following questions:

- I.** The Bond Equivalent yield of the same Commercial Paper shall be approximately.....
- | | |
|-----------|------------|
| (a) 2.51% | (b) 10.05% |
| (c) 7.53% | (d) 11.05% |
- II.** The Effective Interest Rate per annum of same CP shall approximately be.....
- | | |
|------------|------------|
| (a) 10.44% | (b) 10.05% |
| (c) 2.51% | (d) 11.05% |
- III.** The total cost of funds to the company shall approximately be.....
- | | |
|------------|------------|
| (a) 11.27% | (b) 11.85% |
| (c) 12.24% | (d) 10.88% |



SECURITY VALUATION

IV. Which of the following instruments cannot be used by a bank to meet its short-term funding requirements?

- (a) Call/Notice Money
- (b) Commercial Paper
- (c) Certificate of Deposit
- (d) Repurchase Agreement (Repo)

V. The period of Commercial Paper ranges from.....

- (a) 14 days to 364 days
- (b) 3 months to 6 months
- (c) 7 days to 1 year
- (d) 1 year to 3 years

Answer: Case Scenario – 05

- I. (b) 10.05%**
- II. (a) 10.44%**
- III. (c) 12.24%**
- IV. (b) Commercial Paper**
- V. (c) 7 days to 1 year**

Case Scenario – 06

Following Financial data are available for PQR Ltd. for the financial year ending 2023:

	(₹ in lakh)
8% Debentures	125
10% Bonds (2022)	50
Equity Shares (₹ 10 each)	100
Reserves and Surplus	300
Total Assets	600
Assets Turnovers ratio	1.1
Effective interest rate	8%
Effective tax rate	40%
Operating margin	10%
Dividend payout ratio	16.67%
Current market Price of Share	₹14
Required rate of return of investors	15%

From the information given above, choose the correct answer to the following questions:

I. 10% Bonds must have issued in the month of.....

- (a) May 2022
- (b) June 2022
- (c) July 2022
- (d) August 2022

II. Amount of retained earning for the financial year 2023 approximately is.....

- (a) 52.00 lakh
- (b) 31.20 lakh
- (c) 26.00 lakh
- (d) 5.20 lakh

III. Return on Equity (ROE) of PQR Ltd. is.....

- (a) 15.00%
- (b) 6.50%



- (c) 10.00% (d) 7.80%

IV. Sustainable Growth Rate of PQR Ltd. shall be approximately.....

- (a) 15.00% (b) 6.50%
(c) 10.00% (d) 7.80%

V. Fair price of share of PQR Ltd. using Dividend Discount Model shall be approximately.....

- (a) 10 (b) 14
(c) 6.12 (d) 6.51

Answer: Case Scenario – 06

- I. (b) June 2022**
II. (c) 26.00 lakh
III. (d) 7.80%
IV. (b) 6.50%
V. (d) 6.51

Case Scenario – 07

Z Ltd. paid a dividend of ₹ 5 for the current year. The dividend is expected to grow at 25% for the next 6 years and at 10% per annum thereafter. The return of government bond is 13% per annum and market return is expected to be around 20%. The correlation between market return and Z Ltd. share return is 0.3733. The standard

deviation of market return and Z Ltd. shares is 12% and 18% respectively.

Round off to two decimal places.

From the information given above, choose the correct answer to the following questions:

MULTIPLE CHOICE QUESTIONS

I. What is the expected return of Z Ltd shares?

- (a) 15% (b) 23.92%
(c) 16.92% (d) 16.5%

II. The intrinsic value of Z Ltd. shares approximately is.....

- (a) ₹ 156.75 (b) ₹ 303.14
(c) ₹ 349.62 (d) ₹ 341.30

III. If current market price of the shares is ₹ 315 than stock is.....

- (a) Over valued
(b) Under valued
(c) Fairly valued
(d) Cannot be determined

(EXAM NOVEMBER 2024)



SECURITY VALUATION

Answer: Case Scenario – 07

- I. (c) 16.92%
- II. (a) ₹ 156.75
- III. (a) Over valued

Case Scenario – 08

The following information is available in respect of Bond A and Bond B.

	Bond A	Bond B
Face value, redeemable at par	₹1000	₹ 1000
Coupon rate, payable annually (%)	6%	10%
Time to maturity (years)	5	3

An investor has the portfolio consisting of 75% of Bond A and 25% of Bond B. The current YTM's prevailing in the market is 10%.

Year (n) :	1	2	3	4	5
PVIF (10%,n) :	0.9091	0.8264	0.7513	0.6830	0.6209

From the information given above, choose the correct answer to the following questions:

MULTIPLE CHOICE QUESTIONS

- I. The price and duration of the Bond B shall be approximatelyrespectively.
- (a) ₹ 826.43 and 2.49 (b) ₹ 1,000 and 2.74
- (c) ₹ 924.85 and 2.74 (d) ₹ 1,000 and 2.49

- II. The price and duration of the Bond A shall be approximatelyrespectively.
- (a) ₹ 848.34 and 4.41
- (b) ₹ 811.09 and 4.38
- (c) ₹ 1,227.44 and 4.41
- (d) ₹ 658.15 and 3.90
- III. The price sensitivity of the portfolio approximately is.....
- (a) -4.03
- (b) -2.49
- (c) -3.63
- (d) -3.98

(EXAM NOVEMBER 2025)

Answer: Case Scenario – 08

- I. (b) ₹ 1,000 and 2.74
- II. (a) ₹ 848.34 and 4.41
- III. (d) - 3.98

Case Scenario – 09

XYZ Ltd., a medium-sized company in the renewable energy sector, is experiencing steady sales growth. The company's management, however, is concerned about balancing rapid growth with long-term



sustainability. In the past year, XYZ's growth objectives have led to aggressive expansion plans, but management now realizes that such growth might not be financially sustainable in the long run. This raises concerns about how to maintain the company's financial health while meeting its ambitious growth targets.

The CFO of XYZ Ltd. highlights the importance of Sustainable Growth Rate (SGR).

The company now needs to ensure that its operational and financial policies align with its growth goals. XYZ must avoid expanding too quickly, which could strain its financial resources and lead to excessive borrowing. Moreover, management must also consider the long-term implications of resource consumption, particularly in the renewable energy industry, where sustainability is key to both current and future stakeholders.

XYZ Ltd. also realizes that it needs to focus on building its growth capability alongside its growth strategy. Without the necessary infrastructure and financial planning in place, the company's efforts to achieve long-term, sustainable growth could be in jeopardy. Furthermore, the company is aware of the risks of relying too much on external financing and recognizes the need for a balance between maintaining sufficient equity and minimizing debt.

Given the importance of these considerations, XYZ's management team must now review their growth strategy and financial policies to ensure they are consistent with the firm's sustainable growth objectives.

From the information given above, choose the correct answer to the following questions:

I. The concept of Sustainable Growth Rate introduced by.....

- (a) Harry Markowitz
- (b) William Sharpe
- (c) Black Scholes
- (d) Robert C. Higgins

II. The Sustainable Growth Rate (SGR) represents.....

- (a) the rate at which the company can grow by issuing more equity.
- (b) the maximum rate of growth in sales that can be achieved without borrowing additional funds.
- (c) the growth rate determined by market demand for XYZ's products.
- (d) the rate of growth determined by inflationary pressures.

III. According to the case scenario the risk associated with growing too quickly is that.....

- (a) the company might not be able to retain competent staff.
- (b) the company could face liquidity issues due to over-expansion.
- (c) the company's stock price might decline.
- (d) it could reduce the company's market share.



IV. Which of the following twin cornerstones are necessary for XYZ Ltd. to achieve sustainable growth?

- (a) Market conditions and competition.
- (b) Growth capability and growth strategy
- (c) Product innovation and marketing strategy.
- (d) Cost-cutting measures and increased sales.

V. In an inflationary condition if creditors require that XYZ Ltd.'s historical cost debt-to-equity ratio stay constant, the inflation.....

- (a) Reduces the need for external financing.
- (b) Increases the sustainable growth rate by lowering costs.
- (c) Lowers the sustainable growth rate.
- (d) It has no effect on the company's growth rate.

(MTP July – 2025)

Answer: Case Scenario – 09

I. (d) Robert C. Higgins

II. (b) the maximum rate of growth in sales that can be achieved without borrowing additional funds.

III. (b) the company could face liquidity issues due to overexpansion.

IV. (b) Growth capability and growth strategy

V. (c) Lowers the sustainable growth rate.

Case Scenario – 10

ABC Ltd. is planning to expand its business and therefore raising fund by issuing a convertible bond of ₹ 10 crore. An investor “Mr. X” is interested to invest in the bond of ABC Ltd. Mr. X has following data related to the convertible bond.

The data given below relates to a convertible bond:

Face Value	₹ 250
Coupon Rate	12%
No. of shares per bond	20
Market price of share	₹ 12
Straight value of bond	₹ 235
Market price of convertible bond	₹ 265
Maturity	5 Years

You, being an expert of the matter, are required to answer his questions. Select the most appropriate alternative:

I. The percentage of downside risk of the bond is approximately.....

- (a) 10.42%
- (b) 6.38%
- (c) 2.13%
- (d) 12.77%

II. The conversion premium in percentage term of the bond is.....

- (a) 12.77%
- (b) 10.42%
- (c) 2.18%
- (d) 13.45%



SECURITY VALUATION

III. The conversion parity price of the stock is.....

- (a) ₹ 11.75 (b) ₹ 12.00
(c) ₹ 13.25 (d) ₹ 12.50

IV. If he wants a yield of 15% the maximum price he should be ready to pay for is.....

- (a) 217.41 (b) 224.81
(c) 240.00 (d) 232.32

(MTP March – 2024)

Answer: Case Scenario – 10

- I. (d) 12.77%
II. (b) 10.42%
III. (c) ₹ 13.25
IV. (b) 224.81

Case Scenario – 11

Following Financial data are available for PQR Ltd. for the financial year ending 2023:

Events	(₹ in lakhs)
8% Debentures	125
10% Bonds (2022)	50

Equity Shares (₹ 10 each)	100
Reserves and Surplus	300
Total Assets	600
Assets Turnovers ratio	1.1
Effective interest rate	8%
Effective tax rate	40%
Operating margin	10%
Dividend payout ratio	16.67%
Current market Price of Share	₹ 14
Required rate of return of investors	15%

From the information given above, choose the correct answer to the following questions:

I. Amount of retained earnings for the financial year 2023 approximately is.....

- (A) ₹ 26.00 lakh (B) ₹ 5.20 lakh
(C) ₹ 52.00 lakh (D) ₹ 31.20 lakh

II. 10% Bonds must have been issued in the month of.....

- (A) July 2022 (B) June 2022
(C) August 2022 (D) May 2022

III. Fair price of share of PQR Ltd. using Dividend Discount Model shall be approximately.....

- (A) ₹ 6.12 (B) ₹ 6.51



(C) ₹ 10

(D) ₹ 14

IV. Sustainable Growth Rate of PQR Ltd. shall be approximately

(A) 10.00%

(B) 6.50%

(C) 15.00%

(D) 7.80%

V. Return on Equity (ROE) of PQR Ltd. is.....

(A) 7.80%

(B) 6.50%

(C) 10.00%

(D) 15.00%

(MTP March: 2025)**Answer: Case Scenario – 11****I. (A) ₹ 26.00 lakh****II. (B) June 2022****III. (B) ₹ 6.51****IV. (B) 6.50%****V. (A) 7.80%****Case Scenario – 12**

Zenith Capital, a boutique investment firm, manages portfolios for high-net-worth individuals (HNIs). Their lead portfolio manager, Mr. R, has been closely analyzing market trends to optimize returns for their fixed-income portfolio. Over the past few months, he has observed fluctuations in interest rates and anticipates a significant shift in the near future.

To optimise returns, Mr. R is considering three different investment strategies for clients having ₹ 10 crore of fund and are interested in Fixed Income Portfolio. Each strategy is designed to align with the client's risk appetite and future liquidity needs.

Strategy A: Investing the entire ₹ 10 crore in a single bond with a 7-year maturity to match a specific financial obligation in the future.

Strategy B: Allocating ₹ 5 crore in short-term bonds (1-year maturity) and ₹ 5 crore in long term bonds (7-year maturity) to balance risk and return.

Strategy C: Spreading the ₹ 10 crore investment equally across bonds with maturities of 1 to 5 years to ensure periodic liquidity.

Meanwhile, Mr. R is also considering forecasting models to predict interest rate movements. He is evaluating economic indicators such as inflation, historical rate trends, and a combination of multiple economic factors to enhance the firm's forecasting accuracy.

Mr. R suggested Strategy B for Mr. H (a HNI) having a sum of ₹ 10 crore for investment in Fixed Income Portfolio. As per the strategy half amount on fund is proposed to be invested in 7-year bonds yielding 8% per annum and balance in 1-year short term bond yielding 6% per annum. Interest on these bonds is compounded annually.

Based on the above case scenario, choose the correct answer to the following questions:

I. What is the primary objective of an active bond portfolio strategy?

(A) To maintain a fixed return irrespective of market conditions



SECURITY VALUATION

- (B) To outperform the market by making informed investment decisions
- (C) To minimize volatility and ensure steady returns
- (D) To invest in government bonds only

II. If any HNI follows Strategy A, thenof fixed-income portfolio strategy is being followed.

- (A) Barbell Strategy
- (B) Ladder Strategy
- (C) Bullet Strategy
- (D) Duration Matching

III. In the Barbell Strategy, the funds are typically allocated.....

- (A) by making entire investment in bonds with the same maturity period.
- (B) by dividing investment equally between short-term and long-term bonds.
- (C) by dividing equal amount in bonds with different maturity periods.
- (D) by investing only in short-term bonds.

IV. In the ladder Strategy, the funds are typically allocated.....

- (A) by making entire investment in bonds with the same maturity period.
- (B) by dividing investment equally between short-term and long-term bonds.
- (C) by dividing equal amount in bonds with different maturity periods.
- (D) by investing only in short-term bonds.

V. It is expected that interest rate in coming 8 years are expected to fall by 25 bps each year and if Mr. H does not withdraw any amount from the Fund during these 7 years the total value of the investment at the end of the 7th year shall be approximately.....

- | | |
|--------------------|--------------------|
| (A) ₹ 15.036 crore | (B) ₹ 15.721 crore |
| (C) ₹ 15.739 crore | (D) ₹ 15.829 crore |

(MTP APRIL: 2025)

Answer: Case Scenario – 12

- I. (B) To outperform the market by making informed investment decisions**
- II. (C) Bullet Strategy**
- III. (B) by dividing investment equally between short-term and long-term bonds.**
- IV. (C) by dividing equal amount in bonds with different maturity periods.**
- V. (B) ₹ 15.721 crore**



Case Scenario – 13

Short Bank Ltd. need funds for a period of 7 days. To meet this financial need, on 20th September, 2025, Short Bank Ltd. entered into an agreement with Long Bank Ltd. under which, Short Bank Ltd. will sell 8% GOI Bonds @ 6% p.a. for ₹ 5 crores (Face Value) with initial margin 2%. Each Bond Face Value is ₹ 1,00,000.

The maturity of this 8% GOI Bond is 31st December, 2029, originally issued on 1st January, 2025. Interest payable annually. The clean price of the bond is ₹ 99,420.

Note: Assume 360 days in a year.

From the information given above, choose the correct answer to the Question No. 1 to 3:

- I. The arrangement entered by Long Bank Ltd. is _____ and that by Short Bank Ltd. is _____.
 - (A) Repo, Reverse Repo
 - (B) Lending, Repo
 - (C) Reverse Repo, Borrowing
 - (D) Reverse Repo, Repo

- II. Accrued Interest and dirty price of the bond as on 20th September, 2025 will approximately be ₹_____ and ₹_____ respectively.
 - (A) ₹ 5,822 and ₹ 1,05,242
 - (B) ₹ 5,788 and ₹ 93,632

(C) ₹ 7,954 and ₹ 1,07,374

(D) ₹ 7,954 and ₹ 91,466

- III. The proceeds of the 1st Leg of the transaction shall be approximately ₹_____ and the 2nd Leg proceeds of the transaction shall be ₹_____.

(A) ₹ 5,15,52,000 and ₹ 5,16,12,150

(B) ₹ 5,15,68,580 and ₹ 5,16,28,743

(C) ₹ 5,15,52,000 and ₹ 5,61,12,150

(D) ₹ 5,51,52,000 and ₹ 5,61,12,150

(EXAM JANUARY – 2026)

Answer Case Scenario – 13

- I. (D) Reverse Repo, Repo
- II. (A) ₹ 5,822 and ₹ 1,05,242
- III. (B) ₹ 5,15,68,580 and ₹ 5,16,28,743

Case Scenario – 14

XYZ Ltd. is planning to expand its business and therefore raising fund by issuing a Convertible Bond of ₹ 10 crore. An investor “Mr. A” is interested to invest in the bond of XYZ Ltd. Mr. A has following data related to the Convertible Bond.

The data given below relates to a convertible bond:



Face value	₹ 1,000
Coupon rate	12%
No. of shares per bond	20
Market price of share	₹ 48
Straight value of bond	₹ 940
Market price of convertible bond	₹ 1,060
Maturity	5 Years

From the information given above, choose the correct answer to the following questions:

- I.** The conversion value of the convertible bond is.....
- (a) ₹ 960 (b) ₹ 1000
- (c) ₹ 1060 (d) ₹ 940
- II.** The conversion premium of the bond in absolute terms is.....
- (a) ₹ 60 (b) ₹ 100
- (c) ₹ 120 (d) ₹ 80
- III.** The floor value of the convertible bond is represented by.....
- (a) Market price of share
- (b) Conversion value

- (c) Straight value of bond
- (d) Face value of bond

- IV.** The upside potential of the convertible bond primarily depends on.....
- (a) Coupon rate of the bond
- (b) Face value of the bond
- (c) Market price movement of the equity shares
- (d) Remaining maturity of the bond
- V.** By approximately what percentage the market price of the share should rise so that the investor is indifferent between buying the share from the market and taking conversion route.
- (a) 12.77% (b) 10.42%
- (c) 5.11% (d) 9.23%

Answer Case Scenario – 14

- I.** (a) ₹ 960
- II.** (b) ₹ 100
- III.** (c) Straight value of bond
- IV.** (c) Market price movement of the equity shares

V. (b) 10.42%**Case Scenario – 15**

A Ltd. paid a dividend of ₹ 5 for the last year. The dividend is expected to grow at 25% for the next 6 years and at 10% per annum thereafter. The standard deviation of market return and A Ltd.'s share is 12% and 18% respectively. The correlation between market return and A Ltd. share return is 0.3733. The return of government bond is 13% per annum and market return is expected to be around 20%.

From the information given above, choose the correct answer to the following questions:

- I.** The beta (β) of A Ltd.'s share is closest to.....
- (a) 0.42 (b) 0.56
(c) 0.67 (d) 0.84
- II.** The equity risk premium as per the given data is.....
- (a) 5% (b) 6%
(c) 7% (d) 8%
- III.** The valuation approach can be used to compute the intrinsic value of A Ltd. shares is.....
- (a) Single-stage Dividend Discount Model
(b) CAPM valuation model

(c) Two-stage Dividend Discount Model

(d) Price–Earnings valuation model

IV. If the market price of A Ltd. share is higher than its intrinsic value, the share is said to be.....

(a) Under-valued (b) Over-valued

(c) Fairly valued (d) Speculative

Answer Case Scenario – 15**I. (b) 0.56****II. (c) 7%****III. (c) Two-stage Dividend Discount Model****IV. (b) Over-valued****Case Scenario – 16**

XYZ Ltd. needs funds for a short tenure. Some functional level manager suggested about the bank credit/ overdraft option. On conforming from Finance Department, it was found that company exhausted its credit limits due to meeting recent contingency fund requirements. Then CA X, CFO suggested the idea of floating Commercial papers by XYZ Ltd.

Accordingly, XYZ Ltd. is planning to issue Commercial Paper (CP), the details of which is given below:

Issue Price of CP ₹ 97,550

Face Value ₹ 1,00,000



Maturity Period 3 Months

Issue Expense

Brokerage	0.15% for 3 months
Rating charges	0.50% p.a.
Stamp Duty	0.175% for 3 months

From the information given above, choose the correct answer to the following questions:

- I.** The Bond Equivalent yield for an investor of the same Commercial Paper shall be approximately.....
- (a) 2.51% (b) 10.05%
- (c) 7.53% (d) 11.05%
-
- II.** The Effective Interest Rate per annum of same CP shall approximately be.....
- (a) 10.44% (b) 10.05%
- (c) 2.51% (d) 11.05%
-
- III.** Based on effective interest rate the total annual cost of funds to the company shall approximately be.....
- (a) 11.27% (b) 11.85%
- (c) 12.24% (d) 10.88%

- IV.** Which of the following instruments cannot be used by a bank to meet its short-term funding requirements?
- (a) Call/Notice Money
- (b) Commercial Paper
- (c) Certificate of Deposit
- (d) Repurchase Agreement (Repo)
-
- V.** The period of Commercial Paper ranges from.....
- (a) 14 days to 364 days
- (b) 3 months to 6 months
- (c) 7 days to 1 year
- (d) 1 year to 3 years

Answer Case Scenario – 16

- I. (b) 10.05%**
- II. (a) 10.44%**
- III. (c) 12.24%**
- IV. (b) Commercial Paper**
- V. (c) 7 days to 1 year**





Student Notes



Student Notes



12

ADVANCED CAPITAL BUDGETING

PART 1: INFLATION IN CAPITAL BUDGETING

Question – 01

A firm has projected the following cash flows from a project under evaluation:

Year	₹ lakhs
0	(70)
1	30
2	40
3	30

The above cash flows have been made at expected prices after recognizing inflation. The firm's cost of capital is 10%. The expected annual rate of inflation is 5%.

Show how the viability of the project is to be evaluated.

(SM TYK – 17)

Solution:**Nominal Cash Flows & Nominal Discounting Rate**

$$\begin{aligned} \text{NDR} &= [(1.10 \times 1.05) - 1] \times 100 \\ &= 15.5\% \end{aligned}$$

$$\begin{aligned} \text{NPV} &= \frac{30}{(1.155)^1} + \frac{40}{(1.155)^2} + \frac{30}{(1.155)^3} - 70 \\ &= 5.429 \text{ lakh} \end{aligned}$$

Since NPV is positive hence project is viable.

Question – 02

KLM Ltd. requires ₹ 15,00,000 for a new project.

Useful life of project is 3 years.

Salvage value - NIL. Depreciation is ₹ 5,00,000 p.a.

Given below are projected revenues and costs (excluding depreciation) ignoring inflation:

Year	1	2	3
Revenues in ₹	10,00,000	13,00,000	14,00,000
Costs in ₹	5,00,000	6,00,000	6,50,000

Applicable tax rate is 35%. Assume nominal cost of capital to be 14% (after tax). The inflation rates for revenues and costs are as under:

Year	Revenues %	Costs %
1	9	10
2	8	9
3	6	7

PVF at 14%, for 3 years = 0.877, 0.769 and 0.675

Show amount to the nearest rupee in calculations.



ADVANCED CAPITAL BUDGETING

You are required to calculate net present value of the project.

(SM TYK – 19)

Solution:

अगर Revenue & Cost का Inflation rate अलग अलग है तो NCF & NDR ही हमें लेना होगा।

Nominal CF

	1	2	3
Revenue	10,00,000 (1.09) =10,90,000	13,00,000 (1.09) (1.08) = 15,30,360	14,00,000 (1.09) (1.08) (1.06) = 17,46,965
(-) Cost	5,00,000 (1.10) 5,50,000	6,00,000 (1.10) (1.09) 7,19,400	6,50,000 (1.10) (1.09) (1.07) 8,33,905
CFBT – (i)	5,40,000	8,10,960	9,13,060
(-) Depreciation	5,00,000	5,00,000	5,00,000
PBT	40,000	3,10,960	4,13,060
Tax @ 35% – (ii)	14,000	1,08,836	1,44,571
CFAT (i – ii)	5,26,000	7,02,124	7,68,489
(x) PVF (14%)	0.877	0.769	0.675

PVCI = ₹ 15,19,965

(-) PVCO = ₹ 15,00,000

NPV = ₹ 19,965

Since NPV is positive hence project should be accepted.

Question – 03

Determine NPV of the project with the following information:

Initial Outlay of project	₹ 40,000
Annual revenues (Without inflation)	₹ 30,000
Annual costs excluding depreciation (Without inflation)	₹ 10,000
Useful life	4 years
Salvage value	Nil
Tax Rate	50%
Cost of Capital (Including inflation premium of 10%)	12%

Solution:

Alternative 1: RCF & RDR

CFAT

Sales = 30,000

(-) cost = 10,000

CFBT (i) = 20,000

(-) Dep. $\left(\frac{40,000}{4}\right)$ = 10,000

PBT = 10,000

Tax @ 50% (ii) = 5,000

CFAT (i – ii) = 15,000

RDR = $\left[\frac{1.12}{1.10} - 1\right] \times 100 = 1.82\%$

NPV = $(15000 \times 3.824) = \underline{40,000}$



$$= 17,360$$

Alternative 2: NCF & NDR

Since inflation rate is 10% a year, real cash flows may be stated in nominal cash flows as follows:

Nominal Cash Flow = (1 + Inflation Rate) Real Cash Flows

Year	Real Cash Flows	Nominal Cash Flows
1	15,000	$15,000 \times 1.10 = 16,500$
2	15,000	$15,000 \times (1.10)^2 = 18,150$
3	15,000	$15,000 \times (1.10)^3 = 19,965$
4	15,000	$15,000 \times (1.10)^4 = 21,962$

NPV using nominal discounting rate 12%

$$\frac{16,500}{(1.12)^1} + \frac{18,150}{(1.12)^2} + \frac{19,965}{(1.12)^3} + \frac{21,962}{(1.12)^4} - 40,000$$

$$= ₹ 14,732 + ₹ 14,469 + ₹ 14,211 + ₹ 13,957 - ₹ 40,000$$

$$= ₹ 17,369 \text{ (Approx)}$$

Question – 04

BC Ltd. is contemplating on buying a new machine at ₹ 70,00,000 with an additional working capital requirement of ₹ 10,00,000. The machine is expected to have an economic useful life of 5 years, with no salvage value. The company follows the straight line method of depreciation and same is accepted for tax purposes. The machine is expected to generate an incremental increase in the before tax cash operating income of ₹ 25,00,000 (in real terms) per year for a period of 5 years. The relevant tax rate is 35%. Inflation is expected to be 6% per year and the firm's cost of capital in real term is 10% per year. Assuming that the working capital requirement will remain unchanged throughout the period, in spite of inflation.

Advise the company whether the machine should be purchased or not.

Show your NPV calculation in real term.

PV Factor at 10% & 6% are as under –

PV Factor	1	2	3	4	5
At 10%	0.909	0.826	0.751	0.683	0.621
At 6%	0.943	0.890	0.840	0.792	0.747

(Exam May – 2025) (6 Marks)

Solution:

Working Notes:

(1) Cash Outflow (Initial Outlay) = ₹ 70,00,000 + ₹ 10,00,000 = ₹ 80,00,000

(2) Cash Flow After Tax and Present Value

Particulars	(₹)
Incremental cash operating income	25,00,000
Less: Taxes (0.35)	8,75,000
CFAT	16,25,000
Cum. PV Factor for 5 years at 10%	3.790
Present Value	61,58,750

(3) PV of tax shield due to Depreciation

Tax saving due to Depreciation per year	14,00,000
Tax rate	35%
Tax saving per year for five years	4,90,000

PV of tax shield due to Depreciation

PART 2: RISK IN CAPITAL BUDGETING

(I) STATISTICAL TECHNIQUES

Question – 05

Shivam Ltd. is considering two mutually exclusive projects A and B. Project A costs ₹ 36,000 and project B ₹ 30,000. You have been given below the net present value probability distribution for each project.

Project A		Project B	
NPV Estimates (₹)	Probability	NPV Estimates (₹)	Probability
15,000	0.2	15,000	0.1
12,000	0.3	12,000	0.4
6,000	0.3	6,000	0.4
3,000	0.2	3,000	0.1

- Compute the expected net present values of projects A and B.
- Compute the risk attached to each project i.e. standard deviation of each probability distribution.
- Compute the profitability index of each project.
- Which project do you recommend? State with reasons.

(SM TYK – 06)

Solution:

- Expected NPV**

Project A

Years	Tax Saving (Nominal)	Inflation Factor at 6%	Real Tax saving	PVF @ 10%	PV (₹)
1	4,90,000	0.943	4,62,070	0.909	4,20,021.63
2	4,90,000	0.890	4,36,100	0.826	3,60,218.60
3	4,90,000	0.840	4,11,600	0.751	3,09,111.60
4	4,90,000	0.792	3,88,080	0.683	2,65,058.64
5	4,90,000	0.747	3,66,030	0.621	2,27,304.63
Present value of tax shield due to depreciation					15,81,715.10

- PV of release of Working Capital

	(₹)
Release of Working Capital at the end of 5 th year	10,00,000
Inflation factor at 6% at the end of 5 th year	0.747
Cash inflow in real terms	7,47,000
PVF @ 10% at the end of 5 th year	0.621
Present value of inflow	4,63,887

Calculation of NPV

Particulars	Present Value (₹)
Initial Outlay	(80,00,000)
Present Value of CFAT	61,58,750
Present Value tax shield on depreciation	15,81,715.10
Present Value release of working	4,63,887
	2,04,352.10

Recommendation: The Company should purchase the machine as the NPV of real cash flow is positive.

$$= (15,000 \times 0.2) + (12,000 \times 0.3) + (6,000 \times 0.3) + (3,000 \times 0.2)$$

$$= 9,000$$

Project B

$$= (15,000 \times 0.1) + (12,000 \times 0.4) + (6,000 \times 0.4) + (3,000 \times 0.1)$$

$$= 9,000$$

(ii) Standard Deviation

Project A

$$\sigma_x = \sqrt{(15,000 - 9,000)^2 0.2 + (12,000 - 9,000)^2 0.3 + (6,000 - 9,000)^2 0.3 + (3,000 - 9,000)^2 0.2}$$

$$= 4,450$$

Project B

$$\sigma_B = \sqrt{(15,000 - 9,000)^2 0.1 + (12,000 - 9,000)^2 0.4 + (6,000 - 9,000)^2 0.4 + (3,000 - 9,000)^2 0.1}$$

$$\sigma_B = 3,795$$

	A	B
NPV	9,000	9,000
S.D.	4,450	3,795

(iii) Profitability Index

$$PI = \frac{PVCI}{PVCO}$$

$$A = \frac{36,000 + 9,000}{36,000} = 1.25$$

$$B = \frac{30,000 + 9,000}{30,000} = 1.30$$

(iv) Coefficient of Variation = $\frac{\sigma}{\bar{x}}$

$$A = \frac{4,450}{9,000} = 0.49$$

$$B = \frac{3,795}{9,000} = 0.42$$

Project B should be accepted due to lower risk (C.V.)

Question – 06

KLM Ltd., is considering taking up one of the two projects-Project-K and Project-So Both the projects having same life require equal investment of ₹ 80 lakhs each. Both are estimated to have almost the same yield. As the company is new to this type of business, the cash flow arising from the projects cannot be estimated with certainty. An attempt was therefore, made to use probability to analyze the pattern of cash flow from other projects during the first year of operations. This pattern is likely to continue during the life of these projects. The results of the analysis are as follows:

Project K		Project S	
Cash Flow (in ₹)	Probability	Cash Flow (in ₹)	Probability
11	0.10	09	0.10
13	0.20	13	0.25
15	0.40	17	0.30
17	0.20	21	0.25
19	0.10	25	0.10



Required:

- (i) Calculate variance, standard deviation and co-efficient of variance for both the projects.
- (ii) Which of the two projects is riskier?

(SM TYK – 04)

Solution:

Calculation of Variance & SD, EV.

Project K

Expected Cash Flows

$$= (11 \times 0.10) + (13 \times 0.20) + (15 \times 0.40) + (17 \times 0.20) + (19 \times 0.10)$$

$$= 15$$

$$\sigma^2 = (11-15)^2 \cdot 0.10 + (13-15)^2 \cdot 0.20 + (15-15)^2 \cdot 0.4 + (17-15)^2 \cdot 0.20 + (19-15)^2 \cdot 0.10$$

$$= 4.8$$

$$\sigma = \sqrt{4.8} = 2.19.$$

$$C.V. = \frac{\sigma}{\bar{x}} = \frac{2.19}{15} = 0.146$$

Project S

Expected Cash Flows

$$= (9 \times 0.10) + (13 \times 0.25) + (17 \times 0.30) + (21 \times 0.25) + (25 \times 0.10)$$

$$= 17$$

$$\sigma^2 = (9-17)^2 \cdot 0.10 + (13-17)^2 \cdot 0.25 + (17-17)^2 \cdot 0.3 + (21-17)^2 \cdot 0.25 + (25-17)^2 \cdot 0.10$$

$$= 20.8$$

$$\sigma = \sqrt{20.8} = 4.56$$

$$C.V. = \frac{\sigma}{\bar{x}} = \frac{4.56}{17} = 0.268$$

Project S is riskier as it has higher Coefficient of Variation.

Question – 07

A company is considering Projects X and Y with following information:

Project	Expected NPV (₹)	Standard Deviation
X	1,22,000	90,000
Y	2,25,000	1,20,000

- (i) Which project will you recommend based on the above data?
- (ii) Explain whether your opinion will change, if you use coefficient of variation as a measure of risk.
- (iii) Which measure is more appropriate in this situation and why?

(SM TYK – 03)

Solution:

- (i) **On the basis of NFV project Y is better due to higher NPV.**

On the basis of standard deviation, project x is better due to lower standard deviation.



(ii) **Coefficient of Variation** = $\frac{\sigma}{x}$

$$X = \frac{90,000}{1,22,000} = 0.738$$

$$Y = \frac{1,20,000}{2,25,000} = 0.533$$

Project Y is better due to lower (C.V.)

(iii) However, the NPV method in such conflicting situation is best because the NPV method is in compatibility of the objective of wealth maximization in terms of time value.

(II) CONVENTIONAL TECHNIQUES

(i) Risk Adjusted Discounting Rate

Question – 08

Determine the risk adjusted net present value of the following projects:

	X	Y	Z
Net cash outlays (₹)	2,10,000	1,20,000	1,00,000
Project life	5 years	5 years	5 years
Annual Cash inflow (₹)	70,000	42,000	30,000
Coefficient of variation	1.2	0.8	0.4

The Company selects the risk-adjusted rate of discount on the basis of the coefficient of variation:

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Coefficient of Variation	Risk-Adjusted Rate of Return	P.V. Factor 1 to 5 years At risk adjusted rate of discount
0.0	10%	3.791
0.4	12%	3.605
0.8	14%	3.433
1.2	16%	3.274
1.6	18%	3.127
2.0	22%	2.864
More than 2.0	25%	2.689

(SM TYK – 15)

Solution:

Risk adjusting NPV

Project X

RADR = 16%

$$\text{RANPV} = (70,000 \times 3.274) - 2,10,000 = 19,180$$

Project Y

RADR = 14%

$$= (42,000 \times 3.433) - 1,20,000 = 24,186$$

Project Z

$$\text{RADR} = (30,000 \times 3.605) - 1,00,000 = 8,150$$



ADVANCED CAPITAL BUDGETING

Question – 09

New Projects Ltd. is evaluating 3 projects, P-I, P-II, P-III. Following information is available in respect of these projects:

	P-I	P-II	P-III
Cost	₹ 15,00,000	₹ 11,00,000	₹ 19,00,000
Inflow-Year 1	6,00,000	6,00,000	4,00,000
Year 2	6,00,000	4,00,000	6,00,000
Year 3	6,00,000	5,00,000	8,00,000
Year 4	6,00,000	2,00,000	12,00,000
Risk Index	1.80	1.00	0.60

Minimum required rate of return of the firm is 15% and applicable tax rate is 40%. The risk free interest rate is 10%.

Required:

- (i) Find out the risk-adjusted discount rate (RADR) for these projects.
- (ii) Which project is the best?

(SM TYK – 16)

Solution:

- (i) The risk free rate of interest and risk factor for each of the projects are given. The risk adjusted discount rate (RADR) for different projects can be found on the basis of CAPM as follows:

$$\text{Required Rate of Return} = I_{RF} + (K_0 - I_{RF}) \text{ Risk Factor}$$

$$\text{For P-I : RADR} = 0.10 + (0.15 - 0.10) 1.80 = 19\%$$

$$\text{For P-II : RADR} = 0.10 + (0.15 - 0.10) 1.00 = 15\%$$

$$\text{For P-III : RADR} = 0.10 + (0.15 - 0.10) 0.60 = 13\%$$

- (ii) The three projects can now be evaluated at 19%, 15% and 13% discount rate as follows:

Project P-I

Annual Inflows	₹ 6,00,000
PVAF (19%, 4)	2.639
PV of Inflows (₹ 6,00,000 × 2.639)	₹ 15,83,400
Less: Cost of Investment	<u>₹ 15,00,000</u>
Net Present Value	<u>₹ 83,400</u>

Project P-II

Year	Cash Inflow (₹)	PVF (15%,n)	PV (₹)
1	6,00,000	0.870	5,22,000
2	4,00,000	0.756	3,02,400
3	5,00,000	0.658	3,29,000
4	2,00,000	0.572	<u>1,14,400</u>
Total Present Value			12,67,800
Less: Cost of Investment			<u>11,00,000</u>
Net Present Value			<u>1,67,800</u>

Project P-III

Year	Cash Inflow (₹)	PVF (15%,n)	PV (₹)
1	4,00,000	0.885	3,54,000



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2	6,00,000	0.783	4,69,800
3	8,00,000	0.693	5,54,400
4	12,00,000	0.613	<u>7,35,600</u>
Total Present Value			21,13,800
Less: Cost of Investment			<u>19,00,000</u>
Net Present Value			<u>2,13,800</u>

Project P-III has highest NPV. So, it should be accepted by the firm.

(ii) Certainty Equivalent Approach

Question – 10

The Textile Manufacturing Company Ltd., is considering one of two mutually exclusive proposals, Projects M and N, which require cash outlays of ₹ 8,50,000 and ₹ 8,25,000 respectively. The certainty-equivalent (C.E) approach is used in incorporating risk in capital budgeting decisions. The current yield on government bonds is 6% and this is used as the risk free rate. The expected net cash flows and their certainty equivalents are as follows:

Project M			Project N	
Year-end	Cash Flow ₹	C.E.	Cash Flow ₹	C.E.
1	4,50,000	0.8	4,50,000	0.9
2	5,00,000	0.7	4,50,000	0.8
3	5,00,000	0.5	5,00,000	0.7

Present value factors of ₹ 1 discounted at 6% at the end of year 1, 2 and 3 are 0.943, 0.890 and 0.840 respectively.

Required:

- Which project should be accepted?
- If risk adjusted discount rate method is used, which project would be appraised with a higher rate and why?

(SM TYK – 14)

Solution:

(i) Statement Showing the Net Present Value of Project M

Year end	Cash Flow (₹) (a)	C.E. (b)	Adjusted Cash flow (₹) (c) = (a) × (b)	Present value factor at 6% (d)	Total Present value (₹) (e) = (c) × (d)
1	4,50,000	0.8	3,60,000	0.943	3,39,480
2	5,00,000	0.7	3,50,000	0.890	3,11,500
3	5,00,000	0.5	2,50,000	0.840	<u>2,10,000</u>
Less: Initial Investment					<u>8,60,980</u>
Net Present Value					<u>8,50,000</u> <u>10,980</u>

Statement Showing the Net Present Value of Project N

Year end	Cash Flow (₹) (a)	C.E. (b)	Adjusted Cash flow (₹) (c) = (a) × (b)	Present value factor at 6% (d)	Total Present value (₹) (e) = (c) × (d)
1	4,50,000	0.9	4,05,000	0.943	3,81,915
2	5,00,000	0.8	3,60,000	0.890	3,20,400
3	5,00,000	0.7	3,50,000	0.840	<u>2,94,000</u>
					<u>9,96,315</u>



Less: Initial Investment			8,25,000
Net Present Value			<u>1,71,315</u>

Decision: Since the net present value of Project N is higher, so the project N should be accepted.

- (ii) Certainty - Equivalent (C.E.) Co-efficient of Project M (2.0) is lower than Project N (2.4). This means Project M is riskier than Project N as "higher the riskiness of a cash flow, the lower will be the CE factor". If risk adjusted discount rate (RADR) method is used, Project M would be analyzed with a higher rate.

RADR is based on the premise that riskiness of a proposal may be taken care of, by adjusting the discount rate. The cash flows from a more risky proposal should be discounted at a relatively higher discount rate as compared to other proposals whose cash flows are less risky. Any investor is basically risk averse. However, he may be ready to take risk provided he is rewarded for undertaking risk by higher returns. So, more risky the investment is, the greater would be the expected return. The expected return is expressed in terms of discount rate which is also the minimum required rate of return generated by a proposal if it is to be accepted. Therefore, there is a positive correlation between risk of a proposal and the discount rate.

(III) OTHER TECHNIQUES

(i) Sensitivity Analysis

Question – 11

From the following details relating to a project, analyze the sensitivity of the project to changes in initial project cost, annual cash inflow and cost of capital:

Initial Project Cost (₹)	1,20,000
Annual Cash Inflow (₹)	45,000
Project Life (Years)	4
Cost of Capital	10%

To which of the three factors, the project is most sensitive? (Use annuity factors: for 10% 3.169 and 11% 3.103).

(SM TYK – 10)

Solution:

Calculation of NPV

PV of cash inflows (₹ 45,000 × 3.169)	1,42,605
Initial Project Cost	<u>1,20,000</u>
NPV	<u>22,605</u>

If initial project cost is varied adversely by 10%*

NPV (Revised) (₹ 1,42,605 – ₹ 1,32,000)	₹ 10,605
Change in NPV (₹ 22,605 – ₹ 10,605)/₹ 22,605 i.e.	53.08 %



If annual cash inflow is varied adversely by 10%*

Revised annual inflow	₹ 40,500
NPV (Revised) (₹ 40,500 × 3.169) – (₹ 1,20,000)	(+) ₹ 8,345
Change in NPV (₹ 22,605 – ₹ 8,345)/₹ 22,605	63.08 %

If cost of capital is varied adversely by 10%*

NPV (Revised) (₹ 45,000 × 3.103) – ₹ 1,20,000	(+) ₹19,635
Change in NPV (₹ 22,605 – ₹19,635)/₹ 22,605	13.14 %

Conclusion: Project is most sensitive to 'annual cash inflow'.

*Note: Students may please note that they may assume any other percentage rate other than 10 % say 15%, 20 % 25 % etc.

Question – 12

XYZ Ltd. is considering a project for which the following estimates are available:

	₹
Initial Cost of the project	10,00,000
Sales price/unit	60
Cost/unit	40
Sales volumes	
Year 1	20000 units
Year 2	30000 units
Year 3	30000 units

Discount rate is 10% p.a.

You are required to measure the sensitivity of the project in relation to each of the following parameters:

(a) Sales Price/unit

- (b) Unit cost
- (c) Sales volume
- (d) Initial outlay and
- (e) Project lifetime Taxation may be ignored.

(SM TYK – 09)

Solution:

Approach I

$$NPV = PPCI - PVCO$$

$$= 20 \times \left[\frac{20,000}{(1.10)^1} + \frac{30,000}{(1.10)^2} + \frac{30,000}{(1.10)^3} \right] - 10,00,000$$

$$= 20 \times 65,514.65 - 10,00,000$$

$$= ₹ 3,10,293$$

(a) Selling Price = 10% ↓ [60 × 10% = 6]

$$= 14 \times \left[\frac{20,000}{(1.10)^1} + \frac{30,000}{(1.10)^2} + \frac{30,000}{(1.10)^3} \right] - 10,00,000$$

$$= - 82,795$$

$$\text{Sensitivity} = \frac{3,10,293 - (-82,795)}{3,10,293} \times 100 = 126.68\%$$

(b) Unit Cost 10% ↑

$$= 40 \times 10\% = 4$$

$$= 16 \times 65,514.65 - 10,00,000 = 48,234$$



$$\text{Sensitivity} = \frac{3,10,293 - 48,234}{3,10,293} \times 100 = 84.46\%$$

(c) Sales Volume 10% ↓

$$= 20 \times \left[\frac{18,000}{(1.10)^1} + \frac{27,000}{(1.10)^2} + \frac{27,000}{(1.10)^3} \right] - 10,00,000$$

$$= 1,79,264$$

$$\text{Sensitivity} = \frac{3,10,293 - 1,79,264}{3,10,293} \times 100 = 42.23\%$$

(d) Initial Outlays 10% ↑

$$= 10,00,000 + 10\% = 11,00,000$$

$$\text{NPV} = 20 \times 65,514.65 - 11,00,000$$

$$= 2,10,293$$

$$\text{Sensitivity} = \frac{3,10,293 - 2,10,293}{3,10,293} \times 100 = 32.22\%$$

Project life 33.33% ↓

$$\text{Life} = 3 \text{ Years}$$

$$\text{Life} = 3 - 33.33\%$$

$$= 2 \text{ years}$$

$$\text{NPV} = 20 \times \frac{20,000}{(1.10)^1} + \frac{30,000}{(1.10)^2} - 10,00,000$$

$$= -1,40,496$$

$$\text{Sensitivity} = \frac{3,10,293 - (-1,40,496)}{3,10,293} \times 100 = 145.28\%$$

$$= \frac{145.28}{33.33} \times 10 = 43.59\%$$

Approach II

(a) Selling Price

$$0 = X \times \frac{20,000}{(1.10)^1} + \frac{30,000}{(1.10)^2} + \frac{30,000}{(1.10)^3} - 10,00,000$$

$$65,514.65 \times = 10,00,000$$

$$x = \frac{10,00,000}{65,514.65} = ₹ 15.26$$

$$\text{Contribution} = \text{S.P.} - \text{VC}$$

$$15.26 = x - 40$$

$$x = 55.26$$

$$\text{Sensitivity in selling price} = \frac{60 - 55.26}{60} \times 100 = 7.90\%$$

(b) Sensitivity in Unit Cost

$$\text{Contribution} = 15.26$$

$$\text{Contribution} = \text{S.P.} - \text{VC}$$

$$15.26 = 60 - x$$

$$x = 44.74$$

$$\text{Sensitivity in unit cost} = \frac{44.74 - 40}{40} \times 100$$

$$= 11.85\%$$



(c) Sensitivity in Sales Volume

Alternative 1

Let assume no. of units be x

$$= 20 \times \frac{2x}{(1.10)^1} + \frac{3x}{(1.10)^2} + \frac{3x}{(1.10)^3} - 10,00,000 = 0$$

$$= 20 \times 6.5515 x = 10,00,000$$

$$x = \frac{10,00,000}{20 \times 6.5515} = 7,631.84 \text{ units}$$

$$1 \text{ Year} = 7,631.84 \times 2 = 15,263.68$$

$$2 \text{ Year} = 7,631.84 \times 3 = 22,895.52$$

$$1 \text{ Year} = 7,631.84 \times 3 = 22,895.52$$

$$= 61,054.72$$

$$\text{Sensitivity in units} = \frac{80,000 - 61,054.72}{80,000} \times 100 = 23.68\%$$

Alternative II (Best)

Assume P.V. of units be x

$$= 20 \times x - 10,00,000 = 0$$

$$x = 50,000 \text{ units}$$

$$\text{Sensitivity in units} = \frac{65,514.65 - 50,000}{65,514.65} \times 100 = 23.68\%$$

ICAI

$$= \frac{3,10,293}{13,10,293} \times 100 = 23.68\%$$

(d) Initial Outlays

$$\text{Sensitivity in project cost} = \frac{\text{NPV}}{\text{Initial Outlays}} \times 100$$

$$= \frac{3,10,293}{10,00,000} \times 100 =$$

31.03%

(e) Project Life

Life 3 years NPV = 3,10,293

Life 2 years NPV = -1,40,496

2 years ----- (1,40,496)

3 years ----- 3,10,293

1 year ----- (4,50,789)

$$2 \text{ years} + \left(\frac{1}{4,50,789} \times 1,40,496 \right)$$

2.31 years

$$\text{Sensitivity in life} = \frac{3 - 2.31}{3} \times 100 = 22.97\%$$

Question – 13

Red Ltd. is considering a project with the following Cash flows:



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Years	Cost of Plant	Recurring Cost	Savings
0	10,000		
1		4,000	12,000
2		5,000	14,000

The cost of capital is 9%. Measure the sensitivity of the project to changes in the levels of plant value, running cost and savings (considering each factor at a time) such that the NPV becomes zero. The P.V. factor at 9% are as under:

Year	Factor
0	1
1	0.917
2	0.842

Which factor is the most sensitive to affect the acceptability of the project?

(SM TYK – 11)

Solution:

P.V. of Cash Flows

Year 1	Running Cost	₹ 4,000 × 0.917	= (₹ 3,668)
	Savings	₹ 12,000 × 0.917	= ₹ 11,004
Year 2	Running Cost	₹ 5,000 × 0.842	= (₹ 4,210)
	Savings	₹ 14,000 × 0.842	= ₹ 11,788
			= ₹ 14,914

Year 0	Less: P.V. of Cash Outflow ₹ 10,000 × 1 = ₹ 10,000
	NPV = ₹ 4,914

Sensitivity Analysis

(i) Increase of Plant Value by ₹ 4,914

$$\therefore \frac{4,914}{10,000} \times 100 = 49.14\%$$

(ii) Increase of Running Cost by ₹ 4,914

$$\frac{4,914}{3,668 + 4,210} = \frac{4,914}{7,878} \times 100 = 62.38\%$$

(iii) Fall in Saving by ₹ 4,914

$$\frac{4,914}{11,004 + 11,788} = \frac{4,914}{22,792} \times 100 = 21.56\%$$

Hence, savings factor is the most sensitive to affect the acceptability of the project as in comparison of other two factors a slight % change in this fact shall more affect the NPV than others.

Question – 14

R Ltd. is considering a project with the following Cash flows:

Years	Cost of Plant	Recurring Cost	Savings
0	20,000		
1		8,000	24,000
2		10,000	28,000

The cost of capital is 9%.

Evaluate the sensitivity of the project in respect of all factors except time such that:



- (i) NPV become zero and
(ii) adversely varying factors value by 10%.

The P.V. factor at 9% are as under:

Years	Cost of Plant
0	1
1	0.917
2	0.842

Note: Round off calculation upto 2 decimal points.

(MTP April – 2024)

Solution:

Working Note:

Year 1	Running Cost	₹ 8,000 × 0.917	= (₹ 7,336)
	Savings	₹ 24,000 × 0.917	= ₹ 22,008
Year 2	Running Cost	₹ 10,000 × 0.842	= (₹ 8,420)
	Savings	₹ 28,000 × 0.842	= ₹ 23,576
			= ₹ 29,828
Year 0	Less: P.V. of Cash Outflow	₹ 20,000 × 1	= ₹ 20,000
	NPV		= ₹ 9,828

(i) Sensitivity Analysis (by making NPV Zero)

(1) Increase of Plant Value by ₹ 9,828

$$\therefore \frac{9,828}{20,000} \times 100 = 49.14\%$$

(2) Increase of Running Cost by ₹ 9,828

$$\frac{9,828}{7,336 + 8,420} = \frac{9,828}{15,756} \times 100 = 62.38\%$$

(3) Fall in Saving by ₹ 9,828

$$\frac{9,828}{22,008 + 23,576} = \frac{9,828}{45,584} \times 100 = 21.56\%$$

Hence, savings factor is the most sensitive to affect the acceptability of the project as in comparison of other two factors a slight % change in this fact shall more affect the NPV than others.

(ii) Sensitivity Analysis if there is a variation of 10% in the factors.

- (1) If the initial project cost is varied adversely by say 10%.

$$\text{NPV (Revised)} (\text{₹ } 9,828 - \text{₹ } 2,000) = \text{₹ } 7,828$$

$$\text{Change in NPV} = \frac{\text{₹ } 9,828 - \text{₹ } 7,828}{\text{₹ } 9,828} = 20.35\%$$

- (2) If Annual Running Cost is varied by say 10%.

$$\text{NPV (Revised)} (\text{₹ } 9,828 - \text{₹ } 800 \times 0.917 - \text{₹ } 1000 \times 0.842)$$

$$= \text{₹ } 9,828 - \text{₹ } 733.60 - \text{₹ } 842 = \text{₹ } 8,252.40$$



$$\text{Change in NPV} = \frac{\text{₹ } 9,828 - \text{₹ } 8,252.40}{\text{₹ } 9,828} = 16.03\%$$

(3) If Saving is varied by say 10%.

$$\text{NPV (Revised)} (\text{₹ } 9,828 - \text{₹ } 2400 \times 0.917 - \text{₹ } 2800 \times 0.842)$$

$$= \text{₹ } 9,828 - \text{₹ } 2,200.80 - \text{₹ } 2,357.60 = \text{₹ } 5,269.60$$

$$\text{Change in NPV} = \frac{\text{₹ } 9,828 - \text{₹ } 5,269.60}{\text{₹ } 9,828} \times 100 = 46.38\%$$

Hence, savings factor is the most sensitive to affect the acceptability of the project.

Question – 15

The Easygoing Company Limited is considering a new project with initial investment, for a product “Survival”. It is estimated that IRR of the project is 16% having an estimated life of 5 years.

Financial Manager has studied that project with sensitivity analysis and informed that annual fixed cost sensitivity is 7.8416%, whereas cost of capital (discount rate) sensitivity is 60%.

Other information available are:

Profit Volume Ratio (P/V) is 70%,

Variable cost ₹ 60/- per unit

Annual Cash Flow ₹ 57,500/-

Ignore Depreciation on initial investment and impact of taxation.
Calculate

(i) Initial Investment of the Project

(ii) Net Present Value of the Project

(iii) Annual Fixed Cost

(iv) Estimated annual unit of sales

(v) Break Even Units

Cumulative Discounting Factor for 5 years

8%	9%	10%	11%	12%	13%	14%	15%	16%	17%	18%
3.339	3.890	3.791	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127

(SM TYK – 12)

Solution:

(i) Initial Investment

IRR = 16% (Given)

At IRR, NPV shall be zero, therefore

Initial Cost of Investment = PVAF (16%,5) × Cash Flow (Annual)

$$= 3.274 \times \text{₹ } 57,500 = \text{₹ } 1,88,255$$

(ii) Net Present Value (NPV)

Let Cost of Capital be X, then $\frac{16-X}{X} = 60\%$ X = 10%

Thus NPV of the project

$$= \text{Annual Cash Flow} \times \text{PVAF (10\%, 5)} - \text{Initial Investment}$$



$$= ₹ 57,500 \times 3.791 - ₹ 1,88,255$$

$$= ₹ 2,17,982.50 - ₹ 1,88,255$$

$$= ₹ 29,727.50$$

(iii) Annual Fixed Cost

Alternative I

Let change in the Fixed Cost which makes NPV zero is X.

Then,

$$₹ 29,727.50 - 3.791X = 0$$

$$\text{Thus } X = ₹ 7,841.60$$

Let original Fixed Cost be Y then,

$$Y \times 7.8416\% = ₹ 7,841.60$$

$$Y = ₹ 1,00,000$$

Thus Fixed Cost is equal to ₹ 1,00,000

Alternative II

$$\text{Sensitivity in FC} = \frac{\text{NPV}}{\text{PV of FC}}$$

$$7.8416\% = \frac{29,727.50}{\text{PV of FC}}$$

$$\text{PV of FC} = ₹ 3,79,100$$

$$\text{PV of FC} = \text{Annual FC} \times \text{PVAF}$$

$$3,79,100 = X \times 3.791$$

$$X = ₹ 1,00,000$$

(iv) Estimated Annual Units of Sales

$$\text{Selling price per unit} = \frac{₹ 60}{100\% - 70\%} = ₹ 200$$

$$\frac{\text{Annual Cash Flow} + \text{Fixed Cost}}{\text{P/V Ratio}} = \text{Sales Value}$$

$$\frac{₹ 57,500 + ₹ 1,00,000}{0.70} = ₹ 2,25,000$$

$$\text{Sales in Units} = \frac{₹ 2,25,000}{₹ 200} = 1,125 \text{ units}$$

(v) Break Even Units

$$\frac{\text{Fixed Cost}}{\text{Contribution Per Unit}} = \frac{1,00,000}{140} = 714.285 \text{ units}$$

Question – 16

Unnat Ltd. is considering investing ₹ 50,00,000 in a new machine. The expected life of machine is five years and has no scrap value. It is expected that 2,00,000 units will be produced and sold each year at a selling price of ₹ 30.00 per unit. It is expected that the variable costs to be ₹ 16.50 per unit and fixed costs to be ₹ 10,00,000 per year. The cost of capital of Unnat Ltd. is 12% and acceptable level of risk is 20%.

You are required to measure the sensitivity of the project's net present value to a change in the following project variables:

- (a) Sale Price;
- (b) Sales Volume;

- (c) Variable Cost;
- (d) On further investigation it is found that there is a significant chance that the expected sales volume of 2,00,000 units per year will not be achieved. The sales manager of Unnat Ltd. suggests that sales volumes could depend on expected economic states which could be assigned the following probabilities:

State of Economy	Annual Sales (in Units)	Probability
Poor	1,75,000	0.30
Normal	2,00,000	0.60
Good	2,25,000	0.10

Calculate expected net present value of the project and give your decision whether company should accept the project or not.

(SM TYK – 13)

Solution:

NPV Calculation

$$\text{NPV} = (30 - 16.50) \times 2,00,000 \text{ unit} \times 3.605 - 10,00,000 \times 3.605 - 50,00,000$$

$$= 13.50 \times 7,21,000 - 36,05,000 - 50,00,000$$

$$= ₹ 11,28,500$$

(a) Sales Price

Let assume contribution per unit be x

$$x \times 7,21,000 - 86,05,000 - 50,00,000 = 0$$

$$x = \frac{86,05,000}{7,21,000} = 11.93$$

$$\text{CPU} = \text{SP} - \text{VC}$$

$$11.93 = x - 16.50$$

$$x = ₹ 28.43$$

$$\text{Sensitivity} = \frac{30 - 28.43}{30} \times 100 = 5.23\%$$

Alternative

Let the sale price/Unit be S so that the project would break even with 0 NPV.

$$\therefore ₹ 50,00,000 = [2,00,000 (S - ₹ 16.50) - ₹ 10,00,000] \text{ PVIAF} (12\%, 5)$$

$$₹ 50,00,000 = [2,00,000 S - ₹ 33,00,000 - ₹ 10,00,000] 3.605$$

$$₹ 50,00,000 = [2,00,000S - ₹ 43,00,000] 3.605$$

$$₹ 13,86,963 = 2,00,000S - ₹ 43,00,000$$

$$₹ 56,86,963 = 2,00,000S$$

$$S = ₹ 28.43 \text{ which represents a fall of } (30 - 28.43)/30 \text{ or } 0.0523 \text{ or } 5.23\%$$

(b) Sales Volume

Let assume present value of sales volume be x

$$13.50 x - 36,05,000 - 50,00,000 = 0$$



$$x = \frac{86,05,000}{13.50} = 6,37,407$$

$$\text{Sensitivity} = \frac{7,21,000 - 6,37,407}{7,21,000} \times 100 = 11.59\%$$

Alternative

Let V be the sale volume so that the project would break even with 0 NPV.

$$\therefore ₹ 50,00,000 = [V (₹ 30 - ₹ 16.50) - ₹ 10,00,000] \text{PVIAF} (12\%, 5)$$

$$₹ 50,00,000 = [V (₹ 13.50) - ₹ 10,00,000] \text{PVIAF} (12\%, 5)$$

$$₹ 50,00,000 = [₹ 13.50V - ₹ 10,00,000] 3.605$$

$$₹ 13,86,963 = ₹ 13.50V - ₹ 10,00,000$$

$$₹ 23,86,963 = ₹ 13.50V$$

$$V = 1,76,812 \text{ which represents a fall of } (2,00,000 - 1,76,812)/2,00,000 \text{ or } 0.1159 \text{ or } 11.59\%$$

(c) Variable Cost

Let assume variable cost per unit be x

$$\text{CPU} = \text{SP} - \text{VC}$$

$$11.93 = 30 - x$$

$$x = 18.07$$

$$\text{Sensitivity} = \frac{18.07 - 16.50}{16.50} \times 100 = 9.51\%$$

Alternative

Let the variable cost be V so that the project would break even with 0 NPV.

$$₹ 50,00,000 = [2,00,000 (₹ 30 - V) - ₹ 10,00,000]$$

$$\text{PVIAF}(12\%, 5)$$

$$₹ 50,00,000 = [₹ 60,00,000 - 2,00,000 V - ₹ 10,00,000] 3.605$$

$$₹ 50,00,000 = [₹ 50,00,000 - 2,00,000 V] 3.605$$

$$₹ 13,86,963 = ₹ 50,00,000 - 2,00,000 V$$

$$₹ 36,13,037 = 2,00,000V$$

$$V = ₹ 18.07 \text{ which represents a fall of } (18.07 - 16.50)/16.50 \text{ or } 0.0951 \text{ or } 9.51\%$$

(d) Expected Net Present Value

$$(1,75,000 \times 0.30) + (2,00,000 \times 0.60) + (2,25,000 \times 0.10) = 1,95,000$$

$$\text{NPV} = [1,95,000 \times ₹ 13.50 - ₹ 10,00,000] 3.605 - ₹ 50,00,000$$

$$= ₹ 8,85,163$$

Further NPV in worst and best cases will be as follows:

Worst Case:

$$[1,75,000 \times ₹ 13.50 - ₹ 10,00,000] 3.605 - ₹ 50,00,000 = - ₹ 88,188$$

Best Case:

$[2,25,000 \times ₹ 13.50 - ₹ 10,00,000] 3.605 - ₹ 50,00,000 = ₹ 23,45,188$

Thus, there are 30% chances that the rise will be a negative NPV and 70% chances of positive NPV. Since acceptable level of risk of Unnat Ltd. is 20% and there are 30% chances of negative NPV hence project should not be accepted.

Question – 17

PQ Ltd. expects sales of ₹ 100 lakhs in the year 1. The same will increase by ₹ 20 lakhs per year over the next four years. At the end of 5 years the project would be wound up. The Depreciation will be charged at 20% p.a. on straight line method. The expenses excluding the depreciation will be 40% of the sales. There will be no salvage value of the plant. PQ Ltd. proposes to invest in the plant an amount where the Net Present Value will be Zero.

Corporate Tax rate is 30%.

You are required to calculate the investment which can be made in the plant.

(Exam November – 2024) (8 Marks)

Solution:

Working Notes:

(1) Expected Sales

Year	Expected Sales
1	₹ 100 lakhs
2	₹ 120 lakhs
3	₹ 140 lakhs
4	₹ 160 lakhs
5	₹ 180 lakhs

(2) Expected Expenses Excluding Depreciation

Year	Expenses
1	₹ 40 lakhs
2	₹ 48 lakhs
3	₹ 56 lakhs
4	₹ 64 lakhs
5	₹ 72 lakhs

(3) Cash Inflow from the Project

Let P be the cost of the plant then chargeable depreciation for each year shall be 0.20P. Accordingly, annual cash flow from the project shall be computed as follows:

Year	Expected Sales ₹ lakhs	Exp. ₹ lakhs	Dep. (3)	Profit Before Tax	Tax @ 30%	Profit After Tax
1	100	40	0.20 P	60 – 0.20P	18 – 0.06P	42 – 0.14P
2	120	48	0.20 P	72 – 0.20P	21.6 – 0.06P	50.4 – 0.14P
3	140	56	0.20 P	84 – 0.20P	25.2 – 0.06P	58.8 – 0.14P
4	160	64	0.20 P	96 – 0.20P	28.8 – 0.06P	67.2 – 0.14P
5	180	72	0.20 P	108 – 0.20P	32.4 – 0.06P	75.6 – 0.14P

Year	Profit After Tax	Dep. Added Back	Cash Inflow
1	42 – 0.14P	0.20P	42 + 0.06P
2	50.40 – 0.14P	0.20P	50.40 + 0.06P
3	58.80 – 0.14P	0.20P	58.80 + 0.06P
4	67.20 – 0.14P	0.20P	67.20 + 0.06P



ADVANCED CAPITAL BUDGETING

5	$75.60 - 0.14P$	$0.20P$	$75.60 + 0.06P$
Total			$294 + 0.30P$

Since NPV will be Zero the required comes as follows:

Sum of Cash Inflows – Plant Cost = 0

$$294 + 0.30P - P = 0$$

$$P = 420$$

Thus, the required investment to be made in plant shall be ₹ 420 lakhs.

Alternative solution if a discount rate of 10% is applied, though students may solve the question using a rate other than 10%.

Working Notes:

(1) Expected Sales

Year	Expected Sales
1	₹ 100 lakhs
2	₹ 120 lakhs
3	₹ 140 lakhs
4	₹ 160 lakhs
5	₹ 180 lakhs

(2) Expected Expenses Excluding Depreciation

Year	Expenses
1	₹ 40 lakhs
2	₹ 48 lakhs
3	₹ 56 lakhs
4	₹ 64 lakhs
5	₹ 72 lakhs

(3) Cash Inflow from the Project

Let P be the cost of the plant then chargeable depreciation for each year shall be 0.20P. Accordingly, annual cash flow from the project shall be computed as follows:

Year	Expected Sales ₹ lakhs	Exp. ₹ lakhs	Dep. (3)	Profit Before Tax	Tax @ 30%	Profit After Tax
1	100	40	$0.20P$	$60 - 0.20P$	$18 - 0.06P$	$42 - 0.14P$
2	120	48	$0.20P$	$72 - 0.20P$	$21.6 - 0.06P$	$50.4 - 0.14P$
3	140	56	$0.20P$	$84 - 0.20P$	$25.2 - 0.06P$	$58.8 - 0.14P$
4	160	64	$0.20P$	$96 - 0.20P$	$28.8 - 0.06P$	$67.2 - 0.14P$
5	180	72	$0.20P$	$108 - 0.20P$	$32.4 - 0.06P$	$75.6 - 0.14P$

Year	Profit After Tax	Dep. Added Back	Cash Inflow	PVF @ 10%	PV of Cash Inflow
1	$42 - 0.14P$	$0.20P$	$42 + 0.06P$	0.909	$38.178 + 0.05454P$
2	$50.4 - 0.14P$	$0.20P$	$50.40 + 0.06P$	0.826	$41.6304 + 0.04956P$
3	$58.8 - 0.14P$	$0.20P$	$58.80 + 0.06P$	0.751	$44.1588 + 0.04506P$
4	$67.2 - 0.14P$	$0.20P$	$67.20 + 0.06P$	0.683	$45.8976 + 0.04098P$
5	$75.6 - 0.14P$	$0.20P$	$75.60 + 0.06P$	0.621	$46.9476 + 0.03726P$
Total					$216.8124 + 0.2274P$

Since NPV will be Zero the required comes as follows:

Sum of Cash Inflows – Plant Cost = 0

$$216.8124 + 0.2274P - P = 0$$

$$P = 280.63$$



Thus, the required investment to be made in plant shall be ₹ 280.63 lakhs.

Question – 18

X Ltd. is considering its new project with the following details:

Sr. No.	Particulars	Figures
1	Initial capital cost	₹ 400 Cr.
2	Annual unit sales	5 Cr.
3	Selling price per unit	₹ 100
4	Variable cost per unit	₹ 50
5	Fixed costs per year	₹ 50 Cr.
6	Discount Rate	6%

Required:

- Calculate the NPV of the project.
- Compute the impact on the project's NPV considering a 2.5 per cent adverse variance in each variable. Which variable is having maximum effect?

Consider Life of the project as 3 years.

Solution:

1. Calculation of Net Cash Inflow per Year

	Particulars	Amount (₹)
A	Selling price per unit	100
B	Variable cost per unit	50
C	Contribution per unit (A – B)	50
D	Number of units sold per year	5 Cr.
E	Total Contribution (C × D)	₹ 250 Cr.
F	Fixed cost per year	₹ 50 Cr.
G	Net cash inflow per year (E - F)	₹ 200 Cr.

Calculation of Net Present Value (NPV) of the Project

Year	Year Cash Flow (₹ in Cr.)	PV factor @ 6%	Present Value (PV) (₹ in Cr.)
0	(400.00)	1.000	(400.00)
1	200.00	0.943	188.60
2	200.00	0.890	178.00
3	200.00	0.840	168.00
Net Present Value			134.60

Here, NPV represent the most likely outcomes and not the actual outcomes. The actual outcome can be lower or higher than the expected outcome.

2. Sensitivity Analysis considering 2.5% Adverse Variance in each variable

	Particulars	Base	Initial capital cost increased to ₹ 410 crore	Selling Price per Unit Reduced to ₹ 97.5	Variable Cost Per Unit increased to ₹ 51.25	Fixed Cost Per Unit increased to ₹ 51.25	Units sold per year reduced to 4.875 crore
		(₹)	(₹)	(₹)	(₹)	(₹)	(₹)
A	Selling price per unit	100	100	97.5	100	100	100
B	Variable cost per unit	50	50	50	51.25	50	50
C	Contribution per unit (A – B)	50	50	47.5	48.75	50	50
		(₹in Cr.)	(₹in Cr.)	(₹in Cr.)	(₹in Cr.)	(₹in Cr.)	(₹in Cr.)
D	Number of units sold per year	5	5	5	5	5	4.875

ADVANCED CAPITAL BUDGETING

	(units in Crores)						
E	Total Contribution (C × D)	250	250	237.5	243.75	250	243.75
F	Fixed cost per year	50	50	50	50	51.25	50
G	Net Cash Inflow per year (E – F)	200	200	187.5	193.75	198.75	193.75
H	PV of Net cash Inflow per year (G × 2.673)	534.60	534.60	501.19	517.89	531.26	517.89
I	Initial capital cost	400	410	400	400	400	400
J	NPV (H – I)	134.60	124.60	101.19	117.89	131.26	117.89
	K Percentage Change in NPV	-	-7.43%	-24.82%	-12.41%	-2.48%	-12.41%

The above table shows that by changing one variable at a time by 2.5% (adverse) while keeping the others constant, the impact in percentage terms on the NPV of the project can be calculated. Thus, the change in selling price has the maximum effect on the NPV by 24.82%.

(ii) Scenario Analysis

Question – 19

XYZ Ltd. is considering a project “A” with an initial outlay of ₹ 14,00,000 and the possible three cash inflow attached with the project as follows:

Particulars	Year 1	Year 2	Year 3
Worst case	450	400	700
Most likely	550	450	800
Best case	650	500	900

Assuming the cost of capital as 9%, determine NPV in each scenario. If XYZ Ltd is certain about the most likely result in first two years but uncertain about the third year’s cash flow, analyze what will be the NPV expecting worst scenario in the third year.

Solution:

The possible outcomes will be as follows:

Year	PVF @ 9%	Worst Case		Most likely		Best case	
		Cash Flow	PV	Cash Flow	PV	Cash Flow	PV
		(₹ '000)	(₹ '000)	(₹ '000)	(₹ '000)	(₹ '000)	(₹ '000)
0	1	(1,400)	(1,400)	(1,400)	(1,400)	(1,400)	(1,400)
1	0.917	450	412.65	550	504.35	650	596.05
2	0.842	400	336.80	450	378.90	500	421.00
3	0.772	700	540.40	800	617.60	900	694.80
NPV			- 110.15		100.85		311.85

If XYZ Ltd. is certain about the most likely result in first two years but uncertain about the third year’s cash flow, then, NPV expecting worst case scenario is expected in the third year will be as follows:

$$= - ₹ 14,00,000 + \frac{₹ 5,50,000}{(1 + 0.09)} + \frac{₹ 4,50,000}{(1 + 0.09)^2} + \frac{₹ 7,00,000}{(1 + 0.09)^3}$$

$$= - ₹ 14,00,000 + ₹ 5,04,587 + ₹ 3,78,756 + ₹ 5,40,528$$

$$= ₹ 23,871$$



Question – 20

A firm has an investment proposal, requiring an outlay of ₹ 80,000. The investment proposal is expected to have two years economic life with no salvage value. In year 1, there is a 0.4 probability that cash inflow after tax will be ₹ 50,000 and 0.6 probability that cash inflow after tax will be ₹ 60,000. The probability assigned to cash inflow after tax for the year 2 is as follows:

The cash inflow year 1	₹ 50,000	₹ 60,000
The cash inflow year 2	Probability	Probability
	₹ 24,000 0.2	₹ 40,000 0.4
	₹ 32,000 0.3	₹ 50,000 0.5
	₹ 44,000 0.5	₹ 60,000 0.1

The firm uses a 10% discount rate for this type of investment.

Required:

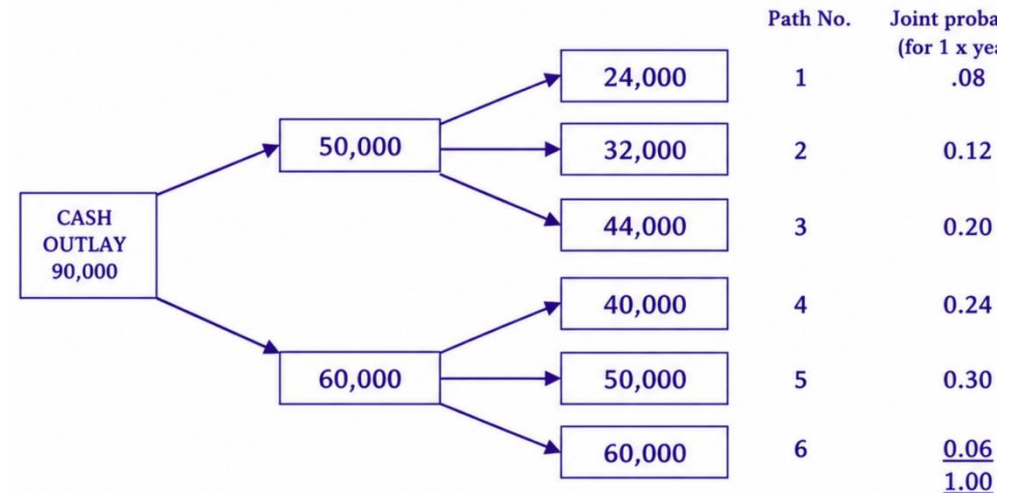
- Construct a decision tree for the proposed investment project and calculate the expected net present value (NPV).
- What net present value will the project yield, if worst outcome is realized? What is the probability of occurrence of this NPV?
- What will be the best outcome and the probability of that occurrence?
- Will the project be accepted?

(Note: 10% discount factor 1 year 0.909; 2 year 0.826)

(SM TYK – 20)

Solution:

- The decision tree diagram is presented in the chart, identifying various paths and outcomes, and the computation of various paths/outcomes and NPV of each path are presented in the following tables:



The Net Present Value (NPV) of each path at 10% discount rate is given below:

Path	Year 1 Cash Flows (₹)	Year 2 Cash Flows (₹)	Total Cash Inflows (PV) (₹)	Cash Inflows NPV (₹)	
1	$50,000 \times 0.909 = 45,450$	$24,000 \times 0.826 = 19,824$	65,274	80,000	(-) 14,726
2	45,450	$32,000 \times 0.826 = 26,432$	71,882	80,000	(-) 8,118
3	45,450	$44,000 \times 0.826 = 36,344$	81,794	80,000	1,794

ADVANCED CAPITAL BUDGETING

4	$60,000 \times 0.909 = 54,540$	$40,000 \times 0.826 = 33,040$	87,580	80,000	7,580
5	54,540	$50,000 \times 0.826 = 41,300$	95,840	80,000	15,840
6	54,540	$60,000 \times 0.826 = 49,560$	1,04,100	80,000	24,100

Statement showing Expected Net Present Value ₹

z	NPV (₹)	Joint Probability	Expected NPV
1	-14,726	0.08	-1,178.08
2	-8,118	0.12	-974.16
3	1,794	0.20	358.80
4	7,580	0.24	1,819.20
5	15,840	0.30	4,752.00
6	24,100	0.06	1,446.00
			6,223.76

- (ii) If the worst outcome is realized the project will yield NPV of – ₹ 14,726. The probability of occurrence of this NPV is 8% and a loss of ₹ 1,178 (path 1).
- (iii) The best outcome will be path 6 when the NPV is at ₹ 24,100. The probability of occurrence of this NPV is 6% and a expected profit of ₹ 1,446.
- (iv) The project should be accepted because the expected NPV is positive at ₹ 6,223.76 based on joint probability.

Question – 21

Skylark Airways is planning to acquire a light commercial aircraft for flying class clients at an investment of ₹ 50,00,000. The expected cash flow after tax for the next three years is as follows:

(₹)

Year 1		Year 2		Year 3	
CFAT	Probability	CFAT	Probability	CFAT	Probability
14,00,000	0.1	15,00,000	0.1	18,00,000	0.2
18,00,000	0.2	20,00,000	0.3	25,00,000	0.5
25,00,000	0.4	32,00,000	0.4	35,00,000	0.2
40,00,000	0.3	45,00,000	0.2	48,00,000	0.1

The Company wishes to take into consideration all possible risk factors relating to airline operations. The company wants to know:

- The expected NPV of this venture assuming independent probability distribution with 6 per cent risk free rate of interest.
- The possible deviation in the expected value.
- How would standard deviation of the present value distribution help in Capital Budgeting decisions?

(SM TYK – 01)

Solution:

(i) Expected NPV

(₹ in lakhs)

Year I			Year II			Year III		
CFA T	P	CF × P	CFA T	P	CF × P	CFA T	P	CF × P
14	0.1	1.4	15	0.1	1.5	18	0.2	3.6
18	0.2	3.6	20	0.3	6.0	25	0.5	12.5
25	0.4	10.0	32	0.4	12.8	35	0.2	7.0
40	0.3	12.0	45	0.2	9	48	0.1	4.8
	$\bar{x}$ or CF	27.0		$\bar{x}$ or CF	29.3			$\bar{x}$ or CF 27.9



ADVANCED CAPITAL BUDGETING

NPV	PV factor @ 6%	Total PV
27	0.943	25.461
29.3	0.890	26.077
27.9	0.840	<u>23.436</u>
	PV of cash inflow	74.974
	Less: Cash outflow	<u>50.000</u>
	NPV	<u>24.974</u>

(ii) Possible deviation in the expected value

Year I				
$X - \bar{X}$	$X - \bar{X}$	$X - \bar{X}^2$	P_1	$X - \bar{X}^2 P_1$
14 - 27	-13	169	0.1	16.9
18 - 27	-9	81	0.2	16.2
25 - 27	-2	4	0.4	1.6
40 - 27	13	169	0.3	<u>50.7</u>
				<u>85.4</u>

$$\sigma_1 \sqrt{85.4} = 9.241$$

Year II				
$X - \bar{X}$	$X - \bar{X}$	$(X - \bar{X})^2$	P_2	$(X - \bar{X})^2 P_2$
15-29.3	-14.3	204.49	0.1	20.449
20-29.3	-9.3	86.49	0.3	25.947
32-29.3	2.7	7.29	0.4	2.916
45-29.3	15.7	246.49	0.2	49.298
				98.61

$$\sigma_2 \sqrt{98.61} = 9.930$$

Year III				
$X - \bar{X}$	$X - \bar{X}$	$(X - \bar{X})^2$	P_3	$(X - \bar{X})^2 P_3$
18-27.9	-9.9	98.01	0.2	19.602
25-27.9	-2.9	8.41	0.5	4.205

35-27.9	7.1	50.41	0.2	10.082
48-27.9	20.1	404.01	0.1	40.401
				74.29

$$\sigma_3 \sqrt{74.29} = 8.619$$

Standard deviation about the expected value:

$$\sigma \sqrt{\frac{85.4}{(1.06)^2} + \frac{98.61}{(1.06)^4} + \frac{74.29}{(1.06)^6}} = 14.3696$$

(iii) Standard deviation is a statistical measure of dispersion; it measures the deviation from a central number i.e. the mean.

In the context of capital budgeting decisions especially where we take up two or more projects giving somewhat similar mean cash flows, by calculating standard deviation in such cases, we can measure in each case the extent of variation. It can then be used to identify which of the projects is least risky in terms of variability of cash flows.

A project, which has a lower coefficient of variation will be preferred if sizes are heterogeneous.

Besides this, if we assume that probability distribution is approximately normal we are able to calculate the probability of a capital budgeting project generating a net present value less than or more than a specified amount.

Question – 22

Project X and Project Y are under the evaluation of XY Co. The estimated cash flows and their probabilities are as below:

Project X : Investment (year 0) ₹ 70 lakhs



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Probability Weights	0.30	0.40	0.30
Years	₹ lakhs	₹ lakhs	₹ lakhs
1	30	50	65
2	30	40	55
3	30	40	45

Project Y: Investment (year 0) ₹ 80 lakhs.

Probability Weighted	Annual cash flows through life
	₹ lakhs
0.20	40
0.50	45
0.30	50

- (a) Which project is better based on NPV, criterion with a discount rate of 10%?
- (b) Compute the standard deviation of the present value distribution and analyze the inherent risk of the projects.

(SM TYK – 05)

Solution:

(a) **Calculation of NPV of XY Co.:**

Project X		Cash flow	PVF	PV
Year				
1	$(30 \times 0.3) + (50 \times 0.4) + (65 \times 0.3)$	48.5	0.909	44.09
2	$(30 \times 0.3) + (40 \times 0.4) + (55 \times 0.3)$	41.5	0.826	34.28
3	$(30 \times 0.3) + (40 \times 0.4) + (45 \times 0.3)$	38.5	0.751	28.91
				<u>107.28</u>
	NPV: $(107.28 - 70.00) =$			(+) <u>37.28</u>

Project Y (For 1-3 Years)

1-3	$(40 \times 0.2) + (45 \times 0.5) + (50 \times 0.3)$ NPV $(113.16 - 80.00) =$	45.5	2.487	<u>113.16</u> (+) <u>33.16</u>
-----	---	------	-------	-----------------------------------

(b) **Calculation of Standard deviation σ**

As per Hiller's model

Project X

Year

$$1 \quad \sqrt{(30 - 48.5)^2 0.30 + (50 - 48.5)^2 0.40 + (65 - 48.5)^2 0.30}$$

$$= \sqrt{185.25} = 13.61$$

$$2 \quad \sqrt{(30 - 41.5)^2 0.30 + (40 - 41.5)^2 0.40 + (55 - 41.5)^2 0.30}$$

$$= \sqrt{95.25} = 9.76$$

$$3 \quad \sqrt{(30 - 38.5)^2 0.30 + (40 - 38.5)^2 0.40 + (45 - 38.5)^2 0.30}$$

$$= \sqrt{35.25} = 5.94$$

Standard Deviation about the expected value

$$= \sqrt{\frac{185.25}{(1+0.10)^2} + \frac{95.25}{(1+0.10)^4} + \frac{35.25}{(1+0.10)^6}}$$

$$= \sqrt{\frac{185.25}{1.21} + \frac{95.25}{1.4641} + \frac{35.25}{1.7716}}$$

$$= \sqrt{153.10 + 65.06 + 19.90}$$



$$= \sqrt{238.06}$$

$$= 15.43$$

Project Y (For 1-3 Years)

$$\sqrt{(40 - 45.5)^2 0.20 + (45 - 45.5)^2 0.50 + (50 - 45.5)^2 0.30}$$

$$= \sqrt{12.25} = 3.50$$

Standard Deviation about the expected value

$$= \sqrt{\frac{12.25}{(1+0.10)^2} + \frac{12.25}{(1+0.10)^4} + \frac{12.25}{(1+0.10)^6}}$$

$$= \sqrt{\frac{12.25}{1.21} + \frac{12.25}{1.4641} + \frac{12.25}{1.7716}}$$

$$= \sqrt{10.12 + 8.37 + 6.91}$$

$$= \sqrt{25.4}$$

$$= 5.03$$

Analysis: Project Y is less risky as its Standard Deviation is less than Project X.

Question – 23

XY Ltd. has under its consideration a project with an initial investment of ₹ 1,00,000. Three probable cash inflow scenarios with their probabilities of occurrence have been estimated as below:

Annual cash inflow (₹)	20,000	30,000	40,000
Probability	0.1	0.7	0.2

The project life is 5 years and the desired rate of return is 20%. The estimated terminal values for the project assets under the three probability alternatives, respectively, are ₹ 0, 20,000 and 30,000.

You are required to:

- Find the probable NPV;
- Find the worst-case NPV and the best-case NPV; and
- State the probability occurrence of the worst case, if the cash flows are perfectly positively correlated over time.

(SM TYK – 08)

Solution:

(i) **NPV based on expected cash flows would be as follows:**

Worst Case:

$$= ₹ 1,00,000 + \frac{₹ 20,000}{(1 + 0.20)^1} + \frac{₹ 20,000}{(1 + 0.20)^2} + \frac{₹ 20,000}{(1 + 0.20)^3} + \frac{₹ 20,000}{(1 + 0.20)^4} + \frac{₹ 20,000}{(1 + 0.20)^5}$$

$$= - ₹ 1,00,000 + ₹ 16,666.67 + ₹ 13,888.89 + ₹ 11,574.07 + ₹ 9,645.06$$

$$+ ₹ 8037.76$$

$$\text{NPV} = - ₹ 40,187.76$$

Base Case:

$$= - ₹ 1,00,000 + \frac{₹ 30,000}{(1 + 0.20)^1} + \frac{₹ 30,000}{(1 + 0.20)^2} + \frac{₹ 30,000}{(1 + 0.20)^3} + \frac{₹ 30,000}{(1 + 0.20)^4}$$

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$$+ \frac{₹ 30,000}{(1 + 0.20)^5} + \frac{₹ 20,000}{(1 + 0.20)^5}$$

$$NPV = -₹ 2,244.08$$

Best Case:

$$= ₹ 1,00,000 + \frac{₹ 40,000}{(1 + 0.20)^1} + \frac{₹ 40,000}{(1 + 0.20)^2} + \frac{₹ 40,000}{(1 + 0.20)^3} + \frac{₹ 40,000}{(1 + 0.20)^4}$$

$$+ \frac{₹ 40,000}{(1 + 0.20)^5} + \frac{₹ 30,000}{(1 + 0.20)^5}$$

$$= - ₹ 1,00,000 + ₹ 33,333.33 + ₹ 27,777.78 + ₹ 23,148.15 + ₹ 19,290.12$$

$$+ ₹ 16,075.10 + ₹ 12,056.33$$

$$NPV = ₹ 31,680.81$$

Expected NPV

$$= (- ₹ 40,187.76 \times 0.1) + (- ₹ 2,244.08 \times 0.7) + (₹ 31,680.81 \times 0.2)$$

$$= ₹ 746.52$$

(ii) For the worst case, the cash flows from the cash flow column farthest on the left are used to calculate NPV

$$= ₹ 1,00,000 + \frac{₹ 20,000}{(1 + 0.20)^1} + \frac{₹ 20,000}{(1 + 0.20)^2} + \frac{₹ 20,000}{(1 + 0.20)^3} + \frac{₹ 20,000}{(1 + 0.20)^4} + \frac{₹ 20,000}{(1 + 0.20)^5}$$

$$= - ₹ 1,00,000 + ₹ 16,666.67 + ₹ 13,888.89 + ₹ 11,574.07 + ₹ 9,645.06 + ₹ 8,037.76$$

$$NPV = - ₹ 40,187.76$$

For the best case, the cash flows from the cash flow column farthest on the right are used to calculate NPV

$$= ₹ 1,00,000 + \frac{₹ 40,000}{(1 + 0.20)^1} + \frac{₹ 40,000}{(1 + 0.20)^2} + \frac{₹ 40,000}{(1 + 0.20)^3} + \frac{₹ 40,000}{(1 + 0.20)^4}$$

$$+ \frac{₹ 40,000}{(1 + 0.20)^5} + \frac{₹ 30,000}{(1 + 0.20)^5}$$

$$= - ₹ 1,00,000 + ₹ 33,333.33 + ₹ 27,777.78 + ₹ 23,148.15 + ₹ 19,290.12$$

$$+ ₹ 16,075.10 + ₹ 12,056.33$$

$$NPV = ₹ 31,680.81$$

(iii) If the cash flows are perfectly dependent, then the low cash flow in the first year will mean a low cash flow in every year. Thus, the possibility of the worst case occurring is the probability of getting ₹20,000 net cash flow in year 1 is 10%.

Question – 24

Following are the estimates of the net cash flows and probability of a new project of M/s X Ltd.:

	Year	P = 0.3	P = 0.5	P = 0.2
Initial investment	0	4,00,000	4,00,000	4,00,000
Estimated net after tax cash inflows per year	1 to 5	1,00,000	1,10,000	1,20,000
Estimated salvage value (after tax)	5	20,000	50,000	60,000

Required rate of return from the project is 10%. Find:

- The expected NPV of the project.
- The best case and the worst case NPVs.



- (iii) The probability of occurrence of the worst case if the cash flows are perfectly dependent overtime and independent overtime.
- (iv) Standard deviation and coefficient of variation assuming that there are only three streams of cash flow, which are represented by each column of the table with the given probabilities.
- (v) Coefficient of variation of X Ltd. on its average project which is in the range of 0.95 to 1.0. If the coefficient of variation of the project is found to be less risky than average, 100 basis points are deducted from the Company's cost of Capital

Should the project be accepted by X Ltd?

(SM TYK – 07)

Solution:

(i) Expected NPV

Worst Case

$$\begin{aligned}\text{ENPV} &= (-)4,00,000 + 1,00,000 \times 3.790 + 20,000 \times 0.621 \\ &= -₹ 8,580/-\end{aligned}$$

Base Case

$$\begin{aligned}\text{ENPV} &= (-)4,00,000 + 1,20,000 \times 3.790 + 60,000 \times 0.621 \\ &= ₹ 92,060/-\end{aligned}$$

Best Case

$$\text{ENPV} = (-)4,00,000 + 1,20,000 \times 3.790 + 60,000 \times 0.621$$

$$= ₹ 92,060/-.$$

Expected NPV

$$\begin{aligned}&= 0.30 \times (-) 8,580 + 0.5 \times 47,950 + 92,060 \times 0.20 \\ &= ₹ 39,813/-\end{aligned}$$

(ii) ENPV of The Worst Case

$$1,00,000 \times 3.790 = ₹ 3,79,000$$

(Students may have 3.791 also the values will change accordingly)

$$20,000 \times 0.621 = ₹ 12,420/-$$

$$\text{ENPV} = (-)4,00,000 + 3,79,000 + 12,420 = (-) ₹ 8,580/-$$

ENPV of the best case

$$\begin{aligned}\text{ENPV} &= (-)4,00,000 + 1,20,000 \times 3.790 + 60,000 \times 0.621 \\ &= ₹ 92,060/-\end{aligned}$$

(iii) (a) Required probability = 0.3

(b) Required probability = $(0.3)^5 = 0.00243$

(iv) The base case NPV = $(-) 4,00,000 + (1,10,000 \times 3.79) + (50,000 \times 0.621)$

$$= ₹ 47,950/-$$

$$\begin{aligned}\text{ENPV} &= 0.30 \times (-) 8,580 + 0.5 \times 47,950 + 92,060 \times 0.20 \\ &= ₹ 39,813/-\end{aligned}$$



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Therefore,

$$\sigma_{ENPV} = \sqrt{0.3(-8,580 - 39,813)^2 + 0.5(47,950 - 39,813)^2 + 0.2(92,060 - 39,813)^2}$$

$$= ₹ 35,800/-$$

Therefore, CV = 35,800/39,813

$$= 0.90$$

(v) Risk adjusted out of cost of capital of X Ltd. = 10% - 1% = 9%.

NPV

Expected Cash Flows

$$1 \text{ to } 5 = (1,00,000 \times 0.3 + 1,10,000 \times 0.5 + 1,20,000 \times 0.2)$$

$$= 1,09,000$$

Expected Salvage Value [5th year]

$$= (20,000 \times 0.3 + 50,000 \times 0.5 + 60,000 \times 0.2)$$

$$= 43,000$$

$$NPV = (1,09,000 \times 3.890) + (43,000 \times 0.650) - 4,00,000$$

$$= ₹ 51,960$$

Therefore, the project should be accepted.

Question – 25

Cyber Company is considering two mutually exclusive projects. Investment outlay of both the projects is ₹ 5,00,000 and each is

expected to have a life of 5 years. Under three possible situations their annual cash flows and probabilities are as under:

Situation	Probabilities	Cash Flow (₹)	
		Project A	Project B
Good	0.3	6,00,000	5,00,000
Normal	0.4	4,00,000	4,00,000
Worse	0.3	2,00,000	3,00,000

The cost of capital is 7 per cent, which project should be accepted? Explain with workings.

(SM TYK – 02)

Solution:

Expected Cash Flows & Standard Deviation

Project A

Expected Cash Flows

$$= (6,00,000 \times 0.3) + (4,00,000 \times 0.4) + (2,00,000 \times 0.3)$$

$$= 4,00,000$$

$$\sigma_x = \sqrt{(6,00,000 - 4,00,000)^2 \cdot 0.3 + (4,00,000 - 4,00,000)^2 \cdot 0.4 + (2,00,000 - 4,00,000)^2 \cdot 0.3}$$

$$= ₹ 1,54,919$$

Project B

$$ENCF = 0.3 (5,00,000) + 0.4 (4,00,000) + 0.3 (3,00,000)$$

$$= 4,00,000$$



$$\sigma^2 = 0.3 (5,00,000 - 4,00,000)^2 + 0.4 (4,00,000 - 4,00,000)^2 + 0.3 (3,00,000 - 4,00,000)^2$$

$$\sigma = \sqrt{6,00,00,00,000}$$

$$\sigma = 77,459.66$$

Expected NPV

$$\begin{aligned} A &= (4,00,000 \times \text{PVAF } 7\% \ 5) - 5,00,000 \\ &= (4,00,000 \times 4.100) - 5,00,000 = 11,40,000 \end{aligned}$$

$$B = (4,00,000 \times 4.100) - 5,00,000 = 11,40,000$$

Project B should be accepted due to lower risk.

Recommendation: NPV in both projects being the same, the project should be decided on the basis of standard deviation and hence project 'B' should be accepted having lower standard deviation, means less risky.

(iii) Simulation

Question – 26

Cost of Machine = ₹ 1,30,000

Discounting Rate = 10% p.a.

Annual Cash Flow		Project Life	
Value (₹)	Probability	Value (Year)	Probability
10,000	0.02	3	0.05
15,000	0.03	4	0.10
20,000	0.15	5	0.30
25,000	0.15	6	0.25

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30,000	0.30	7	0.15
35,000	0.20	8	0.10
40,000	0.15	9	0.03
		10	0.02

Random Number

53479	81115	98036	12217	59526
97344	70328	58116	91964	26240
66023	38277	74523	71118	84892
99776	75723	03172	43112	83086
30176	48979	92153	38416	42436
81874	83339	14988	99937	13213
19839	90630	71863	95053	55532
09337	33435	53869	52769	18801
31151	58295	40823	41330	21093
67619	52515	03037	81699	17106

Take beginning two digit random number. One for annual cash flows & second for project life.

Solution:

Annual Cash Flow				Project Life			
Value (₹)	Probabil-ity	Cumula-tive Probabil-ity	Tow Digit Rando m No.	Value (Year)	Probabi l-ity	Cumula-tive Probabil-ity	Two Digit Rando m No.
10,000	0.02	0.02	00 – 01	3	0.05	0.05	00 – 04
15,000	0.03	0.05	02 – 04	4	0.10	0.15	05 – 14
20,000	0.15	0.20	05 – 19	5	0.30	0.45	15 – 44
25,000	0.15	0.35	20 – 34	6	0.25	0.70	45 – 69
30,000	0.30	0.65	35 – 64	7	0.15	0.85	70 – 84
35,000	0.20	0.85	65 – 84	8	0.10	0.95	85 – 94
40,000	0.15	1.00	85 – 99	9	0.03	0.98	95 – 97
				10	0.02	1.00	98 – 99



Simulation Results

Annual Cash Flow			Project Life			
Run	Random No.	Corres. Value of Annual Cash Flow (1)	Random No.	Corres. Value of Project Life	PVAF @ 10% (2)	NPV (1) × (2) – 1,30,000
1	53	30,000	97	9	5.759	42,770
2	66	35,000	99	10	6.145	85,075
3	30	25,000	81	7	4.868	(8,300)
4	19	20,000	09	4	3.170	(66,600)
5	31	25,000	67	6	4.355	(21,125)
6	81	35,000	70	7	4.868	40,380
7	38	30,000	75	7	4.868	16,040
8	48	30,000	83	7	4.868	16,040
9	90	40,000	33	5	3.791	21,640
10	58	30,000	52	6	4.355	650

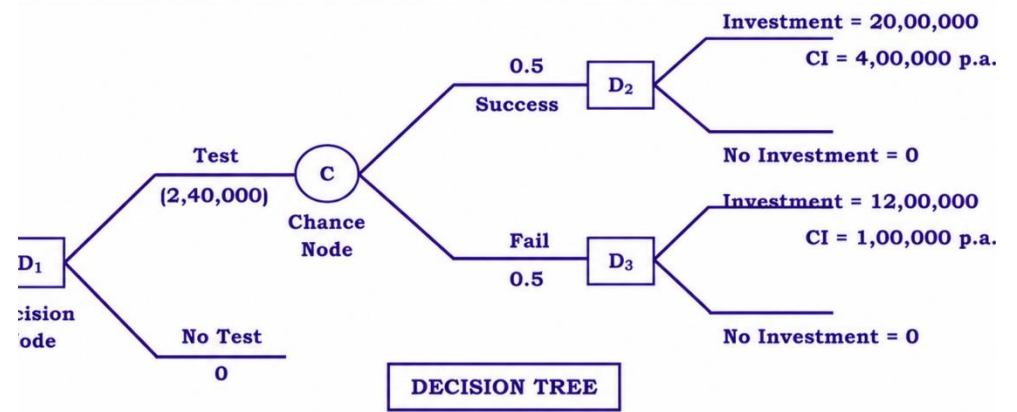
(iv) Decision tree

Question – 27

L & R Limited wishes to develop new virus-cleaner software. The cost of the pilot project would be ₹ 2,40,000. Presently, the chances of the product being successfully launched on a commercial scale are rated at 50%. In case it does succeed. L&R can invest a sum of ₹20 lacs to market the product. Such an effort can generate perpetually, an annual net after tax cash income of ₹4 lacs. Even if the commercial launch fails, they can make an investment of a smaller amount of ₹12 lacs with the hope of gaining perpetually a sum of ₹1 lac. Evaluate the proposal, adopting decision tree approach. The discount rate is 10%.

Solution:

Decision tree diagram is given below:



At Decision Point D₂

Option 1: If investment NPV = $\frac{4,00,000}{10\%} - 20,00,000 = 20,00,000$

Option 2: No investment NPV = 0

No investment is better

At Decision Point D₃

Option 1: Investment NPV = $\frac{1,00,000}{10\%} - 12,00,000 = -2,00,000$

Option 2: No investment NPV = 0

Investment is better due to higher NPV

Calculation of EMV at Point C

$$\begin{aligned} \text{EMV} &= (20,00,000 \times 0.5) + (0 \times 0.5) \\ &= 10,00,000 \end{aligned}$$

At Decision Point D₁

Option 1: Testing NPV = $(10,00,000 - 2,40,000) = ₹ 7,60,000$

Option 2: No Testing NPV = 0

Testing is better due to higher NPV

PART 3: REPLACEMENT DECISION

Question – 28

Company X is forced to choose between two machines A and B. The two machines are designed differently but have identical capacity and do exactly the same job. Machine A costs ₹ 1,50,000 and will last for 3 years. It costs ₹ 40,000 per year to run. Machine B is an 'economy' model costing only ₹ 1,00,000, but will last only for 2 years, and costs ₹ 60,000 per year to run. These are real cash flows. The costs are forecasted in rupees of constant purchasing power. Ignore tax. Opportunity cost of capital is 10 per cent. Which machine company X should buy?

(SM TYK – 24)

Solution:

Calculation of EAPVCO

Machine A

$$PVCO = 1,50,000 + (40,000 \times 2.487)$$

$$= 2,94,474$$

$$EA PVCO = \frac{PVCO}{PVAF} = \frac{2,94,480}{2.487} = 1,00,314$$

Machine B

$$PVCO = 1,00,000 + (60,000 \times 1.735)$$

$$EA PVCO = \frac{2,04,100}{1.735} = 1,17,637$$

Question – 29

A Company named Roby's cube decided to replace the existing Computer system of their organization. Original cost of old system was ₹ 25,000 and it was installed 5 years ago. Current market value of old system is ₹ 5,000. Depreciation of the old system was charged with life of 10 years with Estimated Salvage value as Nil. Depreciation of the new system will be charged with life over 5 years Present cost of the new system is ₹ 50,000. Estimated Salvage value of the new system is ₹1,000. Estimated cost savings with new system is ₹ 5,000 per year. Increase in sales with new system is assumed at 10% per year based on original total sales of ₹ 1,00,000. Company follows straight line method of depreciation. Cost of capital of the company is 10% whereas tax rate is 30%.

Solution:

Calculation of NPV

	Year	PVF (10%)	Amount	P.V.
(A) Cash outflows				
Cost of new machine	0	1,000	50,000	50,000
Sale of old machine	0	1,000	(7,250)	(7,250)
(W.N.1)				(7,250)
				42,750
(B) Cash inflows				
Incremental CFAT	1-5	3.791	12,690	48,108
(W.N.2)	5	0.621	1,000	621



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Incremental [2,000 – 0]	T.V.				
(B)					48,729
NPV (B – A)					5,979

Since NPV is positive, hence old machine should be replaced.

W.N. 1: Sale of Old Machine

Sales consideration = ₹ 5,000(i)

(-) B.V. = ₹ 12,500

Capital loss = ₹ 7,500

Tax saving @ 30% = 2,250(ii)

Net (i + ii) = 7,250

W.N. 2: Incremental CFAT

Increase in sales [₹ 1,00,000 × 10%] = ₹ 10,000

Saving in Exp = ₹ 5,000

Incremental CFAT = ₹ 15,000(i)

(-) Incremental depreciation

[Depreciation of new – Depreciation of old]

$$\left[\left(\frac{50,000 - 1,000}{5} \right) - \frac{25,000}{10} \right] = ₹ 7,300$$

PBT = ₹ 7,700

Tax @ 30% = ₹ 2310(ii)

Incremental CFAT (i + ii) = 12,690

Question – 30

A company has an old machine having book value zero – which can be sold for ₹ 50,000. The company is thinking to choose one from following two alternatives:

- (i) To incur additional cost of ₹ 10,00,000 to upgrade the old existing machine.
- (ii) To replace old machine with a new machine costing ₹ 20,00,000 plus installation cost ₹ 50,000.

Both above proposals envisage useful life to be five years with salvage value to be nil.

The expected after tax profits for the above three alternatives are as under :

Year	Old existing Machine (₹)	Upgraded Machine (₹)	New Machine (₹)
1	5,00,000	5,50,000	6,00,000
2	5,40,000	5,90,000	6,40,000
3	5,80,000	6,10,000	6,90,000
4	6,20,000	6,50,000	7,40,000
5	6,60,000	7,00,000	8,00,000

The tax rate is 40 per cent.

The company follows straight line method of depreciation. Assume cost of capital to be 15 per cent.



P.V.F. of 15%, 5 = 0.870, 0.756, 0.658, 0.572 and 0.497. You are required to advise the company as to which alternative is to be adopted.

(SM TYK – 23)

Solution:

Option 1: Upgraded Machine

Calculation of NPV

	1	2	3	4	5
Upgrade					
PAT	5,50,000	5,90,000	6,10,000	6,50,000	7,00,000
(+) Depreciation	2,00,000	2,00,000	2,00,000	2,00,000	2,00,000
CFAT	7,50,000	7,90,000	8,10,000	8,50,000	9,00,000
(-) CFAT (old)	5,00,000	5,40,000	5,80,000	6,20,000	6,60,000
Incremental CFAT	2,50,000	2,50,000	2,30,000	2,30,000	2,40,000
(x) PVF	0.870	0.756	0.658	0.572	0.497

PVCI = 8,08,680

(-) PVCO = 10,00,000

NPV = (1,91,320)

Option 2: New Machine

	1	2	3	4	5
New Machine					
PAT	6,00,000	6,40,000	6,90,000	7,40,000	8,00,000
(+) Depreciation	4,10,000	4,10,000	4,10,000	4,10,000	4,10,000
20,50,000 5					
CFAT	10,10,000	10,50,000	11,00,000	11,50,000	12,10,000
	0	0	0	0	0

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(-) CFAT (old)	5,00,000	5,40,000	5,80,000	6,20,000	6,60,000
Incremental CFAT	5,10,000	5,10,000	5,20,000	5,30,000	5,50,000
(x) PVF	0.870	0.756	0.658	0.572	0.497

Working Note 1:

Sale of old machine

Sale consideration = 50,000

(-) B.V. = 0

Capital gain = 50,000

Tax @ 40% = 20,000

= 30,000

PVCI = 17,47,930

(-) PVCO = 20,20,000 [20,50,000 – 30,000]

NPV = (2,72,070)

Since NPV is negative in both option, hence old machine without up gradation is better

Question – 31

A machine used on a production line must be replaced at least every four years. Costs incurred to run the machine according to its age are:

Age of the Machine (years)					
	0	1	2	3	4
Purchase price (in ₹)	60,000				
Maintenance (in ₹)		16,000	18,000	20,000	20,000
Repair (in ₹)		0	4,000	8,000	16,000



ADVANCED CAPITAL BUDGETING

Scrap Value (in ₹)		32,000	24,000	16,000	8,000
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Future replacement will be with identical machine with same cost. Revenue is unaffected by the age of the machine. Ignoring inflation and tax, determine the optimum replacement cycle. PV factors of the cost of capital of 15% for the respective four years are 0.8696, 0.7561, 0.6575 and 0.5718.

(SM TYK – 26)

Solution:

Working Notes

First of all, we shall calculate cash flows for each replacement cycle as follows:

One year replacement cycle

₹

Year	Replacement Cost	Maintenance & Repair	Residual Value	Net Cash Flow
0	(60,000)	-	-	(60,000)
1	-	(16,000)	32,000	16,000

Two years replacement cycle

₹

Year	Replacement Cost	Maintenance & Repair	Residual Value	Net Cash Flow
0	(60,000)	-	-	(60,000)
1	-	(16,000)	-	(16,000)
2	-	(22,000)	24,000	2,000

Three years replacement cycle

₹

Year	Replacement Cost	Maintenance & Repair	Residual Value	Net Cash Flow
0	(60,000)	-	-	(60,000)

1	-	(16,000)	-	(16,000)
2	-	(22,000)	-	(22,000)
3	-	(28,000)	16,000	(12,000)

Four years replacement cycle

₹

Year	Replacement Cost	Maintenance & Repair	Residual Value	Net Cash Flow
0	(60,000)	-	-	(60,000)
1	-	(16,000)	-	(16,000)
2	-	(22,000)	-	(22,000)
3	-	(28,000)	-	(28,000)
4	-	(36,000)	8,000	(28,000)

Now we shall calculate NPV for each replacement cycles

Year	PVF @ 15%	1 Year		2 Years		3 Years		4 Years	
		Cash Flows	PV	Cash Flows	PV	Cash Flows	PV	Cash Flows	PV
0	1	-60,000	-60,000	-60,000	-60,000	-60,000	-60,000	-60,000	-60,000
1	0.8696	16,000	13,914	-16,000	-13,914	-16,000	-13,914	-16,000	-13,914
2	0.7561	-	-	2,000	1,512	-22,000	-16,634	-22,000	-16,634
3	0.6575	-	-	-	0	-12,000	-7,890	-28,000	-18,410
4	0.5718	-	-	-	0	0	0	-28,000	-16,010
			-46,086		-72,402		-98,438		-1,24,968

Replacement Cycle		EAC (₹)
1 Year	46,086	52,997
2 Years	72,402	44,536
3 Years	98,438	43,114
4 Years	1,24,968	43,772



Since EAC is least in case of replacement cycle of 3 years hence machine should be replaced after every three years.

Note: Alternatively, Answer can also be computed by excluding initial outflow as there will be no change in final decision.

Question – 32

Company Y is operating an elderly machine that is expected to produce a net cash inflow of ₹ 40,000 in the coming year and ₹ 40,000 next year. Current salvage value is ₹ 80,000 and next year's value is ₹ 70,000. The machine can be replaced now with a new machine, which costs ₹ 1,50,000, but is much more efficient and will provide a cash inflow of ₹ 80,000 a year for 3 years. Company Y wants to know whether it should replace the equipment now or wait a year with the clear understanding that the new machine is the best of the available alternatives and that it in turn be replaced at the optimal point. Ignore tax. Take opportunity cost of capital as 10 per cent. Advise with reasons.

(SM TYK – 25)

Solution:

Calculation of NPV

	Year	PVF (10%)	Replace New		Replace on year	
			Amount	P.V.	Amount	P.V.
Cash outflows						
Cost of new machine	0	1.000	1,50,000	1,50,000	-	-
	1	0.909	-	-	1,50,000	1,36,350
Sale of old machine	0	1.000	(80,000)	(80,000)	-	-
	1	0.909	-	-	(70,000)	(63,630)

ADVANCED CAPITAL BUDGETING

Total (A)				70,000		72,720
Cash inflows						
CFAT	1-3	2.487	80,000	1,98,960	-	-
	1	0.909	-	-	40,000	36,360
	2-4	2.261	-	-	80,000	1,80,880
Total (B)				1,98,960		2,17,240
NPV (B-A)				1,28,960		1,44,520

Machine should be replaced in year 1 due higher NPV.

Question – 33

X Ltd. is a taxi operator. Each taxi cost to company ₹ 4,00,000 and has a useful life of 3 years. The taxi's operating cost for each of 3 years and salvage value at the end of year is as follows:

	Year 1	Year 2	Year 3
Operating Cost	₹ 1,80,000	₹ 2,10,000	₹ 2,38,000
Resale Value	₹ 2,80,000	₹ 2,30,000	₹ 1,68,000

You are required to determine the optimal replacement period of taxi if cost of capital of X Ltd. is 10%.

Solution:

NPV if taxi is kept for 1 Year

$$= - ₹ 4,00,000 + ₹ 1,00,000 (0.909)$$

$$= - ₹ 3,09,100$$

NPV if taxi is kept for 2 Year

$$= - ₹ 4,00,000 - ₹ 1,80,000 \times 0.909 + ₹ 20,000 \times 0.826$$

$$= - ₹ 5,47,100$$



NPV if taxi is kept for 3 Year

$$= - ₹ 4,00,000 - ₹ 1,80,000 \times 0.909 - ₹ 2,10,000 \times 0.826 - ₹ 70,000 \times 0.751$$

$$= - ₹ 7,89,650$$

Since above NPV figures relate to different periods, there are not comparable. to make them comparable we shall use concept of EAC as follows:

EAC of 1 year

$$\frac{3,09,100}{0.909} = ₹ 3,40,044$$

EAC of 2 year

$$\frac{5,47,100}{1.735} = ₹ 3,15,331$$

EAC of 3 year

$$\frac{7,89,650}{2.486} = ₹ 3,17,639$$

Since lowest EAC incur if taxi for 2 year; Hence the optimum replacement cycle to replace taxi in 2 years.

Question – 34

Trouble Free Solutions (TFS) is an authorized service center of a reputed domestic air conditioner manufacturing company. All complaints/service related matters of Air conditioner are attended by this service center. The service center employs a large number of mechanics, each of whom is provided with a motor bike to attend the complaints. Each mechanic travels approximately 40,000 kms per annum. TFS decides to continue its present policy of always buying a new bike for its mechanics but wonders whether the present policy

of replacing the bike every three year is optimal or not. It is of believe that as new models are entering into market on yearly basis, it wishes to consider whether a replacement of either one year or two years would be better option than present three year period. The fleet of bike is due for replacement shortly in near future.

The purchase price of latest model bike is ₹ 55,000. Resale value of used bike at current prices in market is as follows:

Period	₹
1 Year old	35,000
2 Year old	21,000
3 Year old	9,000

Running and Maintenance expenses (excluding depreciation) are as follows:

Year	Road Taxes Insurance etc. (₹)	Petrol Repair Maintenance etc. (₹)
1	3,000	30,000
2	3,000	35,000
3	3,000	43,000

Using opportunity cost of capital as 10% you are required to determine optimal replacement period of bike.

(SM TYK – 27)

Solution:**NPV if Bike is kept for 1 Year**

$$= - ₹ 55,000 + ₹ 2,000 (0.909)$$



ADVANCED CAPITAL BUDGETING

$$= - ₹ 53,182$$

NPV if Bike is kept for 2 Year

$$= - ₹ 55,000 - ₹ 33,000 \times 0.909 - ₹ 17,000 \times 0.826$$

$$= - ₹ 99,039$$

NPV if Bike is kept for 3 Year

$$= - ₹ 55,000 - ₹ 33,000 \times 0.909 - ₹ 38,000 \times 0.826 - ₹ 37,000 \times 0.751$$

$$= - ₹ 1,44,172$$

EAC of 1 year

$$\frac{53,182}{0.909} = ₹ 58,506$$

EAC of 2 year

$$\frac{99,039}{1,735} = ₹ 57,083$$

EAC of 3 year

$$\frac{1,44,172}{2,486} = ₹ 57,993$$

Thus, from above table it is clear that EAC is least in case of 2 years, hence bike should be replaced every two years.

Question – 35

SS Company is considering the replacement of its existing machine with a new machine. The Purchase price of the New machine is ₹ 26 Lakhs and its expected Life is 8 years. The company follows straight-line method of depreciation on the original investment (scrap value

is not considered for the purpose of depreciation). The other expenses to be incurred for the New Machine are as under:

- (a) Installation Charges ₹ 9,000
- (b) Fees paid to the consultant for his advice to buy New Machine ₹ 6,000.
- (c) Additional Working Capital required ₹ 17,000. (will be released after 8 years)

The written down value of the existing machine is ₹ 76,000, and its Cash Salvage Value is ₹ 12,500. The dismantling of this machine would cost ₹ 4,500. The Annual Earnings (before tax but after depreciation) from the New Machine would amount to ₹ 3,15,000. Income tax rate is 35%. The Company's required Rate of Return is 13%.

You are required to advise on the viability of the proposal.

$$PVIF (13\%, 8) = 0.376 \quad PVIFA (13\%, 8) = 4.80$$

(MTP August – 2025)

Solution:

Working Notes:

1. Computation of Annual Depreciation-

Particulars	₹
Purchase Price	26,00,000
Add: 1. Installation Charges	9,000
2. Fees Paid to Consultant for Advice	6,000
Total Cost of New Machine	26,15,000



ADVANCED CAPITAL BUDGETING

Useful Life	8 Years
Annual Depreciation (Total Cost/No. of Years)	3,26,875

2. Computation of Annual Cash Savings-

Particulars	₹
Annual Earnings	3,15,000
Less: Tax @ 35%	1,10,250
Earnings after Tax	2,04,750
Add: Depreciation on New Machine	3,26,875
Annual Cash Saving	5,31,625

3. Tax effect on sale of Old Machine-

Particulars	₹
Proceeds of Sale	12,500
Less: Cost of Removal	4,500
Net Proceeds	8,000
Less: WDV	76,000
Net Loss due to Sale	68,000
Tax savings due to Loss on Sale @ 35%	23,800
Total Cash Inflow due to Sale (₹ 8,000 + ₹ 23,800)	31,800

4. Computation of Net Present Value-

Particulars	Period	Cash Flow (₹)	PVF @ 13%	PV (₹)
(a) Annual Cash inflow after Tax	1-8	5,31,625	4.8	25,51,800
(b) Net Salvage Value of Existing Machine	0	31,800	1.0	31,800

(c) Working Capital Realized	8	17,000	0.376	6,392
Present Value of Cash Inflows				25,89,992
Less: 1. Initial Investment	0	26,15,000	1.0	26,15,000
2. Initial Working Capital	0	17,000	1.0	17,000
NPV of the Proposal				(42,008)

Decision: Since NPV of the project is negative it is not viable.

Question – 36

A & Co. is contemplating whether to replace an existing machine or to spend money on overhauling it. A & Co. currently pays no taxes. The replacement machine costs ₹ 90,000 now and requires maintenance of ₹ 10,000 at the end of every year for eight years. At the end of eight years it would have a salvage value of ₹ 20,000 and would be sold. The existing machine requires increasing amounts of maintenance each year and its salvage value falls each year as follows:

Year	Maintenance (₹)	Salvage (₹)
Present	0	40,000
1	10,000	25,000
2	20,000	15,000
3	30,000	10,000
4	40,000	0

The opportunity cost of capital for A & Co. is 15%.

Required:



ADVANCED CAPITAL BUDGETING

When should the company replace the machine?

(Notes: Present value of an annuity of Re. 1 per period for 8 years at interest rate of 15% : 4.4873; present value of Re. 1 to be received after 8 years at interest rate of 15% : 0.3269).

(SM TYK – 22)

Solution:

A & Co.

Equivalent cost of (EAC) of new machine

		₹
(i)	Cost of new machine now	90,000
	Add: PV of annual repairs @ ₹ 10,000 per annum for 8 years (₹ 10,000 × 4.4873)	<u>44,873</u> 1,34,873
		6,538
	Less: PV of salvage value at the end of 8 years (₹ 20,000 × 0.3269)	<u>1,28,335</u> 28,600
	Equivalent annual cost (EAC) (₹ 1,28,355/4.4873)	

PV of cost of replacing the old machine in each of 4 years with new machine

Scenario	Year	Cash Flow	PV @ 15%	PV
		₹		₹
Replace Immediately	0	(28,600)	1.00	(28,600)
		40,000	1.00	<u>40,000</u>
				<u>11,400</u>
Replace in one year	1	(28,600)	0.870	(24,882)

	1	(10,000)	0.870	(8,700)
	1	25,000	0.870	<u>21,750</u>
				<u>(11,832)</u>
Replace in two years	1	(10,000)	0.870	(8,700)
	2	(28,600)	0.756	(21,622)
	2	(20,000)	0.756	(15,120)
	2	15,000	0.756	<u>11,340</u>
				<u>(34,102)</u>
Replace in three years	1	(10,000)	0.870	(8,700)
	2	(20,000)	0.756	(15,120)
	3	(28,600)	0.658	(18,819)
	3	(30,000)	0.658	(19,740)
	3	10,000	0.658	<u>6,580</u>
				<u>(55,799)</u>
Replace in 4 years	1	(10,000)	0.870	(8,700)
	2	(20,000)	0.756	(15,120)
	3	(30,000)	0.658	(19,740)
	4	(28,600)	0.572	(16,359)
	4	(40,000)	0.572	<u>(22,880)</u>
				<u>(82,799)</u>

Advice: The company should replace the old machine immediately because the PV of cost of replacing the old machine with new machine is least.

RESIDUAL

Question – 37

Jumble Consultancy Group has determined relative utilities of cash flows of two forthcoming projects of its client company as follows:

Cash Flow in ₹	-15,000	-10,000	-4,000	0	15,000	10,000	5,000	1,000
	0	0	0		0	0	0	0
Utilities	-100	-60	-3	0	40	30	20	10



ADVANCED CAPITAL BUDGETING

The distribution of cash flows of project A and Project B are as follows:

Project A

Cash Flow (₹)	-15,000	-10,000	15,000	10,000	5,000
Probability	0.10	0.20	0.40	0.20	0.10

Project B

Cash Flow (₹)	-10,000	-4,000	15,000	5,000	10,000
Probability	0.10	0.15	0.40	0.25	0.10

Which project should be selected and why ?

(SM TYK – 21)

Solution:

Evaluation of project utilizes of Project A and Project B

Project A			
Cash Flow (in ₹)	Probability	Utility	Utility Value
-15,000	0.10	-100	-10
-10,000	0.20	-60	-12
15,000	0.40	40	16
10,000	0.20	30	6
5,000	0.10	20	<u>2</u>
			<u>2</u>

Project B			
Cash Flow (in ₹)	Probability	Utility	Utility Value
-10,000	0.10	-60	-6
-4,000	0.15	-3	-0.45
15,000	0.40	40	16

5,000	0.25	20	5
10,000	0.10	30	<u>3</u>
			<u>17.55</u>

Project B should be selected as its expected utility is more.

Question – 38

D and Co. is a company which installs pipes for supply of oxygen in the hospitals as per their specifications. It is planning to invest ₹ 40 crore in a new facility to convert vans and trucks into ambulance. Each ambulance will be designed and built according to customer requirements. D and Co. expects ambulance production and sales in the first four years of operation to be as follows:

Year	1	2	3	4
Ambulances produced and sold	250	300	450	450

The selling price for an ambulance depends on the van or truck which is converted, the quality of the units installed and the extent of conversion work required. D and Co. has undertaken research into likely sales and costs of different kinds of ambulances which could be selected by customers, as follows:

Ambulance Type	Basic	Standard	Deluxe
Probability of selection	20%	45%	35%
Selling price (₹/unit)	3,00,000	42,00,000	72,00,000
Conversion cost (₹/unit)	23,00,000	29,00,000	40,00,000

Fixed costs of the production facility are expected to depend on the volume of ambulance production as follows:



ADVANCED CAPITAL BUDGETING

Production volume (units/year)	200-299	300-399	400-499
Fixed costs (₹ crore/year)	40	50	55

The applicable tax rate for D and Co. is 28% per year, being settled in the year in which it arises. The company can claim tax allowable depreciation on the cost of the investment on a straight-line basis over ten years.

D and Co. evaluates investment projects using an after-tax discount rate of 11%.

Required:

- (i) Advise the company on the financial viability of the planned investment for the first four years of operation if tax benefit on unabsorbed depreciation for the remaining period will not be available.
- (ii) Advise the company on the financial viability of the planned investment continuing to produce and sell ambulances beyond the first four years if after the fourth year of operation, D and Co. expects to continue to produce and sell 450 ambulances per year for the foreseeable future.

Note: (1) Use PV Factors rounded off upto three decimal points.

(2) Present all calculation in ₹ Crore rounded off upto three decimal points.

(MTP April – 2026)

Solution:

Working Notes:

1. Average selling price

$$= (30,00,000 \times 0.20) + (42,00,000 \times 0.45) + (72,00,000 \times 0.35)$$

$$= ₹ 50,10,000 \text{ per unit}$$

2. Average conversion cost

$$= (23,00,000 \times 0.20) + (29,00,000 \times 0.45) + (40,00,000 \times 0.35)$$

$$= ₹ 31,65,000 \text{ per unit}$$

3. Computation of Sales Income

Year	1	2	3	4
Sales volume (units/year)	250	300	450	450
Average selling price (₹ /unit)	50,10,000	50,10,000	50,10,000	50,10,000
Sales income (₹ crore/year)	125.25	150.30	225.45	225.45

4. Computation of Conversion Cost

Year	1	2	3	4
Sales volume (units/year)	250	300	450	450
Average conversion cost (₹ /unit)	31,65,000	31,65,000	31,65,000	31,65,000
Conversion cost (₹ crore/year)	79.125	94.950	142.425	142.425



5. Tax allowable depreciation = ₹ 40 crore/10 = ₹ 4 crore per year

Benefit of tax allowable depreciation = ₹ 4 crore × 0.28 = ₹ 1.12 crore per year

(i) Calculation of NPV over four years

Year	1	2	3	4
	₹ crore	₹ crore	₹ crore	₹ crore
Sales income	125.25	150.30	225.45	225.45
Conversion cost	(79.125)	(94.95)	(142.425)	(142.425)
Contribution	46.125	55.35	83.025	83.025
Fixed cost	(40)	(50)	(55)	(55)
Before-tax cash flow	6.125	5.35	28.025	28.025
Tax liability at 28%	(1.715)	(1.498)	(7.847)	(7.847)
Tax allowable depreciation benefits	1.120	1.120	1.120	1.120
After-tax cash flow	5.53	4.972	21.298	21.298
Discount at 11%	0.901	0.812	0.731	0.659
Present values	4.983	4.037	15.569	14.035
Total present value of inflows				38.624
Initial investment				40.000
NPV				(1.376)

Advice: If only the first four years of operation are considered, the NPV of the planned investment is negative and so it would not be financially acceptable.

(ii) Ignoring tax allowable depreciation, after-tax cash flow from year five onwards will be:

₹ 28.025 crore – ₹ 7.847 crore = ₹ 20.178 crore per year

Present value of this cash flow in perpetuity

= (₹ 20.178 crore / 0.11) × 0.659 = ₹ 120.885 crore

There would be a further six years of tax benefits from tax allowable depreciation. The present value of these cash flows would be:

₹ 1.12 crore × 2.787 = ₹ 3.121 crore

Increase in NPV of production and sales continuing beyond the first four years would be:

₹ 120.885 crore + ₹ 3.121 crore = ₹ 124.006 crore

In other words, NPV of the planned investment would be:

₹ 122.632 crore (₹ 124.006 crore – ₹ 1.376 crore)

Advice: If production and sales beyond the first four years are considered, the NPV is strongly positive and so the planned investment is financially acceptable.

MULTIPLE CHOICE QUESTIONS**Case Scenario – 01**

XYZ Ltd. is a mid-sized manufacturing company that produces industrial equipment. The company is considering a new investment project—a state-of-the-art automated production line, which is expected to improve production efficiency. The details of the same project are as follows:

	₹
Initial Cost of the project	10,00,000
Sales price/unit	60
Cost/unit	40
Sales volumes	
Year 1	20,000 units
Year 2	30,000 units
Year 3	30,000 units

The applicable discount rate is 10% p.a.

Based on above case scenario answer the following questions:

- I.** Sensitivity analysis helps to identify.....
- the exact profitability of the project
 - the break-even point.
 - the degree to which a change in each variable affects the NPV.
 - the amount of investment required

- II.** The sale price per unit so that the project would break even with zero NPV shall be approximately.....
- ₹ 40.00
 - ₹ 55.26
 - ₹ 60.00.
 - ₹ 44.74
- III.** The cost per unit so that the project would break even with zero NPV shall be approximately.....
- ₹ 40.00
 - ₹ 55.28
 - ₹ 60.00.
 - ₹ 44.74
- IV.** Overallin the sale volume will lead to the project to break even with zero NPV.
- increase of 23.68%
 - fall of 23.68%
 - Increase of 31.03%
 - fall of 31.03%
- V.** A/anin the initial outlay will lead to the project to break even with zero NPV.
- increase of 23.68%
 - fall of 23.68%
 - Increase of 31.03%
 - fall of 31.03 %

(MTP October – 2024 & November – 2025)

Answer: Case Scenario – 01

- I. (c) the degree to which a change in each variable affects the NPV.**
- II. (b) ₹ 55.26**



ADVANCED CAPITAL BUDGETING

- | | | |
|------|-----|---------------------------|
| III. | (d) | ₹ 44.74 |
| IV. | (b) | fall of 23.68% |
| V. | (c) | Increase of 31.03% |

Case Scenario – 02

XYZ Ltd. plans to invest ₹ 8,00,000 in a new unit. The project is expected to have a useful life of 4 years, with no salvage value at the end of its life. The annual depreciation charge for the project is ₹ 2,00,000.

Projected revenues and costs for the project, ignoring inflation, are provided as follows:

Year	Revenues (₹)	Costs (₹)
1	6,00,000	3,00,000
2	7,00,000	4,00,000
3	8,00,000	4,00,000
4	8,00,000	4,00,000

XYZ Ltd. is subject to a corporate tax rate of 60%, and the cost of capital for the project, including inflation premium, is 10%.

Depreciation provides a tax benefit, and inflation rates for revenues and costs over the project's lifespan are as follows:

Year	Revenue Inflation	Cost Inflation
1	10%	12%
2	9%	10%
3	8%	9%

4 7% 8%

From the information given above, choose the correct answer to the following questions:

I. The depreciation tax benefit for the project per year shall be.....

- (a) ₹ 1,20,000 (b) ₹ 1,50,000
(c) ₹ 2,00,000 (d) ₹ 1,80,000

II. The inflation-adjusted revenue in Year 2 shall be.....

- (a) ₹ 7,00,000 (b) ₹ 8,39,300
(c) ₹ 4,92,800 (d) ₹ 5,01,760

III. The total cash inflow in Year 1 after adjusting for inflation and tax benefit on depreciation shall be.....

- (a) ₹ 3,30,000 (b) ₹ 3,36,000
(c) ₹ 2,49,600 (d) ₹ 4,92,800

IV. The inflation-adjusted cost in Year 2 shall be.....

- (a) ₹ 7,00,000 (b) ₹ 8,39,300
(c) ₹ 4,92,800 (d) ₹ 5,01,760

V. The present value of cash inflow for the year 3 shall be approximately.....

- (a) ₹ 213,604 (b) ₹ 226,299
(c) ₹ 226,886 (d) ₹ 239,949

(MTP August – 2025)**Answer: Case Scenario – 02**

- I. (a) ₹ 1,20,000
- II. (b) ₹ 8,39,300
- III. (c) ₹ 2,49,600
- IV. (c) ₹ 4,92,800
- V. (d) ₹ 239,949

Case Scenario – 03

P Ltd is considering a new project with the following details:

Initial project cost	₹ 4,80,000
Annual projected sales	₹ 4,00,000
Annual projected variable cost	₹ 1,60,000
Annual projected fixed cost	₹ 60,000
Project life	4 years
Cost of capital	10% p.a.

Consider Cumm. PVF for 4 years @ 10% = 3.169 and @ 11% = 3.103

Note: (a) Ignore depreciation on initial project cost and taxation.

(b) Calculation up to 2 decimal places.

From the information given above, choose the correct answer to the following question No. I to III:

(EXAM SEPTEMBER - 2025)

- I. If cost of capital is varied adversely by 10%, what is the percentage change in NPV ?
- (a) 11.00% (b) 12.10%
- (c) 13.14% (d) 14.45%
- II. If initial project cost is varied adversely by 10%, what is the percentage change in NPV?
- (a) 42.42% (b) 52.80%
- (c) 53.09% (d) 63.09%
- III. If annual cash inflow is varied adversely by 10%, what is the percentage change in NPV ?
- (a) 33.38% (b) 53.09%
- (c) 63.09% (d) 52.80%

Answer: Case Scenario – 03

- I. (c) 13.14%
- II. (c) 53.09%
- III. (c) 63.09%

Case Scenario – 04

PQR Ltd. is considering two new products A and B, only one of which can be added to its production line. Product A is sure seller. It is certain that 2,00,000 units of product A with the firm's maximum



capacity can be manufactured and sold each year with a contribution margin of ₹ 5 per unit.

Product B with a contribution margin of ₹ 10 per unit is potentially more profitable. However, there is uncertainty about its marketability and following sales forecast has been prepared:

<u>Sales units of B (per annum)</u>	<u>Probability</u>
50,000	0.25
1,00,000	0.50
1,50,000	0.25

Fixed cost per year is ₹ 6,00,000.

From the information given above, choose the correct answer to the following Question No. 13 to 15:

- I.** If Company select product A, the profit of the company is –
- (A) ₹ 2,00,000 (B) ₹ 4,00,000
(C) ₹ 6,00,000 (D) ₹ 10,00,000
- II.** If company select product B and sale 1,40,000 units, the profit of the company is –
- (A) ₹ 4,00,000 (B) ₹ 8,00,000
(C) ₹ 10,00,000 (D) ₹ 12,00,000
- III.** If company select product B, the expected value of profit of the company is –

ADVANCED CAPITAL BUDGETING

- (A) ₹ 2,00,000 (B) ₹ 4,00,000
(C) ₹ 8,00,000 (D) ₹ 10,00,000

(EXAM JANUARY – 2026)

Answer Case Scenario – 04

- I. (B) ₹ 4,00,000**
II. (B) ₹ 8,00,000
III. (B) ₹ 4,00,000

Case Scenario – 05

ABC Ltd. of plans its to invest ₹ 16,00,000 in a new unit. The project is expected to have a useful life of 4 years, with no salvage value at the end of its life. The annual depreciation charge for the project is ₹ 400,000.

Projected revenues and costs for the project, ignoring inflation, are provided as follows:

Year	Revenues (₹)	Costs (₹)
1	12,00,000	6,00,000
2	14,00,000	8,00,000
3	16,00,000	8,00,000
4	16,00,000	8,00,000

ABC Ltd. is subject to a corporate tax rate of 60%, and the cost of capital for the project, including inflation premium, is 10%.



ADVANCED CAPITAL BUDGETING

Depreciation provides a tax benefit, and inflation rates for revenues and costs over the project's lifespan are as follows:

Year	Revenue Inflation	Cost Inflation
1	10%	12%
2	9%	10%
3	8%	9%
4	7%	8%

Based on above information, answer the following questions:

I. The depreciation tax benefit for the project per year shall be_____

- (a) ₹ 300,000
- (b) ₹ 240,000
- (c) ₹ 360,000
- (d) ₹ 400,000

II. The inflation-adjusted revenue in Year 2 shall be_____

- (a) ₹ 16,78,600
- (b) ₹ 14,00,000
- (c) ₹ 10,03,520
- (d) ₹ 9,85,600

III. The total cash inflow in Year 1 after adjusting for inflation and tax benefit on depreciation shall be_____

- (a) ₹ 6,72,000
- (b) ₹ 660,000
- (c) ₹ 985,600
- (d) ₹ 499,200

IV. The inflation-adjusted cost in Year 2 shall be

- (a) ₹ 16,78,600
- (b) ₹ 14,00,000
- (c) ₹ 10,03,520
- (d) ₹ 9,85,600

V. The present value of cash inflow for the year 3 shall be approximately_____

- (a) ₹ 4,52,598
- (b) ₹ 4,27,208
- (c) ₹ 4,79,898
- (d) ₹ 4,53,772

(RTP Sep – 2025)

Answer Case Scenario – 05

- I. (b) ₹ 240,000
- II. (a) ₹ 16,78,600
- III. (d) ₹ 499,200
- IV. (d) ₹ 9,85,600
- V. (c) ₹ 4,79,898



Student Notes



V'Smart Academy

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CA PAVAN KARMELE



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Student Notes



Student Notes



13

INTERNATIONAL FINANCIAL MANAGEMENT

PART I: INTERNATIONAL CAPITAL BUDGETING

Question – 01

ABC Ltd. is considering a project in US, which will involve an initial investment of US \$ 1,10,00,000. The project will have 5 years of life. Current spot exchange rate is ₹ 48 per US \$. The risk free rate in US is 8% and the same in India is 12%. Cash inflow from the project is as follows:

Year	Cash in flow
1	US \$ 20,00,000
2	US \$ 25,00,000
3	US \$ 30,00,000
4	US \$ 40,00,000
5	US \$ 50,00,000

Calculate the NPV of the project using foreign currency approach. Required rate of return on this project is 14%.

(SM TYK – 01)

Solution:

$$\begin{aligned} \text{RADR of USA} &= \left[\left(\frac{1.14}{1.12} \times 1.08 \right) - 1 \right] \times 100 \\ &= 9.9\% \end{aligned}$$

Calculation of NPV

Year	Cash Flow (Million) US\$	PV Factor @ 9.9%	P.V.
1	2.00	0.910	1.820
2	2.50	0.828	2.070
3	3.00	0.753	2.259
4	4.00	0.686	2.744
5	5.00	0.624	3.120
			12.013
		Less: Investment	11.000
		NPV	1.013

Therefore, Rupee NPV of the project is = ₹ (48 × 1.013) Million

= ₹ 48.624 Million

Question – 02

X Ltd., an Indian company, is considering a proposal to make an investment of USD 1,65,00,000 in Latin America. The project will have a life of 5 years. The current spot exchange rate is INR/USD 72. All investments and revenues will occur in USD. The USD and INR risk free rates are 8% and 12% respectively.

The following cash flow is expected from the project.

Year	Cash Inflows (USD)
1	30,00,000
2	37,50,000
3	45,00,000
4	60,00,000
5	75,00,000



Assume required rate of return on the project as 14%.

You are required to calculate:

- The viability of the project using foreign currency approach.
- What will be the impact if there is a withholding tax of 10% applicable on the project.

(Exam January – 2021) (8 Marks)

Solution:

(i) NPV

$$\begin{aligned}\text{RADR of USA} &= \left[\left(\frac{1.14}{1.12} \times 1.08 \right) - 1 \right] \times 100 \\ &= 9.93\%\end{aligned}$$

$$\begin{aligned}\text{NPV} &= (\$ 30,00,000 \times 0.910) + (\$ 37,50,000 \times 0.827) + (\$ 45,00,000 \times 0.753) + (\$ 60,00,000 \times 0.685) + (\$ 75,00,000 \times 0.623) - 1,65,00,000 \\ &= \$ 15,02,250\end{aligned}$$

$$\begin{aligned}\text{NPV in (₹)} &= \$ 15,02,250 \times 72 \\ &= ₹ 10,81,62,000\end{aligned}$$

Since NPV is positive hence project should be accepted.

(ii) Withholding Tax

$$\text{PVC I } (15,02,250 + 1,65,00,000) = 18,00,02,250$$

$$(-) \text{ withholding tax} = 10\%$$

$$\text{PVC I} = \$ 1,62,02,025$$

$$\begin{aligned}(-) \text{ PVCO} &= \$ 1,65,00,000 \\ \text{NPV} &= - \$ 2,97,975\end{aligned}$$

$$\begin{aligned}\text{NPV in (₹)} &= - \$ 2,97,975 \times 72 \\ &= - ₹ 2,14,54,200\end{aligned}$$

NPV is negative project should be rejected.

Question – 03

DK Ltd. is considering as investment proposal in Sri Lanka involving an initial investment of LKR 25 billion. The current spot exchange rate is INR/LKR 0.37. The risk free rate in India is 6% and the same is Sri Lanka is 5.02%. The project will generate a cash flow of LKR 5 billion in the first year. The cash flow will increase by LKR 1 billion each year for the next 4 years. The project will bind up on completion of 5 years with no salvage value.

The required rate of return for the project is 8%

- You are required to find out the investment worth of the project by
 - Home Currency Approach
 - Foreign Currency Approach
- Compare the outcome under both the approaches.

Given :

t	1	2	3	4	5
PVIF (8%, t)	0.92593	0.85734	0.79383	0.75503	0.68058
PVIF (7%, t)	0.93457	0.87344	0.81630	0.76290	0.71299

(Exam December – 2021) (8 Marks)



Solution:

Working Notes :

Calculation of Forward Exchange Rates

End of Year	₹	₹/KR
1	$0.37 \times \frac{1.06}{1.052}$	0.373
2	$0.373 \times \frac{1.06}{1.052}$	0.376
3	$0.376 \times \frac{1.06}{1.052}$	0.380
4	$0.379 \times \frac{1.06}{1.052}$	0.384
5	$0.382 \times \frac{1.06}{1.052}$	0.388

(i) Home Currency Approach

Year	Cash Flow Billion LKR	₹/LKR	Cash Flow Billion ₹	PVF @ 8%	PV Billion ₹
1	5	0.373	1.865	0.92593	1.7269
2	6	0.376	2.256	0.85734	1.9342
3	7	0.380	2.660	0.79383	2.1116
4	8	0.384	3.072	0.73503	2.2580
5	9	0.388	3.492	0.68058	2.3766
					10.4073
Less: Investment	25	0.37			9.2500
				NPV	1.1573

*Alternatively if students have used the PVIF (8%, 4) as given in the question paper then answer NPV would be 1.2188 instead of 1.1573

(ii) Foreign Currency Approach

$$(1 + 0.06) (1 + \text{Risk Premium}) = 1.08$$

$$1 + \text{Risk Premium} = 1.08/1.06 = 1.01887$$

$$\text{Therefore, Risk adjusted LKR Rate} = 1.01887 \times 1.0502 - 1 = 0.07$$

Calculation of NPV

Year	Cash Flow (Billion LKR)	PVF @ 7%	PV (Billion LKR)
1	5	0.93457	4.6729
2	6	0.87344	5.2406
3	7	0.81630	5.7141
4	8	0.76290	6.1032
5	9	0.71299	6.4169
			28.1477
Less: Investment			25.0000
		NPV	3.1477

Thus, Rupee NPV of the project = $0.37 \times 3.1477 = 1.1646$ billion

Decision: NPV is positive in the approach so, project will worth investment.

Question – 04

PQR Ltd. is considering a project in US, which involve an initial investment of ₹ 124.50 Crore. The project will have useful life of 5 years Current spot exchange rate is INR/USD is 83. The risk free rate in US is 4.186% and the same in India is 6.9768%. Cash inflows in USD from the project are as follows:

Year	1	2	3	4	5
Cash Inflow	30,00,000	40,00,000	50,00,000	60,00,000	70,00,000

PQR Ltd. is expecting net surplus of ₹ 1858.08 lakh to be received after closure of the project. There is no salvage value. PQR Ltd. want to take a forward cover to protect itself from exchange rate fluctuations.

N	1	2	3	4	5
PVIF (6.976%, n)	0.935	0.874	0.817	0.764	0.714
PVIF (4.186%, n)	0.959	0.921	0.884	0.849	0.815
PVIF (12%, n)	0.893	0.797	0.712	0.636	0.567
PVIF (15%, n)	0.870	0.756	0.658	0.572	0.497

You are required to recommend the INR/USD rate for the forward cover?

(Exam November – 2024)

Solution:

Let F be the recommended INR/USD rate for the forward cover. Accordingly, year-wise equivalent cash inflows in Indian Rupees shall be as follows:

Year	Cash Inflow in USD Lakh	Cash Inflow in ₹ Lakh
1	30.00	30.00F
2	40.00	40.00F
3	50.00	50.00F
4	60.00	60.00F
5	70.00	70.00F

Now let us compute Net Present Value of project assuming a discount rate of 12% as follows:

Year	PVF @ 12%	Cash Flow in ₹ Lakh	PV in ₹ Lakh
0	1.00	- 12450.00	- 12450.00
1	0.893	30.00F	26.79F
2	0.797	40.00F	31.88F
3	0.712	50.00F	35.60F
4	0.636	60.00F	38.16F
5	0.567	70.00F	39.69F
			172.12F – 12,450

Since expected surplus after closure of the project is ₹ 1,858.08 Lakh, we can compute the value of F as follows:

$$1,858.08 = 172.12F - 12,450$$

$$F = 83.13$$

Thus, for forward cover the rate of ₹ 83.13/ USD is recommended.

Alternatively, if students have assumed discounting rate as 15% then answer will be as follows:

Let F be the recommended INR/USD rate for the forward cover. Accordingly, year-wise equivalent cash inflows in Indian Rupees shall be as follows:

Year	Cash Flow in USD Lakh	Cash Inflow in ₹ Lakh
1	30.00	30.00F
2	40.00	40.00F
3	50.00	50.00F
4	60.00	60.00F
5	70.00	70.00F

Now let us compute Net Present Value of project assuming a discount rate of 15% as follows:

Year	PVF @ 15%	Cash Flow in ₹ Lakh	PV in ₹ Lakh
0	1.00	- 12,450.00	- 12,450.00
1	0.870	30.00F	26.10F
2	0.756	40.00F	30.24F
3	0.658	50.00F	32.90F
4	0.572	60.00F	34.32F
5	0.497	70.00F	34.79F
			158.35F – 12,450

Since expected surplus after closure of the project is ₹ 18,58.08 Lakh, we can compute the value of F as follows:

$$1,858.08 = 158.35F - 12,450$$

$$F = 90.36$$

Thus, for forward cover the rate of ₹ 90.36/ USD is recommended.

Alternative Solution if students have assumed that the discounting rate 15% for the given cash inflows then applicable discounting rates for the project is –

$$(1 + 0.06978) / (1 + \text{Risk Premium}) = (1 + 0.15)$$

$$\text{Or, } 1 + \text{Risk Premium} = 1.15 / 1.06978$$

$$= 1.075$$

$$\text{Therefore, Risk adjusted dollar rate is } = (1.0750 \times 1.04186) - 1$$

$$= 1.1199 - 1$$

$$= 0.12$$

Calculation of NPV

Year	Cash Flow US\$ Lakh	PV Factor at 12%	PV (US\$ Lakh)
1	30.00	0.893	26.79
2	40.00	0.797	31.88
3	50.00	0.712	35.60
4	60.00	0.636	38.16
5	70.00	0.567	39.69
			172.12
Less: Investment			150.00
NPV			22.12

Since PQR Ltd. is expecting a net surplus of ₹ 1,858.08 lakh after the closure of the project the recommended rate of INR/ USD is (₹ 1,858.08 lakh/ USD 22.12 lakh) ₹ 84.00.

Question – 05

XY Limited is engaged in large retail business in India. It is contemplating for expansion into a country of Africa by acquiring a group of stores having the same line of operation as that of India.

The exchange rate for the currency of the proposed African country is extremely volatile. Rate of inflation is presently 40% a year. Inflation in India is currently 10% a year. Management of XY Limited expects these rates likely to continue for the foreseeable future.

Estimated projected cash flows, in real terms, in India as well as African country for the first three years of the project are as follows:

	Year – 0	Year – 1	Year – 2	Year – 3
Cash flows in Indian ₹ (000)	-50,000	-1,500	-2,000	-2,500

Cash flows in African Rands (000)	-2,00,000	+50,000	+70,000	+90,000
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XY Ltd. assumes the year 3 nominal cash flows will continue to be earned each year indefinitely. It evaluates all investments using nominal cash flows and a nominal discounting rate. The present exchange rate is African Rand 6 to ₹ 1.

You are required to calculate the net present value of the proposed investment considering the following:

- African Rand cash flows are converted into rupees and discounted at a risk adjusted rate.
- All cash flows for these projects will be discounted at a rate of 20% to reflect it's high risk.
- Ignore taxation.

	Year - 1	Year - 2	Year - 3
PVIF @ 20%	.833	.694	.579

(SM TYK – 03 & Exam May – 2013) (6 Marks)

Solution:

Step 1: Calculation of Exchange Rate

$$FR = SR \times \frac{1+i}{1+i}$$

$$\text{Year 1 FR} = 6 \times \frac{1.40}{1.10} = 7.6364$$

$$\text{Year 2} = 7.6364 \times \frac{1.40}{1.10} = 9.7191$$

$$\text{Year 3} = 9.719 \times \frac{1.40}{1.10} = 12.3696$$

	0	1	2	3
Real CF (AR)	- 2,00,000	+ 50,000	+ 70,000	+ 90,000
Nominal CF (40%)	- 2,00,000	+ 70,000	+ 1,37,200	+ 2,46,960
Exchange rate (AR/€)	6	7.6364	9.7191	12.3696
Nominal CF (₹) (1)	- 33,333	+ 9167	+14,117	+ 19,965
Real CF (India)	- 50,000	- 1,500	- 2,000	- 2,500
NCF (2)	- 50,000	- 1,650	- 2,420	- 3,328
Total CF (1) + (2)	- 83,333	+ 7517	+ 11,697	+ 16,637
(x) PVF @ 20%	1.000	0.833	0.694	0.579
PV	- 83,333	+ 6,262	+ 8,118	+ 9,633

$$\text{Present Value} = - 59,320$$

$$\text{(+) PV of TV} = \left(\frac{16,637}{20\%} \right) \times 0.579 = 48,164$$

$$\text{NPV} = -11,156 \text{ Reject}$$

Question – 06

XY Limited is engaged in large retail business in India. It is contemplating for expansion into a country of Africa by acquiring a group of stores having the same line of operation as that of India.

The exchange rate for the currency of the proposed African country is extremely volatile. Rate of inflation is presently 40% a year. Inflation in India is currently

10% a year. Management of XY Limited expects these rates likely to continue for the foreseeable future.

Estimated projected cash flows, in nominal terms, in India as well as African country for the first three years of the project are as follows:



	Year – 0	Year – 1	Year – 2	Year – 3
Cash flows in Indian(000)	–2,00,000	–6,600	–10,000	–13,000
Cash flows in African Rands (000)	8,00,000	+2,80,000	+5,50,000	+10,00,000

XY Ltd. assumes the year 3 nominal cash flows will continue to be earned each year indefinitely. It evaluates all investments using nominal cash flows and a nominal discounting rate. The present exchange rate is African Rand 6 to ₹ 1.

You are required to calculate the net present value of the proposed investment considering the fact that the company uses discounting rate of 10.80% to evaluate any project but to reflect high risk of this project it is considering to adjust a risk premium of 8.30%

Note: -

1. Use PV Factors upto 3 decimal points.
2. Use Exchange Rates upto 4 decimal points.
3. Compute final calculation in multiple of ₹ 000 and round off them upto zero.
4. Ignore taxation.

(MTP April – 2025)

Solution:

First, we shall compute the discount rate to calculate the NPV of the project.

$(1 + \text{Risk Premium}) (1 + \text{Normal Discounting Rate}) = (1 + \text{Required Rate})$

$(1.083) (1.108) = (1 + \text{Required Rate})$

Required Rate = 0.20 i.e. 20%

Calculation of NPV

Year	0	1	2	3
Inflation factor in India	1.00	1.10	1.21	1.331
	1.00	1.40	1.96	2.744
Inflation factor in Africa	6.00	7.6364	9.7190	12.3696
Exchange Rate (as per IRP)	-	-6,600	-10,000	-13,000
Cash Flows in ₹ '000	2,00,000			
Nominal (1)		2,80,000	5,50,000	10,00,000
Cash Flows in African Rand '000	8,00,000	36,666	56,590	80,843
	-	30,066	46,590	67,843
Nominal	1,33,333			
In Indian ₹ '000 (2)	-	0.833	0.694	0.579
	3,33,333	25,045	32,333	39,281
Net Cash Flows in ₹ '000 (1) + (2)	1			
PVF @ 20%	-3,33,333			
PV				

NPV of 3 years = -2,36,674 (₹ '000)



$$\text{NPV of Terminal Value} = \frac{67,843}{0.20} \times 0.579 = 1,96,405 \text{ (₹ '000)}$$

$$\begin{aligned} \text{Total NPV of the Project} &= -2,36,674 \text{ (₹ '000)} + 1,96,405 \text{ (₹ '000)} \\ &= -40,269 \text{ (₹ '000)} \end{aligned}$$

Question – 07

XYZ Ltd., a company based in India, manufactures very high quality modern furniture and sells to a small number of retail outlets in India and Nepal. It is facing tough competition. Recent studies on marketability of products have clearly indicated that the customers are now more interested in variety and choice rather than exclusivity and exceptional quality. Since the cost of quality wood in India is very high, the company is reviewing the proposal for import of woods in bulk from Nepalese supplier.

The estimate of net Indian (₹) and Nepalese Currency (NC) cash flows in Nominal terms for this proposal is shown below:

Year	Net Cash Flow (in millions)			
	0	1	2	3
NC	-25.000	2.600	3.800	4.100
Indian (₹)	0	2.869	4.200	4.600

The following information is relevant:

- (i) XYZ Ltd. evaluates all investments by using a discount rate of 9% p.a. All Nepalese customers are invoiced in NC. NC cash flows are converted to Indian (₹) at the forward rate and discounted at the Indian rate.
- (ii) Inflation rates in Nepal and India are expected to be 9% and 8% p.a. respectively. The current exchange rate is ₹ 1 = NC 1.6

Assuming that you are the finance manager of XYZ Ltd., calculate the net present value (NPV) and modified internal rate of return (MIRR) of the proposal.

You may use following values with respect to discount factor for ₹ 1 @9%.

	Present Value	Future Value
Year 1	0.917	1.188
Year 2	0.842	1.090
Year 3	0.772	1

(SM TYK – 05 & Exam November – 2015) (6 Marks)

Solution:

(i) Calculation of NPV

Step 1: Forward Rates

$$1 = \text{NC } 1.60 \times \frac{1.09}{1.08} = 1.6148$$

$$2 = \text{NC } 1.6148 \times \frac{1.09}{1.08} = 1.6298$$

$$3 = \text{NC } 1.6298 \times \frac{1.09}{1.08} = 1.6449$$

Step 2: NPV

	0	1	2	3
CF (NC)	- 25.000	+ 2.600	+ 3.800	+ 4.100
Exchange rate (NC/₹)	1.60	1.6148	1.6298	1.6449
CF (₹)	- 15.625	+ 1.6101	+ 2.3316	+ 2.4926
CF India	0	+ 2.869	+ 4.200	+ 4.600
Total CF	- 15.625	4.4791	6.5316	7.0926
X PVF (9%)	1	0.917	0.842	0.772

PV	- 15.625	+ 4.1073	+ 5.4996	+ 5.4755
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$$\text{NPV} = - 0.5426$$

(ii) Modified IRR

Terminal value

$$(1) \quad 4.4791 (1.09)^2 = 5.3216$$

$$(2) \quad 6.5316 \times 1.09 = 7.1194$$

$$(3) \quad = 7.0926$$

$$\underline{\hspace{1cm}} = 19.53$$

$$15.625 (1 + r)^3 = 19.53$$

$$r = \left[\left(\frac{19.53}{15.625} \right)^{1/3} - 1 \right] \times 100$$

$$= 7.72\%$$

Question – 08

DD Ltd. a company based in India manufactures good quality of leather bags and sells to retail outlets in India and USA. The cost of quality leather in India is very high, the company is reviewing the proposal of importing of leather in bulk from USA supplier. The estimate of net US \$ and Indian ₹ Currency Cash Flows in nominal terms for this proposal is given below:

	Net Cash Flow (in Lakh)			
Year	0	1	2	3
In US \$	(25)	5	7	8
In ₹	0	60	80	90

If not imported cost of leather to be purchased in India (in ₹)	400	450	500	600
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Other information:

- (i) DD Ltd. evaluates all investments by using discount rate of 9% p.a.
- (ii) All US customers are invoiced in US \$. US \$ Cash flows converted into ₹ at the forward rate and discounted at Indian Rate.
- (iii) Inflation in USA and India are expected to be 9% and 8% respectively.
- (iv) The current exchange rate 1 US \$ = ₹ 74

You are required to Calculate Net Present Value and recommend the decision. Present value factor @ 9% are as under:

1 Year	2 Year	3 Year
0.917	0.842	0.772

(Exam December – 2021) (8 Marks)

Solution:

Calculation of Forward Rates

Forward Rate

$$1 \quad = 74 \times \frac{1.08}{1.09} = 73.321$$

$$2 \quad = 73.321 \times \frac{1.08}{1.09} = 72.65$$

$$3 \quad = 72.65 \times \frac{1.08}{1.09} = 71.98$$



INTERNATIONAL FINANCIAL MANAGEMENT

	0	1	2	3
CF (\$)	- 25	5	7	8
Exchange rate (NC/₹)	74	73.321	72.65	71.98
CF (₹)	- 1,850	366.6	508.55	575.84
CF India	-	60	80	90
Cost of leather in India	- 400	- 450	-500	-600
CF	- 2250	- 23.40	88.55	65.84
X PVF (9%)	1.000	0.917	0.842	0.772
	-2250	-21.46	74.56	50.83

NPV = - 2146.07 Reject

Question – 09

A multinational company is planning to set up a subsidiary company in India (where hitherto it was exporting) in view of growing demand for its product and competition from other MNCs. The initial project cost (consisting of Plant and Machinery including installation) is estimated to be US\$ 500 million. The net working capital requirements are estimated at US\$ 50 million. The company follows straight line method of depreciation. Presently, the company is exporting two million units every year at a unit price of US\$ 80, its variable cost per unit being US\$ 40.

The Chief Financial Officer has estimated the following operating cost and other data in respect of proposed project:

- (i) Variable operating cost will be US \$ 20 per unit of production;
- (ii) Additional cash fixed cost will be US \$ 30 million p.a. and project's share of allocated fixed cost will be US \$ 3 million p.a. based on principle of ability to share;
- (iii) Production capacity of the proposed project in India will be 5 million units;

- (iv) Expected useful life of the proposed plant is five years with no salvage value;
- (v) Existing working capital investment for production & sale of two million units through exports was US \$ 15 million;
- (vi) Export of the product in the coming year will decrease to 1.5 million units in case the company does not open subsidiary company in India, in view of the presence of competing MNCs that are in the process of setting up their subsidiaries in India;
- (vii) Applicable Corporate Income Tax rate is 35%, and
- (viii) Required rate of return for such project is 12%.

Assuming that there will be no variation in the exchange rate of two currencies and all profits will be repatriated, as there will be no withholding tax, estimate Net Present Value (NPV) of the proposed project in India.

Present Value Interest Factors (PVIF) @ 12% for five years are as below:

Year	1	2	3	4	5
PVIF	0.8929	0.7972	0.7118	0.6355	0.5674

(SM TYK – 04 & Exam May – 2014) (8 Marks)

Solution:

W.N. 1: CFAT (Millions)

Sales [5 million × \$ 80]	\$ 400
(-) VC [5 million × \$ 20]	\$ 100
(-) additional FC	\$ 30



CFBT	\$ 270	(i)
(-) Dep $\left(\frac{\$ 500}{5}\right)$	\$ 100	
PBT	<u>\$ 170</u>	
Tax @ 33%	\$ 59.5	(ii)
CFAT (i – ii)	\$ 210.5	

CFAT (Export)

Sales	\$ 120	
[1.5 × \$ 80]		
(-) VC (1.5 × \$ 40)	\$ 60	
CFBT	<u>\$ 60</u>	
Tax @ 35%	\$ 21	
CFAT	<u>\$ 39</u>	
Incremental CFAT (\$ 210.50 – \$ 39)	= \$ 171.50	

Calculation of NPV

(Millions)

	Year	PVF	Amount	P.V.
(A) Incremental cash outflow				
Cost of plant	0	1.000	\$ 500	\$ 500
Working capital (\$ 50 - \$15)	0	1.000	\$ 35	\$ 35
Total				\$ 535

(B) Incremental cash inflow				
Incremental CFAT (WN 1)	1-5	3.6048	\$ 171.50	\$ 618.22
WC recovered (Incremental)	5	0.5674	\$ 35	\$ 19.859
Total				\$ 638.079
NPV (B-A)				\$ 103.079

Since NPV is positive hence project should be accepted.

Question – 10

A US company wants to setup a manufacturing plant in India which requires an initial outlay of ₹ 8 Million. It is expected to have a useful life of 5 years with a salvage of ₹ 2 Million. The company follows straight line method of depreciation. To support additional level of activity, investment would require one time additional working capital of ₹ 1 Million.

Since the cost of production lower in India, the variable cost of production would be ₹ 30 per unit. Additional fixed cost per annum is estimated at ₹ 0.5 Million. The company is projecting its annual sales to 80000 units at the price of ₹ 100 per unit. Applicable tax rate to the company is 34% and its cost of capital is 8%.

Inflation rates in US and India are expected to be 8% and 9% respectively. The current exchange rate is ₹ 72 per US Dollar.

Assuming that all profit will be repatriated every year and there will be no withholding taxes, estimate the net present value of the proposed project in India and evaluate its feasibility.

PVF @ 8% for the five years are as under:

Rate	1 Year	2 Year	3 Year	4 Year	5 Year
8%	0.926	0.857	0.794	0.735	0.681

(Exam December – 2021) (8 Marks)

Solution:

Forward Rate

$$1 \text{ year} = ₹ 72 \times \frac{1.09}{1.08} = 72.67$$

$$2 \text{ year} = 72.67 \times \frac{1.09}{1.08} = 73.34$$

$$3 \text{ year} = 73.34 \times \frac{1.09}{1.08} = 74.02$$

$$4 \text{ year} = 74.02 \times \frac{1.09}{1.08} = 74.71$$

$$5 \text{ year} = 74.71 \times \frac{1.09}{1.08} = 75.40$$

W.N. 2: CFAT (₹)

$$\text{Sales (80,000 units} \times 100) = 80,00,000$$

$$\text{VC (80,000} \times 30) = 24,00,000$$

$$\text{FC} = 5,00,000$$

$$\text{CFBT (i)} = 51,00,000$$

$$(-) \text{ Dep } \left(\frac{80,00,000 - 20,00,000}{5} \right) = 12,00,000$$

$$\text{PBT} = 39,00,000$$

$$\text{Tax @ 34\% (ii)} = 13,26,000$$

$$\text{CFAT (i – ii)} = 37,74,000$$

NPV

	0	1	2	3	4	5
Cost of plant	80,00,000	-	-	-	-	-
Working capital	10,00,000	-	-	-	-	10,00,000
Salvage	-	-	-	-	-	20,00,000
CFAT	-	37,74,000	37,74,000	37,74,000	37,74,000	37,74,000
CF (₹)	90,00,000	37,74,000	37,74,000	37,74,000	37,74,000	67,74,000
Exchange rate	72	72.67	73.34	74.02	74.71	75.40
CF (\$)	1,25,000	51,933.40	51,458.96	50,986.22	50,515.32	89,840.85
(x) PVF	1.000	0.926	0.857	0.794	0.735	0.681

$$\text{NPV} = \$ 1,05,984.09 \text{ Accept.}$$

Question – 11

A USA based company is planning to set up a software development unit in India. Software developed at the Indian unit will be bought back by the US parent at a transfer price of US \$10 millions. The unit will remain in existence in India for one year; the software is expected to get developed within this time frame.

The US based company will be subject to corporate tax of 30 per cent and a withholding tax of 10 per cent in India and will not be eligible for tax credit in the US. The software developed will be sold in the US market for US \$ 12.0 millions. Other estimates are as follows:

Rent for fully furnished unit with necessary hardware in India
₹ 15,00,000

Man power cost (80 software professional will be working



for 10 hours each day) ₹ 400 per man hour

Administrative and other costs ₹ 12,00,000

Advise the US Company on the financial viability of the project. The rupee-dollar rate is ₹48/\$.

Note: Assume 365 days a year.

(SM TYK – 02, RTP Nov – 2021 & Exam May – 2017) (8 Marks)

Solution:

Cost of Software in India

Rent	₹ 15,00,000
Manpower (80 × 10 × 400 × 365)	₹ 11,68,00,000
Administration	₹ 12,00,000
Cost	<u>₹ 11,95,00,000</u>

Tax Amount

Sales (\$ 1,00,00,000 × 48)	₹ 48,00,00,000
(-) Cost	₹ 11,95,00,000
Profit	<u>₹ 36,05,00,000</u>
Tax @ 30%	₹ 10,81,50,000

Total Cost of Software

$$= ₹ 11,95,00,000 + 10,81,50,000$$

$$= ₹ 22,76,50,000$$

$$\text{In \$} = \frac{₹ 22,76,50,000}{48}$$

$$= \$ 4.743 \text{ Millions}$$

Software will be sold at \$ 12 Millions & Cost is 4.743 Millions hence Project Should be Accepted.

Question – 12

VK Ltd. is an Indian company which is planning to set up a manufacturing plant through its subsidiary in the small country Farland, (where hitherto it was exporting) in view of growing demand for its product and competition from other MNCs. The currency of Farland is the Farroh (Fr.).

An initial investment of Fr. 80 million in plant and machinery would be required. In addition to that the initial investment in working capital of Fr. 6 million would be also required which shall be financed through a loan from a local bank of Farland, at interest rate of 10% p.a. The working capital shall also be subject to inflation. At the end of 5 years, the subsidiary would be taken over by the Govt. of Farland for a price of Fr. 2 million. The part of the proceeds would be used to pay off the bank loan.

It is expected that subsidiary shall produce Net Cash Flows from Operations of Fr. 30 million per year at current price level over the five-year period, before allowing for Farland inflation of 8% per year. Depreciation on Plant and Machinery shall be charged at 20% per year on straight line basis. As a result of setting up the subsidiary, VK Ltd. expects to lose after-tax export income from Farland of INR 8,00,000 per year in current price terms, before allowing for India inflation of 3%. Profits in Farland are taxed at a rate of 20% after allowing deduction for interest and depreciation. All after-tax cash profits are remitted to the India at the end of each year. Indian tax @ 30% is charged on profit earned, but due to tax treaty between



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Farland and the India the tax paid in Farland is allowed to be set off against any India Tax liability. Taxation is paid in the year in which the liability arises. VK Ltd. requires foreign investments to be discounted at 12%. The current exchange rate is Fr.2.5/INR and the Farroh is expected to depreciate against INR by 5% per year.

Advise should VK Ltd. undertake the investment in Farland or not.

Note:-

1. Present Figures in thousands multiple.
2. Round off all calculations.
3. PVF @12%

Year	1	2	3	4	5
PVF	0.893	0.797	0.712	0.636	0.567

(MTP October – 2023)

Solution:

(i) Forward Rate

$$\begin{aligned}
 \text{Fr/₹} &= 2.5 \\
 1 \text{ year} &= 2.5 \times 1.05 = 2.625 \\
 2 \text{ year} &= 2.625 \times 1.05 = 2.7563 \\
 3 \text{ year} &= 2.7563 \times 1.05 = 2.8941 \\
 4 \text{ year} &= 2.8941 \times 1.05 = 3.0388 \\
 5 \text{ year} &= 3.0388 \times 1.05 = 3.1907
 \end{aligned}$$

CFAT (Fr ‘000’)

	1	2	3	4	5
EBITDA (30,000)	32,400	34,992	37,791	40,815	44,080
(-) Depreciation	16,000	16,000	16,000	16,000	16,000
(-) Interest (6,000 × 10%)	600	600	600	600	600
PBT	15,800	18,392	21,191	24,215	27,480
Tax @ 20%	3,160	3,678	4,238	4,843	5,496
PAT	12,640	14,714	16,953	19,372	21,984
(+) Depreciation	16,000	16,000	16,000	16,000	16,000
CFAT (Fr)	28,640	30,714	32,953	35,372	37,984

WC Requirement

	0	1	2	3	4	5
WC	6,000	6,480	6,998	7,558	8,163	---
Additional WC (Fr)	---	480	518	560	605	---

(“000”)

	0	1	2	3	4	5
Cost (Fr)	(80,000)	---	---	---	---	---
CFAT	---	28,640	30,714	32,953	35,372	37,984
Additional WC	---	(480)	(518)	(560)	(605)	---
Salvage	---	---	---	---	---	2,000
Recovered WC	---	---	---	---	---	2,163
CF (Fr)	-80,000	28,160	30,196	32,393	34,767	42,147
Exchange Rate (Fr/₹)	2.5	2.625	2.7563	2.8941	3.0388	3.1907
CF (₹)	-32,000	10,728	10,955	11,193	11,441	13,209
(-) Tax	---	(602)	(667)	(732)	(797)	(861)
Loss of Export	---	(824)	(849)	(874)	(900)	(927)
CFAT	-32,000	9,302	9,439	9,587	9,744	11,421
PVF	1.000	0.893	0.797	0.712	0.636	0.567



NPV = 3,328

Working Tax in India

	1	2	3	4	5
PBT (Fr)	15,800	18,392	21,191	24,215	27,480
Exchange Rate Fr/₹	2.625	2.7563	2.8941	3.0388	3.1907
PBT (₹)	6,019	6,673	7,322	7,969	8,613
Tax @ 10%	602	667	732	797	861

ICAI SOLUTION:

Working Notes:

(1) Calculation of the project cash flows for VK Ltd.'s subsidiary in Farland

Fr.'000

Year	0	1	2	3	4	5
Cash flow from operating		32,400	34,992	37,791	40,815	44,080
Depreciation		16,000	16,000	16,000	16,000	16,000
Interest		0	0	0	0	0
		600	600	600	600	600
Profit after tax		15,800	18,392	21,191	24,215	27,480
Farland tax		0	2	1	5	0
		3,160	3,678	4,238	4,843	5,496
Profit after tax		12,640	14,714	16,953	19,372	21,984
Add back depreciation		16,000	16,000	16,000	16,000	16,000
		0	0	0	0	0
Initial investment	-	28,640	30,714	32,953	35,372	37,984
Change in W.C.	80,000	0	4	3	2	4
Loan capital	0	-480	-518	-560	-605	-653
Sales on Subsidiary						-
		---	---	---	---	6000

	---					2000
	-	2816	3019	3239	3476	3333
	80000	0	6	3	7	1

(2) Expected Exchange Rates

Year	Rate
0	2.50
1	$2.50 \times 1.05 = 2.63$
2	$2.50 \times (1.05)^2 = 2.76$
3	$2.50 \times (1.05)^3 = 2.89$
4	$2.50 \times (1.05)^4 = 3.04$
5	$2.50 \times (1.05)^5 = 3.19$

(3) Calculation of Tax paid in India

Year	1	2	3	4	5
PBT (fr)	15,800	18,392	21,191	24,215	27,480
Tax @ 10 %	1,580	1,839	2,119	2,422	2,748
Exchange Rate	2.63	2.76	2.89	3.04	3.19
Tax in India (INR' 000)	601	666	733	797	861

Calculation Net Present Value (NPV) for VK Ltd.'s subsidiary at parent company level

Year	0	1	2	3	4	5
Project cash flow (Fr.'000)	-	28,160	30,196	32,393	34,767	33,331
	80,000	2.63	2.76	2.89	3.04	3.19
Exchange Rate (Fr./INR)	2.50	--	--	--	--	--
	-					
Cash Invested from India (INR '000)	32,000	10,707	10,941	11,209	11,437	10,449
	--	601	666	733	797	861

Cash Received in India (INR '000)	- 32,000	10,106 824	10,275 849	10,476 874	10,640 900	9,588 927
Tax in India (INR '000)	- 32,000 1	9,282 0.893 8,289	9,426 0.797 7,513	9,602 0.712 6,837	9,740 0.636 6,195	8,661 0.567 4,911
Lost export after tax (INR '000)	-					
Parent Cash Flow PVF	32,000					
NPV						1,745

Decision: Since NPV of the project is positive it should be accepted.

Question – 13

Mr. Vishwas, a friend of Mr. Pramod who is one of the Directors of Ashirwad Limited, is a citizen of Mauritius. His immediate family members including his parents, born in India are residing in India. He has many friends in different parts of India, due to which he happens to visit India on frequent basis. He along with Mr. Pramod evince interest in setting up business in India and formally incorporate a company to commence their operations. Accordingly, a company is called “Aerious Private Ltd.” got incorporated in Mumbai.

To start with he received a business proposal from one of his friends Nimish a consultant. It is estimated that in equivalent terms the business shall require an initial investment of MUR 100 Million and thereafter MUR 2 Million each year will be needed as working capital fund.

He wished to evaluate whether the business proposal is viable or not. The information related to exchange rate and inflation rate is as follows:

Spot Rate for 1 Mauritian Dollar (MUR) = 1.88 Indian Rupee (INR)

The inflation in India is 6% and in Mauritius is 5%.

It is expected that this inflation rate will remain unchanged for the next 4 years.

INR 8 Crore out of initial investment shall be required for setting up a plant. The useful life of the plant is 4 years. At the end of 4th year estimated salvage value of this plant shall be INR 80 lakhs. Depreciation of the plant shall be charged on the basis of straight-line method.

40 % of the investment shall be through debt funds from Mauritius at the cost of 10% (post tax) while remaining funds shall be arranged by him and his friends. They expect a rate of return of 12% on their funds.

Expected revenues & costs (excluding depreciation) in real term are as under:

Year	1	2	3	4
Revenues (₹ Crore)	6.00	7.00	8.00	8.00
Costs (₹ Crore)	3.00	4.00	4.00	4.00

Assume that applicable tax rate in India is 30%. Since there is Double tax avoidance agreement between India and Mauritius, the company is not required to pay tax in Mauritius if tax has been paid in India.

The applicable inflation rates for revenues & costs are as follows:

Year	Revenues	Costs
1	10%	12%
2	9%	10%
3	8%	9%
4	7%	8%



He wants an expert opinion for the same investment proposal.

Demonstrate whether investment in this project is viable option or not.

Note:

1. Round off calculations upto 4 decimal points.
2. Show INR calculations in Crore and MUR calculations in Million.

Solution:

(i) Forward Rate

$$\begin{aligned}
 1. \quad &= ₹ 1.88 \times \frac{1.06}{1.05} = 1.8979 \\
 2. \quad &= ₹ 1.8979 \times \frac{1.06}{1.05} = 1.9160 \\
 3. \quad &= ₹ 1.9160 \times \frac{1.06}{1.05} = 1.9342 \\
 4. \quad &= ₹ 1.9342 \times \frac{1.06}{1.05} = 1.9526
 \end{aligned}$$

(ii) Nominal Cash Flows

Revenue:

$$\begin{aligned}
 1. \quad &6 \times 1.10 = 6.60 \text{ Cr.} \\
 2. \quad &7 \times 1.10 \times 1.09 = 8.393 \text{ Cr.} \\
 3. \quad &8 \times 1.10 \times 1.09 \times 1.08 = 10.3594 \text{ Cr.} \\
 4. \quad &8 \times 1.10 \times 1.09 \times 1.08 \times 1.07 = 11.0845 \text{ Cr.}
 \end{aligned}$$

Cost:

$$\begin{aligned}
 1. \quad &3 \times 1.12 = 3.36 \text{ Cr.} \\
 2. \quad &4 \times 1.12 \times 1.10 = 4.928 \text{ Cr.} \\
 3. \quad &4 \times 1.12 \times 1.10 \times 1.09 = 5.3715 \text{ Cr.} \\
 4. \quad &4 \times 1.12 \times 1.10 \times 1.09 \times 1.08 = 5.8012 \text{ Cr.}
 \end{aligned}$$

(iii) Calculation of CFAT (₹)

	1	2	3	4
Revenue	6.60	8.393	10.3594	11.0845
(-) Cost	3.36	4.928	5.3715	5.8012
CFBT (i)	3.24	3.465	4.9879	5.2833
(-) Depreciation $\frac{8 - 0.8}{4}$	1.80	1.80	1.80	1.80
PBT	1.44	1.665	3.1879	3.4833
Tax @ 30% (ii)	0.4320	0.4995	0.9564	1.0450
CFAT [(i) - (ii)]	2.808	2.9655	4.0315	4.2383

(iv) Calculation of Working Capital (₹)

$$\begin{aligned}
 1^{\text{st}} \text{ Year} \quad &\text{MUR } 2\text{m} \times 1.8979 = 0.3796 \text{ Cr.} \\
 2^{\text{nd}} \text{ Year} \quad &\text{MUR } 2\text{m} \times 1.9160 = 0.3832 \text{ Cr.} \\
 3^{\text{rd}} \text{ Year} \quad &\text{MUR } 2\text{m} \times 1.9342 = 0.3868 \text{ Cr.} \\
 4^{\text{th}} \text{ Year} \quad &\text{MUR } 2\text{m} \times 1.9526 = 0.3905 \text{ Cr.}
 \end{aligned}$$

$$\text{Total Working Capital} = 1.5401 \text{ Cr.}$$

(v) WACC

$$\begin{aligned}
 \text{WACC} &= (0.4 \times 10) + (0.6 \times 12) \\
 &= 11.2\%
 \end{aligned}$$



(vi) Calculation of NPV

	0	1	2	3	4
Initial investment	-18.80	---	---	---	---
Working capital	---	-0.3796	- 0.3832	- 0.3868	- 0.3905
WC recovered	---	---	---	---	+1.5401
Salvage	---	---	---	---	+0.8000
CFAT	---	2.808	2.9655	4.0315	4.2383
Total (₹)	-18.80	2.4284	2.5823	3.6447	6.1879
Exchange rate	1.88	1.8979	1.9160	1.9342	1.9526
Total CF (MUR)	-100	12.7952	13.4776	18.8434	31.6906
PVT @ 11.2%	1	0.8993	0.8087	0.7273	0.6540

NPV = - 43.1635 Reject

Question – 14

A proposed foreign investment involves creation of a plant with an annual output of 1 million units. The entire production will be exported at a selling price of USD 10 per unit.

At the current rate of exchange dollar cost of local production equals to USD 6 per unit. Dollar is expected to decline by 10% or 15%. The change in local cost of production and probability from the expected current level will be as follows:

Decline in value of USD (%)	Reduction in local cost of production (USD/unit)	Probability
0	-	0.4
10	0.30	0.4

15	0.15 Additional reduction	0.2
----	---------------------------	-----

The plant at the current rate of exchange will have a depreciation of USD 1 million annually. Assume local Tax rate as 30%.

You are required to find out:

- Annual Cash Flow After Tax (CFAT) under all the different scenarios of exchange rate.
- Expected value of CFAT assuming no repatriation of profits.
- Viability of the investment proposal assuming an initial investment of USD 25 million on plant and working capital with a required rate of return of 11% on investment and on the basis of CFAT arrived under option (ii). The CFAT will grow @ 3% per annum in perpetuity.

(Exam January – 2021) (8 Marks)

Solution:

(i) Calculation of CFAT

	I	II	III
Sales	1,00,00,000	1,00,00,000	1,00,00,000
(10,00,000 × 10)	60,00,000	57,00,000	55,50,000
(-) Cost of Production	40,00,000	43,00,000	44,50,000
CFBT (i) - Depreciation	10,00,000	9,00,000	8,50,000
PBT	30,00,000	34,00,000	36,00,000
Tax @ 30%(ii)	9,00,000	10,20,000	10,80,000
CFAT (i – ii)	31,00,000	32,80,000	33,70,000

(ii) Expected CFAT

$$= (31,00,000 \times 0.4) + (32,80,000 \times 0.4) + (33,70,000 \times 0.2)$$

$$= \$ 32,26,000$$

(iii) Calculation of NPV

$$\text{PVCI} = \frac{\text{CFAT}_1}{K_e - g}$$

$$= \frac{\$ 32,26,000 (1.03)}{0.11 - 0.03}$$

$$= \$ 4,15,34,750$$

$$\text{NPV} = 4,15,34,750 - \$ 25,00,000$$

$$= \$ 1,65,34,750$$

Since NPV is positive hence Project should be accepted.

PART II: ADR & GDR**Question – 15**

Odessa Limited has proposed to expand its operations for which it requires funds of \$ 15 million, net of issue expenses which amount to 2% of the issue size. It proposed to raise the funds through a GDR issue. It considers the following factors in pricing the issue:

- (i) The expected domestic market price of the share is ₹ 300
- (ii) 3 shares underly each GDR
- (iii) Underlying shares are priced at 10% discount to the market price

- (iv) Expected exchange rate is ₹ 60/\$

You are required to compute the number of GDR's to be issued and cost of GDR to Odessa Limited, if 20% dividend is expected to be paid with a growth rate of 20%.

(Exam Nov – 2014) (8 Marks)

Solution:

$$\text{Net Issue Size} = \$15 \text{ million}$$

$$\text{Gross Issue} = \frac{\$15 \text{ million}}{0.98} = \$15.306 \text{ million}$$

$$\text{Issue Price per GDR in ₹ } (300 \times 3 \times 90\%) \quad \text{₹ 810}$$

$$\text{Issue Price per GDR in \$ } (\text{₹ 810} / \text{₹ 60}) \quad \$13.50$$

$$\text{Dividend Per GDR (D1)} = ₹ 2 \times 3 = \quad ₹ 6$$

* Assumed to be on based on Face Value of ₹ 10 each share.

$$\text{Net Proceeds Per GDR} = ₹ 810 \times 0.98 = ₹ 793.80$$

- (a) Number of GDR to be issued

$$\frac{\$15.306 \text{ million}}{\$13.50} = 1.1338 \text{ million}$$

- (b) Cost of GDR to Odessa Ltd

$$K_e = \frac{6.00}{793.80} + 0.20 = 20.76\%$$

Question – 16

M/s. Raghu Ltd. is interested in expanding its operation and planning to install manufacturing plant at US. It requires 8.82



million USD (net of issue expenses/ floatation cost) to fund the proposed project. GDRs are proposed to be issued to finance this project. The estimated floatation cost of GDRs is 2%.

Additional information:

- (i) Expected market price of share at the time of issue of GDR is ₹ 360 (Face Value ₹ 100)
- (ii) Each GDR will represent two underlying Shares.
- (iii) The issue shall be priced at 10% discount to the market price.
- (iv) Expected exchange rate is INR/USD 72.
- (v) Dividend is expected to be paid at the rate of 20% with growth rate of 12%.

Requirement:

- (1) You, as a financial consultant, are required to compute the number of GDRs to be issued and cost of the GDR.
- (2) What is your suggestion if the company receives an offer from a US Bank willing to provide an equivalent loan with an interest rate of 12%?
- (3) How much company can save by choosing the option as recommended by you?

(RTP May-2022, MTP April-2022 & Exam July - 2021) (8 Marks)

Solution:

(1) No. of GDR & Cost of GDR

$$\text{Gross Issues} = \frac{\$ 8.82 \text{ m}}{0.98}$$

$$= 9 \text{ Million}$$

$$\text{Price of GDR} = 360 \times 20 \times 90\%$$

$$= ₹ 648$$

$$= \frac{648}{72} = \$ 9$$

$$\text{No. of GDR} = \frac{\$9 \text{ million}}{\$9} = 1 \text{ million}$$

$$\text{Cost of GDR} = \frac{₹ 100 \times 20\% \times 2}{648 \times 98\%} + 0.12$$

$$= 18.30\%$$

(2) Since equivalent loan Interest is 12% i.e. less than Cost of GDR (18.30%) then it is better to accept US Bank Offer

(3) Savings

$$= 18.30\% - 12\%$$

$$= 6.30\%$$

Question – 17

MITU Ltd. wants to expand business outside India. For the project installation US funds \$ 14.775 Million are required. Company wants to raise money by issue of GDRs.

Following information is available:

- (1) 7 shares shall underly each GDR.
- (2) GDR shall be priced at 7% discount to market price.
- (3) Market price of share is ₹ 500 (Face Value ₹ 100) per share.



- (4) Expected exchange rate is \$1 = ₹ 81.3750.
- (5) Dividend expected to be paid is 15% with growth rate 10%.
- (6) Flotation Cost of GDR is 1.5%.

Required:

Compute the number of GDRs to be issued and cost of the GDR to the company.

(Note: Calculate in lacs with four decimals.)

(Exam Jan. – 2026)

Solution:

Net Issue Size = \$ 14.775 million

Gross Issue = $\frac{\$ 14.775 \text{ million}}{0.985} = \$ 15 \text{ million}$

Issue Price per GDR in ₹ (500 × 7 × 93%) ₹ 3,255

Issue Price per GDR in \$ (₹ 3255 / ₹ 81.3750) \$ 40

Dividend Per GDR (D1) (₹ 15 × 7) ₹ 105

Net Proceeds Per GDR (₹ 3255 × 0.985) ₹ 3,206.18

(a) Number of GDR to be issued = $\frac{\$ 15 \text{ million}}{\$ 40}$
 = 0.375 million / 3.75 Lakhs / 3,75,000

(b) Cost of GDR to X Ltd. $K_e = \frac{105.00}{3206.18} + 0.10 = 13.27\%$

PART III: ADJUSTED PRESENT VALUE**Question – 18**

XYZ Ltd. is presently all equity financed. The directors of the company have been evaluating investment in a project which will require ₹ 270 lakhs capital expenditure on new machinery. They expect the capital investment to provide annual cash flows of ₹ 42 lakhs indefinitely which is net of all tax adjustments. The discount rate which it applies to such investment decisions is 14% net.

The directors of the company believe that the current capital structure fails to take advantage of tax benefits of debt, and propose to finance the new project with undated perpetual debt secured on the company's assets. The company intends to issue sufficient debt to cover the cost of capital expenditure and the after tax cost issue.

The current annual gross rate of interest required by the market on corporate undated debt of similar risk is 10%. The after tax costs of issue are expected to be ₹ 10 lakh. Company's tax rate is 30%.

You are required to calculate The adjusted present value of the investment.

Solution:**Calculation of Adjusted Present Value of Investment (APV)**

Adjusted PV

= Base Case PV + PV of financing decisions associated with the project

Base Case NPV for the project:



$$(-) ₹ 270 \text{ lakhs} + (₹ 42 \text{ lakhs} / 0.14) = (-) ₹ 270 \text{ lakhs} + ₹ 300 \text{ lakhs}$$

$$= ₹ 30$$

$$\text{Issue costs} = ₹ 10 \text{ lakhs}$$

$$\begin{aligned} \text{Thus, the amount to be raised} &= ₹ 270 \text{ lakhs} + ₹ 10 \text{ lakhs} \\ &= ₹ 280 \text{ lakhs} \end{aligned}$$

$$\begin{aligned} \text{Annual tax relief on interest payment} &= ₹ 280 \times 0.1 \times 0.3 \\ &= ₹ 8.4 \text{ lakhs in perpetuity} \end{aligned}$$

$$\begin{aligned} \text{The value of tax relief in perpetuity} &= ₹ 8.4 \text{ lakhs} / 0.1 \\ &= ₹ 84 \text{ lakhs} \end{aligned}$$

Therefore, APV

$$= \text{Base case PV} - \text{Issue Costs} + \text{PV of Tax Relief on debt interest}$$

$$= ₹ 30 \text{ lakhs} - ₹ 10 \text{ lakhs} + 84 \text{ lakhs}$$

$$= ₹ 104 \text{ lakhs}$$

Question – 19

The Management of a multinational company TL Ltd. is engaged in construction of Infrastructure Project. A proposal to construct a Toll Road in Nepal is under consideration of the Management.

The following information is available:

- ❖ The initial investment will be in purchase of equipment costing USD 250 lakhs. The economic life of the equipment is 10 years. The depreciation on the equipment will be charged on straight line method.

- ❖ EBIDTA to be collected from the Toll Road is projected to be USD 33 lakhs per annum for a period of 20 years.
- ❖ To encourage investment Nepalese government is offering a 15-year term loan of USD 150 lakhs at an interest rate of 6 per cent per annum. The interest is to be paid annually. The loan will be repaid at the end of 15 year in one tranche.
- ❖ The required rate of return for the project under all equity financing is 12 per cent per annum.
- ❖ Post tax cost of debt is 5.6 per cent per annum.
- ❖ Corporate Tax Rate is 30 per cent.
- ❖ All cash Flows will be in USD.

You are required to advise the management of TL Ltd. on the viability of the proposal by using Adjusted Net Present Value method. Ignore inflation.

Given

$$\text{PVIFA} (12\%, 10) = 5.650, \text{PVIFA} (12\%, 20) = 7.469, \text{PVIFA} (8\%, 15) = 8.559, \text{PVIF} (8\%, 15) = 0.315.$$

Note: Make calculations in USD Lakhs and round off them upto 3 decimal points.

(RTP November – 2024)



Solution:

(i) Net Present Value (All Equity Financed) – Base NPV

Particulars	Period	USD Lakhs	PVF @ 12%	PV (USD Lakhs)
Initial Investment	0	(250.00)	1.000	(250.000)
EBIDTA	1 to 20	33.00	7.469	246.477
Tax	1 to 20	(9.90)	7.469	(73.943)
Depreciation	1 to 10	(25.00)		
Tax Saving on Dep.	1 to 10	7.50	5.650	42.375
NPV				(35.091)

(ii) Present Value of Impact of Financing by Debt

Particulars	Period	USD Lakhs	PVF @ 8%	PV (USD Lakhs)
Tax Saving on Interest	1 to 15	2.70	8.559	23.109

Adjusted Present Value of the Project

Base NPV + PV of Tax Shield on Interest

= - US\$ 35.091 + US \$ 23.109 lakh

= - US\$ 11.982 lakh

Advise: Since APV is negative, TL Ltd. should not accept the project.



Student Notes

14

BUSINESS VALUATION

PART I: ECONOMIC VALUE ADDED (EVA) & MARKET VALUE ADDED**Question – 01**

The following data pertains to XYZ Inc. engaged in software consultancy business as on 31 December 2010.

	(\$ Million)
Income from consultancy	935.00
EBIT	180.00
Less: Interest on Loan	18.00
EBT	162.00
Tax @ 35%	56.70
	105.30

Balance Sheet		(\$ Million)	
Liabilities	Amount	Assets	Amount
Equity Stock (10 million share @ \$ 10 each)	100	Land and Building	200
Reserves & Surplus	325	Computers & Software's	295
Loans	180	Current Assets:	
Current Liabilities	180	Debtors	150
		Bank	100
		Cash	40
	785		785

With the above information and following assumption you are required to compute

- Economic Value Added
- Market Value Added.

Assuming that:

- WACC is 12%.
- The share of company currently quoted at \$ 50 each

(SM TYK – 14)

Solution:**(a) Determination of Economic Value Added (EVA)**

	\$ Million
EBIT	180.00
Less: Taxes @ 35%	63.00
Net Operation Profit after Tax	117.00
Less: Cost of Capital Employed [W.No.1]	72.60
Economic value added	44.40

(b) Determination of Market Value Added (MVA)

	\$ Million
Market value of Equity Stock [W.no.2]	500
Equity Fund [W.no.3]	425



BUSINESS VALUATION

Market Value Added

75

Working Notes:

(1) Total Capital Employed

Equity Stock	\$ 100 Million
Reserve and Surplus	\$ 325 Million
Loan	<u>\$ 180 Million</u>
	<u>\$ 605 Million</u>

WACC 12%

Cost of Capital employed \$ 605 Million × 12% \$72.60 Million

(2) Market Price per equity share (A) \$ 50

No. of equity share outstanding (B) 10 Million

Market value of equity stock (A) × (B) \$ 500 Million

(3) Equity Fund

Equity Stock	\$ 100 Million
Reserves & Surplus	<u>\$ 325 Million</u>
	<u>\$ 425 Million</u>

Question – 02

RST Ltd.'s current financial year's income statement reported its net income after tax as ₹ 25,00,000. The applicable corporate income tax rate is 30%.

Following is the capital structure of RST Ltd. at the end of current financial year:

	₹
Debt (Coupon rate = 11%)	40 lakhs
Equity (Share Capital + Reserves & Surplus)	125 lakhs
Invested Capital	165 lakhs

Following data is given to estimate cost of equity capital:

Equity Beta of RST Ltd.	1.36
Risk –free rate i.e. current yield on Govt. bonds	8.5%
Average market risk premium (i.e. Excess of return on market portfolio over risk-free rate)	9%

Required:

- Estimate Weighted Average Cost of Capital (WACC) of RST Ltd.; and
- Estimate Economic Value Added (EVA) of RST Ltd.

(SM TYK – 10)

Solution:

Cost of Equity as per CAPM

$$\begin{aligned}k_e &= R_f + \beta \times \text{Market Risk Premium} \\&= 8.5\% + 1.36 \times 9\% \\&= 8.5\% + 12.24\% \\&= 20.74\%\end{aligned}$$



Cost of Debt

$$k_d = 11\%(1 - 0.30) = 7.70\%$$

WACC

$$\begin{aligned}(k_0) &= k_e \times \frac{E}{E+D} + k_d \times \frac{D}{E+D} \\ &= 20.74 \times \frac{125}{165} + 7.70 \times \frac{40}{165} \\ &= 15.71 + 1.87 \\ &= 17.58\%\end{aligned}$$

$$\begin{aligned}\text{Taxable Income} &= ₹ 25,00,000 / (1 - 0.30) \\ &= ₹ 35,71,429 \text{ or } ₹ 35.71 \text{ lakhs}\end{aligned}$$

$$\begin{aligned}\text{Operating Income} &= \text{Taxable Income} + \text{Interest} \\ &= ₹ 35,71,429 + ₹ 4,40,000 \\ &= ₹ 40,11,429 \text{ or } ₹ 40.11 \text{ lacs}\end{aligned}$$

$$\begin{aligned}\text{EVA} &= \text{EBIT} (1 - \text{Tax Rate}) - \text{WACC} \times \text{Invested Capital} \\ &= ₹ 40,11,429 (1 - 0.30) - 17.58\% \times ₹ 1,65,00,000 \\ &= ₹ 28,08,000 - ₹ 29,00,700 \\ &= - ₹ 92,700\end{aligned}$$

Question – 03

Following is the information of M/s. DY Ltd. for the year ending 31/03/2021:

BUSINESS VALUATION

Particulars	
Sales	₹ 1000 Lakh
Operating Expenses Including Interest	₹ 620 Lakh
8% Debentures	₹ 250 Lakh
Equity Share Capital (Face Value of ₹ 10 each)	₹ 250 Lakh
Reserves and Surplus	₹ 250 Lakh
Market Value of DY Ltd.	₹ 900 Lakh
Corporate Tax Rate	30%
Risk Free Rate of Return	7%
Market Rate of Return	12%
Equity Beta	1.4

You are required to

- Calculate Weighted Average Cost of Capital of DY Ltd.
- Calculate Economic Value Added
- Calculate Market Value Added

(Exam December – 2021) (8 Marks)

Solution:

(i) Calculation of Weighted Average Cost of Capital

$$\text{Equity } (250 + 250) = 500$$

$$\text{Debt} = 250$$

$$\begin{aligned}K_e &= R_f + (R_m - R_f) \beta \\ &= 7 + (12 - 7) 1.4 \\ &= 14\%\end{aligned}$$

$$K_d = I (1 - t)$$



$$= 8 (1 - 0.30)$$

$$= 5.6\%$$

$$\text{WACC} = \frac{(500 \times 14) + (250 \times 5.6)}{750}$$

$$= 11.2\%$$

(ii) Calculation of Economic Value Added

NOPAT

$$\text{Sales} = ₹ 1,000 \text{ lacs}$$

$$(-) \text{ Operating Cost } [620 - 20] = ₹ 600 \text{ lacs}$$

$$\text{EBIT} = ₹ 400 \text{ lacs}$$

$$(-) \text{ Tax @ } 30\% = 120$$

$$\text{NOPAT} = 280$$

$$\text{EVA} = \text{NOPAT} - C/E \times \text{WACC}$$

$$= 280 - 750 \times 11.2\%$$

$$= 196$$

(iii) Calculation of Market Value Added

$$\text{MVA} = \text{Market Value of Firm} - \text{Book Value of Firm}$$

$$= 900 - 750$$

$$= 150 \text{ lacs}$$

Question – 04

Delta Ltd.'s current financial year's income statement reports its net income as ₹ 15,00,000. Delta's marginal tax rate is 40% and its interest expense for the year was ₹ 15,00,000. The company has ₹ 1,00,00,000 of invested capital, of which 60% is debt. In addition, Delta Ltd. tries to maintain a Weighted Average Cost of Capital (WACC) of 12.6%.

- (i) Compute the operating income or EBIT earned by Delta Ltd. in the current year.
- (ii) What is Delta Ltd.'s Economic Value Added (EVA) for the current year?
- (iii) Delta Ltd. has 2,50,000 equity shares outstanding. According to the EVA you computed in (ii), how much can Delta pay in dividend per share before the value of the company would start to decrease? If Delta does not pay any dividends, what would you expect to happen to the value of the company?

(SM TYK – 13)**Solution:**

$$(i) \quad \text{Taxable income} = \text{Net Income} / (1 - 0.40)$$

$$\text{or, Taxable income} = ₹ 15,00,000 / (1 - 0.40) = ₹ 25,00,000$$

$$\text{Again, taxable income} = \text{EBIT} - \text{Interest}$$

$$\text{or, EBIT} = \text{Taxable Income} + \text{Interest}$$

$$= ₹ 25,00,000 + ₹ 15,00,000$$

$$= ₹ 40,00,000$$

$$(ii) \quad \text{EVA} = \text{EBIT} (1 - T) - (\text{WACC} \times \text{Invested capital})$$



$$= ₹ 40,00,000 (1 - 0.40) - (0.126 \times ₹ 1,00,00,000)$$

$$= ₹ 24,00,000 - ₹ 12,60,000$$

$$= ₹ 11,40,000$$

$$\begin{aligned} \text{(iii) EVA Dividend} &= ₹ 11,40,000 / 2,50,000 \\ &= ₹ 4.56 \end{aligned}$$

If Delta Ltd. does not pay a dividend, we would expect the value of the firm to increase because it will achieve higher growth, hence a higher level of EBIT. If EBIT is higher, then all else equal, the value of the firm will increase.

Question – 05

With the help of the following information of Jatayu Limited compute the Economic Value Added:

Capital Structure	Equity capital ₹ 160 Lakhs Reserves and Surplus ₹140lakhs 10% Debentures ₹ 400 lakhs
Cost of equity	14%
Financial Leverage	1.5 times
Income Tax Rate	30%

(SM TYK – 09)

Solution:

$$\text{Financial Leverage} = \text{PBIT} / \text{PBT}$$

$$1.5 = \text{PBIT} / (\text{PBIT} - \text{Interest})$$

BUSINESS VALUATION

$$1.5 = \text{PBIT} / (\text{PBIT} - 40)$$

$$1.5 (\text{PBIT} - 40) = \text{PBIT}$$

$$1.5 \text{ PBIT} - 60 = \text{PBIT}$$

$$1.5 \text{ PBIT} - \text{PBIT} = 60$$

$$0.5 \text{ PBIT} = 60$$

$$\text{or } \text{PBIT } 0.5 = \frac{60}{0.5} = ₹ 120 \text{ lakhs}$$

$$\text{NOPAT} = \text{PBIT} - \text{Tax} = ₹ 120 \text{ lakhs} (1 - 0.30) = ₹ 84 \text{ lakhs.}$$

Weighted Average Cost of Capital (WACC)

$$= 14\% \times (300/700) + (1 - 0.30) \times (10\%) \times (400/700) = 10\%$$

$$\text{EVA} = \text{NOPAT} - (\text{WACC} \times \text{Total Capital})$$

$$\text{EVA} = ₹ 84 \text{ lakhs} - 0.10 \times ₹ 700 \text{ lakhs}$$

$$\text{EVA} = ₹ 14 \text{ lakhs}$$

Question – 06

The following information is given for 3 companies that are identical except for their capital structure:

	Orange	Grape	Apple
Total invested capital	1,00,000	1,00,000	1,00,000
Debt/assets ratio	0.8	0.5	0.2
Shares outstanding	6,100	8,300	10,000
Pre tax cost of debt	16%	13%	15%
Cost of equity	26%	22%	20%
Operating Income (EBIT)	25,000	25,000	25,000



BUSINESS VALUATION

The tax rate is uniform 35% in all cases.

- (i) Compute the Weighted average cost of capital for each company.
- (ii) Compute the Economic Valued Added (EVA) for each company.
- (iii) Based on the EVA, which company would be considered for best investment? Give reasons.
- (iv) If the industry PE ratio is 11x, estimate the price for the share of each company.
- (v) Calculate the estimated market capitalization for each of the Companies.

(SM TYK – 12 & MTP April – 2022)

Solution:

(i) Working for calculation of WACC

	Orange	Grape	Apple
Total debt	80,000	50,000	20,000
Post tax Cost of debt	10.40%	8.45%	9.75%
Equity fund	20,000	50,000	80,000

WACC

Orange: $(10.4 \times 0.8) + (26 \times 0.2) = 13.52\%$

Grape: $(8.45 \times 0.5) + (22 \times 0.5) = 15.225\%$

Apple: $(9.75 \times 0.2) + (20 \times 0.8) = 17.95\%$

(ii)

	Orange	Grape	Apple
WACC	13.52	15.225	17.95
EVA = [EBIT (1-T) (WACC × Invested Capital)]	2,730	1,025	-1,700

- (iii) Orange would be considered as the best investment since the EVA of the company is highest and its weighted average cost of capital is the lowest

(iv) Estimated Price of each company shares

	Orange	Grape	Apple
EBIT (₹)	25,000	25,000	25,000
Interest (₹)	12,800	6,500	3,000
Taxable Income (₹)	12,200	18,500	22,000
Tax 35% (₹)	4,270	6,475	7,700
Net Income (₹)	7,930	12,025	14,300
(÷) Shares	6,100	8,300	10,000
EPS (₹)	1.30	1.45	1.43
Stock price (EPS × PE Ratio)	14.30	15.95	15.73

Since the three entities have different capital structures they would be exposed to different degrees of financial risk. The PE ratio should therefore be adjusted for the risk factor.

(v) Market Capitalization

Estimated Stock Price (₹)	14.30	15.95	15.73
No. of shares	6,100	8,300	10,000
Estimated Market Cap (₹)	87,230	1,32,385	1,57,300



Question – 07

Herbal Gyan is a small but profitable producer of beauty cosmetics using the plant Aloe Vera. This is not a high-tech business, but Herbal's earnings have averaged around ₹ 12 lakh after tax, largely on the strength of its patented beauty cream for removing the pimples.

The patent has eight years to run, and Herbal has been offered ₹ 40 lakhs for the patent rights. Herbal's assets include ₹ 20 lakhs of working capital and ₹ 80 lakhs of property, plant, and equipment. The patent is not shown on Herbal's books. Suppose Herbal's cost of capital is 15 percent. What is its Economic Value Added (EVA)?

(SM TYK – 15)**Solution:**

EVA = Income earned – (Cost of capital × Total Investment)

Total Investments

Particulars	Amount
Working Capital	₹ 20 lakhs
Property, Plant, and equipment	₹ 80 lakhs
Patent rights	₹ 40 lakhs
Total	₹ 140 lakhs

Cost of Capital 15%

EVA = ₹ 12 lakh – (0.15 × ₹ 140 lakhs)

= ₹ 12 lakh – ₹ 21 lakh

= -₹ 9 lakh

Thus, Herbal Gyan has a negative EVA of ₹ 9 lakhs.

Question – 08

Constant Engineering Ltd. has developed a high tech product which has reduced the Carbon emission from the burning of the fossil fuel. The product is in high demand. The product has been patented and has a market value of ₹ 100 Crore, which is not recorded in the books. The Net Worth (NW) of Constant Engineering Ltd. is ₹ 200 Crore. Long term debt is ₹ 400 Crore. The product generates a revenue of ₹ 84 Crore. The rate on 365 days Government bond is 10 percent per annum. Market portfolio generates a return of 12 percent per annum. The stock of the company moves in tandem with the market. Calculate Economic Value added of the company.

(SM TYK – 16 & Exam May – 2018) (5 Marks)**Solution:**

EVA = Income Earned – (Cost of Capital × Total Investment)

Total Investments

	Amount (₹ Crore)
Net Worth	200.00
Long Term Debts	400.00
Patent Rights	100.00
Total	700.00

$$WACC (k_0) = k_e \times \frac{E}{E + D} + k_d \times \frac{D}{E + D}$$

$$= 12 \times \frac{300}{700} + 10 \times \frac{400}{700}$$

$$= 5.14\% + 5.71\%$$

$$= 10.85\%$$

EVA = Profit Earned – WACC × Invested Capital



$$= ₹ 84 \text{ crore} - 10.85\% \times ₹ 700 \text{ crore}$$

$$= ₹ 8.05 \text{ crore}$$

Question – 09

Compute EVA of A Ltd. with the following information:

All Figure are in ₹ lac

Profit & Loss Statement		Balance Sheet	
Revenue	1,000	PPE	1,000
Direct Costs	-390	Current Assets	300
Selling, General & Admin. Exp. (SGA)	-200		1,300
EBIT	410	Equity	700
Interest	-10	Reserves	100
EBT	400	Non-Current	100
Tax Expense	-120	Borrowings	400
EAT		Current Liabilities & Provisions	
	280		1,300

Assume Bad Debts provision of ₹ 20 lac is included in the SGA, and same amount is reduced from the trade receivables in current assets.

Also assume that the pre-tax Cost of Debt is 12%, Tax Rate is 30% and Cost of Equity (i.e. shareholder's expected return) is 8.45%.

Solution:**(1) WACC**

$$\text{Equity } (700 + 100) = 800$$

$$\text{Debt} = 100$$

$$= 900$$

$$K_e = 8.45\%$$

$$K_d = 12 (1 - 0.30)$$

$$= 8.40\%$$

$$\text{WACC} = \frac{(800 \times 8.45) + (100 \times 8.40)}{900}$$

$$= 8.44\%$$

(2) NOPAT

$$\text{EBIT} = 410$$

$$(-)\text{Tax @ } 30\% = 123$$

$$\text{NOPAT} = 287$$

$$(+)\text{ Provision} = 20$$

$$\text{Adjusted NOPAT} = 307$$

(3) Adjusted Capital Invested

$$\text{Equity} = 800$$

$$\text{Debt} = 100$$

$$= 900$$

$$(+)\text{ Provision} = 20$$

$$= 920$$

$$\text{EVA} = 307 - 920 \times 8.44\%$$

$$= ₹ 229.35 \text{ lacs}$$



PART II: VALUATION OF BUSINESS

Question – 10

Following information are available in respect of XYZ Ltd. which is expected to grow at a higher rate for 4 years after which growth rate will stabilize at a lower level:

Base year information:

Revenue	- ₹ 2,000 crores
EBIT	- ₹ 300 crores
Capital expenditure	- ₹ 280 crores
Depreciation	- ₹ 200 crores

Information for high growth and stable growth period are as follows:

	High Growth	Stable Growth
Growth in Revenue & EBIT	20%	10%
Growth in Capital Expenditure and Depreciation	20%	Capital expenditure are offset by depreciation
Risk Free Rate	10%	9%
Equity Beta	1.15	1
Market Risk Premium	6%	5%
Pre Tax Cost of Debt	13%	12.86%
Debt Equity Ratio	1 : 1	2 : 3

For all time, working capital is 25% of revenue and corporate tax rate is 30%. What is the value of the firm?

(SM TYK – 06 & MTP March – 2022)

Solution:

Working Note 1: WACC

High Growth Period

$$K_e = 10 + 6 \times 1.15 = 16.9\%$$

$$K_d = 13 (1 - 0.30) = 9.1\%$$

$$K_0 = (16.9 \times 0.5) + (9.1 \times 0.5) = 13\%$$

Stable Growth Period

$$K_e = 9 + 5 \times 1 = 14\%$$

$$K_d = 12.86 (1 - 0.30) = 9\%$$

$$K_0 = (14 \times 3/5) + (8 \times 2/5) = 12\%$$

Working Note 2: FCFF

	1	2	3	4	5
NOPAT = EBIT (1 - t)	252	302.40	362.88	435.456	479.00
= 300 (1 - 0.30)					
= 210					
(-) [C.E – Depreciation] = 80	96	115.20	138.24	165.89	-
(-) Δ in Working Capital (W.N. 3)	100	120	144	172.80	103.68
FCFF	56	67.20	80.64	96.766	375.32

$$TV_3 = \frac{FCFF_4}{K_0 - g}$$



BUSINESS VALUATION

$$= \frac{375.32}{0.12 - 0.10} = 18,766 \text{ Cr.}$$

$$\begin{aligned} V_B (13\%) &= (56 \times 0.885) + (67.20 \times 0.783) + (80.64 \times 0.693) + \\ &\quad (96.766 \times 0.613) + (18,766 \times 0.613) \\ &= 11,720.937 \text{ Cr.} \end{aligned}$$

Working Note 3: Change in Working Capital

	0	1	2	3	4	5
Revenue	2,000	2,400	2,880	3,456	4,147.20	4,561.92
Working Capital @ 25%	500	600	720	864	1036.80	1140.48
Δ WC	-	100	120	144	172.80	103.68

Question – 11

Following information is given in respect of WXY Ltd., which is expected to grow at a rate of 20% p.a. for the next three years, after which the growth rate will stabilize at 8% p.a. normal level, in perpetuity.

	For the year ended March 31, 2014
Revenues	₹ 7,500 Crores
Cost of Goods Sold (COGS)	₹ 3,000 Crores
Operating Expenses	₹ 2,250 Crores
Capital Expenditure	₹ 750 Crores
Depreciation (included in Operating Expenses)	₹ 600 Crores

During high growth period, revenues & Earnings before Interest & Tax (EBIT) will grow at 20% p.a. and capital expenditure net of depreciation will grow at 15% p.a. From year 4 onwards, i.e. normal growth period revenues and EBIT will grow at 8% p.a. and

incremental capital expenditure will be offset by the depreciation. During both high growth & normal growth period, net working capital requirement will be 25% of revenues.

The Weighted Average Cost of Capital (WACC) of WXY Ltd. is 15%. Corporate Income Tax rate will be 30%. Required:

Estimate the value of WXY Ltd. using Free Cash Flows to Firm (FCFF) & WACC methodology.

The PVIF @ 15 % for the three years are as below:

Year	t ₁	t ₂	t ₃
PVIF	0.8696	0.7561	0.6575

(SM TYK – 08)

Solution:

Determination of forecasted Free Cash Flow of the Firm (FCFF) (₹ in crores)

	Yr. 1	Yr.2	Yr.3	Terminal Year
Revenue	9000.00	10800.00	12960.00	13996.80
COGS	3600.00	4320.00	5184.00	5598.72
Operating Expenses	1980.00	2376.00	2851.20	3079.30
Depreciation	720.00	864.00	1036.80	1119.74
EBIT	2700.00	3240.00	3888.00	4199.04
Tax @ 30%	810.00	972.00	1166.40	1259.71
EAT	1890.00	2268.00	2721.60	2939.33
Capital Exp. – Dep.	172.50	198.38	228.60	-
Δ Working Capital	375.00	450.00	540.00	259.20
Free Cash Flow (FCF)	1342.50	1619.62	1953.47	2680.13



* Excluding Depreciation. Present Value (PV) of FCFF during the explicit forecast period is:

FCFF(₹ in crores)	PVF @ 13%	PV (₹ in crores)
1342.50	0.8696	1167.44
1619.62	0.7561	1224.59
1953.47	0.6575	1284.41
		3676.44

PV of the terminal, value is:

$$\frac{2,680.13}{0.15-0.08} \times \frac{1}{(1.15)^3} = ₹ 38,287.57 \text{ Crore} \times 0.6575 = ₹ 25,174.08 \text{ Crore}$$

The value of the firm is :

$$₹ 3,676.44 \text{ Crores} + ₹ 25,174.08 \text{ Crores} = ₹ 28,850.52 \text{ Crores}$$

Question – 12

The valuation of Hansel Limited has been done by an investment analyst. Based on an expected free cash flow of ₹ 54 lakhs for the following year and an expected growth rate of 9 percent, the analyst has estimated the value of Hansel Limited to be ₹ 1800 lakhs. However, he committed a mistake of using the book values of debt and equity.

The book value weights employed by the analyst are not known, but you know that Hansel Limited has a cost of equity of 20 percent and post tax cost of debt of 10 percent. The value of equity is thrice its book value, whereas the market value of its debt is nine-tenths of its book value. What is the correct value of Hansel Ltd ?

(SM TYK – 06)

Solution:

Calculation of K_0 [Book Value]

Let assumed existing K_0 be x

$$V_F = \frac{FCFF_1}{K_0 - g}$$

$$1,800 = \frac{54}{x - 0.09}$$

$$1,800 \times x - 162 = 54$$

$$x = \frac{54 + 162}{1,800}$$

$$= 0.12 \text{ or } 12\%$$

Book Value Weights of Equity & Debt

$$(K_e \times W_E) + K_d \times (1 - W_E) = 12$$

$$20 W_E + 10 - 10 W_E = 12$$

$$10 W_E = 2$$

$$W_E = 2/10$$

$$W_E = 0.20$$

$$W_d = 0.8$$

Market Value Weights of Equity & Debt

$$\text{Market value of equity} = 0.2 \times 3 = 0.6$$

$$\text{Debt} = 0.8 \times \frac{9}{10} = 0.72$$



$$K_o = \frac{(20 \times 0.6) + (10 \times 0.72)}{1.32} = 1.32$$

$$= 14.545\%$$

Correct Value of Firm

$$V_F = \frac{54}{0.14545 - 0.09} = 973.83 \text{ lacs}$$

Question – 13

ABC (India) Ltd., a market leader in printing industry, is planning to diversify into defense equipment businesses that have recently been partially opened up by the GOI for private sector. In the meanwhile, the CEO of the company wants to get his company valued by a leading consultants, as he is not satisfied with the current market price of his scrip.

He approached consultant with a request to take up valuation of his company with the following data for the year ended 2009:

Share Price	₹ 66 per share
Outstanding debt	1934 lakh
Number of outstanding shares	75 lakh
Net income (PAT)	17.2 lakh
EBIT	245 lakh
Interest expenses	218.125 lakh
Capital expenditure	234.4 lakh

BUSINESS VALUATION

Depreciation	234.4 lakh
Working capital	44 lakh
Growth rate 8% (from 2010 to 2014)	
Growth rate 6% (beyond 2014)	
Free cash flow	240.336 lakh (year 2014 onwards)

The capital expenditure is expected to be equally offset by depreciation in future and the debt is expected to decline by 30% in 2014.

Required:

Estimate the value of the company and ascertain whether the ruling market price is undervalued as felt by the CEO based on the foregoing data. Assume that the cost of equity is 16%, and 30% of debt repayment is made in the year 2014.

Solution:

As per Firm Cash Flow Approach

(i) Computation of Tax Rate

EBIT	= ₹ 245 lakh
Interest	= ₹ 218.125 lakh
PBT	= ₹ 26.875 lakh
PAT	= ₹ 17.2 lakh
Tax paid	= ₹ 9.675 lakh



BUSINESS VALUATION

$$\begin{aligned}\text{Tax rate} &= ₹ 9.675 / 26.875 \\ &= 0.36 = 36\%\end{aligned}$$

(ii) Computation for Increase in Working Capital

$$\begin{aligned}\text{Working capital (2009)} &= ₹ 44 \text{ lakh} \\ \text{Increase in 2010} &= ₹ 44 \times 0.08 \\ &= ₹ 3.52 \text{ lakh}\end{aligned}$$

It will continue to increase @ 8% per annum.

(iii) Weighted Average Cost of Capital

$$\begin{aligned}\text{Present Debt} &= ₹ 1934 \text{ lakh} \\ \text{Interest Cost} &= ₹ 218.125 \text{ lakh} / ₹ 1934 \\ &= 11.28 \% \\ \text{Equity Capital} &= 75 \text{ lakh} \times ₹ 66 \\ &= ₹ 4950 \text{ lakh}\end{aligned}$$

$$\begin{aligned}K_c &= \frac{4,950}{1,934 + 4,950} \times 16\% + \frac{1,934}{1,934 + 4,950} \times 11.28(1-0.36) \\ &= 11.51 + 2.028 = 13.54\end{aligned}$$

(iv) As capital expenditure and depreciation are equal, they will not influence the free cash flows of the company.

(v) Computation of Free Cash Flows upto 2012

Year	2010	2011	2012	2013	2014
	₹ lakh	₹ lakh	₹ lakh	₹ lakh	₹ lakh
EBIT (1-t)	169.344	182.89	197.52	213.22	230.39
Increase working capital	3.52	3.80	4.10	4.43	4.78
Debt repayment	-	-	-	-	1934 × 0.30 = 580.2
Free cash flows	165.824	179.09	193.41	208.89	-354.59
PVF @ 13.54%	0.8807	0.7757	0.6832	0.6017	0.53
PV of free cash flow @ 13.54%	146.04	138.92	132.14	125.69	-187.93

(vi) Cost of Capital (2014 Onwards)

$$\text{Debt} = 0.7 \times ₹ 1934 = ₹ 1,353.80 \text{ lakh}$$

$$\text{Equity} = ₹ 4950 \text{ lakh}$$

$$\begin{aligned}K_c &= \frac{4,950}{4,950 + 1,353.80} \times 16\% + \frac{1,353.80}{4,950 + 1,353.80} \times 11.28(1-0.36) \\ &= 12.56 + 1.55\% \\ &= 14.11\%\end{aligned}$$

(vii) Continuing Value

$$\begin{aligned}&\frac{240.336}{0.1411 - 0.06} \times (1/1.1354)^5 \\ &= ₹ 1,570.556 \text{ lakh}\end{aligned}$$

(a) Value of the firm

$$\begin{aligned}&= \text{PV of free cash flows upto 2014} + \text{continuing value} \\ &= ₹ 354.86 \text{ lakh} + ₹ 1,570.556 \text{ lakh}\end{aligned}$$



= ₹ 1,925.416 lakh

(b) Value per share

= (Value of Firm – Value of Debt)/Number of Shares

= (₹ 1,925.416 lakh – ₹ 1,353.80 lakh)/75 lakh

= ₹ 7.622 < ₹ 66 (present market price)

Question – 14

Eagle Ltd. reported a profit of ₹ 77 lakhs after 30% tax for the financial year 2011-12. An analysis of the accounts revealed that the income included extraordinary items of ₹ 8 lakhs and an extraordinary loss of ₹10 lakhs. The existing operations, except for the extraordinary items, are expected to continue in the future. In addition, the results of the launch of a new product are expected to be as follows:

	₹ In Laksh
Sales	70
Material Cost	20
Labour Cost	12
Fixed Cost	10

You are required to:

- Calculate the value of the business, given that the capitalization rate is 14%.
- Determine the market price per equity share, with Eagle Ltd.'s share capital being comprised of 1,00,000 13% preference shares of ₹ 100 each and 50,00,000 equity shares of ₹ 10 each and the P/E ratio being 10 times.

(SM TYK – 02)

Solution:**(i) Computation of Business Value**

	(₹ Lakhs)
Profit before tax $\frac{77}{1 - 0.30}$	110
Less: Extraordinary income	(8)
Add: Extraordinary losses	10
	112
Profit from new product (₹ Lakhs)	
Sales	70
Less: Material costs	20
Labour costs	12
Fixed costs	10 (42)
	28
	140.00
Less: Taxes @ 30%	42.00
Future Maintainable Profit after taxes	98.00
Relevant Capitalization Factor	0.14
Value of Business (₹ 98/0.14)	700

(ii) Determination of Market Price of Equity Share

Future maintainable profits (After Tax)	₹ 98,00,000
Less: Preference share dividends 1,00,000 shares of ₹ 100 @ 13%	₹ 13,00,000
Earnings available for Equity Shareholders	₹ 85,00,000
No. of Equity Shares	50,00,000
Earning per share = $\frac{₹ 85,00,000}{50,00,000}$	₹ 1.70
PE Ratio	10
Market price per share	₹ 17



Question – 15

Sun Ltd. recently made a profit of ₹ 200 crore and paid out ₹ 80 crore (slightly higher than the average paid in the industry to which it pertains). The average PE ratio of this industry is 9. The estimated beta of Sun Ltd. is 1.2. As per Balance Sheet of Sun Ltd., the shareholder's fund is ₹ 450 crore and number of shares is 10 crore. In case the company is liquidated, building would fetch ₹ 200 crore more than book value and stock would realize ₹ 50 crore less.

The other data for the industry is as follows:

Projected Dividend Growth	4%
Risk Free Rate of Return	6%
Market Rate of Return	11%

Calculate the valuation of Sun Ltd. using

- P/E Ratio
- Dividend Growth Model
- Book Value
- Net Realizable Value

(RTP May – 2021)

Solution:

- $\text{₹ } 200 \text{ crore} \times 9 = \text{₹ } 1800 \text{ crore}$
- $$K_e = 6\% + 1.2 (11\% - 6\%) = 12\%$$

$$= \frac{80 \text{ Crore} \times 1.04}{0.12 - 0.04} = \text{₹ } 1,040 \text{ crore}$$
- ₹ 450 crore

- $\text{₹ } 450 \text{ crore} + \text{₹ } 200 \text{ crore} - \text{₹ } 50 \text{ crore} = \text{₹ } 600 \text{ crore}$

Question – 16

ABC Co. is considering a new sales strategy that will be valid for the next 4 years. They want to know the value of the new strategy. Following information relating to the year which has just ended, is available:

Income Statement	₹
Sales	20,000
Gross margin (20%)	4,000
Administration, Selling & distribution expense (10%)	2,000
PBT	2,000
Tax (30%)	600
PAT	1,400
Balance Sheet Information	
Fixed Assets	8,000
Current Assets	4,000
Equity	12,000

If it adopts the new strategy, sales will grow at the rate of 20% per year for three years. From 4th year onward Cash Flow will be stabilized. The gross margin ratio, Assets turnover ratio, the Capital structure and the income tax rate will remain unchanged.

Depreciation would be at 10% of net fixed assets at the beginning of the year.

The Company's target rate of return is 15%.

Determine the incremental value due to adoption of the strategy.

(SM TYK – 03 & RTP May – 2020)



Solution:

Pre-Strategy Value

$$\text{Pre-Strategic Value} = \frac{1,400 \text{ Cr.}}{15\%} = ₹ 9,333 \text{ Cr.}$$

Post-Strategic Value

Projected B/s

	1	2	3	4
FA (8,000)	9,600	11,520	13,824	13,824
CA (4,000)	4,800	5,760	6,912	6,912
Total Asset	14,400	17,280	20,736	20,736
Equity	14,400	17,280	20,736	20,736

Working Note 1:

Calculation of Depreciation

	1	2	3	4
Opening WDV	8,000	9,600	11,520	13,824
(-) Depreciation	800	960	1,152	1,382
Balance	7,200	8,640	10,368	12,442
(-) Closing WDV	9,600	11,520	13,824	13,824
Capital Expenditure	2,400	1,880	3,456	1,382

FCFF

	1	2	3	4
PAT (1,400)	1,680	2,016	2,419	2,419
(+) Depreciation	800	960	1,152	1,382
CFAT	2,480	2,976	3,571	3,801

BUSINESS VALUATION

(-) Capital Expenditure	2,400	2,880	3,456	1,382
(-) Δ Working Capital	800	960	1,152	0
FCFF	- 720	- 864	-1,037	2,419

$$TV_3 = \frac{2,419}{15\%} = 16,127$$

$$V_B = (-720 \times 0.870) + (-864 \times 0.756) + (-1,037 \times 0.658) + (-16,127 \times 0.658)$$

$$= ₹ 8,650$$

Value of Strategy

$$= 8,650 - 9,333$$

$$= - 683$$

Not financed viable.

Question – 17

Following details are available for X Ltd.

Income Statement for the year ended 31st March, 2018

Particulars	Amount
Sales	40,000
Gross Profit	12,000
Administrative Expenses	6,000
Profit Before tax	6,000
Tax @ 30%	1,800
Profit After Tax	4,200

Balance sheet as on 31st March, 2018



Particulars	Amount
Fixed Assets	10,000
Current Assets	6,000
Total Assets	16,000
Equity Share Capital	15,000
Sundry Creditors	1,000
Total Liabilities	16,000

The Company is contemplating for new sales strategy as follows :

- Sales to grow at 30% per year for next four years.
- Assets turnover ratio, net profit ratio and tax rate will remain the same.
- Depreciation will be 15% of value of net fixed assets at the beginning of the year.
- Required rate of return for the company is 15%

Evaluate the viability of new strategy

(Exam November – 2018) (12 Marks)

Solution:

Projected Balance Sheet					
	Year 1	Year 2	Year 3	Year 4	Year 5
Fixed Assets (25% of Sales)	13,000	16,900	21,970	28,561.00	28,561.00
Current Assets (15% of Sales)	7,800	10,140	13,182	17,136.00	17,136.00
Total Assets	20,800	27,040	35,152	45,697.60	45,679.60

BUSINESS VALUATION

Equity (37.5% of sales)	19,500	25,350	32,955	42,841.50	42,841.50
Sundry Creditors (2.5% of Sales)	1,300	1,690	2,197	2,856.10	2,856.10
Total Liabilities	20,800	27,040	35,152	45,697.60	45,697.60

Projected Cash Flows:-

	Year 1	Year 2	Year 3	Year 4	Year 5
Sales	52,000	67,600	87,880.00	1,14,244.00	1,14,244.00
PBT (15% of sales)	7,800	10,140	13,182.00	17,136.60	17,136.60
PAT (10.5% of sales)	5,460	7,098	9,227.40	11,995.62	11,995.62
Depreciation	1,500	1,950	2,535.00	3,295.50	4,284.15
Addition to Fixed Assets	4,500	5,850	7,605.00	9,886.50	4,284.15
increase in Net Current Assets	1,500	1,950	2,535.00	3,295.50	--
Operating cash flow	960	1,248	1,622.40	2,109.12	11,995.62

Projected Cash Flows:-

Present value of Projected Cash Flows:-

Cash Flows	PVF at 15%	PV
960	0.870	835.20
1248	0.756	943.49
1622.40	0.658	1067.54
2109.12	0.572	<u>1206.42</u>
		4,052.65

Residual Value = 11,995.62/0.15

= 79,970.80



Present value of Residual value = $79,970.80 \times \text{PVF} (15\%, 4)$

$$= 79,970.80 \times 0.572$$

$$= 45,743.30$$

Total shareholders' value = $45743.30 + 4052.65$

$$= 49795.95$$

Pre-strategy value = $4200/0.15$

$$= 28,000$$

$\therefore$ Value of strategy = $49795.95 - 28,000$

$$= 21795.95$$

Conclusion: The strategy is financially viable.

COMPARABLE METHOD

Question – 18

XY Ltd., a Cement manufacturing Company has hired you as a financial consultant of the company. The Cement Industry has been very stable for some time and the cement companies SK Ltd. & AS Ltd. are similar in size and have similar product market mix characteristic. Use comparable method to value the equity of XY Ltd. In performing analysis, use the following ratios:

- (i) Market to book value
- (ii) Market to replacement cost
- (iii) Market to sales

(iv) Market to Net Income

The following data are available for your analysis:

(Amount in ₹)

	SK Ltd	AS Ltd.	XY Ltd.
Market Value	450	400	
Book Value	400	300	250
Replacement Cost	600	550	500
Sales	550	450	500
Net Income	18	16	14

(Exam November – 2019) (5 Marks)

Solution:

Estimation of Reties

Sl. No.	Particulars	SK Ltd	AS Ltd.	Average
(i)	Market to Book Value	$\left(\frac{450}{400}\right) = 1.125$	$\left(\frac{400}{300}\right) = 1.333$	1.2290
(ii)	Market to Replacement Cost	$\left(\frac{450}{600}\right) = 0.750$	$\left(\frac{450}{550}\right) = 0.727$	0.7385
(iii)	Market to Sales	$\left(\frac{450}{550}\right) = 0.818$	$\left(\frac{400}{450}\right) = 0.889$	0.8535
(iv)	Market to Net Income	$\left(\frac{450}{18}\right) = 25$	$\left(\frac{400}{16}\right) = 25$	25

Application of Ratios to XY Ltd.

Sl. No	Particulars	XY Ltd. (₹)	Average	Indicative Value of XY Ltd. (₹)
(i)	Book Value	250	1.2290	$250 \times 1.2290 = 307.25$
(ii)		500	0.7385	$500 \times 0.7385 = 369.25$



(iii)	Replacement	500	0.8535	$500 \times 0.8535 = 426.75$
(iv)	Cost	14	25	$14 \times 25 = 350.00$
	Sales			
	Net Income			
Average				₹ 363.31

Value of XY Ltd. according to the comparable method is ₹ 363.31

CHOP SHOP APPROACH OR BREAKUP VALUE APPROACH

Question – 19

Using the chop-shop approach (or Break-up value approach), assign a value for Cranberry Ltd. whose stock is currently trading at a total market price of €4 million. For Cranberry Ltd, the accounting data set forth three business segments: consumer wholesale, retail and general centers. Data for the firm's three segments are as follows:

Business Segment	Segment Sales	Segment Assets	Segment Operating Income
Wholesale	€225,000	€600,000	€75,000
Retail	€720,000	€500,000	€150,000
General	€ 2,500,000	€4,000,000	€700,000

Industry data for “pure-play” firms have been compiled and are summarized as follows:

Business Segment	Capitalization/Sales	Capitalization/Assets	Capitalization/Operating Income
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BUSINESS VALUATION

Wholesale	0.85	0.7	9
Retail	1.2	0.7	8
General	0.8	0.7	4

Solution:

Business Segment	Capital-to-Sales	Segment Sales	Theoretical Values
Wholesale	0.85	€2,25,000	€ 1,91,250
Retail	1.2	€7,20,000	€ 8,64,000
General	0.8	€ 25,00,000	€ 20,00,000
Total Value			€ 30,55,250

Business Segment	Capital-to-Assets	Segment Assets	Theoretical Values
Wholesale	0.7	€ 6,00,000	€ 4,20,000
Retail	0.7	€ 5,00,000	€ 3,50,000
General	0.7	€ 40,00,000	€ 28,00,000
Total Value			€ 35,70,000

Business Segment	Capital-to-Operating Income	Operating Income	Theoretical Values
Wholesale	9	€ 75,000	€ 6,75,000
Retail	8	€ 1,50,000	€ 12,00,000
General	4	€ 7,00,000	€ 28,00,000
Total Value			€ 46,75,000

$$\text{Average theoretical value} = \frac{30,55,250 + 35,70,000 + 46,75,000}{3} = 37,66,750$$

Average theoretical value of Cranberry Ltd. = €37,66,750



FCFE APPROACH**Question – 20**

Calculate the value of share from the following information:

Profit after tax of the company	₹ 290 crores
Equity capital of company	₹ 1,300 crores
Par value of share	₹ 40 each
Debt ratio of company (Debt/ Debt + Equity)	27%
Long run growth rate of the company	8%
Beta 0.1; risk free interest rate	8.7%
Market returns	10.3%
Capital expenditure per share	₹ 47
Depreciation per share	₹ 39
Change in Working capital	₹ 3.45 per share

(RTP May – 2020)**Solution:**

(FCFE Approach)

$$\text{PAT} = 290$$

$$\text{No. of shares} = \frac{1,300}{40} = 32.50 \text{ Cr.}$$

$$\text{EPS} = \frac{290}{32.50} = ₹ 8.923$$

Calculation of FCFE per shares

$$\text{EPS} = 8.923$$

(-) Capital expenditure per share (Net of Depreciation)

Only equity contribution

$$(47 - 39) (1 - 0.27) = ₹ 5.84$$

$$(-) \Delta \text{WC} = 3.45 (1 - 0.27) = ₹ 2.5185$$

$$\text{FCFE per share} = 0.5645$$

$$\text{Value of equity per share} = \frac{\text{FCFE}_1}{K_e - g}$$

$$K_e = 8.7 + (10.3 - 8.7) 0.1$$

$$= 8.86\%$$

$$\text{Value per share} = \frac{0.5645 (1.08)}{0.0886 - 0.08}$$

$$= ₹ 70.89$$

Question – 21

Calculate the value of one equity share of X Ltd. from the following information:

Profit of the company (Before Tax)	₹ 8,000 crores
Equity share capital of the company	₹ 19,000 crores
No. of equity shares	380 crores
Long run growth rate of the company	7%
Risk free rate of return	9.50%
Beta of the company	0.1
Market risk premium	3.10%



Total capital expenditure	₹ 20,140 crores
Chargeable depreciation	₹ 17,100 crores
Total increase in working capital	₹ 1,755.60 crores
New debt to be issued for funding	₹ 2,062.108 crores
Tax rate	30%

Note: All calculation to rounded off upto 4 decimal points and final value of equity share to be rounded off upto 2 decimal points.

(MTP October – 2024)

Solution:

Profit After Tax (PAT) or Net Income

$$= ₹ 8,000 \text{ crores } (1 - 0.30)$$

$$= ₹ 5,600 \text{ crores}$$

Free Cash Flow to Equity (FCFE)

$$= \text{Net Income} - \text{Capital Expenditures} + \text{Depreciation} -/+ \text{Change in Net Working Capital} + \text{New Debt Issued} - \text{Debt Repayments} + \text{Net issue of Preference Shares} - \text{Preference Share Dividends}$$

Free Cash Flow to Equity (FCFE)

$$= ₹ 5,600 \text{ crores} - ₹ 20,140 \text{ crore} + ₹ 17,100 \text{ crore} - ₹ 1,755.60 \text{ crore} + ₹ 2,062.108 \text{ crore}$$

$$= ₹ 2,866.508 \text{ crore}$$

Cost of Equity

$$= R_f + \beta (R_m - R_f) \text{ or } R_f + \beta \text{ Market Risk Premium}$$

$$= 9.50\% + 0.1 \times 3.10\%$$

$$= 9.81\%$$

$$\text{Value of Equity} = \frac{\text{FCFE} (1 + g)}{K_e - g}$$

$$= \frac{2,866.508 \text{ crore } (1.07)}{0.0981 - 0.07}$$

$$= \frac{3,067.1636 \text{ crore}}{0.0281}$$

$$= ₹ 1,09,151.7295 \text{ crore}$$

$$\text{Value of One Equity Share} = \frac{1,09,151.7295 \text{ crore}}{380 \text{ crore}}$$

$$= ₹ 287.24$$

Alternatively, it can also be calculated by using per share basis as follows:

$$\text{FCFE per share} = \frac{\text{FCFE}}{\text{No. of Equity Shares}}$$

$$= \frac{2,866.508 \text{ crore}}{380 \text{ crore}} = ₹ 7.5434$$

$$\text{Value of per Equity Share} = \frac{\text{FCFE} (1 + g)}{K_e - g}$$

$$= \frac{7.5434(1.07)}{0.0981 - 0.07}$$

$$= \frac{8.0714}{0.0281}$$

$$= ₹ 287.24$$



Question – 22

ABC Ltd's share is currently traded at the price of ₹ 192.50 per share. Mr. Roni is planning to purchase the shares of the company. For this purpose, he has taken the services of a financial analyst to know whether the price of ABC Ltd. is fairly priced. The analyst has assembled the following information:

- The before-tax required rates of return on ABC Ltd. debt, preferred stock, and common stock are 8.60%, 11%, and 13%, respectively.
- The company's target capital structure is 20% debt, 30% preferred stock and 50% Common stock.
- The market value of the company's debt is ₹ 275 million and its preferred stock are valued at 120 million.
- ABC Ltd's free cash flow to the firm (FCFF) for the year just ended is ₹ 125 million. FCFF is expected to grow at a constant rate of 8% for the foreseeable future.
- The tax rate is 30%.
- ABC Ltd. has 20 million outstanding common shares.

You are required to —

- As a financial analyst, on the basis of value per share, advise Mr. Roni whether he should purchase the shares of the company at market price or not.
- Assume, we are to get same value of equity as calculated in (i) for using FCFE approach, calculate free cash flow to the equity (FCFE) for the year just ended, if FCFE is expected to grow at a constant rate of 8.50% for the foreseeable future.

Calculation up to 2 decimal points.

(Exam September – 2025)

Solution:

Working Notes:

(I) Calculation of WACC

$$= 8.60\% (1 - 0.30) \times 20\% + 11\% \times 30\% + 13\% \times 50\%$$

$$= 1.20\% + 3.30\% + 6.50\% = 11\%$$

(II) Value of Firm Based on FCFF

$$= \frac{\text{₹ 125 Millions (1.08)}}{0.11 - 0.08} = \frac{\text{₹ 135 Million}}{0.03} = \text{₹ 4500 Million}$$

- To decide whether the value of share is justified let us compute the value per share based on FCFF as follows:

Value of Firm	₹ 4500
Less: Value of Company's Debt	Million
Less: Value of Company's Preferred Stock	₹ 275 Million
Value of Equity Shares	₹ 120 Million
No. of Equity Shares	₹ 4105
Value of Per Equity Share	Million
	20 Million
	₹ 205.25

Advise: Mr. Roni should purchase share at this price as it is underpriced.

(ii) Computation of Free Cash Flow to Equity

Value of one Equity Share as per FCFF ₹ 205.25



Accordingly, by using Growth Model formula we can find the FCFE per share as follows:

$$205.25 = \frac{\text{FCFF} (1.085)}{0.13 - 0.085}$$

FCFE per share = ₹ 8.51

No. of Equity Shares outstanding = 20 million

FCFE of the ABC Ltd. shall be
 ₹ 8.51 × 20 million = ₹ 170.20 million

Alternatively, this calculation can be made on the total capital instead of per share basis as follows:

$$4,105 = \frac{\text{FCFF} (1.085)}{0.13 - 0.085}$$

FCFE = ₹ 170.25 million

MISCELLANEOUS

Question – 23

Given below is the Balance Sheet of S Ltd. as on 31.3.2008:

Liabilities	₹ (in lakh)	Assets	₹ (in lakh)
Share capital (share of ₹ 10)	100	Land and building	40
Reserves and surplus	40	Plant and machinery	80
Long Term Debts	30	Investments	10
		Stock	20
	<u>170</u>	Debtors	15
		Cash at bank	5
			<u>170</u>

You are required to work out the value of the Company's, shares on the basis of Net Assets method and Profit-earning capacity (capitalization) method and arrive at the fair price of the shares, by considering the following information:

- Profit for the current year ₹ 64 lakhs includes ₹4 lakhs extraordinary income and ₹ 1 lakh income from investments of surplus funds; such surplus funds are unlikely to recur.
- In subsequent years, additional advertisement expenses of ₹ 5 lakhs are expected to be incurred each year.
- Market value of Land and Building and Plant and Machinery have been ascertained at ₹ 96 lakhs and ₹ 100 lakhs respectively. This will entail additional depreciation of ₹ 6 lakhs each year.
- Effective Income-tax rate is 30%.
- The capitalization rate applicable to similar businesses is 15%.

Solution:

	₹ lakh
Net Assets Method	
Assets: Land & Buildings	96
Plant & Machinery	100
Investments	10
Stocks	20
Debtors	15
Cash & Bank	<u>5</u>
Total Assets	246
Less: Long Term Debts	<u>30</u>
Net Assets	<u>216</u>



BUSINESS VALUATION

Value per share	
(a)	Number of shares $\frac{1,00,00,000}{10} = 10,00,000$
(b)	Net Assets $\frac{2,16,00,000}{10,00,000} = ₹ 21.6$

Fair Price = $\frac{21.60 + 19.40}{2} = \frac{41.00}{2}$	₹ 20.50
--	---------

Question – 24

You are interested in buying some equity stocks of RK Ltd. The company has 3 divisions operating in different industries. Division A captures 10% of its industries sales which is forecasted to be ₹ 50 crore for the industry. Division B and C captures 30% and 2% of their respective industry's sales, which are expected to be ₹ 20 crore and ₹ 8.5 crore respectively. Division A traditionally had a 5% net income margin, whereas divisions B and C had 8% and 10% net income margin respectively. RK Ltd. has 3,00,000 shares of equity stock outstanding, which sell at ₹ 250.

The company has not paid dividend since it started its business 10 years ago. However from the market sources you come to know that RK Ltd. will start paying dividend in 3 years time and the pay-out ratio is 30%. Expecting this dividend, you would like to hold the stock for 5 year. By analyzing the past financial statements, you have determined that RK Ltd.'s required rate of return is 18% and that P/E ratio of 10 for the next year and on ending P/E ratio of 20 at the end of the fifth year are appropriate.

Required:

- (i) Would you purchase RK Ltd. equity at this time based on your one year forecast?
- (ii) If you expect earnings to grow @ 15% continuously, how much are you willing to pay for the stock of RK Ltd ?

Ignore taxation.

PV factors are given below :

Profit-earning Capacity Method	₹ lakh	
Profit before tax		64.00
Less: Extraordinary income	4.00	
Investment income (not likely to recur)	1.00	5.00
		59.00
Less: Additional expenses in forthcoming years		
Advertisement	5.00	
Depreciation	6.00	11.00
Expected earnings before taxes		48.00
Less: Income-tax @ 30%		14.40
Future maintainable profits (after taxes)		33.60

Value of business

$$\text{Capitalization factor} = \frac{33.60}{0.15} = 224$$

$$\text{Less: Long term Debts} = \frac{30}{194}$$

$$\text{Value per share} = \frac{1,94,00,000}{10,00,000} = ₹ 19.40$$

Fair Price of Share	₹
Value as per Net Assets Method	21.60
Value as per Profit earning capacity (Capitalization) method	19.40



Years	1	2	3	4	5
PVIF@ 18%	0.847	0.718	0.609	0.516	0.437

(Exam November – 2019 & MTP March – 2021)

Solution:

Working Notes:

Computation of Earnings per Share (EPS)

Particulars		Amount (₹)
Margin of Division A	(₹ 50 crore × 10% × 5%)	25,00,000
Margin of Division B	(₹ 20 crore × 30% × 8%)	48,00,000
Margin of Division C	(₹ 8.5 crore × 2% × 10%)	1,70,000
		74,70,000
No. of Equity Shares		3,00,000
EPS		₹ 24.90

(i) Market Price based on One Year Forecast

Expected Market Price at the end of the year = ₹ 24.90 × 10
= ₹ 249

PV of the Expected Price = ₹ 249 × 0.847 = ₹ 210.90

I would **NOT** like to purchase the share as the expected market price of shares is less than its current price of ₹ 250.

(ii) If Earning is expected to grow @ 15%

BUSINESS VALUATION

Year	EPS (₹)	Dividend (₹)	PVF@18%	PV (₹)
1	28.64	---	0.847	---
2	32.93	---	0.718	---
3	37.87	11.36	0.609	6.92
4	43.55	13.07	0.516	6.74
5	50.08	15.02	0.437	6.56
				20.22

Share Price after 5 years = $\frac{15.02 (1.15)}{0.18 - 0.15} = ₹ 575.77$

PV of the Market Price after 5 years = ₹ 575.77 × 0.437 = ₹ 251.61

Total PV of Inflows = ₹ 20.22 + ₹ 251.61 = ₹ 271.83

Thus, the maximum price I would be willing to pay for the share shall be ₹ 271.83.

Question – 25

There is a privately held company X Pvt. Ltd. that is operating into the detail space, and is now scouting for angel investors. The unleveraged beta based on the industry in which it operates is 1.8, and the average debt to equity ratio of X Pvt. Ltd. is hovering at 40:60. The rate of return provided by risk free GOI Bonds is 5%. The rate of market return for the industry is 11%. The FCFs for the next 3 years are as follows:

	Year 1	Year 2	Year 3
Free Cash Flows (₹ Crores)	10	12	15

The pre-tax cost of debt is 12%. Assume a tax regime of 30%.



Determine the potential value to be placed for X Pvt. Ltd. based on above-mentioned FCFs.

Note: Use PVF and round off calculations upto 3 decimal points.

(MTP September – 2024)

Solution:

To compute the value of A Ltd. first, we shall calculate WACC of the company. Since its share is not trading in the market, we shall use proxy beta to calculate the cost of equity. Since the unlevered beta of the industry is 1.8 the levered beta of the company will be:

$$= 1.8[1 + (1 - 0.3) \cdot 40/60]$$

$$= 2.64$$

The Cost of equity in accordance with CAPM

$$= r(f) + \beta (R_m - R_f)$$

$$= 5\% + 2.64 (11\% - 5\%)$$

$$= 20.84\%$$

The WACC = Cost of Equity + Cost of Debt

$$= 20.84 (60/100) + 12.0 (1 - 0.3) (40/100)$$

$$= 15.864$$

Finally, the free cash flows can be discounted at the WACC obtained above as under –

BUSINESS VALUATION

	Year 1	Year 2	Year 3
Free Cash Flows	10	12	15
Discount Factor	0.863	0.745	0.643
PVs of Cash Flows	8.63	8.94	9.645
Value of X Pvt. Ltd. (₹ Crores)			27.215

PART III: GEARING OF BETA

Question – 26

The total market value of the equity share of O.R.E. company is ₹ 60,00,000 and the total value of the debts is ₹ 40,00,000. The treasury estimate that the beta of the stock is currently 1.5 and that the expected risk premium on the market is 10%. The treasury bill rate is 8%.

Required:

- (1) What is the beta of the company's existing portfolio of assets?
- (2) Estimate the company's cost of capital and the discount rate for an expansion of the company's business.

Solution:

(1) Asset Beta

$$\begin{aligned}
 B_A &= B_E \times \frac{E}{E+D} \\
 &= 1.5 \times \frac{60,00,000}{1,00,00,000}
 \end{aligned}$$



$$= 0.9$$

(2) Cost of Capital

$$K_e = R_f + MRP \times B_E$$

$$= 8 + 10 \times 1.5 = 23\%$$

$$K_d = R_f = 8\%$$

$$WACC = \frac{(60,00,000 \times 23) + (40,00,000 \times 8)}{1,00,00,000}$$

$$= 17\%$$

If company expand same business with same risk then discount rate 17% should be used.

Question – 27

Equity of KGF Ltd. (KGFL) is ₹ 410 Crores, its debt, is worth ₹ 170 Crores. Printer Division segments value is attributable to 74%, which has an Asset Beta (β_p) of 1.45, balance value is applied on Spares and Consumables Division, which has an Asset Beta (β_{sc}) of 1.20 KGFL Debt beta (β_D) is 0.24.

You are required to calculate:

- (i) Equity Beta (β_E),
- (ii) Ascertain Equity Beta (β_E), if KGF Ltd. decides to change its Debt Equity position by raising further debt and buying back of equity to have its Debt Equity Ratio at 1.90. Assume that the present Debt Beta (β_{D1}) is 0.35 and any further funds raised by way of Debt will have a Beta (β_{D2}) of 0.40.
- (iii) Whether the new Equity Beta (β_E) justifies increase in the value of equity on account of leverage?

Solution:

$$\text{KGF Ltd. [E 410 + D 170]} = ₹ 580 \text{ Cr.}$$

$$B_D = 0.24$$

Printer (74%)**SCD (26%)**

$$B_P = 1.45$$

$$B_P = 1.20$$

B_A of KGF Ltd.

$$B_A = (1.45 \times 74\%) + (1.20 \times 26\%)$$

$$= 1.385$$

(i) Equity Beta of KGFL

$$B_A = \left(B_E \times \frac{E}{E+D} \right) + \left(B_D \times \frac{D}{E+D} \right)$$

$$1.385 = \left(B_E \times \frac{410}{580} \right) + \left(0.24 \times \frac{170}{580} \right)$$

$$1.385 = 0.7069 B_E + 0.0703$$

$$B_E = \frac{1.385 - 0.0703}{0.7069}$$

$$= 1.86$$

(ii) B_E of KGF Ltd.

$$\text{Equity} = 580 \text{ Cr.} \times \frac{1}{2.9}$$

$$= 200 \text{ Cr.}$$

$$\text{Debt} = 580 \text{ Cr.} \times \frac{1.9}{2.9}$$



BUSINESS VALUATION

$$= 380 \text{ Cr.}$$

$$\text{Present Debt} = 170$$

$$B_D = 0.35$$

$$\text{Further Debt} = 210$$

$$B_D = 0.40$$

$$1.385 = \left(B_E \times \frac{200}{580} \right) + \left(0.35 \times \frac{170}{580} \right) + \left(0.40 \times \frac{210}{580} \right)$$

$$1.385 = 0.3448 B_E + 0.1026 + 0.1448$$

$$B_E = \frac{1.385 - 0.1026 - 0.1448}{0.3448}$$

$$= 3.299$$

- (iii) Since B_E increased due to increase in debt, hence risk of equity also increase.

Question – 28

ZIO is a small-to-medium-sized privately held company specializing in electrical equipment manufacturing and is seeking additional investors. Below are key financial indicators to assist in evaluating the investment potential:

- * Break-even Achieved: The Company has reached its break-even point this year.
- * EBITA: ₹ 110 Lakh, including an extraordinary gain of ₹ 16 Lakh.
- * Pending Adjustments: ₹ 38 Lakh in preliminary sales promotion costs are yet to be written off.
- * Unlevered Beta: 1.5 (based on the industry benchmark).

- * Capital Structure: Debt-to-Equity Ratio of 30:70
- * Risk-Free Rate: 6% (based on liquid bonds).
- * Market Rate of Return: 12% (internal industry assessment).
- * Equity Value (EV): The EV is to be taken at a multiple of 8 on EBITDA.
- * The pre-tax cost of debt is 12.45% and assume a tax regime of 30%

The Future Cash Flows (FCFs) for the next three years are as follows

	Year 1	Year 2	Year 3
Future cash flows (₹ in Lakh)	150	200	220

Future cash flows are discounted at Weighted Average Cost of Capital (WACC)

PV Factor at 15% & 14% are as under-

	1	2	3
PV Factor at 15%	0.870	0.756	0.658
PV Factor at 14%	0.877	0.769	0.675

Calculation upto 2 decimal places.

	Y1	Y2	Y3
Future Cash Flows	100	120	150

You are required to calculate potential value to be placed on ZIO Company.

(Exam May – 2025) (7 Marks)



Solution:

The levered beta of the company will be

$$1.5[1 + (1 - 0.3) \times 30/70] = 1.95$$

The adjusted EBITDA would be

$$₹ 110 \text{ Lakh} - ₹ 16 \text{ Lakh} - ₹ 38 \text{ Lakh} = ₹ 56 \text{ Lakh}$$

The EV will be multiple of 8 on the ₹ 56 Lakh obtained above = ₹ 448 Lakh

The Cost of equity in accordance with CAPM = $R_f + \beta (R_m - R_f)$

$$= 6\% + 1.95 (12\% - 6\%) = 17.70\%$$

The WACC = Cost of Equity + Cost of Debt

$$= 17.70 (70/100) + 12.45 (1-0.3) (30/100)$$

$$= 15.00\%$$

Finally, the future cash flows can be discounted at the WACC obtained above as under –

	Y1	Y2	Y3
Future Cash Flows	150	200	220
Discount Factor	0.870	0.756	0.658
PVs of Cash Flows	130.50	151.20	144.76
Value of the Firm			426.46

Question – 29

There is a privately held company X Pvt. Ltd that is operating into the retail space, and is now scouting for angel investors. The details pertinent to valuing X Pvt. Ltd are as follows –

The company has achieved break even this year and has an EBITDA of ₹ 90 crore. The unleveraged beta based on the industry in which it operates is 1.8, and the average debt to equity ratio is hovering at 40:60. The rate of return provided by risk free liquid bonds is 5%. The EV is to be taken at a multiple of 5 on EBITDA. The accountant has informed that the EBITDA of ₹ 90 crore includes an extraordinary gain of ₹ 10 crore for the year, and a potential write off of preliminary sales promotion costs of ₹ 20 crore are still pending. The internal assessment of rate of market return for the industry is 11%. The FCFs for the next 3 years are as follows:

(₹ crore)

The post-tax cost of debt is 8.40%. Assume a tax regime of 30%.

What is the potential value to be placed on X Pvt. Ltd?

Note: While PV Factors values to be rounded off to 3 decimal points the other calculations to be rounded off to 2 decimal points.

(MTP March – 2025)

Solution:

The levered beta of the company will be $1.8 [1 + (1 - 0.3) \times 40/60]$
= 2.64

The adjusted EBITDA would be ₹ 90 crore – ₹ 10 crore – ₹ 20 crore = ₹ 60 crore

The EV will be multiple of 5 on the 60 obtained above = ₹ 300 crore

The Cost of equity in accordance with CAPM

$$= R_f + \beta (R_m - R_f)$$



$$= 5\% + 2.64 (11\% - 5\%) = 20.84\%$$

The WACC = Cost of Equity + Cost of Debt

$$= 20.84 (60/100) + 8.40 (40/100) = 15.864$$

Finally, the future cash flows can be discounted at the WACC obtained above as under –

	Y1	Y2	Y3
Future Cash flows (₹ crore)	100	120	150
Discount factor (₹ crore)	0.863	0.745	0.643
PVs of cash flows (₹ crore)	86.30	89.40	96.45
Value of Firm (₹ crore)			272.15

Question – 30

ABC, a large business house is planning to sell its wholly owned subsidiary KLM. Another large business entity XYZ has expressed its interest in making a bid for KLM. XYZ expects that after acquisition the annual earning of KLM will increase by 10%.

Following information, ignoring any potential synergistic benefits arising out of possible acquisitions, are available:

- Profit after tax for KLM for the financial year which has just ended is estimated to be ₹ 10 crore.
- KLM's after-tax profit has an increasing trend of 7% each year and the same is expected to continue.
- Estimated post tax market return is 10% and risk-free rate is 4%. These rates are expected to continue.
- Corporate tax rate is 30%.

BUSINESS VALUATION

	XYZ	ABC	Proxy entity for KLM in the same line of business
No. of shares	100	80	---
Current share price	lakhs	lakhs	---
Dividend pay out	₹ 287	₹ 375	50%
Debt: Equity at market values	40%	50%	1:4
P/E ratio	1 : 2	1:3	12
Equity beta	10	13	1:1
	1	1:1	

Assume that gearing level of KLM to be the same as for ABC and a debt beta is zero.

You are required :

- To calculate appropriate cost of equity for KLM based on the data available for the proxy entity.
- A range of values for KLM both before and after any potential synergistic benefits to XYZ of the acquisition.
- Compute the market value of KLM as a part of ABC.

Note: Round off calculation up to 2 decimal and compute figure in ₹ crores.

(RTP May – 2025)

Solution:

- To calculate cost of equity for KLM first we shall calculate β of KLM as follows:

β (equity un-gearred for the proxy company)



BUSINESS VALUATION

$$= 1.1 \times 4 / [4 + (1 - 0.3)] = 0.94$$

$$0.94 = \beta \text{ equity geared} \times 3 / [3 + (1 - 0.3)]$$

$$\beta \text{ equity geared} = 1.16$$

$$\begin{aligned} \text{Cost of equity} &= 0.04 + 1.16 \times (0.10 - 0.04) \\ &= 10.96\% \end{aligned}$$

- (b) Based on the data available range of valuation can be computed using P/E and dividend-based valuation approach.

(i) P/E valuation (Based on earning of ₹ 10 Crore)

	Using proxy entity's P/E	Using XYZ's P/E
Pre synergistic value	$= 12 \times ₹ 10 \text{ Crore}$ $= ₹ 120 \text{ Crore}$	$= 10 \times ₹ 10 \text{ Crore}$ $= ₹ 100 \text{ Crore}$
Post synergistic value	$= 12 \times ₹ 10 \text{ Crore}$ $\times 1.1$ $= ₹ 132 \text{ Crore}$	$= 10 \times ₹ 10 \text{ Crore}$ $\times 1.1$ $= ₹ 110 \text{ Crore}$

(ii) Divided valuation model

	Based on 50% pay-out	Based on 40% pay-out
Pre synergistic value	$= \frac{0.5 \times 10 \times 1.07}{0.1096 - 0.07}$ $= ₹ 135.10 \text{ Crore}$	$= \frac{0.4 \times 10 \times 1.07}{0.1096 - 0.07}$ $= ₹ 108.08 \text{ Crore}$
Pro synergistic value	$= \frac{0.5 \times 10 \times 1.1 \times 1.07}{0.1096 - 0.07}$ $= ₹ 148.61 \text{ Crore}$	$= \frac{0.4 \times 10 \times 1.1 \times 1.07}{0.1096 - 0.07}$ $= ₹ 118.89 \text{ Crore}$

(iii) Range of valuation

	P/E Based	Dividend Based
Pre synergistic	₹ 100 Crore	₹ 135.10
Post synergistic	₹ 110 Crore	₹ 148.61

(c) Market Price

Although no information is available about the value of KLM, it may be possible to calculate a market value based on proportion of earnings of ABC that is generated by KLM.

Market value of ABC = 80 Lakh Shares $\times$ ₹ 375 = ₹ 300 Crore

Post-tax earnings of ABC = ₹ 300 crore / 13 = ₹ 23.08 Crore

If market value of ABC is allocated to KLM in the proportion of relative earning of KLM to that of ABC, KLM would have a market value of ₹ 300 crore $\times$ [10/23.08] = ₹ 129.98 Crore.

KLM's Post Tax earning = ₹ 10 Crore.

If ABC's P/E ratio is applied to it, the market value of KLM becomes ₹ 10 Crore $\times$ 13 = ₹ 130 Crore.

Question – 31

STR Ltd.'s current financial year's income statement reported its net income after tax as ₹ 50 Crore.

Following is the capital structure of STR Ltd. at the end of current financial year:

	₹
Debt (Coupon rate = 11%)	80 Crore
Equity (Share Capital + Reserves & Surplus)	250 Crore
Invested Capital	330 Crore

Following data is given to estimate cost of equity capital:



Asset Beta of STR Ltd.	1.11
Risk free rate of return	8.5%
Average market risk premium	9%

The applicable corporate income tax rate is 30%.

Estimate Economic Value added (EVA) of STR Ltd. in ₹ lakh.

(RTP November – 2021 & MTP October – 2020)

Solution:

B_E

$$1.11 = B_E \times \frac{250}{250 + 80(1 - 0.30)}$$

$$1.11 = B_E \times \frac{250}{306}$$

$$B_E = 1.359$$

$$K_e = 8.5 + 9 \times 1.359 = 20.73\%$$

$$K_d = 11(1 - 0.30) = 7.7\%$$

$$K_0 = \left(20.73 \times \frac{250}{330}\right) + \left(7.7 \times \frac{80}{330}\right)$$

$$= 17.57\%$$

NOPAT

$$EBIT = \left[\frac{50}{(1 - 0.30)} + 8.80\right] = 80.2286 \text{ Cr.}$$

(-) Tax @ 30%

$$\text{NOPAT} = 56.1600 \text{ Cr.}$$

$$\begin{aligned} \text{EVA} &= \text{NOPAT} - C/E \times \text{WACC} \\ &= 56.1600 - 330 \times 17.57\% \\ &= 1.821 \text{ Cr.} \end{aligned}$$

Question – 32

PN Limited submits the following details for the financial year ended on 31st March 2025:

Number of Equity Shares	1,50,000
Current market price per share	12
10% Debts	₹ 2,00,000
Cash and Cash Equivalents	₹ 5,00,000
Gross Profit	₹ 12,00,000
Indirect Expenses (Excluding Depreciation & Interest)	₹ 5,00,000
Depreciation	₹ 30,000
Risk-free rate of return	7%
Market rate of return	16%
Beta of the Company	0.8
Applicable Tax Rate	20%

On the basis of above details, you are required to calculate the following :

- (i) Cost of Equity of the company using CAPM.



BUSINESS VALUATION

- (ii) Earnings Per Share (EPS) of the company.
- (iii) Equity Value of the company if applicable EBIDTA multiple is 4.
- (iv) Enterprise Value of the company.
- Calculation up to 2 decimal points.

(Exam September – 2025)

Solution:

(i) Cost of Equity using CAPM

$$7\% + 0.8(16\% - 7\%) = 14.20\%$$

(ii) Earnings per Share (EPS) (₹)

Gross Profit	12,00,000
less: Indirect Expenses	5,00,000
EBIDTA	7,00,000
Less: Depreciation	30,000
	6,70,000
Less: Interest on Debt (10% on ₹ 2,00,000)	20,000
	6,50,000
Less: Tax @ 20%	1,30,000
Profit After Tax (PAT)	5,20,000
Number of Equity Shares	1,50,000
Earnings Per Share (EPS)	3.47

(iii) Equity Value of the Company

EBITDA	₹ 7,00,000
EBITDA multiple	4
Capitalized Value	₹ 28,00,000
Less: Outstanding Debts	₹ 2,00,000
Equity Value	₹ 26,00,000

(iv) Enterprise Value of Company

Number of Equity Shares	1,50,000
Current Market Price (CMP)	12
Market Capitalization	₹ 18,00,000
Add: Outstanding Debts	₹ 2,00,000
Less: Cash and Cash Equivalent	₹ 5,00,000
Enterprise Value	15,00,000



MULTIPLE CHOICE QUESTIONS**Case Scenario – 01**

During one business meeting at XYZ Ltd., one of the member pointed out that while evaluating the performance of any company one should not only see its Operating Income but should also analyze its Capital structure as well. Weighted Average Cost of Capital changes on the basis of capital structure keeping all other factors unchanged.

He presented data relating to 3 companies Alpha Ltd., Beta Ltd. and Gama Ltd. whose operating Income are equal, but their capital structure is different.

The following information relating to these 3 companies is as follows:
(in ₹ '000)

	Alpha Ltd.	Beta Ltd.	Gama Ltd.
Total invested capital	20,00,000	20,00,000	20,00,000
Debt/Assets ratio	0.8	0.5	0.2
Shares outstanding	61,000	83,000	1,00,000
Pre tax Cost of Debt	16%	13%	15%
Cost of Equity	26%	22%	20%
Operating Income (EBIT)	5,00,000	5,00,000	5,00,000

The Tax rate is uniform 35% in all cases. The industry PE ratio is 11X.

Based on above case scenario, choose the most appropriate answer of the following:

- I. The weighted average cost of capital of Alpha Ltd. shall approximately be
- (a) 13.520% (b) 15.225%
- (c) 17.950% (d) 18.000%
- II. The Economic Valued Added (EVA) for Beta Ltd. is.....

- (a) ₹ 54600 Thousand
- (b) ₹ 20500 Thousand
- (c) (-)₹ 34000 Thousand
- (d) ₹ 21500 Thousand

- III. The price per share of Gama Ltd. shall be

- (a) ₹ 28.60 (b) ₹ 31.90
- (c) ₹ 31.46 (d) ₹ 29.45

- IV. The estimated market capitalization for Alpha Ltd. is.....

- (a) ₹ 26,47,700 Thousand
- (b) ₹ 31,46,000 Thousand
- (c) ₹ 17,44,600 Thousand
- (d) ₹ 23,73,800 Thousand

- V. Earning per share of Beta Ltd. is.....

- (a) ₹ 2.60 (b) ₹ 2.90
- (c) ₹ 2.86 (d) ₹ 2.15

(MTP April – 2024 & August – 2025)

Answer: Case Scenario – 01

- I. (a) 13.520%
- II. (b) ₹ 20500 Thousand
- III. (c) ₹ 31.46
- IV. (c) ₹ 17,44,600 Thousand
- V. (d) ₹ 2.15

Case Scenario – 02

During one business meeting at ABC Ltd., one of the members presented data relating to 3 companies X Ltd., Y Ltd. and Z Ltd. whose operating Income are equal, but their capital structure is different.

(In ₹ '000)

	X Ltd.	Y Ltd.	Z Ltd.
Total invested capital	40,00,000	40,00,000	40,00,000
Debt/Assets ratio	0.8	0.5	0.2
Shares outstanding	1,22,000	1,66,000	2,00,000
Post tax cost of debt	10.40%	8.45%	9.75%
Cost of equity	26%	22%	20%
Operating income (EBIT)	10,00,000	10,00,000	10,00,000

The Tax rate is uniform 35% in all cases. The industry PE ratio is 10X.

From the information given above, choose the correct answer to the following questions:

- I.** The weighted average cost of capital of Y Ltd. shall be approximately
- (a) 13.52%
(b) 15.23%
(c) 17.95%
(d) 18.00%
- II.** The Economic Valued Added (EVA) of X Ltd. is.....
- (a) ₹ 1,09,200 Thousand
(b) ₹ 1,71,600 Thousand
(c) ₹ 2,82,000 Thousand
(d) ₹ 3,91,000 Thousand
- III.** The price per share of Z Ltd. shall be

- (a) ₹ 28.60
(b) ₹ 31.90
(c) ₹ 31.46
(d) ₹ 29.45

- IV.** The estimated market capitalisation of Y Ltd. is.....

- (a) ₹ 80,29,900 Thousand
(b) ₹ 48,10,000 Thousand
(c) ₹ 52,91,000 Thousand
(d) ₹ 59,41,650 Thousand

- V.** Earning per share of X Ltd. approximately is.....

- (a) ₹ 2.60
(b) ₹ 2.90
(c) ₹ 2.86
(d) ₹ 2.15

Answer Case Scenario – 02

- I. (b) 15.23%**
II. (a) ₹ 1,09,200 Thousand
III. (a) ₹ 28.60
IV. (b) ₹ 48,10,000 Thousand
V. (a) ₹ 2.60



Student Notes



Student Notes



15

MERGER

PART I: MERGER

STOCK DEALQuestion – 01

A Ltd., a listed company, is considering merger of B Ltd. which is also a listed company, with itself by means of a stock swap (exchange). B Ltd. has agreed to a plan under which A Ltd. will offer the current market value of B Ltd.'s shares.

Additional Information:

Particulars	A Ltd.	B Ltd.
Earnings after tax (₹)	10,00,000	2,50,000
Number of shares outstanding	4,00,000	2,00,000
Current market price (₹) per share	50	20

On the basis of above information, you are required to calculate the following:

- What is the pre-merger Earnings per Share (EPS) and P/E ratio of both the companies?
- If B Ltd.'s P/E is 10, what is its current market price per share? What is the exchange ratio? What will A Ltd.'s post-merger EPS be?
- What must the exchange ratio be for A Ltd.'s Pre-merger and Post-merger EPS to be the same?

(Exam November – 2019) (8 Marks)

Solution:**(i) Before Merger**

	A Ltd.	B Ltd.
Earning after tax (₹)	10,00,000	2,50,000
No. of shares outstanding	4,00,000	2,00,000
EPS	₹ 2.50	₹ 1.25
Current Market Price/Share	₹ 50	₹ 20
P/E Ratio	20	16

(ii) If B Ltd.'s P/E Ratio is 10

Then, its Current Market Price = $10 \times ₹ 1.25 = ₹ 12.50$

Exchange Ratio = $12.50 : 50$ i.e. 1 share of A Ltd. for every 4 shares of B Ltd.

No. of shares to be issued = 50,000

A Ltd. Post-Merger EPS

Post-Merger Earning $(10,00,000 + 2,50,000)$ ₹ 12,50,000

No. of Equity Shares after Merger $(4,00,000 + 50,000)$
4,50,000

EPS ₹ 2.78

(iii) Calculation of Exchange Ratio for A Ltd.'s pre-merger and post-merger EPS to be the same

= Total earnings/Pre-merger EPS of A Ltd.

= ₹ 12,50,000/₹ 2.50

= 5,00,000 shares

Now, number of shares to be issue to B Ltd.

= 5,00,000 – 4,00,000

= 1,00,000 shares

Therefore, the share exchange ratio is 1,00,000 : 2,00,000 or 1:2. It means for every two shares in B Ltd., one share should be issued from A Ltd.

Question – 02

MK Ltd. is considering acquiring NN Ltd. The following information is available:

Company	Earning after Tax (₹)	No. of Equity Shares	Market Value Per Share (₹)
MK Ltd.	60,00,000	12,00,000	200.00
NN Ltd.	18,00,000	3,00,000	160.00

Exchange of equity shares for acquisition is based on current market value as above. There is no synergy advantage available.

- Find the earning per share for company MK Ltd. after merger, and
- Find the exchange ratio so that shareholders of NN Ltd. would not be at a loss.

(SM TYK – 03 & MTP October – 2020)

Solution:

- Earning per share of company MK Ltd after merger:-**

Exchange ratio 160 : 200 = 4 : 5.

that is 4 shares of MK Ltd. for every 5 shares of NN Ltd.

∴ Total number of shares to be issued = $4/5 \times 3,00,000$
= 2,40,000 Shares.

∴ Total number of shares of MK Ltd. and NN Ltd.

= 12,00,000 (MK Ltd.) + 2,40,000 (NN Ltd.)

= 14,40,000 Shares

Total profit after tax = ₹ 60,00,000 MK Ltd.

= ₹ 18,00,000 NN Ltd.

= ₹ 78,00,000

∴ EPS. (Earning Per Share) of MK Ltd. after merger

₹ 78,00,000 / 14,40,000 = ₹ 5.42 per share

- To find the exchange ratio so that shareholders of NN Ltd. would not be at a Loss:**

Present earning per share for company MK Ltd.

= ₹ 60,00,000 / 12,00,000 = ₹ 5.00

Present earning per share for company NN Ltd.

= ₹ 18,00,000 / 3,00,000 = ₹ 6.00



MERGER

∴ Exchange ratio should be 6 shares of MK Ltd. for every 5 shares of NN Ltd.

∴ Shares to be issued to NN Ltd.

$$= 3,00,000 \times 6/5 = 3,60,000 \text{ shares}$$

Now, total No. of shares of MK Ltd. and NN Ltd.

$$= 12,00,000 \text{ (MK Ltd.)} + 3,60,000 \text{ (NN Ltd.)}$$

$$= 15,60,000 \text{ shares}$$

∴ EPS after merger = ₹ 78,00,000 / 15,60,000 = ₹ 5.00 per share

Total earnings available to shareholders of NN Ltd. after merger

$$= 3,60,000 \text{ shares} \times ₹ 5.00 = ₹ 18,00,000.$$

This is equal to earnings prior merger for NN Ltd.

∴ Exchange ratio on the basis of earnings per share is recommended.

Question – 03

Cauliflower Limited is contemplating acquisition of Cabbage Limited. Cauliflower Limited has 5 lakh shares having market value of ₹ 40 per share while Cabbage Limited has 3 lakh shares having market value of ₹ 25 per share. The EPS for Cabbage Limited and Cauliflower Limited are ₹ 3 per share and ₹ 5 per share respectively. The managements of both the companies are discussing two alternatives for exchange of shares as follows:

(i) In proportion to relative earnings per share of the two companies.

(ii) 1 share of Cauliflower Limited for two shares of Cabbage Limited.

Required:

- (a) Calculate the EPS after merger under both the alternatives.
- (b) Show the impact on EPS for the shareholders of the two companies under both the alternatives.

(RTP November – 2021)

Solution:

(i) **Exchange ratio in proportion to relative EPS**

Company	Existing No. of shares	EPS	Total earnings
Cauliflower Ltd.	5,00,000	5.00	25,00,000
Cabbage Ltd.	3,00,000	3.00	<u>9,00,000</u>
Total earnings			<u>34,00,000</u>

No. of shares after merger 5,00,000 + 1,80,000 = 6,80,000

Note: 1,80,000 may be calculated as = $\left(3,00,000 \times \frac{3.00}{5.00}\right)$

$$\text{EPS for Cauliflower Ltd. after merger} = \frac{34,00,000}{6,80,000} = 5.00$$

Impact on EPS

	₹
<u>Cauliflower Ltd. 's shareholders</u>	
EPS before merger	5.00
EPS after merger	<u>5.00</u>
Increase/ Decrease in EPS	<u>0.00</u>



MERGER

Cabbage Ltd. 's shareholders	
EPS before merger	<u>3.00</u>
EPS after the merger $5.00 \times 3/5$	<u>3.00</u>
Increase/ Decrease in EPS	<u>0.00</u>

(ii) Merger effect on EPS with share exchange ratio of 1:2

Total earnings after merger	₹ 34,00,000
No. of shares post merger	
$5,00,000 + 1,50,000 (0.5 \times 3,00,000)$	6,50,000
EPS $(34,00,000 \div 6,50,000)$	₹ 5.23

Impact on EPS

	₹
Cauliflower Ltd. 's shareholders	
EPS before merger	5.00
EPS after merger	<u>5.23</u>
Increase/ Decrease in EPS	<u>0.23</u>
Cabbage Ltd. 's shareholders	
EPS before merger	<u>3.000</u>
EPS after the merger 5.23×0.5	<u>2.615</u>
Decrease in EPS	<u>0.385</u>

Question – 04

ABC Ltd. is a company operating in the software industry. It is considering the acquisition of XYZ Ltd. which is also into software industry. The following information are available for the companies:

	ABC Ltd.	XYZ Ltd.
Earnings after tax (₹)	9,00,000	2,40,000
Number of equity shares	1,50,000	60,000
P/E ratio (no. of times)	14	10

ABC Ltd. is planning to offer a premium of 25% over the market price of XYZ Ltd. Required:

- What is the swap ratio based on current market price?
- Find the number of shares to be issued by ABC Ltd. to the shareholders of XYZ Ltd.
- Compute the new EPS of ABC Ltd. after merger and comment on the impact of merger.
- Determine the market price of the share when P/E ratio remains unchanged.
- Compute the market price when P/E declines to 12 and comment on the results. Figures are to be rounded off to 2 decimals.

(Exam November – 2019) (8 Marks)

Solution:

Computation of Market Price of the Shares

Particulars	ABC Ltd.	XYZ Ltd.
EAT	₹ 9,00,000	₹ 2,40,000
No. of Equity Shares	1,50,000	60,000
EPS	₹ 6.00	₹ 4.00
P/E Ratio	14.00	10.00
Market Price Per Share	₹ 84.00	₹ 40.00

(i) Exchange Ratio based on Current Market Price

$$\text{Exchange ratio } 40 \times 1.25:84 = 50:84$$



MERGER

or that is 50 shares of ABC Ltd. for every 84 shares of XYZ Ltd.

25 shares of ABC Ltd. for every 42 shares of XYZ Ltd.

(ii) No. of Shares to be issued

$\frac{50}{84} = 0.596$ i.e. 0.60 share for 1 share of XYZ Ltd.

$60,000 \times 0.60 = 36,000$

(iii) Computation of EPS and Impact after Merger

Total earnings after merger	₹ 11,40,000
No. of shares post merger (1,50,000 + 36,000)	1,86,000
EPS	6.13

Impact on EPS

For ABC Ltd.'s shareholders	₹
EPS before merger	6.00
EPS after merger	<u>6.13</u>
Increase in EPS	<u>0.13</u>
For XYZ Ltd.'s Shareholders	
EPS before merger	4.00
Equivalent EPS after the merger (6.13×0.6)	<u>3.68</u>
Decrease in EPS	<u>0.32</u>

Thus, with the proposed merger while the EPS for shareholders of ABC Ltd. will improve and EPS for shareholders of XYZ Ltd. will be decreased.

(iv) Market Price of Share after Merger

New EPS	₹ 6.13
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PE Ratio	14
New Price of the Share (₹ 6.13 × 14)	₹ 85.82

(v) New Market Price of share if PE Ratio falls to 12

New EPS	₹ 6.13
PE Ratio	12
New Price of the Share (₹ 6.13 × 12)	₹ 73.56

Gain/ loss from the Merger to the shareholders of ABC Ltd.

Market Price of Share	₹ 73.56
Market Price of Share before Merger	₹ 84.00
Loss from the merger (per share)	₹ 10.44

Gain/ loss from the Merger to the shareholders of XYZ Ltd.

Equivalent Market Price of Share (73.56 × 0.6)	₹ 44.14
Market Price of Share before Merger	₹ 40.00
Gain from the merger (per share)	₹ 4.14

Comments: With the merger there is a decrease in the market price of shares for the shareholders of ABC Ltd and a gain for shareholders of XYZ Ltd.

Question – 05

B Ltd. Wants to acquire S Ltd. and has offered a swap ratio of 2:3 (2 shares for every 3 share of S Ltd.). Following information is available:

Particulars	B Ltd.	S Ltd.
Profit after tax (in ₹)	21,00,000	4,50,000
Equity shares outstanding (Nos.)	6,00,000	1,80,000
EPS (₹)	3.5	2.5
PE Ratio	10 times	7 times



Price quoting per share on BSE before the merger announcement (₹)	35.00	17.50
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Required:

- (i) The number of equity shares to be issued by B Ltd. for acquisition of S Ltd.
- (ii) What is the EPS of B Ltd. after the acquisition?
- (iii) Determine the equivalent earnings per share of S Ltd. and calculate per share gain or loss to shareholders of S Ltd.
- (iv) What is the expected market price per share of B Ltd. after the acquisition, assuming its PE Multiple remains unchanged?
- (v) Determine the market value of the merged firm.
- (vi) After the announcement of merger, price of shares of S Ltd. rose by 10% on BSE. Mr. X, an investor, having 10,000 shares of S Ltd. is having another investment opportunity, which yields annual return of 14% is seeking your advice whether he needs to offload the shares in the market or accept the shares from B Ltd.

(RTP May – 2022)**Solution:**

- (i) **The number of shares to be issued by B Ltd.:**

The exchange ratio is 2:3

$$\text{So, new shares} = 1,80,000 \times \frac{2}{3} = 1,20,000 \text{ shares.}$$

- (ii) **EPS of B Ltd. after acquisition:**

Total Earnings	(₹ 21,00,000 + ₹ 4,50,000)	₹ 25,50,000
No. of Shares	(6,00,000 + 1,20,000)	7,20,000
EPS	(₹ 25,50,000/7,20,000)	₹ 3.5416 or 3.54

- (iii) **Equivalent EPS of S Ltd. and gain/loss to shareholders:**

Equivalent EPS of S Ltd. ($₹ 3.54 \times \frac{2}{3}$)	₹ 2.36
Less: EPS before merger	2.50
Loss	(0.14)

- (iv) **New market price of B Ltd. (P/E remaining unchanged):**

Present P/E Ratio of B Ltd.	10 times
Expected EPS after merger	₹ 3.54
Expected Market Price ($₹ 3.54 \times 10$)	₹ 35.40

- (v) **Market value of merged firm:**

Total number of shares	7,20,000
Expected market price	₹ 35.40
Total value ($7,20,000 \times 35.40$)	₹ 2,54,88,000

- (vi)

a) Equivalent EPS of S Ltd.	₹ 2.36
b) BSE price per share before merger announcement	₹ 17.50
c) After the merger announcement 10% increase in price of shares.	₹ 1.75
d) Present market price of share (b + c)	₹ 19.25
e) Return on market price per share (a/d)	12.26

As Mr. X is having another opportunity to earn 14% and expected return on S Ltd.'s share is 12.26%, it is advisable to offload in market.

Question – 06

Alfa Ltd. wants to acquire Beta Ltd. and has offered a swap ratio of 1:2 (0.5 shares for every one share of Beta Ltd.) Following information is provided:

	Alfa Ltd.	Beta Ltd.
Profit after tax (₹)	18,00,000	3,60,000
Equity shares outstanding (Nos.)	6,00,000	1,80,000
EPS (₹)	3	2
PE Ratio	10 times	7 times
Market price per share (₹)	30	14

- (i) You are required to determine:
- The number of equity shares to be issued by Alfa Ltd. for acquisition of Beta Ltd.
 - The EPS of Alfa Ltd. after the acquisition.
 - The equivalent earnings per share of Beta Ltd.
 - The expected market price per share of Alfa Ltd. * after the acquisition, if PE increases to 12 times.
 - The market value of the merged firm.
- (ii) If you are the shareholder of Beta Ltd. and holding 100 shares, will you be interested to sell your stake? Why?

(Exam November – 2022) (8 Marks)

Solution:

- (i) (a) **The number of shares to be issued by Alfa Ltd.:**

The Exchange ratio is 0.5

So, new Shares = $1,80,000 \times 0.5 = 90,000$ shares.

- (b) **EPS of Alfa Ltd. after acquisition:**

Total Earnings (₹ 18,00,000 + ₹ 3,60,000) ₹ 21,60,000

No. of Shares (6,00,000 + 90,000) 6,90,000

EPS (₹ 21,60,000)/6,90,000 ₹ 3.13

- (c) **Equivalent EPS of Beta Ltd.:**

No. of new Shares 0.5

EPS ₹ 3.13

Equivalent EPS (₹ 3.13 × 0.5) ₹ 1.57 or ₹ 1.56

- (d) **New Market Price of Alfa Ltd. (P/E = 12):**

Revised P/E Ratio of Alfa Ltd. 12 times

Expected EPS after merger ₹ 3.13

Expected Market Price (₹ 3.13 × 12) ₹ 37.56

- (e) **Market Value of merged firm:**

Total number of Shares 6,90,000

Expected Market Price ₹ 37.56

Total value (6,90,000 × 37.56) ₹ 2,59,16,400

- (ii) Present market Value of share of Beta Ltd. (100 × ₹ 14) ₹ 1,400



MERGER

Revised market price of each share of Alfa Ltd. after Merger

- (i) Decompose the share price of both the companies into EPS & P/E components. Also segregate their EPS figures into Return On Equity (ROE) and Book Value/Intrinsic Value per share components.
- (ii) Estimate future EPS growth rates for both the companies.
- (iii) Based on expected operating synergies, R Ltd. estimated that the intrinsic value of S Ltd. Equity share would be ₹ 25 per share on its acquisition. You are required to develop a range of justifiable Equity Share Exchange ratios that can be offered

by R Ltd. to the shareholders of S Ltd. Based on your analysis on parts (i) and (ii), would you expect the negotiated terms to be closer to the upper or the lower exchange ratio limits and why?

(SM TYK – 27)

Solution:

(i) Determination of EPS, P/E Ratio, ROE and BVPS of R Ltd. & S Ltd.

	R Ltd.	S Ltd.
EAT (₹)	5,33,000	2,49,600
N	2,00,000	1,60,000
EPS (EAT ÷ N)	2.665	1.56
Market Price Per Share	50	20
PE Ratio (MPS/EPS)	18.76	12.82
Equity Fund (Equity Value)	24,00,000	16,00,000
BVPS (Equity Value ÷ N)	12	10
ROE (EAT ÷ EF) or	0.2221	0.156
ROE (EAT ÷ EF) × 100	22.21%	15.60%

(ii) Determination of Growth Rate of EPS of R Ltd. & S Ltd.

	R Ltd.	S Ltd.
Retention Ratio (1-D/P Ratio)	0.80	0.70
Growth Rate (ROE × Retention Ratio) or	0.1777	0.1092
Growth Rate (ROE × Retention Ratio) × 100	17.77%	10.92%

(iii) Justifiable equity share exchange ratio

(a) Market Price Based = $\frac{MPS_S}{MPS_R}$

$= ₹ 20 / ₹ 50$

$= 0.40:1$ (lower limit)

(b) Intrinsic Value Based = ₹ 25 / ₹ 50

$= 0.50:1$ (max. limit)

Since R Ltd. has higher EPS, PE, ROE and higher growth expectations the negotiated term would be expected to be closer to the lower limit, based on existing share price.

Question – 08

X Ltd. is studying the possible acquisition of Y Ltd. by way of merger. The following data are available in respect of both the companies.

Particulars	X Ltd.	Y Ltd.
Market Capitalization (₹)	75,00,000	90,00,000
Gross Profit Ratio	20%	20%
Inventory Turnover Ratio	5 times	4 times
Debtor Turnover Ratio	3 times	5 times
12% Debenture (₹)	10,00,000	-
10% Debenture (₹)	-	14,40,000
No. of Equity Shares	1,00,000	60,000
Operating Expenses	86%	78%
Corporate Tax Rate	30%	30%
Closing Stock (₹)	15,00,000	5,00,000
Debtors (₹)	10,00,000	8,00,000

You are required to calculate:

- (i) Swap ratio based on EPS & MPS respectively as weightage of 40% and 60%.
- (ii) Post Merger EPS
- (iii) Post Merger market price assuming same PE Ratio of X Ltd.
- (iv) Post Merger gain or loss in EPS.

(MTP May – 2020)

Solution:

Working Notes:

(1) Inventory Turnover Ratio = $\frac{\text{COGS}}{\text{Closing Stock}}$

X Ltd.

$$5 = \frac{\text{COGS}}{15,00,000}$$

$$\text{COGS} = ₹ 75,00,000$$

Gross Profit Ratio = 20% means COGS is 80% of Sales, then

$$\text{Sales} = \frac{75,00,000 \times 100}{80}$$

$$= ₹ 93,75,000$$

Y Ltd.

$$4 = \frac{\text{COGS}}{5,00,000}$$

$$\text{COGS} = ₹ 20,00,000$$

$$\text{Sales} = \frac{20,00,000 \times 100}{80}$$

$$= ₹ 25,00,000$$

Statement of Profit

	X Ltd.	Y Ltd.
Sales	93,75,000	25,00,000
Less: Operating Exp.	80,62,500	19,50,000
EBIT	13,12,500	5,50,000
Less: Interest EBT	1,20,000	1,44,000
Less: Tax @ 30%	11,92,500	4,06,000
EAT	3,57,750	1,21,800
	8,34,750	2,84,200

(2)

	X Ltd.	Y Ltd.
No. of Shares	1,00,000	60,000
EPS (EAT/ No. of Shares)	8,34,750/1,00,000 = ₹ 8.34	2,84,200/60,000 = ₹ 4.74
Market Price Share (Market Capitalization/ No. Shares)	75,00,000/1,00,000 = ₹ 75	90,00,000/60,000 = ₹ 150
PE Ratio (MPS/ EPS)	75/ 8.34 = 8.99	150/ 4.74 = 31.65

(i) Swap Ratio = $\frac{\text{Target Co.}}{\text{Acquirer Co.}}$

	Acquirer Co. X Ltd.	Target Co. Y Ltd.	Weight
EPS	8.34	4.74	0.40
MPS	75	150	0.60

EPS	$\frac{4.74}{8.34} \times 0.40 =$	0.227
MPS	$\frac{150}{75} \times 0.60 =$	1.200
		1.427

(ii) Post Merger EPS

$$= \frac{\text{EAT}_X + \text{EAT}_Y}{\text{No. of Shares of Both Cos.}}$$

$$= \frac{8,34,750 + 2,84,200}{1,00,000 + (60,000 \times 1.427)}$$

$$= \frac{11,18,950}{1,85,620}$$

$$= 6.03$$

(iii) Post Merger market price assuming same PE of X Ltd.

$$\text{MPS} = \text{PE} \times \text{EPS}$$

$$= 8.99 \times 6.03$$

$$= ₹ 54.21$$

(iv) Gain or Loss to the share holders

	Pre-Merger EPS	Post Merger EPS
X Ltd.	₹ 8.34	₹ 6.03
Y Ltd.	₹ 4.74	₹ 6.03 × 1.427 = ₹ 8.60

While Shareholders of X Ltd. will lose EPS of ₹ 2.31 (₹ 8.34 – ₹ 6.03) per share the shareholders of Y Ltd. stands to gain EPS of ₹ 3.86 (₹ 8.60 – ₹ 4.74) per share.

Question – 09

T Ltd. and E Ltd. are in the same industry. The former is in negotiation for acquisition of the latter. Important information about the two companies as per their latest financial statements is given below:

	T Ltd.	E Ltd.
₹ 10 Equity shares outstanding	12 Lakhs	6 Lakhs
Debt:		
10% Debentures (₹ Lakhs)	580	--
12.5% Institutional Loan (₹ Lakhs)	--	240

MERGER

Earning before interest, depreciation and tax (EBIDAT) (₹ Lakhs)	400.86	115.71
Market Price/share (₹)	220.00	110.00

T Ltd. plans to offer a price for E Ltd., business as a whole which will be 7 times EBIDAT reduced by outstanding debt, to be discharged by own shares at market price.

E Ltd. is planning to seek one share in T Ltd. for every 2 shares in E Ltd. based on the market price. Tax rate for the two companies may be assumed as 30%.

Calculate and show the following under both alternatives - T Ltd.'s offer and E Ltd.'s plan:

- Net consideration payable.
- No. of shares to be issued by T Ltd.
- EPS of T Ltd. after acquisition.
- Expected market price per share of T Ltd. after acquisition.
- State briefly the advantages to T Ltd. from the acquisition.

Note: Calculations (except EPS) may be rounded off to 2 decimals in lakhs.

(SM TYK – 21 & Exam November – 2018) (12 Marks)

Solution:

As per T Ltd.'s Offer

	₹ In lakhs
(i) Net Consideration Payable	809.97
7 times EBIDAT, i.e. 7 × ₹ 115.71 lakh	<u>240.00</u>



	Less: Debt	<u>569.97</u>
(ii)	No. of shares to be issued by T Ltd ₹ 569.97 lakh/₹ 220 (rounded off) (Nos.)	2,59,000
(iii)	EPS of T Ltd after acquisition Total EBIDT (₹ 400.86 lakh + ₹ 115.71 lakh) Less: Interest (₹ 58 lakh + ₹ 30 lakh) Less: 30% Tax Total earnings (NPAT) Total no. of shares outstanding (12 lakh + 2.59 lakh) EPS (₹ 300 lakh/ 14.59 lakh)	516.57 <u>88.00</u> 428.57 <u>128.57</u> <u>300.00</u> 14.59 lakh 20.56
(iv)	Expected Market Price: Pre-acquisition P/E multiple: EBIDAT (₹ in lakhs) Less: Interest (580 × 10/100)(₹ in lakhs) EBT Less: 30% Tax (₹ in lakhs) EAT (₹ in lakhs) (÷) No. of shares (lakhs) EPS Hence, PE multiple $\frac{220}{20}$ Expected market price after acquisition (₹ 20.56 × 11)	400.86 <u>58.00</u> 342.86 <u>102.86</u> 240.00 12.00 ₹ 20.00 11 $\frac{220}{20}$ ₹ 226.16

As per E Ltd's Plan

	₹ In lakhs
(i) Net consideration payable 6 lakhs shares × ₹ 110	660
(ii) No. of shares to be issued by T Ltd ₹ 660 lakhs ÷ ₹ 220	3 lakh
(iii) EPS of T Ltd after Acquisition NOPAT (as per earlier calculations) Total no. of shares outstanding (12 lakhs + 3 lakhs) Earning Per Share (EPS) ₹ 300 lakh/15 lakh	300.00 15 lakh ₹ 20.00
(iv) Expected Market Price (₹ 20 × 11)	220.00

(v) Advantages of Acquisition to T Ltd.

Since the two companies are in the same industry, the following advantages could accrue:

- Synergy, cost reduction and operating efficiency.
- Better market share.
- Avoidance of competition

Question – 10

C Ltd. and P Ltd. both companies operating in the same industry decided to merge and form a new entity S Ltd. The relevant financial details of the two companies prior to merger announcement are as follows:

	C Ltd.	P Ltd.
Annual Earnings after Tax (₹ lakh)	10,000	5,800
No. Shares Outstanding (lakh)	4,000	1,000
PE Ratio (No. of Times)	8	10

The merger will be affected by means of stock swap (exchange) of 3 shares of C Ltd. for 1 share of P Ltd.

After the merger it is expected that due to synergy effects, Annual Earnings (Post Tax) are expected to be 8% higher than sum of the earnings of the two companies individually. Further, it is expected that P/E Ratio of S Ltd. shall be average of P/E Ratios of two companies before the merger.

Evaluate the extent to which shareholders of P Ltd. will be benefitted per share from the proposed merger.

(MTP April – 2021)

Solution:

	C Ltd.	P Ltd.
EAT	10,000 lacs	5,800 lacs
÷ No.	4000	1000
EPS	2.50	5.80
(×) P/E Ratio	8	10
MPS	₹ 20	₹ 58

No. of shares to be issued = 1,000 shares × 3/1

= 3000 shares

$$\begin{aligned}\text{Post Merger EPS} &= \frac{(10,000 + 5,800) (1.08)}{4,000 + 3,000} \\ &= 2.438\end{aligned}$$

$$\begin{aligned}\text{Post Merger MPS} &= 2.438 \times 9 \\ &= ₹ 21.94\end{aligned}$$

$$\text{Avg P/E Ratio} = \frac{8 + 10}{2} = 9$$

Gain or Loss to the Shareholders of P Ltd.

MPS before Merger	₹ 58
MPS after Merger (21.94 × 3)	₹ 65.82
Gain =	<u>₹ 7.82</u>

Question – 11

Reliable Industries Ltd. (RIL) is considering a takeover of Sunflower Industries Ltd. (SIL). The particulars of 2 companies are given below:

Particulars	Reliable Industries Ltd	Sunflower Industries Ltd.
Earnings After Tax (EAT)	₹ 20,00,000	₹ 10,00,000
Equity shares O/s	10,00,000	10,00,000
Earnings per share (EPS)	2	1
PE Ratio (Times)	10	5

Required:

- What is the market value of each Company before merger?
- Assume that the management of RIL estimates that the shareholders of SIL will accept an offer of one share of RIL for four shares of SIL. If there are no synergic effects, what is the market value of the Post-merger RIL? What is the new price per share? Are the shareholders of RIL better or worse off than they were before the merger?

MERGER

- (iii) Due to synergic effects, the management of RIL estimates that the earnings will increase by 20%. What are the new post-merger EPS and Price per share? Will the shareholders be better off or worse off than before the merger?

(SM TYK – 24)

Solution:

(i) Market value of Companies before Merger

Particulars	RIL	SIL
EPS	₹ 2	Re. 1
P/E Ratio	10	5
Market Price Per Share	₹ 20	₹ 5
Equity Shares	10,00,000	10,00,000
Total Market Value	2,00,00,000	50,00,000

(ii) Post Merger Effects on RIL

	₹
Post-merger earnings	30,00,000
Exchange Ratio (1:4)	
No. of equity shares o/s (10,00,000 + 2,50,000)	12.50,000
EPS: 30,00,000/12,50,000	2.4
PE Ratio	10
Market Value 10 × 2.4	24
Total Value (12,50,000 × 24)	3,00,00,000

Gains From Merger:	₹
Post-Merger Market Value of the Firm	3,00,00,000
Less: Pre-Merger Market Value	
RIL 2,00,00,000	
<u>SIL 50,00,000</u>	<u>2,50,00,000</u>
Total gains from Merger	<u>50,00,000</u>

Apportionment of Gains between the Shareholders:

Particulars	RIL (₹)	SIL (₹)
Post-Merger Market Value:		
10,00,000 × 24	2,40,00,000	--
2,50,000 × 24	-	60,00,000
Less: Pre-Merger Market Value	2,00,00,000	50,00,000
Gains from Merger:	40,00,000	10,00,000

Thus, the shareholders of both the companies (RIL + SIL) are better off than before

(iii) Post-Merger Earnings:

Increase in Earnings by 20%

New Earnings: ₹ 30,00,000 × (1 + 0.20) ₹ 36,00,000

No. of equity shares outstanding: 12,50,000

EPS (₹ 36,00,000/12,50,000) ₹ 2.88

PE Ratio 10

Market Price Per Share: = ₹ 2.88 × 10 ₹ 28.80



∴ Shareholders will be better-off than before the merger situation.

Question – 12

H Ltd. agrees to buy over the business of B Ltd. effective 1st April, 2012. The summarized Balance Sheets of H Ltd. and B Ltd. as on 31st March 2012 are as follows:

Balance sheet as at 31st March, 2012 (In Crores of Rupees)

Liabilities:	H. Ltd	B. Ltd.
Paid up Share Capital		
-Equity Shares of ₹100 each	350.00	
-Equity Shares of ₹10 each		6.50
Reserve & Surplus	950.00	25.00
Total	1,300.00	31.50
Assets:		
Net Fixed Assets	220.00	0.50
Net Current Assets	1,020.00	29.00
Deferred Tax Assets	60.00	2.00
Total	1,300.00	31.50

H Ltd. proposes to buy out B Ltd. and the following information is provided to you as part of the scheme of buying:

- The weighted average post tax maintainable profits of H Ltd. and B Ltd. for the last 4 years are ₹ 300 crores and ₹ 10 crores respectively.
- Both the companies envisage a capitalization rate of 8%.
- H Ltd. has a contingent liability of ₹ 300 crores as on 31st March, 2012.

- H Ltd. to issue shares of ₹ 100 each to the shareholders of B Ltd. in terms of the exchange ratio as arrived on a Fair Value basis. (Please consider weights of 1 and 3 for the value of shares arrived on Net Asset basis and Earnings capitalization method respectively for both H Ltd. and B Ltd.)

You are required to arrive at the value of the shares of both H Ltd. and

B Ltd. under:

- Net Asset Value Method
- Earnings Capitalization Method
- Exchange ratio of shares of H Ltd. to be issued to the shareholders of B Ltd. on a Fair value basis (taking into consideration the assumption mentioned in point 4 above.)

(SM TYK – 17)

Solution:**(i) Net Asset Value**

H Ltd.	$\frac{₹ 1300 \text{ Crores} - ₹ 300 \text{ Crores}}{₹ 3.50 \text{ Crores}} = ₹ 285.71$
B Ltd.	$\frac{₹ 31.50 \text{ Crores}}{0.65 \text{ Crores}} = ₹ 48.46$

(ii) Earning Capitalization Value

H Ltd.	$\frac{₹ 300 \text{ Crores} / 0.08}{₹ 3.50 \text{ Crores}} = ₹ 1,071.43^*$
B Ltd.	$\frac{₹ 10 \text{ Crores} / 0.08}{0.65 \text{ Crores}} = ₹ 192.31$

MERGER

*Alternatively, Contingent Liability can also be deducted from this Valuation.

(iii) Fair Value

H Ltd.	$\frac{₹285.71 \times 1 + ₹1,071.43 \times 3}{4} = ₹ 875$
B Ltd.	$\frac{₹48.46 \times 1 + ₹192.31 \times 3}{4} = ₹ 156.3475$

Exchange ratio ₹ 156.3475/₹ 875 = 0.1787

H Ltd should issue its 0.1787 share for each share of B Ltd.

Note: In above solution it has been assumed that the contingent liability will materialize at its full amount.

Question – 13

AB Ltd., is planning to acquire and absorb the running business of XY Ltd. The valuation is to be based on the recommendation of merchant bankers and the consideration is to be discharged in the form of equity shares to be issued by AB Ltd. As on 31.3.2006, the paid up capital of AB Ltd. consists of 80 lakhs shares of ₹ 10 each. The highest and the lowest market quotation during the last 6 months were ₹ 570 and ₹ 430. For the purpose of the exchange, the price per share is to be reckoned as the average of the highest and lowest market price during the last 6 months ended on 31.3.06.

XY Ltd.'s Balance Sheet as at 31.3.2006 is summarized below:

	₹ lakhs
Sources	
Share Capital	
20 lakhs equity shares of ₹10 each fully paid	200
10 lakhs equity shares of ₹10 each, ₹5 paid	50
Loans	100
Total	350
Uses	
Fixed Assets (Net)	150
Net Current Assets	200
	350

An independent firm of merchant bankers engaged for the negotiation, have produced the following estimates of cash flows from the business of XY Ltd.:

Year ended	By way of	₹ lakhs
31.3.07	after tax earnings for equity	105
31.3.08	Do	120
31.3.09	Do	125
31.3.10	Do	120
31.3.11	Do	100
	Terminal Value estimate	200

It is the recommendation of the merchant banker that the business of XY Ltd. may be valued on the basis of the average of (i) Aggregate of discounted cash flows at 8% and (ii) Net assets value. Present value factors at 8% for years

1-5: 0.93 0.86 0.79 0.74 0.68

You are required to:

(a) Calculate the total value of the business of XY Ltd.



- (b) The number of shares to be issued by AB Ltd.; and
- (c) The basis of allocation of the shares among the shareholders of XY Ltd.

(SM TYK – 26)**Solution:**

Price/share of AB Ltd. for determination of number of shares to be issued

$$= (\text{₹ } 570 + \text{₹ } 430)/2 = \text{₹ } 500$$

Value of XY Ltd based on future cash flow capitalization $(105 \times 0.93) + (120 \times 0.86) + (125 \times 0.79) + (120 \times 0.74) \times (300 \times 0.68)$		
Value of XY Ltd based on net assets	₹ lakhs	592.40
Average value $(592.40 + 250)/2$	₹ lakhs	250.00
No. of shares in AB Ltd to be issued ₹ $4,21,20,000/500$		421.20
Basis of allocation of shares	Nos.	84,240
Fully paid equivalent shares in XY Ltd. $(20 + 5)$ lakhs		25,00,000
Distribution to fully paid shareholders $84,240 \times 20/25$		67,392
Distribution to partly paid shareholders $(84,240 - 67,392)$		16,848

Question – 14

Snake Ltd. is taking over Lizard Ltd, both are listed companies. The PE Ratio of Lizard Ltd. has been low as 4 and high as 7 and is currently 5. Lizard Ltd.'s previous year EPS was ₹ 3.40 and current expected EPS this year to be ₹ 4.00.

Determine the different range of values of shares using P/E Model.

(MTP Nov – 2021)**Solution:**

The range of values using P/E Ratio and EPS either historic or projected are as follows.

EPS	Value (₹)	P/E Ratio	Value	Value of Shares
Historic	3.40	Lowest	4	13.60
Historic	3.40	Current	5	17.00
Historic	3.40	Highest	7	23.80
Expected	4.00	Lowest	4	16.00
Expected	4.00	Current	5	20.00
Expected	4.00	Highest	7	28.00

Question – 15

B Ltd. is a highly successful company and wishes to expand by acquiring other firms. Its expected high growth in earnings and dividends is reflected in its PE ratio of 17. The Board of Directors of B Ltd. has been advised that if it were to take over firms with a lower PE ratio than it own, using a share-for-share exchange, then it could increase its reported earnings per share. C Ltd. has been suggested as a possible target for a takeover, which has a PE ratio of 10 and 1,00,000 shares in issue with a share price of ₹ 15. B Ltd. has 5,00,000 shares in issue with a share price of ₹ 12.

Calculate the change in earnings per share of B Ltd. if it acquires the whole of C Ltd. by issuing shares at its market price of ₹12. Assume the price of B Ltd. shares remains constant.

(SM TYK – 01 & RTP November – 2018)

Solution:

Total market value of C Ltd is = $1,00,000 \times 15$ = ₹ 15,00,000

PE ratio (given) = 10

Therefore, earnings = ₹ 15,00,000 / 10
= ₹ 1,50,000

Total market value of B Ltd. is = $5,00,000 \times ₹ 12$ = ₹ 60,00,000

PE ratio (given) = 17

Therefore, earnings = ₹ 60,00,000 / 17
= ₹ 3,52,941

The number of shares to be issued by B Ltd.

₹ 15,00,000 ÷ 12 = 1,25,000

Total number of shares of B Ltd = $5,00,000 + 1,25,000$
= 6,25,000

The EPS of the new firm is = $(₹ 3,52,941 + ₹ 1,50,000) / 6,25,000$
= ₹ 0.80

The present EPS of B Ltd is = ₹ 3,52,941 / 5,00,000

= ₹ 0.71

So the EPS of B will increase from Re. 0.71 to ₹ 0.80 as a result of merger

Question – 16

The following information is provided related to the acquiring Firm Mark Limited and the target Firm Mask Limited:

	Firm Mark Limited	Firm Mask Limited
Earning after tax (₹)	2,000 lakhs	400 lakhs
Number of shares outstanding	200 lakhs	100 lakhs
P/E ratio (times)	10	5

Required:

- What is the Swap Ratio based on current market prices?
- What is the EPS of Mark Limited after acquisition?
- What is the expected market price per share of Mark Limited after acquisition, assuming P/E ratio of Mark Limited remains unchanged?
- Determine the market value of the merged firm.
- Calculate gain/loss for shareholders of the two independent companies after acquisition

(SM TYK – 07)

Solution:

Particulars Mark Ltd.

Mask Ltd.

EPS ₹ 2,000 Lakhs / 200 lakhs ₹ 400 lakhs / 100 lakhs



MERGER

= ₹ 10

₹ 4

Market Price ₹ 10 × 10 = ₹ 100

₹ 4 × 5 = ₹ 20

(i) The Swap ratio based on current market price is

$$\text{₹ } 20 / \text{₹ } 100 = 0.2$$

or 1 share of Mark Ltd. for 5 shares of Mask Ltd.

No. of shares to be issued = 100 lakh × 0.2

= 20 lakhs.

(ii) EPS after merger = $\frac{2,000 \text{ lakhs} + 400 \text{ lakhs}}{200 \text{ lakhs} + 20 \text{ lakhs}}$

= ₹ 10.91

(iii) Expected market price after merger assuming P/E 10 times.

$$= \text{₹ } 10.91 \times 10$$

$$= \text{₹ } 109.10$$

(iv) Market value of merged firm

$$= \text{₹ } 109.10 \text{ market price} \times 220 \text{ lakhs shares}$$

$$= 240.02 \text{ crores}$$

(v) Gain from the merger

Post merger market value of the merged firm ₹ 240.02 crores

Less: Pre-merger market value

Mark Ltd. 200 Lakhs × ₹ 100 = 200 crores

Mask Ltd. 100 Lakhs × ₹ 20 = 20 crores

₹ 220.00 crores

Gain from merger

₹ 20.02 crores

Appropriation of gains from the merger among shareholders:

	Mark Ltd.	Mask Ltd
Post merger value	218.20 crores	21.82 crores
Less: Pre-merger market value	200.00 crores	20.00 crores
Gain to Shareholders	18.20 crores	1.82 crores

Question – 17

You have been provided the following Financial data of two companies:

	Krishna Ltd.	Rama Ltd.
Earnings after taxes	₹ 7,00,000	₹ 10,00,000
No. of Equity shares(outstanding)	2,00,000	4,00,000
EPS	3.5	2.5
P/E ratio	10 times	14 times
Market price per share	₹ 35	₹ 35

Company Rama Ltd. is acquiring the company Krishna Ltd., exchanging its shares on a one-to-one basis for company Krishna Ltd. The exchange ratio is based on the market prices of the shares of the two companies.

Required:

- What will be the EPS subsequent to merger?
- What is the change in EPS for the shareholders of companies Rama Ltd. and Krishna Ltd.?
- Determine the market value of the post-merger firm. PE ratio is likely to remain the same.



- (iv) Ascertain the profits accruing to shareholders of both the companies.

(SM TYK – 11)**Solution:**

(i)	Exchange Ratio	1:1
	New Shares to be issued	2,00,000
	Total shares of Rama Ltd. (4,00,000 + 2,00,000)	6,00,000
	Total earnings (₹ 10,00,000 + ₹ 7,00,000)	₹ 17,00,000
	New EPS (₹ 17,00,000/6,00,000)	₹ 2.83
(ii)	Existing EPS of Rama Ltd.	₹ 2.50
	Increase in EPS of Rama Ltd (₹ 2.83 – ₹ 2.50)	₹ 0.33
	Existing EPS of Krishna Ltd.	₹ 3.50
	Decrease in EPS of Krishna Ltd. (₹ 3.50 – ₹ 2.83)	₹ 0.67
(iii)	P/E ratio of new firm (expected to remain same)	14 times
	New market price (14 × ₹ 2.83)	₹ 39.62
	Total No. of Shares	6,00,000
	Total market Capitalization (6,00,000 × ₹ 39.62)	₹ 2,37,72,000
	Existing market capitalization (₹ 70,00,000 + ₹ 1,40,00,000)	<u>₹ 2,10,00,000</u>
	Total gain	<u>₹ 27,72,000</u>

- (iv)

	Rama Ltd.	Krishna Ltd	Total
No. of shares after merger	4,00,000	2,00,000	6,00,000
Market price	₹ 39.62	₹ 39.62	₹ 39.62
Total Mkt. Values	₹ 1,58,48,000	₹ 79,24,000	₹ 2,37,72,000
Existing Mkt. values	<u>₹ 1,40,00,000</u>	<u>₹ 70,00,000</u>	<u>₹ 2,10,00,000</u>
Gain to share holders	₹ 18,48,000	₹ 9,24,000	₹ 27,72,000

or ₹ 27,72,000 ÷ 3 = ₹ 9,24,000 to Krishna Ltd. and ₹ 18,48,000 to Rama Ltd. (in 2 : 1 ratio)

CASH DEAL**Question – 18**

The CEO of a company thinks that shareholders always look for EPS. Therefore, he considers maximization of EPS as his company's objective. His company's current Net Profits are ₹ 80.00 lakhs and P/E multiple is 10.5. He wants to buy another firm which has current income of ₹ 15.75 lakhs & P/E multiple of 10.

What is the maximum exchange ratio which the CEO should offer so that he could keep EPS at the current level, given that the current market price of both the acquirer and the target company are ₹ 42 and ₹ 105 respectively?

If the CEO borrows funds at 15% and buys out Target Company by paying cash, how much cash should he offer to maintain his EPS? Assume tax rate of 30%.

(SM TYK – 05 & MTP – 2020)

Solution:**(i)**

	Acquirer Company	Target Company
Net Profit	₹ 80 lakhs	₹ 15.75 lakhs
PE Multiple	10.50	10.00
Market Capitalization	₹ 840 lakhs	₹ 157.50 lakhs
Market Price	₹ 42	₹ 105
No. of Shares	20 lakhs	1.50 lakhs
EPS	₹ 4	₹ 10.50

$$\text{Maximum Exchange Ratio} = \frac{10.50}{4} = 2.625:1$$

Thus, for every one share of Target Company 2.625 shares of Acquirer Company.

(ii) Let X lakhs be the amount paid by Acquirer company to Target Company. Then to maintain same EPS i.e. ₹ 4 the number of shares to be issued will be:

$$\frac{(80 \text{ lakhs} + 15.75 \text{ lakhs}) - 0.70 \times 15\% \times X}{20 \text{ lakhs}} = 4$$

$$\frac{95.75 - 0.105 X}{20} = 4$$

$$X = ₹ 150 \text{ lakhs}$$

Thus, ₹ 150 lakhs shall be offered in cash to Target Company to maintain same EPS.

Question – 19

The equity shares of XYZ Ltd. are currently being traded at ₹ 24 per share in the market. XYZ Ltd. has total 10,00,000 equity shares

outstanding in number; and promoters' equity holding in the company is 40%.

PQR Ltd. wishes to acquire XYZ Ltd. because of likely synergies. The estimated present value of these synergies is ₹ 80,00,000.

Further PQR feels that management of XYZ Ltd. has been over paid. With better motivation, lower salaries and fewer perks for the top management, will lead to savings of ₹ 4,00,000 p.a. Top management with their families are promoters of XYZ Ltd. Present value of these savings would add ₹ 30,00,000 in value to the acquisition.

Following additional information is available regarding PQR Ltd.:

Earnings per share	:	₹ 4
Total number of equity shares outstanding	:	15,00,000
Market price of equity share	:	₹ 40

Required:

- What is the maximum price per equity share which PQR Ltd. can offer to pay for XYZ Ltd.?
- What is the minimum price per equity share at which the management of XYZ Ltd. will be willing to offer their controlling interest?

(SM TYK – 31)**Solution:**

- Calculation of maximum price per share at which PQR Ltd. can offer to pay for XYZ Ltd.'s share



MERGER

Market Value (10,00,000 × ₹ 24)	₹ 2,40,00,000
Synergy Gain	₹ 80,00,000
Saving of Overpayment	₹ 30,00,000
	₹ 3,50,00,000
Maximum Price (₹ 3,50,00,000 / 10,00,000)	₹ 35

- (ii) Calculation of minimum price per share at which the management of XYZ Ltd.'s will be willing to offer their controlling interest

Value of XYZ Ltd.'s Management Holding (40% of 10,00,000 × ₹ 24)	₹ 96,00,000
Add: PV of loss of remuneration to top management	₹ 30,00,000
	₹ 1,26,00,000
	4,00,000
No. of Shares	
Minimum Price (₹ 1,26,00,000 / 4,00,000)	₹ 31.50

FREE FLOAT MARKET CAPITALIZATION

Question – 20

The following information relating to the acquiring Company Abhiman Ltd. and the target Company Abhishek Ltd. are available. Both the Companies are promoted by Multinational Company, Trident Ltd. The promoter's holding is 50% and 60% respectively in Abhiman Ltd. and Abhishek Ltd.:

	Abhiman Ltd.	Abhishek Ltd.
Share Capital (₹)	200 lakh	100 lakh
Free Reserve and Surplus (₹)	800 lakh	500 lakh
Paid up Value per share (₹)	100	10
Free float Market Capitalization (₹)	400 lakh	128 lakh
P/E Ratio (times)	10	4

Trident Ltd. is interested to do justice to the shareholders of both the Companies. For the swap ratio weights are assigned to different parameters by the Board of Directors as follows:

Book Value	25%
EPS (Earning per share)	50%
Market Price	25%

- (a) What is the swap ratio based on above weights?
- (b) What is the Book Value, EPS and expected Market price of Abhiman Ltd. after acquisition of Abhishek Ltd. (assuming P.E. ratio of Abhiman Ltd. remains unchanged and all assets and liabilities of Abhishek Ltd. are taken over at book value).
- (c) Calculate:
- Promoter's revised holding in the Abhiman Ltd.
 - Free float market capitalization.
 - Also calculate No. of Shares, Earning per Share (EPS) and Book Value (B.V.), if after acquisition of Abhishek Ltd., Abhiman Ltd. decided to :
 - Issue Bonus shares in the ratio of 1 : 2; and



(2) Split the stock (share) as ₹ 5 each fully paid.

(SM TYK – 20)

Solution:

(a) Swap Ratio

	Abhiman Ltd.	Abhishek Ltd.
Share Capital	200 lakh	100 lakh
Free Reserve	<u>800 lakh</u>	<u>500 lakh</u>
Total	<u>1000 lakh</u>	<u>600 lakh</u>
No. of Shares	2 lakh	10 lakh
Book Value per share	₹500	₹ 60
Promoter's holding	50%	60 %
Non promoter's holding	50%	40 %
Free Float Market Cap.	400 lakh	128 lakh
Total market Cap.	800 lakh	320 lakh
No. of Shares	2 lakh	10 lakh
Market Price	₹ 400	₹ 32
P/E Ratio	10	4
EPS	40	8
Profits (₹ 2 × 40 lakh)	80 lakh	-
(₹ 8 × 10 lakh)	-	₹ 80 lakh

Calculation of Swap Ratio

Book Value	$\frac{60}{500} = 0.12 \times 25\%$	0.03
EPS	$\frac{8}{40} = 0.20 \times 50\%$	0.10
Market Price	$\frac{32}{400} = 0.08 \times 25\%$	<u>0.02</u>
	Total	<u>0.15</u>

Swap ratio is for every one share of Abhishek Ltd., to issue 0.15 shares of Abhiman Ltd. Hence total no. of shares to be issued.

$$10 \text{ Lakh} \times 0.15 = 1.50 \text{ lakh shares}$$

(b) Book Value, EPS & Market Price

Total No of Shares	= 2 Lakh + 1.5 Lakh
	= 3.5 Lakh
Total Capital	= ₹ 200 Lakh + ₹ 150 Lakh
	= ₹ 350 Lakh
Reserves	= ₹ 800 Lakh + ₹ 450 Lakh
	= ₹ 1,250 Lakh
BVPS	= $\frac{\text{₹ 350 Lakh} + \text{₹ 1,250 Lakh}}{3.5 \text{ Lakh}}$
	= ₹ 457.14 per share
Post Merger EPS	= $\frac{\text{Total Profit}}{\text{No. of Share}}$
	= $\frac{\text{₹ 80 Lakh} + \text{₹ 80 Lakh}}{3.5 \text{ Lakh}}$
	= $\frac{\text{₹ 160 Lakh}}{3.5}$
	= ₹ 45.71

Expected Market Price EPS(₹ 45.71) × P/E Ratio (10) = ₹ 457.10

(c) (i) Promoter's holding



Promoter's Revised Abhiman 50% i.e. 1.00 Lakh shares

Holding Abhishek 60% i.e. 0.90 Lakh shares

Total 1.90 Lakh shares

Promoter's % = $1.90/3.50 \times 100 = 54.29\%$

(ii) Free Float Market Capitalization

Free Float Market = $(3.5 \text{ Lakh} - 1.9 \text{ Lakh}) \times ₹ 457.10$

Capitalization = ₹ 731.36 Lakh

(iii) (a) & (b)

Revised Capital ₹ 350 Lakh + ₹ 175 Lakh = ₹ 525 Lakh

No. of shares before
Split (F.V ₹ 100) 5.25 Lakh

No. of Shares after
Split (F.V. ₹ 5) $5.25 \times 20 = 105 \text{ Lakh}$

EPS $160 \text{ Lakh} / 105 \text{ Lakh} = 1.523$

Book Value Cap. ₹ 525 Lakh + ₹ 1075 Lakh

No. of Shares = 105 Lakh

= ₹ 15.238 per share

Question – 21

Intel Ltd., promoted by a Trans National Company, is listed on the stock exchange.

The value of the floating stock is ₹ 45 crores. The Market Price per Share (MPS) is ₹ 150.

The capitalization rate is 20 percent.

The promoters holding is to be restricted to 75 per cent as per the norms of listing requirement. The Board of Directors have decided to fall in line to restrict the Promoters' holding to 75 percent by issuing Bonus Shares to minority shareholders while maintaining the same Price Earnings Ratio (P/E).

You are required to calculate:

- (i) Bonus Ratio;
- (ii) MPS after issue of Bonus Shares; and
- (iii) Free float Market capitalization after issue of Bonus Shares

(Exam May – 2018) (8 Marks)

Solution:

1. No. of Bonus Shares to be Issued:

Free Float Capitalization = ₹ 45 crore

Market Price Per Share = ₹ 150

Shares of Minority = $\frac{₹ 4,500 \text{ lacs}}{₹ 150} = 30 \text{ lacs}$

Minority Share Holding $(100\% - 80\%^*) = 20\%$

Hence Total shares = $\frac{30 \text{ lacs}}{0.20} = 150 \text{ lacs}$

Promoters holding 80%, = 120 lacs shares

Shares remains the same, but holding % to be taken as 75%



$$\text{Hence Total shares} = \frac{120 \text{ lacs}}{0.75} = 160 \text{ lacs}$$

$$\text{Shares of Minority} = 160 \text{ lacs} - 120 \text{ lacs} = 40 \text{ lacs}$$

Bonus 10 lacs for 30 lacs i.e. 1 shares for 3 shares held.

2. Market Price after Bonus Issue:

Let us compute PE with given k_e as follows:

$$PE = \frac{1}{k_e} = \frac{1}{0.20} = 5$$

Market Price Given = ₹ 150

Hence EPS will be (₹ 150/5) = ₹ 30

Total No. of shares before bonus issue = 150 lacs

Accordingly, Total PAT shall be (₹ 30 × 150 lacs) = ₹ 4,500 lacs

Total No. of shares after bonus issue = 150 lacs + 10 lacs = 160 lacs

EPS after Bonus Issue = ₹ 4,500 lacs/160 lacs = ₹ 28.125

Market Price After Bonus Issue = ₹ 28.125 × 5 = ₹ 140.63

3. Free Float Capitalization after Bonus Issue

₹ 140.63 × 40 lacs = ₹ 5,625.20 lacs i.e. ₹ 56.252 crore

Note: Since the information regarding the promoters' holding is missing in the question, above solution is based on assumption of promoter's holding as 80%. However, student can assume any % other than 80% and solve the question accordingly.

MERGER OF BANKS

Question – 22

Bank 'R' was established in 2005 and doing banking in India. The bank is facing DO OR DIE situation. There are problems of Gross NPA (Non Performing Assets) at 40% & CAR/CRAR (Capital Adequacy Ratio/ Capital Risk Weight Asset Ratio) at 4%. The net worth of the bank is not good. Shares are not traded regularly. Last week, it was traded @ ₹ 8 per share. RBI Audit suggested that bank has either to liquidate or to merge with other bank.

Bank 'P' is professionally managed bank with low gross NPA of 5%. It has Net NPA as 0% and CAR at 16%. Its share is quoted in the market @ ₹ 128 per share. The board of directors of bank 'P' has submitted a proposal to RBI for take over of bank 'R' on the basis of share exchange ratio.

The Balance Sheet details of both the banks are as follows:

	Bank 'R' Amt. in ₹ lacs	Bank 'P' Amt. In ₹ lacs
Paid up share capital (F.V. ₹ 10 each)	140	500
Reserves & Surplus	70	5,500
Deposits	4,000	40,000
Other liabilities	890	2,500
Total Liabilities	5,100	48,500
Cash in hand & with RBI	400	2,500
Balance with other banks	-	2,000
Investments	1,100	15,000
Advances	3,500	27,000
Other Assets	100	2,000
Total Assets	5,100	48,500



MERGER

It was decided to issue shares at Book Value of Bank 'P' to the shareholders of Bank 'R'. All assets and liabilities are to be taken over at Book Value.

For the swap ratio, weights assigned to different parameters are as follows:

Gross NPA	30%
CAR	20%
Market price	40%
Book value	10%

- What is the swap ratio based on above weights?
- How many shares are to be issued?
- Prepare Balance Sheet after merger.
- Calculate CAR & Gross NPA % of Bank 'P' after merger.

Solution:

(a) Swap Ratio

Gross NPA	5 : 40 i.e	$5/40 \times 30\% =$	0.0375
CAR	4 : 16 i.e	$4/16 \times 20\% =$	0.0500
Market Price	8 : 128 i.e	$8/128 \times 40\% =$	0.025
Book Value Per Share	15 : 120 i.e	$15/120 \times 10\% =$	0.0125

Thus, for every 1 share of Bank 'R' 0.125 share of Bank 'P' shall be issued.

(b) No. of Equity Shares to be Issued:

$$\frac{\text{₹ 140 lac}}{\text{₹ 10}} \times 0.125 = 1.75 \text{ lac shares}$$

(c) Balance Sheet after Merger

Calculation of Capital Reserve

Book Value of Shares	₹ 210.00 lac
Less: Value of Shares issued	₹ 17.50 lac
Capital Reserve	₹ 192.50 lac

Balance Sheet

	₹ lac		₹ lac
Paid up Share Capital	517.50	Cash in Hand & RBI	2,900.00
Reserves & Surplus	5,500.00	Balance with other	2,000.00
Capital Reserve	192.50	banks	16,100.00
Deposits	44,000.00	Investment	30,500.00
Other Liabilities	3,390.00	Advances	2,100.00
	53,600.00	Other Assets	53,600.00

(d) Calculation CAR & Gross NPA % of Bank 'P' after Merger

$$\text{CAR/CRWAR} = \frac{\text{Total Capital}}{\text{Risky Weighted Assets}}$$

	Bank 'R'	Bank 'P'	Merged
CAR (Given)	4%	16%	
Total Capital	₹ 210 lacs	₹ 6000 lacs	₹ 6210 lacs
Risky Weighted Assets	₹ 5250 lacs	₹ 37500 lacs	₹ 42750 lacs

$$\text{Car} = \frac{\text{₹ 6,210 lacs}}{\text{₹ 42,750 lacs}} = 14.53\%$$

$$\text{GNPA Ratio} = \frac{\text{Gross NPA}}{\text{Gross Deposits}} \times 100$$

	Bank 'R'	Bank 'P'	Merged
GNPA (Given)	0.40	0.5	
	$0.40 = \frac{\text{GNPA}_R}{\text{₹ 3,500 lacs}}$	$0.05 = \frac{\text{GNPA}_S}{\text{₹ 27,000 lacs}}$	



Gross NPA	₹ 1400lac	₹ 1350 lac	₹ 2750 lacs
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MINIMUM & MAXIMUM EXCHANGE RATIO (Conn & Nielson Model)

Question – 23

ABC Ltd. is intending to acquire XYZ Ltd. by way of merger and the following information is available in respect of these companies:

	ABC Ltd.	XYZ Ltd.
Total Earnings (E) (in lakh)	₹ 1,200	₹400
Number of outstanding shares (S) (in lakh)	400	200
Price earnings ratio (P/E)	8	7

- (a) Determine the maximum exchange ratio acceptable to the shareholders of ABC Ltd., if the P/E ratio of the combined firm is expected to be 8?
- (b) Determine the minimum exchange ratio acceptable to the shareholders XYZ Ltd., if the P/E ratio of the combined firm is expected to be 10?

Note: Make calculation in lakh multiples and compute ratio upto 4 decimal points.

(RTP May – 2021)

Solution:

	ABC	XYZ
EAT	1,200 lacs	400 lacs
(÷) No.	400	200
EPS	₹ 3	₹ 2

(×) P/E ratio	8	7
MPS	₹ 24	₹ 14

(a) Maximum Exchange Ratio

Let assume maximum Exchange Ratio be x at which post merger MPS be ₹ 24

$$\left(\frac{1,200 + 400}{400 + 200x} \right) \times 8 = ₹ 24$$

$$12,800 = 9,600 + 4,800x$$

$$x = \frac{12,800 - 9,600}{4,800} = 0.6667:1$$

(b) Minimum Exchange Ratio

Let assume maximum Exchange Ratio be x at which post merger Equivalent MPS should be ₹ 14

$$\left(\frac{1,200 + 400}{400 + 200x} \right) \times 10 \times x = ₹ 14$$

$$16,000x = 5,600 + 2,800x$$

$$13,200x = 5,600$$

$$x = \frac{5,600}{13,200} = 0.4242:1$$

Question – 24

Longitude Limited is in the process of acquiring Latitude Limited on a share exchange basis. Following relevant data are available:

		Longitude Limited	Latitude Limited
Profit after Tax (PAT)	₹ in	120	80
Number of Shares	Lakhs	15	16
Earning per Share (EPS)	Lakhs	8	5
Price Earnings Ratio (P/E Ratio)	₹	15	10
(Ignore Synergy)			

You are required to determine:

- (i) Pre-merger Market Value per Share, and
- (ii) The maximum exchange ratio Longitude Limited can offer without the dilution of
 - (1) EPS and
 - (2) Market Value per Share Calculate Ratio/s up to four decimal points and amounts and number of shares up to two decimal points.

(SM TYK – 13)

Solution:

(i) Pre Merger Market Value of per Share

P/E Ratio × EPS

Longitude Ltd. ₹ 8 × 15 = ₹ 120.00

Latitude Ltd. ₹ 5 × 10 = ₹ 50.00

(ii) (1) Maximum exchange ratio without dilution of EPS

$$\begin{aligned}\text{Swap Ratio} &= \frac{5}{8} \\ &= 0.625:1\end{aligned}$$

(2) Maximum exchange ratio without dilution of market price per share

$$\begin{aligned}\text{Swap Ratio} &= \frac{50}{120} \\ &= 0.4169:1\end{aligned}$$

Question – 25

XYZ Ltd. wants to purchase ABC Ltd. by exchanging 0.7 of its share for each share of ABC Ltd. Relevant financial data are as follows:

Equity shares outstanding	10,00,000	4,00,000
EPS (₹)	40	28
Market price per share (₹)	250	160

- (i) Illustrate the impact of merger on EPS of both the companies.
- (ii) The management of ABC Ltd. has quoted a share exchange ratio of 1:1 for the merger. Assuming that P/E ratio of XYZ Ltd. will remain unchanged after the merger, what will be the gain from merger for ABC Ltd.?
- (iii) What will be the gain/loss to shareholders of XYZ Ltd.?
- (iv) Determine the maximum exchange ratio acceptable to shareholders of XYZ Ltd.

(SM TYK – 08, Exam November – 2019 & December – 2021) (8 Marks)

MERGER

Solution: Working Notes

(a)

	XYZ Ltd.	ABC Ltd.
Equity shares outstanding (Nos.)	10,00,000	4,00,000
EPS	₹ 40	₹ 28
Profit	₹ 400,00,000	₹ 112,00,000
PE Ratio	6.25	5.71
Market price per share	₹ 250	₹ 160

(b) **EPS after merger**

No. of shares to be issued (4,00,000 × 0.70)	2,80,000
Existing Equity shares outstanding	10,00,000
Equity shares outstanding after merger	12,80,000
Total Profit (₹ 400,00,000 + ₹ 112,00,000)	₹ 512,00,000
EPS (₹ 512,00,000 / 12,80,000)	₹ 40

(i) **Impact of merger on EPS of both the companies**

	XYZ Ltd.	ABC Ltd.
EPS after Merger	₹ 40	₹ 28
EPS before Merger	₹ 40	₹ 28*
	Nil	Nil

* ₹ 40 × 0.70

(ii) **Gain from the Merger if exchange ratio is 1: 1**

No. of shares to be issued	4,00,000
Existing Equity shares outstanding	10,00,000
Equity shares outstanding after merger	14,00,000

Total Profit (₹ 400,00,000 + ₹ 112,00,000)	₹ 512,00,000
EPS	₹ 36.57
Market Price of Share (₹ 36.57 × 6.25)	₹ 228.56
Market Price of Share before Merger	₹ 160.00
Impact (Increase/ Gain)	₹ 68.56

(iii) **Gain/ loss from the Merger to the shareholders of XYZ Ltd.**

Market Price of Share	₹ 228.56
Market Price of Share before Merger	₹ 250.00
Loss from the merger (per share)	₹ 21.44

(iv) **Maximum Exchange Ratio acceptable to XYZ Ltd. shareholders**

$$\text{Swap Ratio} = \frac{28}{40}$$

$$= 0.70:1$$

PART II: DEMERGER

Question – 26

The following information is relating to Fortune India Ltd. having two division, viz. Pharma Division and Fast Moving Consumer Goods Division (FMCG Division). Paid up share capital of Fortune India Ltd. is consisting of 3,000 Lakhs equity shares of Re. 1 each. Fortune India Ltd. decided to de-merge Pharma Division as Fortune Pharma Ltd. w.e.f. 1.4.2009. Details of Fortune India Ltd. as on 31.3.2009 and of Fortune Pharma Ltd. as on 1.4.2009 are given below:



MERGER

Particulars	Fortune Pharma Ltd. (₹)	Fortune India Ltd. (₹)
Outside Liabilities		
Secured Loans	400 lakh	3,000 lakh
Unsecured Loans	2,400 lakh	800 lakh
Current Liabilities & Provisions	1,300 lakh	21,200 lakh
Assets		
Fixed Assets	7,740 lakh	20,400 lakh
Investments	7,600 lakh	12,300 lakh
Current Assets	8,800 lakh	30,200 lakh
Loans & Advances	900 lakh	7,300 lakh
Deferred tax/Misc. Expenses	60 lakh	(200) lakh

Board of Directors of the Company have decided to issue necessary equity shares of Fortune Pharma Ltd. of Re. 1 each, without any consideration to the shareholders of Fortune India Ltd. For that purpose following points are to be considered:

- Transfer of Liabilities & Assets at Book value.
- Estimated Profit for the year 2009-10 is ₹ 11,400 Lakh for Fortune India Ltd. & ₹ 1,470 lakhs for Fortune Pharma Ltd.
- Estimated Market Price of Fortune Pharma Ltd. is ₹ 24.50 per share.
- Average P/E Ratio of FMCG sector is 42 & Pharma sector is 25, which is to be expected for both the companies.

Calculate:

- The Ratio in which shares of Fortune Pharma are to be issued to the shareholders of Fortune India Ltd.
- Expected Market price of Fortune India (FMCG) Ltd.

- Book Value per share of both the Companies immediately after Demerger.

(SM TYK – 22)

Solution:

Share holders' funds

Particulars	Fortune India Ltd.	Fortune Pharma Ltd.	Fortune India (FMCG) Ltd.
Assets	70,000	25,100	44,900
Outside Liabilities	25,000	4,100	20,900
Net worth	45,000	21,000	24,000

- Calculation of Shares of Fortune Pharma Ltd. to be issued to shareholders of Fortune India Ltd.**

	Fortune Pharma Ltd.
Estimated Profit (₹ in lakhs)	1,470
Estimated market price (₹)	24.5
Estimated P/E	25
Estimated EPS (₹)	0.98
No. of shares lakhs	1,500

Hence, Ratio is 1 share of Fortune Pharma Ltd. for 2 shares of Fortune India Ltd.

or for 0.50 share of Fortune Pharma Ltd. for 1 share of Fortune India Ltd.

- Expected market price of Fortune India (FMCG) Ltd.**

	Fortune India (FMCG) Ltd.
Estimated Profit (₹ in lakhs)	11,400
No. of equity shares (₹ in lakhs)	3,000



Estimated EPS (₹)	3.8
Estimated P/S	42
Estimated Market price (₹)	159.60

3. Book value per share

	Fortune Pharma Ltd.	Fortune India (FMCG) Ltd.
Net worth (₹ in lakhs)	21,000	24,000
No. of shares (₹ in lakhs)	1,500	3,000
Book value of shares	₹ 14	₹ 8

TRUE COST OF ACQUISITION**Question – 27**

ABC Company is considering acquisition of XYZ Ltd. which has 1.5 crores shares outstanding and issued. The market price per share is ₹ 400 at present. ABC's average cost of capital is 12%. Available information from XYZ indicates its expected cash accruals for the next 3 years as follows:

Year	₹ Crore
1	250
2	300
3	400

Calculate the range of valuation that ABC has to consider. (PV factors at 12% for years 1 to 3 respectively: 0.893, 0.797 and 0.712).

(SM TYK – 01 & MTP April – 2021)**Solution:****VALUATION BASED ON MARKET PRICE**

Market Price per share ₹ 400

Thus value of total business is (₹ 400 × 1.5 Cr.) ₹ 600 Cr.

VALUATION BASED ON DISCOUNTED CASH FLOW

Present Value of cash flows

(₹ 250 cr × 0.893) + (₹ 300 cr. × 0.797) + (₹ 400 cr. × 0.712) =

₹ 747.15 Cr.

Value of per share (₹ 747.15 Cr. / 1.5 Cr) ₹ 498.10 per share

RANGE OF VALUATION

	Per Share ₹	Total ₹ Cr.
Minimum	400.00	600.00
Maximum	498.10	747.15

Question – 28

Elrond Limited plans to acquire Doom Limited. The relevant financial details of the two firms prior to the merger announcement are:

	Elrond Limited	Doom Limited
Market price per share	₹ 50	₹ 25
Number of outstanding shares	20 lakhs	10 Lakhs

The merger is expected to generate gains, which have a present value of ₹ 200 lakhs. The exchange ratio agreed to is 0.5.



What is the true cost of the merger from the point of view of Elrond Limited?

(SM TYK – 02)

Solution:

Market Value of Firm after Merger

$$V_f = (20 \times 50) + (10 \times 25) + 200 = ₹ 1,450$$

Value of Doom Ltd.

$$\text{After Merger } \left(\frac{1,450}{25} \times 5 \right) = 290 \text{ Lakh}$$

$$\text{Pre merger Market Value} = 250 \text{ Lakh}$$

$$\text{True cost of acquisition} = 40 \text{ Lakh}$$

Question – 29

AFC Ltd. wishes to acquire BCD Ltd. The shares issued by the two companies are 10,00,000 and 5,00,000 respectively:

- (i) Calculate the increase in the total value of BCD Ltd. resulting from the acquisition on the basis of the following conditions:

Current expected growth rate of BCD Ltd. 7%

Expected growth rate under control of AFC Ltd., (without any additional capital investment and without any change in risk of operations) 8%

Current Market price per share of AFC Ltd. ₹ 100

Current Market price per share of BCD Ltd. ₹ 20

Expected Dividend per share of BCD Ltd. ₹ 0.60

- (ii) On the basis of aforesaid conditions calculate the gain or loss to shareholders of both the companies, if AFC Ltd. were to offer one of its shares for every four shares of BCD Ltd.
- (iii) Calculate the gain to the shareholders of both the Companies, if AFC Ltd. pays ₹ 22 for each share of BCD Ltd., assuming the P/E Ratio of AFC Ltd. does not change after the merger. EPS of AFC Ltd. is ₹ 8 and that of BCD is ₹ 2.50. It is assumed that AFC Ltd. invests its cash to earn 10%.

(SM TYK – 25)

Solution:

(i) Increase in Total Value of BCD

$$K_e = \frac{D_1}{P_0} + g = \frac{₹ 0.60}{20} + 0.07 = 0.10 \text{ or } 10\%$$

$$P_0 = \frac{₹ 0.60}{0.10 - 0.08} = ₹ 30$$

$$\text{Value of synergy} = (30 - 20) \times 5,00,000 = ₹ 50,00,000$$

(ii) Gain or Loss due to Merger

$$\text{No. of shares to be issued} = \left(5,00,000 \times \frac{1}{4} \right) = 1,25,000 \text{ shares}$$

MPS after merger

$$\text{Value of AFC Ltd before Merger } (10,00,000 \times 100) = ₹ 1,000 \text{ L.}$$

$$\text{Value of BCD before Merger } (5,00,000 \times ₹ 20) = ₹ 100 \text{ Lacs}$$

$$\text{Value of synergy} = ₹ 50 \text{ Lacs}$$

$$V_B \text{ after Merger} = ₹ 1,150 \text{ Lacs}$$



MERGER

÷ No. of shares after merger = 11.25 Lacs

MPS after Merger = ₹ 102.22

Gain or Loss

	AFC	BCD
MPS before merger	₹ 100	₹ 20
MPS after merger	₹ 102.22	25.55 (102.22 × 0.25)
Gain	2.22	5.55

(iii) Gain/Loss

EAT of BCD (5,00,000 × 2.50) = ₹ 12,50,000

(-) Opportunity Cost = ₹ 11,00,000

EAT = ₹ 1,50,000

EAT (AFC) (10,00,000 × 8) = ₹ 80,00,000

Post Merger EPS = $\frac{80,00,000 + 1,50,000}{10,00,000}$ = ₹ 8.15

Post Merger MPS = 8.15 × 12.5 = 101.875

Gain to AFC [101.875 - 100] = ₹ 1.875

Gain to BCD [₹ 22 - ₹20] = ₹ 2

Pre merger P/E Ratio

P/E = $\frac{₹ 100}{8}$ = 12.5

Question – 30

Long Ltd., is planning to acquire Tall Ltd., with the following data available for both the companies:

	Long Ltd.	Tall Ltd.
Expected EPS	₹ 12	₹ 5
Expected DPS	₹ 10	₹ 3
No. of Shares	30,00,000	18,00,000
Current Market Price of Share	₹ 180	₹ 50

As per an estimate Tall Ltd., is expected to have steady growth of earnings and dividends to the tune of 6% per annum. However, under the new management the growth rate is likely to be enhanced to 8% per annum without additional investment.

You are required to:

- Calculate the net cost of acquisition by Long Ltd., if ₹ 60 is paid for each share of Tall Ltd.
- If the agreed exchange ratio is one share of Long Ltd., for every three shares of Tall Ltd., in lieu of the cash acquisition as per (i) above, what will be the net cost of acquisition?
- Calculate Gain from acquisition.

(Exam July – 2021 & RTP May – 2026) (8 Marks)

Solution:

(1) Net Cost of Acquisition

Cash paid [18,00,000 × 60] = ₹ 1,08,00,000

(-) Market value of Tall Ltd. [18,00,000 × 50] = ₹ 9,00,00,000



MERGER

Net Cost of Acquisition = ₹ 18,00,000

(ii) Net Cost of Acquisition (Stock Deal)

Value of synergy

$$K_e = \frac{3}{50} + 0.06 = 0.12$$

$$P_0 = \frac{3}{0.12-0.08} = ₹ 75$$

$$\begin{aligned}\text{Value of synergy} &= (75 - 50) \times 18,00,000 \\ &= ₹ 4,50,00,000\end{aligned}$$

(iii) Post Merger MPS

$$V_A (30 \text{ lacs} \times 180) = ₹ 5,400 \text{ lacs}$$

$$V_B (18 \text{ lacs} \times 50) = ₹ 900 \text{ lacs}$$

$$V_S = ₹ 450 \text{ lacs}$$

$$\text{Market Value} = ₹ 6,750 \text{ lacs}$$

$$\text{Value of Tall after Merger} \left(\frac{6,750}{36} \times 6 \right) = 1,125$$

$$\text{Value of Tall before Berger} = 900$$

$$\text{Cost of Acquisition} = 225 \text{ lacs}$$

$$\text{No.} = 18,00,000 \times \frac{1}{3} = 6,00,000$$

Question – 31

P Ltd. is considering take-over of R Ltd. by the exchange of four new shares in P Ltd. for every five shares in R Ltd. The relevant financial

details of the two companies prior to merger announcement are as follows:

	P Ltd	R Ltd
Profit before Tax (₹ Crore)	15	13.50
No. of Shares (Crore)	25	15
P/E Ratio	12	9

Corporate Tax Rate 30%

You are required to determine:

- Market value of both the company.
- Value of original shareholders.
- Price per share after merger.
- Effect on share price of both the company if the Directors of P Ltd. expect their own pre-merger P/E ratio to be applied to the combined earnings.

(SM TYK – 14)

Solution:

	P Ltd.	R Ltd.
Profit before Tax (₹ in crore)	15	13.50
Tax 30% (₹ in crore)	<u>4.50</u>	<u>4.05</u>
Profit after Tax (₹ in crore)	<u>10.50</u>	<u>9.45</u>



MERGER

Earning per Share (₹)	$\frac{10.50}{25} = ₹ 0.42$	$\frac{9.45}{15} = ₹ 0.63$
Price of Share before Merger (EPS × P/E Ratio)	$₹ 0.42 \times 12 = ₹ 5.04$	$₹ 0.63 \times 9 = ₹ 5.67$

(i) ∴ Market Value of company

P Ltd.	$= ₹ 5.04 \times 25 \text{ Crore}$	$= ₹ 126 \text{ crore}$
R Ltd.	$= ₹ 5.67 \times 15 \text{ Crore}$	$= ₹ 85.05 \text{ crore}$
Combined	$= ₹ 126 + ₹ 85.05$	$= ₹ 211.05 \text{ Crores}$

After Merger

	P Ltd.	R Ltd.
No. of Shares	25 crores	$15 \times \frac{4}{5} = 12 \text{ Crores}$
Combined Shares	$= 37 \text{ Crores}$	
% of Combined Equity Owned	$\frac{25}{37} \times 100 = 67.57\%$	$\frac{12}{37} \times 100 = 32.43\%$

(ii) ∴ Value of Original Shareholders

P Ltd.	R Ltd.
$₹ 211.05 \text{ crore} \times 67.57\%$	$₹ 211.05 \text{ crore} \times 32.43\%$
$= ₹ 142.61$	$= ₹ 68.44$

Alternatively, it can also be computed as follows:

Combined Value of Entity	211.05 crore
No. of shares after Merger	37 crore
Value of Per Share	₹ 5.70405
Value of P Ltd. Shareholders (25 crores × ₹ 5.70405)	₹ 142.60 crore
Value of R Ltd. Shareholders (12 crores × ₹ 5.70405)	₹ 68.45 crore

(iii) ∴ Price per Share after Merger

$$\text{EPS} = \frac{19.95 \text{ Crore}}{37 \text{ Crore}} = ₹ 0.539 \text{ per share}$$

$$\text{P/E Ratio} = 12$$

$$\text{Market Value Per Share} = ₹ 0.539 \times 12 = ₹ 6.47$$

$$\text{Total Market Value} = ₹ 6.47 \times 37 \text{ crore} = ₹ 239.39 \text{ crore}$$

$$\text{Price of Share} = \frac{\text{Market Value}}{\text{Number of Shares}} = \frac{239.39 \text{ crore}}{37 \text{ Crore}} = ₹ 6.47$$

(iv) Effect on Share Price

P Ltd.

$$\text{Gain/loss (-) per share} = ₹ 6.47 - ₹ 5.04 = ₹ 1.43$$

$$\text{i.e. } \frac{6.47 - 5.04}{5.04} \times 100 = 0.284 \text{ or } 28.4\%$$

∴ Share price would rise by 28.4%

R Ltd.

$$6.47 \times \frac{4}{5} = ₹ 5.18$$

$$\text{Gain/loss (-) per share} = ₹ 5.18 - ₹ 5.67 = (- ₹ 0.49)$$



MERGER

$$\text{i.e. } \frac{5.18 - 5.67}{5.67} \times 100 (-) = 0.0864 \text{ or } (-) 8.64\%$$

∴ Share Price would decrease by 8.64%.

Question – 32

M/s. Vasavi Ltd. is considering the takeover of M/s. SKPD Ltd. by the exchange of five new shares in M/s. Vasavi Ltd. for every eight shares in M/s. SKPD Ltd. The relevant financial details of the two companies prior to merger announcement are as follows:

Particulars	M/s. Vasavi Ltd.	M/s. SKPD Ltd.
Profit before tax (₹ crore)	18	20.8
No. of shares (in crore)	20	18
P/E ratio	11	8

Corporate tax rate 30%.

You are required to determine:

- Market value of both the companies
- Value of original share holders
- Price per share after merger
- Effect on share price of both the companies. If the directors of Vasavi Ltd. expect their own pre-merger P/E ratio to be applied to the combined earnings.

(Exam May – 2022) (8 Marks)

Solution:

	M/s Vasavi Ltd.	M/s SKPD Ltd.
Profit before Tax (₹ in crore)	18.00	20.80
Tax 30% (₹ in crore)	5.40	6.24
	12.60	14.56

Profit after Tax (₹ in crore) Earnings per Share	$\frac{12.60}{20} = ₹ 0.63$	$\frac{14.56}{18} = ₹ 0.81$
Price per Share before Merger (EPS × P/E Ratio)	$₹ 0.63 \times 11 = ₹ 6.93$	$₹ 0.81 \times 8 = ₹ 6.48$

a. Market Value of company

M/s Vasavi Ltd. = ₹ 6.93 × 20 Crore = ₹ 138.60 crore

M/s SKPD Ltd. = ₹ 6.48 × 18 Crore = ₹ 116.64 crore

b. Value of Original Shareholders

After Merger

	M/s Vasavi Ltd.	M/s SKPD Ltd.
No. of Shares	20 crores	$18 \times \frac{5}{8} = 11.25$ crores
Combined	31.25 crores	
% of Combined Equity Owned	$\frac{20}{31.25} \times 100 = 64.00\%$	$\frac{11.25}{31.25} \times 100 = 36.00\%$
Value of Original Shareholders	$₹ 255.24 \text{ crore} \times 64.00\%$ = ₹ 163.35 crores	$₹ 255.24 \text{ crore} \times 36\%$ = ₹ 91.89 crores

c. Price per Share after Merger

$$\text{EPS} = \frac{₹ 27.16 \text{ Crore}}{31.25 \text{ Crore}} = ₹ 0.87 \text{ per share}$$

P/E Ratio = 11

Market value per share = ₹ 0.87 × 11 = ₹ 9.57



d. Effect on Share Price

M/s Vasavi Ltd.

Gain/loss (-) per share = ₹ 9.57 – ₹ 6.93 = ₹ 2.64

$$\text{i.e. } \frac{9.57 - 6.93}{6.93} \times 100 = 0.381 \text{ or } 38.10\%$$

∴ Share price would increase by 38.10%

M/s SKPD Ltd.

$$9.57 \times \frac{5}{8} = ₹ 5.98$$

Gain/loss (-) per share = ₹ 5.97 – ₹ 6.48 = (₹ 0.51)

$$\text{i.e. } \frac{5.97 - 6.48}{6.48} \times 100 = (0.0787) \text{ or } (-7.87\%)$$

∴ Share Price would decrease by 7.87%.

Question – 33

Yes Ltd. wants to acquire No Ltd. and the cash flows of Yes Ltd. and the merged entity are given below:

Year	1	2	3	4	5
Yes Ltd.	175	200	320	340	350
Merged Entity	400	450	525	590	620

Earnings would have witnessed 5% constant growth rate without merger and 6% with merger on account of economies of operations after 5 years in each case. The cost of capital is 15%.

The number of shares outstanding in both the companies before the merger is the same and the companies agree to an exchange ratio of 0.5 shares of Yes Ltd. for each share of No Ltd.

PV factor at 15% for years 1-5 are 0.870, 0.756; 0.658, 0.572, 0.497 respectively.

You are required to:

- Compute the Value of Yes Ltd. before and after merger.
- Value of Acquisition and
- Gain to shareholders of Yes Ltd.

(SM TYK – 16)**Solution:****(i) Working Notes:**

Present Value of Cash Flows (CF) upto 5 years

Year End	CF of Yes Ltd. (₹ lakhs)	PVF @15%	PV of CF (₹ lakhs)	CF of Merged Entity (₹ lakhs)	PV of CF of Merged Entity (₹ lakhs)
1	175	0.870	152.25	400	348.00
2	200	0.756	151.20	450	340.20
3	320	0.658	210.56	525	345.45
4	340	0.572	194.48	590	337.48
5	350	0.497	<u>173.95</u>	620	<u>308.14</u>
			<u>882.44</u>		<u>1679.27</u>

PV of cash flows of Yes Ltd. after the forecast period



$$TV_5 = \frac{CF_5 (1 + g)}{K_e - g} = \frac{350 (1 + 0.05)}{0.15 - 0.05} = \frac{367.50}{0.10} = ₹ 3,675 \text{ lakhs}$$

$$PV \text{ of } TV_5 = ₹ 3675 \text{ lakhs} \times 0.497 = ₹ 1,826.475 \text{ lakhs}$$

PV of Cash Flows of Merged Entity after the forecast period

$$TV_5 = \frac{CF_5 (1 + g)}{K_e - g} = \frac{620 (1 + 0.06)}{0.15 - 0.06} = \frac{657.20}{0.09} = ₹ 7,302.22 \text{ lakhs}$$

$$PV \text{ of } TV_5 = ₹ 7302.22 \text{ lakhs} \times 0.497 = ₹ 3,629.20 \text{ lakhs}$$

Value of Yes Ltd.

	Before merger (₹ lakhs)	After merger (₹ lakhs)
PV of CF (1-5 years)	882.440	1679.27
Add: PV of TV ₅	<u>1826.475</u>	<u>3629.20</u>
	<u>2708.915</u>	<u>5308.47</u>

(ii) Value of Acquisition

= Value of Merged Entity – Value of Yes Ltd.

$$= ₹ 5,308.47 \text{ lakhs} - ₹ 2,708.915 \text{ lakhs}$$

$$= ₹ 2,599.555 \text{ lakhs}$$

(iii) Gain to Shareholders of Yes Ltd.

$$\text{Share of Yes Ltd. in merged entity} = ₹ 5,308.47 \text{ lakhs} \times \frac{1}{1.5}$$

$$= ₹ 3,538.98 \text{ lakhs}$$

Gain to shareholder

= Share of Yes Ltd. in merged entity – Value of Yes Ltd. before merger

$$= ₹ 3,538.98 \text{ lakhs} - ₹ 2,708.915 = ₹ 830.065 \text{ lakhs}$$

CAPITAL RESTRUCTURING

Question – 34

The following is the Balance-sheet of Grape Fruit Company Ltd as at March 31st, 2019.

Liabilities	(₹ in lakhs)	Assets	(₹ in lakhs)
Equity shares of ₹ 100 each	600	Land and Building	200
14% preference shares of ₹ 100/- each	200	Plant and Machinery	300
13% Debentures	200	Furniture and Fixtures	50
Debenture interest accrued and payable	26	Inventory	150
Loan from bank	74	Sundry debtors	70
Trade creditors	340	Cash at bank	130
		Preliminary expenses	10
		Cost of issue of debentures	5
		Profit and Loss account	525
	1440		1440

The Company did not perform well and has suffered sizable losses during the last few years. However, it is felt that the company could be nursed back to health by proper financial restructuring. Consequently the following scheme of reconstruction has been drawn up:

- (i) Equity shares are to be reduced to ₹ 25/- per share, fully paid up;
- (ii) Preference shares are to be reduced (with coupon rate of 10%) to equal number of shares of ₹ 50 each, fully paid up.
- (iii) Debenture holders have agreed to forgo the accrued interest due to them. In the future, the rate of interest on debentures is to be reduced to 9 percent.
- (iv) Trade creditors will forego 25 percent of the amount due to them.
- (v) The company issues 6 lakh of equity shares at ₹ 25 each and the entire sum was to be paid on application. The entire amount was fully subscribed by promoters.
- (vi) Land and Building was to be devalued at ₹ 450 lakhs, Plant and Machinery was to be written down by ₹ 120 lakhs and a provision of ₹15 lakhs had to be made for bad and doubtful debts.

Required:

- (a) Show the impact of financial restructuring on the company's activities.
- (b) Prepare the fresh balance sheet after the reconstructions is completed on the basis of the above proposals.

Solution:**Impact of Financial Restructuring**

- (i) Benefits to Grape Fruit Ltd.

- (a) Reduction of Liabilities payable

₹ in lakhs	
Reduction in equity share capital (6 lakh shares × ₹ 75 per share)	450
Reduction in preference share capital (2 lakh shares × ₹ 50 per share)	100
Waiver of outstanding debenture Interest	26
Waiver from trade creditors (₹ 340 lakhs × 0.25)	<u>85</u>
	<u>661</u>
(b) Revaluation of Assets	
Appreciation of Land and Building (₹ 450 lakhs – ₹ 200 lakhs)	<u>250</u>
Total (A)	<u>911</u>

- (ii) Amount of ₹ 911 lakhs utilized to write off losses, fictitious assets and over – valued assets.

Writing off profit and loss account	525
Cost of issue of debentures	5
Preliminary expenses	10
Provision for bad and doubtful debts	15
Revaluation of Plant and Machinery (₹ 300 lakhs – ₹ 180 lakhs)	120
Total (B)	<u>675</u>
Capital Reserve (A) – (B)	<u>236</u>

- (iii) Balance sheet of Grape Fruit Ltd. as 31st March 2011 (after re-construction)

Liabilities	Amt.	Assets		Amt.
12 lakhs equity shares of ₹ 25/- each	300	Land & Building		450
10% Preference shares of ₹ 50/- each	100	Plant & Machinery		180
Capital Reserve	236	Furniture & Fixtures		50
	200	Inventory	7	150
		Sundry debtors		55

9% debentures	74	Prov. for Doubtful	-	280
Loan from Bank	255	Debts	<u>1</u>	
Trade Creditors		Cash-at-Bank	<u>5</u>	
	1165	(Balancing figure)*		1165

*Opening Balance of ₹ 130/- lakhs + Sale proceeds from issue of new equity shares ₹ 150/- lakhs.

RESIDUAL

Question – 35

Mr. X, a financial analyst, intends to value the business of PQR Ltd. in terms of the future cash generating capacity. He has projected the following after tax cash flows :

Year :	1	2	3	4	5
Cash flows (₹ in lakh)	1,760	480	640	860	1,170

It is further estimated that beyond 5th year, cash flows will perpetuate at a constant growth rate of 8% per annum, mainly on account of inflation. The perpetual cash flow is estimated to be ₹ 10,260 lakh at the end of the 5th year.

Required:

- What is the value of the firm in terms of expected future cash flows, if the cost of capital of the firm is 20%.
- The firm has outstanding debts of ₹ 3,620 lakh and cash/bank balance of ₹ 2,710 lakh. Calculate the shareholder value per share if the number of outstanding shares is 151.50 lakh.
- The firm has received a takeover bid from XYZ ltd. of ₹ 225 per share. Is it a good offer?

[Given: PVIF at 20% for year 1 to Year 5: 0.833, 0.694, 0.579, 0.482, 0.402]

(Exam November – 2019) (8 Marks)

Solution:

(i) Value of Firm

Year	Cash Flow (₹ in lakhs)	PVF	PV (₹ in lakhs)
1	1760	0.833	1466.08
2	480	0.694	333.12
3	640	0.579	370.56
4	860	0.482	414.52
5	1170	0.402	470.34
PV of Cash flows upto year 5			3054.62

If PV of Terminal Value is considered with the growth rate (at the end of 5th year)

$$= \frac{10,260 (1 + 0.08)}{0.20 - 0.08} = \frac{11,080.80}{0.12} = ₹ 92,340 \text{ lakh}$$

Now, PV (at the beginning of the year)

$$= ₹ 92,340 \times 0.402 = ₹ 37,120.68 \text{ Lakhs}$$

So, Present Value of the firm

$$= ₹ 3,054.62 + ₹ 37,120.68 = ₹ 40,175.30 \text{ Lakhs}$$

(ii) Value per Share

$$= \text{Value of Firm} - \text{Value of Debt} / \text{No of shares}$$

$$= (40,175.30 - 3,620) / 151.50 = ₹ 241.29$$

MERGER

- (iii) Takeover bid of ₹ 225 per share seems to be not a good offer as it is lesser than the intrinsic value i.e. value per share of ₹ 241.29.

Question – 36

ICL is proposing to take over SVL with an objective to diversify. ICL's profit after tax (PAT) has grown @ 18 per cent per annum and SVL's PAT is grown @ 15 per cent per annum. Both the companies pay dividend regularly. The summarized Profit & Loss Account of both the companies are as follows:

₹ in Crores

Particulars	ICL	SVL
Net Sales	4,545	1,500
PBIT	2,980	720
Interest	750	25
Provisions for Tax	1,440	445
PAT	790	250
Dividends	235	125

	ICL		SVL	
Fixed Assets				
Land & Building (Net)	720		190	
Plant & Machinery (Net)	900		350	
Furniture & Fixtures (Net)	30	1,650	10	550
Current Assets		775		580
Less: Current Liabilities				
Creditors	230		130	
Overdrafts	35		10	
Provision for Tax	145		50	
Provision for dividends	60	470	50	240
Net Assets		1,955		890
Paid up share capital (₹ 10 per share)	250		125	
	1,050	1,300	660	785
Reserve and Surplus		655		105

Borrowing		1,955		890
Capital Employed				

Market price share (₹)	52	75
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ICL's Land & Buildings are stated at current prices. SVL's Land & Buildings are revalued three years ago. There has been an increase of 30 per cent per year in the value of Land & Buildings.

SVL is expected to grow @ 18 per cent each year, after merger.

ICL's Management wants to determine the premium on the shares over the current market price which can be paid on the acquisition of SVL.

You are required to determine the premium using:

- Net Worth adjusted for the current value of Land & Buildings plus the estimated average profit after tax (PAT) for the next five years.
- The dividend growth formula.
- ICL will push forward which method during the course of negotiations?

Period (t)	1	2	3	4	5
FVIF (30%, t)	1.300	1.690	2.197	2.856	3.713
FVIF (15%,t)	1.15	2.4725	3.9938	5.7424	7.7537

(Exam November – 2020) (8 Marks)



Solution:

(i) Value per Share

Net worth = 785

(+) Revaluation in L & B

[190 (1.30)³ – 190]

[190 × 2.197 – 190] 227.43

Net Worth ₹ 1,012.43 Cr.

(+) $\left(\frac{250 \times 7.7535}{5}\right)$ ₹ 387.685 Cr.

Value of SVL ₹ 1,400.115 Cr.

÷ No. of shares 12.50 Cr.

Value per share ₹ 112

Premium = $\frac{112-75}{75} \times 100 = 49.33\%$

(ii) Dividend Growth Model

$$K_e = \frac{D_1}{P_0} + g$$

$$D_0 = \frac{125}{12.5} = ₹ 10$$

$$K_e = \frac{10 (1.15)}{75} + 0.15$$

$$= 0.3033 \text{ or } 30.33\%$$

$$P_0 = \frac{10 (1.18)}{0.3033 - 0.18} = ₹ 95.70$$

$$\text{Premium} = \frac{97.70 - 75}{75} \times 100 = 27.60\%$$

(iii) ICL will push forward dividend growth model, due to lower premium.

Question – 37

M/s. Roly Ltd. wants to acquire M/s. Poly Ltd. The following is the Balance Sheet of Poly Ltd. as on 31st March, 2020:

Liabilities	₹	Assets	₹
Equity Capital (₹ 10 per share)	10,00,000	Cash	20,000
Retained Earnings	3,00,000	Debtors	50,000
12% Debentures	3,00,000	Inventories	2,00,000
Creditors and other liability	3,20,000	Plant & Machinery	16,50,000
Total	19,20,000	Total	19,20,000

Shareholders of Poly Ltd. will get one share of Roly Ltd. at current Market price of ₹ 20 for every two shares. External liabilities are expected to be settled at a discount of ₹ 20,000. Sundry debtors and Inventories are expected to realize ₹ 2,00,000.

Poly Ltd. will run as an independent unit. Cash Flow After Tax is expected to be ₹ 4,00,000 per annum for next 6 years. Assume the disposal value of the plant after 6 years will be ₹ 1,50,000.

Poly Ltd. requires a return of 14%

n	1	2	3	4	5	6
PVIF (14%, n)	0.877	0.769	0.675	0.592	0.519	0.456

Advise the Board of Directors on the financial feasibility of the Proposal.

(Exam Jan – 2021) (8 Marks)

Solution:

Calculation of Purchase Consideration

	₹
Issue of Share 50,000 × ₹ 20	10,00,000
External Liabilities settled	3,00,000
12% Debentures	3,00,000
	16,00,000
Less: Realization of Debtors and Inventories	2,00,000
Cash	20,000
	13,80,000

Net Present Value

= PV of Cash Inflow + PV of Demerger of Roly Ltd. – Cash Outflow

= ₹ 4,00,000 PVAF(14%,6) + ₹ 1,50,000 PVF(14%, 6) – ₹ 13,80,000

= ₹ 4,00,000 × 3.888 + ₹ 1,50,000 × 0.456 – ₹ 13,80,000

= ₹ 15,55,200 + ₹ 68,400 – ₹ 13,80,000

= ₹ 2,43,600

Since NPV of the decision is positive it is advantageous to acquire Poly Ltd.

Question – 38

Simple Ltd. and Dimple Ltd. are planning to merge. The total value of the companies are dependent on the fluctuating business conditions. The following information is given for the total value (debt + equity) structure of each of the two companies.

MERGER

Business Condition	Probability	Simple Ltd. ₹ Lacs	Dimple Ltd. ₹ Lacs
High Growth	0.20	820	1050
Medium Growth	0.60	550	825
Slow Growth	0.20	410	590

The current debt of Dimple Ltd. is ₹ 65 lacs and of Simple Ltd. is ₹ 460 lacs.

Calculate the expected value of debt and equity separately for the merged entity.

(SM TYK – 15)

Solution:

Compute Value of Equity

Simple Ltd.

(₹ in Lacs)

	High Growth	Medium Growth	Slow Growth
Debit + Equity	820	550	410
Less: Debt	460	460	460
Equity	360	90	-50

Since the Company has limited liability the value of equity cannot be negative therefore the value of equity under slow growth will be taken as zero because of insolvency risk and the value of debt is taken at 410 lacs. The expected value of debt and equity can then be calculated as:



Simple Ltd.

(₹ in Lacs)

	High Growth		Medium Growth		Slow Growth		Expected Value
	Prob.	Value	Prob.	Value	Prob.	Value	
Debt	0.20	460	0.60	460	0.20	410	450
Equity	0.20	360	0.60	90	0.20	0	126
		820		550		410	576

Dimple Ltd.

(₹ in Lacs)

	High Growth		Medium Growth		Slow Growth		Expected Value
	Prob.	Value	Prob.	Value	Prob.	Value	
Equity	0.20	985	0.60	760	0.20	525	758
Debt	0.20	65	0.60	65	0.20	65	65
		1050		825		590	823

Expected Values

(₹ in Lacs)

Equity		Debt	
Simple Ltd.	126	Simple Ltd.	450
Dimple Ltd.	758	Dimple Ltd.	65
	884		515

Question – 39

R Ltd. and S Ltd. operating in same industry are not experiencing any rapid growth but providing a steady stream of earnings. R Ltd.'s management is interested in acquisition of S. Ltd. due to its excess plant capacity. Share of S Ltd. is trading in market at ₹ 3.20 each. Other data relating to S Ltd. is as follows:

Balance Sheet of S Ltd

Liabilities	Amount (₹)	Assets	Amount (₹)
Current Liabilities	1,59,80,000	Current Assets	2,48,75,000
Long Term Liabilities	1,28,00,000	Other Assets	94,00,000
Reserve & Surplus	2,79,95,000	Property	
Share Capital		Plants & Equipment	3,45,00,000
(80 Lakhs shares of ₹ 1.5 each)	1,20,00,000		
Total	6,87,75,000	Total	6,87,75,000

Particulars	R Ltd. (₹)	S Ltd. (₹)	Combined Entity (₹)
Profit after Tax	86,50,000	49,72,000	1,21,85,000
Residual Net Cash Flows per year	90,10,000	54,87,000	1,85,00,000
Required return on equity	13.75%	13.05%	12.5%

You are required to compute the following:

- Minimum price per share S Ltd. should accept from R Ltd.
- Maximum price per share R Ltd. shall be willing to offer to S Ltd.
- Floor Value of per share of S Ltd., whether it shall play any role in decision for its acquisition by R Ltd.

(Exam May – 2019) (8 Marks)

Solution:

- Calculation of Minimum price per share S Ltd. should accept from R Ltd.**

$$\text{Value of S Ltd.} = \frac{\text{Residual Cash Flow}}{K_e - g} = \frac{54,87,000}{0.1305 - 0} = ₹ 4,20,45,977$$

$$\text{Value per share of S Ltd.} = \frac{4,20,45,977}{80,00,000} = ₹ 5.26$$

$$\text{Book Value of per share of S Ltd.} = \frac{3,99,95,000}{80,00,000} = ₹ 4.99 \text{ or } ₹ 5$$

Therefore, the minimum price per share S Ltd. should accept from R Ltd. is ₹ 5 (current book value)

(ii) Calculation of Maximum price per share R Ltd. shall be willing to offer to S Ltd.

$$\text{Value of R Ltd.} = \frac{\text{Residual Cash Flow}}{K_e - g} = \frac{90,10,000}{0.1375 - 0} = ₹ 6,55,27,273$$

$$\text{Value of Combined entity} = \frac{1,85,00,000}{0.125-0} = ₹ 14,80,00,000$$

Value of synergy

= Value of Combined entity – Individual values of R Ltd. and S Ltd.

$$= ₹ 14,80,00,000 - (₹ 4,20,45,977 + ₹ 6,55,27,273)$$

$$= ₹ 4,04,26,750$$

Maximum price per share R Ltd. shall be willing to offer to S Ltd. shall be computed as follows:

$$= \frac{\text{Value of S Ltd. as per Residual cash flows} + \text{Synergy benefits}}{\text{No. of Shares}}$$

$$= \frac{4,20,45,977 + 4,04,26,750}{80,00,000} = ₹ 10.31$$

(iii) Floor value of per share of S Ltd shall be ₹ 3.20 (current market price) and it shall not play any role in decision for the acquisition of S Ltd. as it is lower than its current book value.

MULTIPLE CHOICE QUESTIONS

Case Scenario – 01

SM Limited has a market capitalization of ₹ 3,000 crore and the current earnings per share (EPS) is ₹ 200 with a price earnings ratio (PER) of 15. The Board of directors is considering a proposal to buy back 20% of the shares at a premium which can be supported by the financials of the company. The Board expects post buy back market price per share (MPS) of ₹ 3057. Post buy back PER will remain same. The company proposes to fund the buy back by availing 8% bank loan since available resources are committed for expansion plans.

Applicable income tax rate is 30%.

Based on above Case Scenario, select the most appropriate alternative.

- I. The number of shares proposed to be bought back is.....

(a) 12 lakhs	(b) 15 lakhs
(c) 20 lakhs	(d) 22 lakhs

- II. The interest amount which can be paid for availing the bank loan

(a) ₹ 5,280.00 Lakhs	(b) ₹ 5,575.00 Lakhs
(c) ₹ 4,865.00 Lakhs	(d) ₹ 6,485.00 Lakhs

MERGER

III. The loan amount to be raised and

- (a) ₹ 55,650 Lakhs (b) ₹ 62,300 Lakhs
(c) ₹ 66,000 Lakhs (d) ₹ 72,450 Lakhs

IV. The premium per share paid over the current MPS

- (a) ₹ 200 (b) ₹ 250
(c) ₹ 300 (d) ₹ 350

V. % premium over CMP shall be

- (a) 12% (b) 14%
(c) 10% (d) 15%

Answer: Case Scenario – 01

- I. (c) 20 lakhs
II. (a) ₹ 5,280.00 Lakhs
III. (c) ₹ 66,000 Lakhs
IV. (c) ₹ 300
V. (c) 10%

Case Scenario – 02

AES Ltd. wants to acquire DNF Ltd. and has offered a swap ratio of 1:2 (0.5 shares for every one share of DNF Ltd.). Following information is provided:

	ASE Ltd.	DNF Ltd.
Profit after tax	₹ 36,00,000	₹ 7,20,000
Equity shares outstanding (Nos.)	12,00,000	3,60,000
PE Ratio	10 times	7 times
Market price per share	₹ 30	₹ 14

On the basis of above information, choose the most appropriate answer to the following questions:

I. The number of equity shares to be issued by AES Ltd. for acquisition of DNF Ltd. would be.....

- (a) 1,68,000 (b) 1,80,000
(c) 2,40,000 (d) 3,00,000

II. The EPS of AES Ltd. after the acquisition would be.....

- (a) ₹ 2 (b) ₹ 3
(c) ₹ 3.13 (d) ₹ 4.00

III. The equivalent earnings per share of DNF Ltd. would be.....

- (a) ₹ 1 (b) ₹ 1.50
(c) ₹ 1.57 (d) ₹ 2.00



IV. If AES Ltd. PE multiple remains unchanged then its expected market price per share after the acquisition would be.....

- (a) ₹ 14 (b) ₹ 30
(c) ₹ 31.30 (d) ₹ 40.00

(MTP March – 2024)

Answer: Case Scenario – 02

- I. (b) 1,80,000
II. (c) ₹ 3.13
III. (c) ₹ 1.57
IV. (c) ₹ 31.30

Case Scenario – 03

P Ltd is studying the possible acquisition Q Ltd. by way of merger .the following are available:

Firm	After-Tax Earnings	No. of Equity Shares	Market Price Per Share	Book Value Per Share
P Ltd.	₹ 10,00,000	2,00,000	₹ 75	₹ 210
Q Ltd.	₹ 3,00,000	50,000	₹ 60	₹ 105

The merger shall be gone through by exchange of equity share and the exchange ratio is set according to different weights assigned to different basis as mentioned below :-

EPS	50%
Market price	25%
Book value	25%

I. The swap ratio based on assigned weight shall be_____

- Answer-1 : 0.825 Answer-2 : 0.925
Answer-3: 0.952 Answer-4: 0.752

II. Based on swap ratio as per weights the total number of share issue by P Ltd to Q Ltd. shall be_____

- Answer-1 : 46250 Answer-2 : 41250
Answer-3: 47600 Answer-4: 37600

III. Post merger the EPS of the P Ltd shall be _____

- Answer-1 : 5.39 Answer-2 : 5.25
Answer-3: 5.28 Answer-4: 5.47

IV. In case Q Ltd. wants to be sure that its EPS is not diminished by the merger, the relevant exchange ratio to achieve the same objective should be_____

- Answer-1 : 0.83 Answer-2 : 1.20
Answer-3: 1.30 Answer-4: 1.10

(MTP November – 2025)**Answer: Case Scenario – 03**

- I. Answer-2: 0.925**
II. Answer-1: 46250
III. Answer-3: 5.28
IV. Answer-2: 1.20

Case Scenario – 04

The company X Ltd. proposes to take over Y Ltd. The chief executive of the company thinks that shareholders always look for the earnings per share. Therefore, he considers maximization of the earnings per share as his company's objective. The following information is available in respect of X Ltd. and Y Ltd.

	X Ltd.	Y Ltd.
Net Profit	₹ 80 Lakh	₹ 15.75 Lakh
P/E ratio	10.50	10.00
Current market price per share	₹ 42	₹ 85

From the information given above, choose the correct answer to the following questions:

- I.** If the company borrows funds @ 15% rate of interest and buys out Target Company by paying cash, how much it should offer to maintain its EPS assuming tax rate @30%.
- (a) 210 Lakhs (b) 315 Lakhs
(c) 150 Lakhs (d) 0 Lakhs
- II.** Maximum exchange ratio which the company should offer so that the company could keep EPS at current level is.....

- (a) 1:0.952 (b) 1:2.125
(c) 1:2.023 (d) 1:0.196

- III.** No. of shares to be issued by X Ltd. shall be approximately.....

- (a) 3.9375 Lakhs (b) 1.7639 Lakhs
(c) 3.7485 Lakhs (d) 0.3631 Lakhs

(EXAM NOVEMBER 2024)**Answer: Case Scenario – 04**

- I. (c) 150 Lakhs**
II. (b) 1:2.125
III. (a) 3.9375 Lakhs

Case Scenario – 05

Mr. Ramesh, a 40-year-old investor, has invested ₹ 10,00,000 in an actively managed Equity Mutual Fund. The fund has an Expense Ratio of 2.50% and follows the Nifty 50 Index as its benchmark. Upon analyzing the Fund details, he comes across the concept of Tracking Error (TE) and finds out that the same Fund has a Tracking Error (TE) of 3.20%.

A few months later, Mr. Ramesh receives a notification that the Fund has implemented Side Pocketing. The Fund has an exposure of 15% of his investment in a debt instrument of XYZ Ltd, a company facing a severe financial crisis. Since XYZ Ltd has defaulted on its



payments, the Fund Manager has moved this portion into a side pocket.

Following the decision of Fund Manager, Mr. Ramesh decides to reconsider any of the following option:

1. Should he stay invested in this Fund and wait for the Side-Pocketed assets to recover?
2. Should he switch to a Passive Index Fund that has a lower Tracking Error and lower Expense Ratio
3. Should he redeem his remaining liquid holdings and invest in a better-performing actively Managed Fund?

Based on the above scenario and given his current situation, choose the most appropriate answer for the following multiple-choice questions:

- I.** Is it necessary for investors to pay close attention to the Expense Ratio of a Mutual Fund because.....
- (A) a high expense ratio can significantly reduce net returns over time.
- (B) a higher expense ratio always guarantees better fund performance.
- (C) the expense ratio only matters in the first year of investment.
- (D) funds with higher expense ratios are always risk-free.
- II.** The Fund has been in replicating return on Nifty 50.

- (A) Successful (B) Unsuccessful
- (C) Can't say (D) Data is insufficient

- III.** After the decision of Fund Manager for side-pocketing the equivalent portion of Mr. Ramesh's investment shall_____

- (A) remains illiquid until the Fund Manager decides to sell it or the company recovers.
- (B) be immediately written off, and the Mr. Ramesh loses that portion.
- (C) be returned to Mr. Ramesh in proportion to his holdings.
- (D) be moved into a different Mutual Fund Scheme with no risk.

- IV.** If Mr. Ramesh switches to a Passive Index Fund with an expense ratio of 0.8%, then he will save annually compared to his current Expense Ratio of 2.50%?

- (A) ₹ 8,000 (B) ₹ 10,000
- (C) ₹ 17,000 (D) ₹ 18,000

- V.** The advantage for Mr. Ramesh to switch over to a Passive Index Fund shall be_____

- (A) lower expense ratio and lower tracking error.



- (B) guaranteed recovery of side-pocketed assets.
- (C) higher risk exposure compared to active funds.
- (D) avoiding capital gains tax on redemption.

(MTP March: 2025)**Answer: Case Scenario – 05**

I. (A) a high expense ratio can significantly reduce net returns over time.

II. (B) Unsuccessful

III. (C) be returned to Mr. Ramesh in proportion to his holdings.

IV. (C) ₹ 17,000

V. (A) lower expense ratio and lower tracking error.

Case Scenario – 06

ABC Ltd. is planning to acquire XYZ Ltd. The following information is available in this respect :

Company	No. of Shares	Market Price Per Share(₹)	P/E Ratio
ABC Ltd.	12,00,000	16	20 times
XYZ Ltd.	4,00,000	20	10 times

ABC Ltd. plans to acquire the whole of XYZ Ltd. by issuing shares at its market price of ₹ 16, It is expected that the price of ABC Ltd. share will remain constant after this acquisition. Purchase Consideration is ₹80,00,000

Calculation up to 2 decimal places.

From the information given above, choose the correct answer to the following question No. I to 27 :

(EXAM SEPTEMBER - 2025)

I. P/E Ratio of ABC Ltd. after acquisition is

- (a) 20 times (b) 15.38 times
- (c) 10 times (d) 14.55 times

II. EPS of ABC Ltd. & XYZ Ltd. before acquisition are respectively

- (a) ₹ 0.8 & ₹ 2.0 (b) ₹ 2.0 & ₹ 0.8
- (c) ₹ 0.8 & ₹ 1.04 (d) ₹ 1.04 & ₹ 0.8

III. EPS of ABC Ltd. after acquisition is

- (a) ₹ 0.84 (b) ₹ 0.96
- (c) ₹ 1.04 (d) ₹ 1.14

Answer: Case Scenario – 06

I. (b) 15.38 times

II. (a) ₹ 0.8 & ₹ 2.0

III. (c) ₹ 1.04

Case Scenario – 07

Ujwal Bank Ltd. (UBL) and Suraksha Bank Ltd. (SBL) are Scheduled Banks to merge.

UBL is strong Private Sector Bank with stable capital adequacy, while SBL has negative CRAR due to heavy NPAs.



Data of both the Banks is as follows:

Particulars	UBL	SBL
Book Value per share (₹)	50	25
Market Price per share (₹)	200	50
CRAR %	12	(-) 2
NPA %	2	12
No. of shares in thousands	50,000	20,000
Price Earning Ratio (PE Ratio)	20	10

Weights for swap ratio are Book Value per share 20%, Market Price per share 40%, CRAR (%) 20% and balance for NPA %.

From the information given above, choose the correct answer to the Question No. 10 to 12:

- I.** The swap ratio based on information given shall be for 1 share of UBL _____ shares of SBL.
- (A) 1.07 (B) 0.20
(C) 0.86 (D) 1.73
- II.** Based on swap ratio total number of shares issued by UBL to SBL shall be _____ (in Thousands).
- (A) 21,400 shares (B) 24,000 shares
(C) 17,200 shares (D) 4,000 shares
- III.** Post merger Earning Per Share (EPS) of UBL shall be ₹ _____
- (A) ₹ 11.11 (B) ₹ 12.50
(C) ₹ 8.50 (D) ₹ 10.00

(EXAM JANUARY – 2026)

Answer Case Scenario – 07

- (B) 0.20
(D) 4,000 shares
(A) ₹ 11.11

Case Scenario – 08

A Ltd. wants to acquire D Ltd. and has offered a swap ratio of 1:2 (0.5 shares for every one share of D Ltd.). Following information is provided:

	A Ltd.	D Ltd.
Profit after tax	₹ 72,00,000	₹ 14,40,000
Equity shares outstanding (Nos.)	24,00,000	7,20,000
PE Ratio	5 times	3.5 times
Market price per share	₹ 15	₹ 7

From the information given above, choose the correct answer to the following questions:

- I.** The number of equity shares to be issued by A Ltd. for acquisition of D Ltd. would be.....
- (a) 3,36,000 (b) 3,60,000
(c) 4,80,000 (d) 6,00,000
- II.** The EPS of A Ltd. after the acquisition would be.....
- (a) ₹ 2 (b) ₹ 3
(c) ₹ 3.13 (d) ₹ 4.00



III. The equivalent earnings per share of D Ltd. would be.....

- (a) ₹ 1 (b) ₹ 1.50
(c) ₹ 1.57 (d) ₹ 2.00

IV. If A Ltd. PE multiple remains unchanged then market capitalization of A Ltd. after the acquisition would be.....

- (a) ₹ 50.40 Lakh (b) ₹ 360.00 Lakh
(c) ₹ 431.94 Lakh (d) ₹ 400.00 Lakh

Answer Case Scenario – 08

- I. (b) 3,60,000**
II. (c) ₹ 3.13
III. (c) ₹ 1.57
IV. (c) ₹ 431.94 Lakh

Case Scenario – 09

The company ABC Ltd. proposes to take over XYZ Ltd. The chief executive of the company thinks that shareholders always look for the earnings per share. Therefore, he considers maximization of the earnings per share as his company's objective. The following information is available in respect of ABC Ltd. and XYZ Ltd.

	ABC Ltd.	XYZ Ltd.
Net Profit (Post Tax)	₹ 400 Lakh	₹ 315 Lakh
P/E Ratio	10.50	10.00
Current market price per share	₹ 840	₹ 1,700

The company is planning to borrow funds @ 15% rate of interest and buys out the target company by paying cash.

Assume tax rate @30%.

From the information given above, choose the correct answer to the following questions:

- I.** The total market value of equity of XYZ Ltd. is approximately.....
(a) ₹ 3150 lakh (b) ₹ 3400 lakh
(c) ₹ 4200 lakh (d) ₹ 16800 lakh
- II.** Suppose if XYZ Ltd. is borrowing funds at a high interest rate to finance acquisition it will affect its EPS mainly due to.....
(a) increase in operating profit.
(b) increase in interest burden.
(c) increase in market price of shares.
(d) increase in tax liability.
- III.** Suppose if ABC Ltd. is offering about ₹ 4030 Lakh to XYZ Ltd. for the proposed acquisition it will result in.....
(a) EPS accretion
(b) EPS dilution
(c) No Change
(d) Risk free EPS

Answer Case Scenario – 09

- I. (a) ₹ 3150 lakh**



II. (b) increase in interest burden.**III. (b) EPS dilution****Case Scenario – 10**

Mr. X on 1st July 2021, during the initial offer of some Mutual Fund invested in 10,000 units having face value of ₹ 10 for each unit. On 31st March 2022, the dividend paid by the M.F. was 10% and Mr. X found that his annualized yield was 153.33%. On 31st December 2023, 20% dividend was given. On 31st March 2024, Mr. X redeemed all his balance of 11,296.11 units when his annualized yield was 73.52%.

From the information given above, choose the correct answer to the following questions:

I. NAV as on 31/03/2022 shall be approximately.....

- | | |
|-------------|-------------|
| (a) ₹ 19.50 | (b) ₹ 20.50 |
| (c) ₹ 21.50 | (d) ₹ 22.50 |

II. Total number of units as on 31/03/2022 shall be approximately.....

- | | |
|--------------------|--------------------|
| (a) 10487.80 units | (b) 12585.65 units |
| (c) 9465.35 units | (d) 11575.40 units |

III. Dividend as on 31/03/2023 shall be

- | | |
|----------------|----------------|
| (a) ₹ 20625.50 | (b) ₹ 20870.45 |
| (c) ₹ 20975.60 | (d) ₹ 21565.75 |

IV. NAV as on 31/03/2023 shall be approximately.....

- | | |
|-----------|-----------|
| (a) 24.65 | (b) 24.85 |
| (c) 25.95 | (d) 26.45 |

V. NAV as on 31/03/2024 shall be approximately.....

- | | |
|-----------|-----------|
| (a) 20.50 | (b) 25.95 |
| (c) 26.75 | (d) 27.20 |

Answer Case Scenario – 10**I. (b) ₹ 20.50****II. (a) 10487.80 units****III. (c) ₹ 20975.60****IV. (c) 25.95****V. (c) 26.75****Student Notes**



Student Notes



Student Notes



16

CA FINAL AFM EXAM MAY/2026 ATTEMPT

PART – I (MCQ PORTION)

Case Scenario – I

The Government of India issues a 5-year Sovereign Gold Bond (SGB) at an issue price of ₹ 5,000 per gram. An investor Mr. JK has purchased 200 grams SGB under this scheme. The bond carries a coupon rate of 2.5% per annum, calculated as simple interest on the initial investment amount, payable annually. At the end of five years, the bond will be redeemed at the prevailing market price of gold. The annual interest received is taxable at the investor's applicable marginal income tax rate. Any capital gain arising on redemption is taxable at 20% after considering the benefit of indexation. Inflation rate during the holding period is assumed to be 4% per annum.

From the information given above, choose the correct answer to the Question Nos. 1 to 3:

(Exam May – 2026)

1. **An investor Mr. JK purchases 200 grams of SGB at ₹ 5,000 per gram. The investor falls in the 30% tax bracket. Inflation is 4% per annum. What are the post-tax annual interest income and the approximate real post-tax return (%) on initial investment?**
 - (A) ₹ 25,000 and 2.25%
 - (B) ₹ 17,500 and –2.25%
 - (C) ₹ 17,500 and 1.75%
 - (D) ₹ 17,500 and –1.75%
2. **After 5 years, gold price becomes ₹ 6,800 per gram. Inflation is constant at 4% annually. What is the indexed cost of acquisition and taxable capital gain to Mr. JK?**
 - (A) Indexed cost ₹ 12,00,000 ; Taxable gain ₹ 1,60,000
 - (B) Indexed cost ₹ 11,69,860 ; Taxable gain ₹ 1,90,140
 - (C) Indexed cost ₹ 12,16,653 ; Taxable gain ₹ 1,03,343
 - (D) Indexed cost ₹ 12,16,653 ; Taxable gain ₹ 1,43,347
3. **Assume SGB redeems at ₹ 6,800 per gram and Mr. JK is in 30% tax bracket. What is the net amount received after capital gains tax (20% with indexation)?**
 - (A) ₹ 13,56,331
 - (B) ₹ 13,16,996



(C) ₹ 13,31,331

(D) ₹ 13,28,000

Case Scenario – II

NewChem Ltd., a leading chemical manufacturing firm based in Telangana, is evaluating the installation of a new-generation solar power plant to meet the energy needs of its primary manufacturing unit. This initiative is part of NewChem's long-term sustainability and cost-reduction goals.

The total cost of installing the solar plant is ₹ 3.20 crore. Based on the current state electricity tariffs, the plant is projected to generate savings in electricity expenses of ₹ 25 lakh per year in perpetuity.

However, due to regulatory uncertainty, a new state government is expected to take office in one year and it is anticipated that electricity tariffs will be revised. Based on industry analysis, the annual savings from the solar plant could change to one of two possibilities:

- Scenario 1 (Low Savings): Savings could decrease to ₹ 15 lakh per year in perpetuity.
- Scenario 2 (High Savings): Savings could increase to ₹ 40 lakh per year in perpetuity.

The company's Weighted Average Cost of Capital (WACC) is 9.5% and the current risk-free rate based on 10-year government bonds is 6.5%.

From the information above, choose the correct answer to Questions 4 to 6:

4. **NewChem Ltd. plans to invest ₹ 3.20 crore today in a solar power plant generating annual electricity savings of ₹ 25 lakh in perpetuity. The WACC is 9.5%. After one year, savings may change to ₹ 40 lakh (high scenario) or ₹ 15 lakh (low scenario) due to tariff revisions.**

Calculate NPV if invested today and NPV after one year in both scenarios (in ₹ crore).

(A) Today: -0.45; High: 1.01; Low: -1.62

(B) Today: -0.57; High: 1.21; Low: -1.42

(C) Today: -0.57; High: 1.01; Low: -1.62

(D) Today: -0.63; High: 0.95; Low: -1.58

5. **For NewChem's ₹ 3.20 crore solar plant (WACC 9.5%), after one year savings will be ₹ 40 lakh or ₹ 15 lakh perpetuity. Risk-free rate is 6.5%. Using risk-neutral valuation, compute returns in both scenarios and risk-neutral probability of high scenario (%).**

(A) High: 31.56%; Low: -50.63%; Prob: 69.5%

(B) High: 28.45%; Low: -48.25%; Prob: 65.2%

(C) High: 31.56%; Low: -50.63%; Prob: 73.8%

(D) High: 35.20%; Low: -52.15%; Prob: 69.5%



6. **NewChem can invest ₹ 3.20 crore in solar plants today or wait one year, when savings will be ₹ 40 lakh or ₹15 lakh perpetuity (WACC 9.5%, risk-free rate 6.5%). Using risk-neutral probabilities, find PV of option to wait one year, discounted at risk-free rate and WACC. (₹ lakh).**
- (A) PV at RF rate: ₹ 70.23 lakh; WACC: ₹ 68.50 lakh
 (B) PV at RF rate: ₹ 68.50 lakh; WACC: ₹ 70.23 lakh
 (C) PV at RF rate: ₹ 44.15 lakh; WACC: ₹ 42.30 lakh
 (D) PV at RF rate: ₹ 19.53 lakh; WACC: ₹ 18.98 lakh

Case Scenario – III

XYZ Ltd. is professionally engaged in providing consultancy services related to corporate mergers and acquisitions. It is on an expansion mode and decided to hire freshly qualified Chartered Accountants for the post of “M&A Consultant”. For selection of the candidate, each of the candidates needs to pass a written test. Following are some of the questions asked in the test to identify the concepts related to M&A.

You are an examinee in written test paper therefore, you are required to choose the correct option to the Questions Nos. 7 to 9:

(Exam May – 2026)

7. **AB Ltd., the acquiring company may issue substantial amount of convertible debentures to its existing shareholders to be converted at a future date when it faces takeover threat. The tactic used by AB Ltd. is called_____.**

- (A) White Knight
 (B) Poison Put
 (C) Poison Pill
 (D) White Squire
8. **Identify the correct meaning from among the following options when X Ltd., a target company adopts “Greenmail” tactics to defend itself from hostile takeover.**
- (A) When X Ltd. issues bonds that encourage the holder to cash in at higher prices.
 (B) When X Ltd. offers hefty compensations to its managers if they get ousted due to takeover.
 (C) When X Ltd. makes a counter bid for the acquirer company.
 (D) When X Ltd. offers the acquirer a higher price for its shares than market price.
9. **R Ltd. is a parent company and J Ltd. is its subsidiary company. J Ltd. is growing at a faster pace than R Ltd. and also carries higher valuations than other businesses owned by R Ltd. To unlock the value of J Ltd., and generate cash, a strategy which R Ltd. may take is to make J Ltd. go public through an IPO but keep a controlling stake in the newly traded subsidiary. Such tactic adopted by R Ltd. is called _____.**
- (A) Split-up
 (B) Carve Out
 (C) Sell-Off
 (D) Spin-Off

Case Scenario – IV

Mr. X a portfolio manager invests ₹ 25,00,000 in stock Alpha and Beta in ratio of 60 : 40. The daily standard deviation of stock Alpha



is 1.20% and stock Beta is 0.90%. The correlation between Alpha and Beta is 0.6. Mr. X wanted to measure VAR for confidence level 99% and 90% level. (Z score: 99% confidence level = 2.33 and 90% confidence level = 1.28) Assume 252 trading days in a year and mean return is zero.

From the information given above, choose the correct answer to Questions Nos. 10 to 12:

(Exam May – 2026)

10. **What is the approximate annual Value at Risk (VaR) of the portfolio in rupee terms at the given 99% confidence level?**

- (A) ₹ 57,027
- (B) ₹ 5,43,162
- (C) ₹ 9,05,271
- (D) ₹ 7,94,080

11. **What is the approximate diversification benefit (per day) of the portfolio while you calculate amount of VaR at 90% confidence level?**

- (A) ₹ 11,540
- (B) ₹ 23,060
- (C) ₹ 34,580
- (D) ₹ 3,252

12. **If the correlation between Stock Alpha and Stock Beta increases from 0.60 to 0.80, what will be the impact on annual Value at Risk (VaR) at the 99% confidence level?**

- (A) VaR increase by ₹ 48,085
- (B) VaR decrease by ₹ 48,085
- (C) VaR increase by ₹ 3,029
- (D) VaR decrease by ₹ 3,029

Case Scenario – V

Green Agro Exports Ltd. entered a 3-month forward contract on January 1, 2026 to buy USD 2,00,000 (import payment due April 1, 2026).

USD-INR Bank Quotes:

Date	Spot Bid	Spot Ask	1M Fwd Bid	1M Fwd Ask	2M Fwd Bid	2M Fwd Ask	3M Fwd Bid	3M Fwd Ask
01.01.2026	83.20	83.55	-	-	-	-	83.60	84.00
01.02.2026	83.80	84.15	-	-	83.95	84.35	-	-
01.03.2026	83.40	83.75	83.70	84.05	84.10	84.45	-	-
01.04.2026	84.20	84.50	84.30	84.65	-	-	-	-

From the information given above, choose the correct answer to the Questions No. 13 to 15:

(Exam May – 2026)



13. **Green Agro Exports Ltd. entered 3-month USD 2,00,000 forward on Jan 1, 2026. On March 1, 2026, what is maturity settlement cost vs. net extra cost of closing original and rebooking 1-month forward?**
- (A) Maturity ₹ 1,68,00,000; Extra ₹ 60,000
 (B) Maturity ₹ 1,68,00,000; Extra ₹ 10,000
 (C) Maturity ₹ 1,68,00,000; Extra ₹ 70,000
 (D) Maturity ₹ 1,67,20,000; Extra ₹ 70,000
14. **If import obligation is settled early on March 1, 2026, what total cash outflow is required that day to close forward contract and make import payment?**
- (A) ₹ 16,74,000
 (B) ₹ 16,81,000
 (C) ₹ 16,87,000
 (D) ₹ 16,89,000
15. **Green Agro Exports Ltd. cancels the import and closes USD 2,00,000 forward contract (Jan 1, 2026; 84.00 Ask). What are the mark-to-market gain/(loss) on following dates?**
- (A) Feb 1: ₹ 10,000; Mar 1: ₹ 60,000; Apr 1: ₹ 40,000
 (B) Feb 1: ₹ 10,000; Mar 1: ₹ 60,000; Apr 1: ₹ 40,000
 (C) Feb 1: ₹ 30,000; Mar 1: ₹ 60,000; Apr 1: ₹ 40,000
 (D) Feb 1: ₹ 10,000; Mar 1: ₹ 30,000; Apr 1: ₹ 20,000

PART – II (DESCRIPTIVE PORTION)**Question – 01(a)**

Bharat Infrastructure Ltd. (BIL), a premier Indian Engineering firm, has been invited to establish and operate a high-speed data center infrastructure in the Republic of Valoria. The initial investment required for the project is 4,000 million Valors (VL).

Under the agreement, BIL will sell the project back to the Valorian government after 4 years for a guaranteed sale price of 8,000 million VL.

During the four-year operational period, BIL will provide technical maintenance and security protocols and earn annual fee of 80 million VL payable at the end of each year.

To hedge the foreign exchange risk associated with the initial capital investment, BIL's investment bankers have proposed a Currency Swap arrangement with the following terms:

- BIL will swap the principal amount of 4,000 million VL for Indian Rupees immediately (Year 0) at today's spot rate.
- The swap will be reversed at Year 4, with both parties re-exchanging the same principal amounts at the same spot rate locked in at inception.
- The bank will charge an annual facilitation fee of 0.30% calculated on the ₹ principal amount, payable at the end of each year.



The currency swap covers only the initial investment and terminal principal. BIL's policy is to hedge all other foreign currency cash flows using forward contracts at the respective forward rates prevailing in the market today.

Period	Spot	1 Year Forward	2 Year Forward	3 Year Forward	4 Year Forward
Exchange Rate (VL/₹)	50.00	53.00	56.18	59.55	63.12
PV Factor @ 14%	1.000	0.877	0.769	0.675	0.592

Cumulative PVFA for 4 years at 14% = 2.913

Assume a risk-adjusted discount rate of 14% per annum. Ignore taxation.

You are required to evaluate the project's financial viability using the Net Present Value (NPV) method, with all cash flows expressed in Indian Rupees (₹ million).

(Exam May – 2026)

Question – 01(b)

XYZ Ltd. recently paid a dividend of ₹ 40. The company is entering a 5-year transition phase where its current 15% growth rate will decline linearly to a stable perpetual rate.

Financial Parameters

- Transition Period: 5 years, declining linearly from Year 1.
- Current Data: ROE of 20% and retention ratio of 25%.

- Terminal Data (Year 6+): ROE of 12% and retention ratio of 50%.
- Market Data: Required return of 13% and Market Price of ₹ 850.

You are required to:

- Estimate the intrinsic value per share using the H-Model.
- Given the current market price of ₹ 850 per share, assess whether the stock is undervalued or overvalued and calculate the percentage deviation from intrinsic value.

(Exam May – 2026)

Question – 01(c)

MT Ltd. has approached you for an additional loan of ₹ 20 crores to fund its aggressive expansion plans. The company's current sustainable growth rate is 10%, but it aims to grow at 16% annually. Its debt-to-equity ratio is already at the maximum permissible limit of 2.5:1. Additionally, the industry is experiencing 5% inflation, which is increasing the company's asset financing requirements.

As a credit analyst, you are required to explain the key concerns regarding this loan request and suggest two alternative courses of action available to the company to achieve its growth objectives sustainably.

(Exam May – 2026)



Question – 02(a)

Mr. RJ owns two securities X and Y priced according to a two-factor Arbitrage Pricing Model. The factor sensitivities and expected returns are as follows:

Particulars	Security X	Security Y
Beta with Factor 1 (β_1)	1.2	0.8
Beta with Factor 2 (β_2)	0.5	1.4
Expected actual return	14%	16%

Additional Information

- Risk-free rate = 6%
- Risk premium for Factor 1 = 4%
- Risk premium for Factor 2 = 5%

Required:

- Evaluate whether Security X and Security Y are correctly priced under the two-factor Arbitrage Pricing Model.
- Mr. RJ has ₹ 1,00,000 available and intends to construct a long-short portfolio by taking a long position in Security A and a short position in Security B. You are required to analyse and determine the amount to be invested in Security A and the amount to be short in Security B if the desired portfolio sensitivities are 1.40 with respect to Factor 1 and 0.05 with respect to Factor 2.

(Exam May – 2026)**Question – 02(b)**

On 1 July, ALPS Company shares trade at ₹ 2,000. An investor enters the following option: **(3-month maturity)**:

Option	Type	Strike (₹)	Premium (₹)
A	Call	1920	130
B	Call	2000	85
C	Call	2080	45
D	Put	1920	40
E	Put	2000	95
F	Put	2080	155

Contract size = 100 shares.

Risk-free rate = 6% p.a. Continuous compounding not required and the stock does not pay dividends.

Ignore transaction costs. Assume that investor chooses any one option.

Required:

- Assess the ITM, ATM, OTM of each option A to option F at initiation and compute the intrinsic and time value embedded in the premiums.
- In the next 3 months expiry price may increase by 150 or may decrease 150. Under each expiry scenario, examine which options will be rationally exercised and appraise the profit or loss per contract.

(Exam May – 2026)

Question – 02(c)

The technology start-up, Fincorp Pvt. Ltd., is in its early growth stage and is evaluating multiple sources of financing including Angel Investment, Venture Capital (VC), Convertible Notes, SAFE and IPO.

Evaluate the following statements and state whether they are **True or False**, giving brief justification in each case:

- (i) In early-stage start-ups, valuation is primarily based on discounted cash flow (DCF) using stable historical earnings.
- (ii) Venture capital financing typically involves staged funding to reduce agency problems and mitigate investment risk.
- (iii) Bootstrapping reduces dilution risk but may limit growth due to capital constraints.
- (iv) Angel investors generally invest at a later stage than venture capitalists and demand stronger control rights.

(Exam May – 2026)

Question – 03(a)

An Indian company is investing in a 2-year project in Europe (EURO).

Initial Investment = € 4,000,000

Annual after-tax operating cash flow = € 2,500,000

Life = 2 years

Unlevered cost of capital = 12%

Spot rate = ₹ 90/€

Corporate tax rate (India) = 30%

Financing Options: 50% Debt Required

Option	Borrowing Currency	Interest Rate (p.a.)
A	India (INR)	9%
B	Euro (EUR)	5%
C	USA (USD)	3%

Additional Data

Spot USD/INR = ₹ 75

Spot EUR/USD = 1.20

Interest rate in USA = 3% p.a.

Interest rate in Europe = 5% p.a.

Assume: Interest Rate Parity holds and no transaction cost.

Required

- (i) Identify which option is the cheapest and why.
- (ii) Compute Adjusted Present Value (APV) under cheapest option.

Present Value Factor	1 st Year	2 nd Year
12%	0.893	0.797



5%

0.952

0.907

(Exam May – 2026)**Question – 03(b)**

A mutual fund offers two plans: Dividend Plan and Bonus Plan. An investor invests ₹ 12,00,000 on 1 April, 2025 in the ratio 60:40 respectively. Initial NAV of both plans is ₹ 48.

Entry load is 2% and exit load is 1% (on redemption value).

Under the Dividend Plan, a dividend of ₹ 5 per unit is declared on 30 September, 2025 when NAV is ₹ 56. After declaration, NAV becomes ₹ 51. The dividend is reinvested at ₹ 51. Closing NAV on 31 March, 2026 is ₹ 54.

Under the Bonus Plan, a bonus issue of 1:4 is declared on 30 September, 2025 when NAV is ₹ 60. After bonus adjustment, NAV becomes ₹ 45. Closing NAV on 31 March, 2026 is ₹ 52.

Ignore taxation.

You are required to:

Evaluate the total return and Holding Period Return under each plan after considering loads and assess which plan provides higher effective return.

(Exam May – 2026)**Question – 03(c)**

Transfer of NPAs through securitization improves balance sheet health but may involve regulatory and valuation challenges.” Comment on the statement above.

(Exam May – 2026)**Question – 04(a)**

A portfolio manager is analysing four equity securities to construct an optimal risky portfolio using Sharpe’s Single Index Model. The securities are traded in an efficient market and follow CAPM assumptions.

The following information is available:

Security	Expected Return $E(R_i)$	Beta (β_i)	Residual Variance (σ_{ei}^2)
A	17%	1.4	0.0225
B	15%	1.1	0.0196
C	13%	0.9	0.0144
D	11%	0.6	0.01

Additional Information

Risk-free rate (R_f) = 6%

Market expected return (R_m) = 14%

Market variance (σ_m^2) = 0.040

Cut-off Rate (C^*) = 7.95

You are required to:



(Exam May – 2026)

- (i) Given the cut-off rate, identify the securities to be included in the optimal portfolio.
- (ii) Compute the proportion of investment in each selected security.
- (iii) Using CAPM, compute the expected return and value of each selected security.
- (iv) Why is the portfolio constructed using Sharpe's Model consistent with CAPM assumptions?

(Exam May – 2026)**Question – 04(b)**

Consider the recent performance of the closed ended fund.

Period	NAV (₹)	Premium/Discount %
0	20	0
1	22.5	-5
2	19.5	2.3
3	21	-3.2
4	22.3	4

Required:

- (i) Evaluate the average return per period based on Geometric Mean for an investor who bought 1000 shares of the closed fund at the invitation and then sold her position at the end of period 4.
- (ii) Geometric growth rate in NAV over the same period and interpret the results.

Question – 04(c)

Explain the Elliot Wave Theory of Technical Analysis.

OR

ABC Ltd. has a ₹ 100 Crores floating-rate loan, reset annually for 4 years.

It enters into a collar strategy to hedge against interest rate risk.

Cap Rate = 9.5%

Expected Interest rates & Discounting Factors:

Year	Expected Rate	Discounting Factor
1	7.8%	0.96
2	8.6%	0.92
3	9.2%	0.88
4	10.6%	0.84

You are required to identify a Floor Rate that makes the collar a zero cost collar. (Ignore Volatility)

(Exam May – 2026)**Question – 05(a)**

A “momentum-driven” investment consultant is pitching Lumina Cloud Systems (LCS) to your client. The consultant claims LCS is a bargain because its Cash Flow from Operations (CFO), when capitalized at the prime lending rate of 9%, yields a value of ₹ 425 per share. LCS currently trades at ₹ 240 per share.



You are required to perform a Two-Stage FCFE Valuation to verify these claims.

Financial Data & Assumptions:

- Net Income (Base Year): ₹ 180 Crores.
- High-Growth Stage (Years 1-2): Net Income will grow by 25% annually.
- Net Investment in Operating Assets: This is projected to be ₹ 240 Crores in Year 1 and ₹ 280 Crores in Year 2. (Note: This represents capital expenditure less depreciation plus the increase in working capital).
- Debt Policy: The Company finances 30% of its net investment in operating assets through new debt.
- Cost of Equity: The risk-free rate is 6%, the company's Beta is 1.20 and the Equity Risk Premium (ERP) is 5%.
- Stable Stage (Year 3 onwards): Net investment in operating assets will drop to 20% of Net Income each year.
- Outstanding Shares: 60 Crores.
- Present Value Factor:

	1 st Year	2 nd Year	3 rd Year
12%	0.893	0.797	0.7118

Requirements:

- (i) Calculate the total Present Value of the FCFE for the high-growth period (Stage 1).

- (ii) If the current market price of ₹ 240.00 is considered "fair" by the market, calculate the implied sustainable growth rate for the terminal stage (Year 3 onwards).

(Exam May – 2026)

Question – 05(b)

Zenith Commercial Bank's ALCO is evaluating the re-pricing risk for the 0-90 days' time bucket. The following quarterly data is available:

Financial Data:

- Rate Sensitive Assets (RSAs): ₹ 12,500 Crores
- Rate Sensitive Liabilities (RSLs): ₹ 15,800 Crores
- Quarterly Net Interest Income (NII): ₹ 450 Crores

Risk Policy:

The bank's Earnings at Risk (EaR) limit is 8% of quarterly NII. Hedging is mandatory if the potential NII impact in any scenario exceeds this threshold.

You are required to:

- (i) Identify the Gap Position, (Positive/Negative) and determine which of the following scenarios is favourable for the bank's NII, providing a brief justification:
- Scenario A: Repo rate increase of 50 bps.
 - Scenario B: Repo rate decrease of 25 bps.



- (ii) Calculate the Earnings at Risk (EaR) in ₹ Crores and as a percentage of NII for both scenarios. Based on the bank's risk policy, advise whether management should activate hedging strategies or not.

(Exam May – 2026)**Question – 05(c)**

Briefly justify why transaction exposure is generally considered more critical than translation exposure from a financial risk management perspective.

(Exam May – 2026)**Question – 06(a)**

Arotech Industries Ltd. (A), a listed engineering major, are evaluating the acquisition of MedGen Labs Pvt. Ltd. (M), an unlisted pharmaceutical laboratory. Financial parameters as of 31 March, 2025 are as follows:

Particulars	Arotech Industries (A)	MedGen Labs (M)
Profit After Tax (PAT)	₹ 6,000 crore	₹ 450 crore
No. of Equity Shares	500 crore	5 crore
P/E Ratio	30x	1.2 × Industry Avg P/E
Market Price/Standalone Fair Value	₹ 360.00	To be determined

Additional Information:

- Combined Entity: Post-merger PAT is estimated at ₹ 6,850 crore (including synergies). The expected P/E of the combined entity is 34x.
- Synergies Value: The total synergy gain (increase in combined entity value over the sum of standalone values) is capped at ₹ 34,000 crore.
- Settlement Offers (Per 1 share of MedGen):
 - Offer 1 (Cash): ₹ 4,000 (Premium of 5.82% over the standalone fair value of MedGen).
 - Offer 2 (Hybrid Equity): ₹ 1,000 Cash + Arotech shares (X) (Premium of 21.69% over standalone fair value).
 - Offer 3 (Hybrid Debt): ₹ 1,150 Cash + 5 Convertible Bonds of Arotech (Premium of 25.66% over standalone fair value).
- Bond Terms (Arotech): Face Value ₹ 500, 4-year term, 10% discount rate. Each bond converts into 2 Arotech shares or redeems at ₹ 680.24.

You are required to:

- Determine the annual post-merger synergies (PAT basis) and MedGen's standalone P/E ratio.
- Calculate the Industry Average P/E and MedGen's Standalone Fair Value per share.
- Solve for the number of Arotech shares (X) offered in Offer 2.



- (iv) Rank all three offers from MedGen's perspective assuming Arotech's share price has fallen to and remains at ₹ 330 at the time of settlement/conversion.

Question – 06(b)

The current price of F Ltd. is ₹ 120 and a European option with an exercise price of ₹ 115 will expire in 50 days. The annual continuously compounded risk free rate of interest is 8%. The standard deviation of annual returns of F Ltd. is 30%.

Value of $(d_1) = 0.5375$; Year = 365 days and value of $e^{-rt} = 0.9891$.

Using Black-Scholes Model, you are required to calculate:

- (i) The Hedge Ratio
- (ii) The probability that price in spot market on expiration would be higher than the exercise price of the call option.
- (iii) Value of call and put options.

Cumulative Area Tables extract.

	0.00	0.01	0.02	0.03	0.04	0.05
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422

**Student Notes**

NORMAL PROBABILITY DISTRIBUTION TABLE

Normal Probability Distribution Table

Number of Standard Deviations from mean (Z)	Area to the Left of Right (One Tail)	Number of standard Deviations from Mean (Z)	Area to the lft or Right (One Tail)
0.00	0.5000	1.55	0.0606
0.05	0.4801	1.60	0.0548
0.10	0.4602	1.65	0.0495
0.15	0.4404	1.70	0.0446
0.20	0.4207	1.75	0.0401
0.25	0.4013	1.80	0.0359
0.30	0.3821	1.85	0.0322
0.35	0.3632	1.90	0.0287
0.40	0.3446	1.95	0.0256
0.45	0.3264	2.00	0.0228
0.50	0.3085	2.05	0.0202
0.55	0.2912	2.10	0.0179
0.60	0.2743	2.15	0.0158
0.65	0.2578	2.20	0.0139
0.70	0.2420	2.25	0.0122
0.75	0.2264	2.30	0.0107
0.80	0.2119	2.35	0.0094
0.85	0.1977	2.40	0.0082
0.90	0.1841	2.45	0.0071
0.95	0.1711	4.50	0.0062
1.00	0.1557	2.55	0.0054
1.05	0.1469	2.60	0.0047
1.10	0.3570	2.65	0.0040
1.15	0.1251	2.70	0.0035



NORMAL PROBABILITY DISTRIBUTION TABLE

1.20	0.1151	2.75	0.0030
1.25	0.1056	2.80	0.0026
1.30	0.0986	2.85	0.0022
1.35	0.0885	2.90	0.0019
1.40	0.0808	2.95	0.0016
1.45	0.0735	3.00	0.0013
1.50	0.0668		



Student Notes



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